VIGNAN'S
Foundation for Science, Technology & Research
(Deemed to be UNIVERSITY)
-Estd. u/s 3 of UGC Act 1956

R22 Academic Regulations

In Compliance with NEP 2020



Prof. P. NAGABHUSHAN
B.E., M.Tech., Ph.D., FIE, FIETE, FIAPS.
Vice Chancellor, VFSTR
Life time Professor, IIIT – Allahabad, Prayagraj
(Formerly: Director, IIIT-Allahabad 2017-22)

VIGNAN'S
Foundation for Science, Technology & Research
(Deemed to be UNIVERSITY)
-Estd. u/s 3 of UGC Act 1956

PREFACE

'You are born to Blossom' – What an inspiring title the book authored by APJ Abdul Kalam and Arun K Tiwari carries. The journey to blossom has got to be heralded by education. The purpose of education is to ensure that the 'Life Blossoms'. Earning a degree and getting a placement should be the just happening things, and should not become the only celebrated goals for education. In the book cited above, Honourable Kalam, Former President of India, underscores that "The scheme of civil society depends on Educating young people to become enlightened citizens and adults who are responsible, thoughtful and enterprising"

VIGNAN aims to seed these concepts in every learner who transits through this temple of learning. The doctrine of VIGNAN entitled R-22 contains the principles of policies laid down by the University, to realize the spirit of "Blossoming the lives" providing a foundation-strong professional education on the ethos of 'Creative learning for Critical thinking and Critically analysing for Creative decision making'. Certainly, our University is one of the earliest Universities, in fact the University is a trend setting one in completely internalising the concepts of the policies brought out in National Education Policy (New Educational Policy) NEP-2020, and inculcating the spirit in R-22. The R-22 document articulates the Academic Regulations of the University, which is being presented now and shall be inforce with immediate effect from the academic year 2022-23, not only for those who have joined in 2022, also the aspirants of 2021-22 are enabled into the navigation.

R-22 presents a novel design for the academic pursuit, making an exploratory cross disciplinary traversal for a learner who should find learning both holistic and experiential. The learner is ensured to enjoy the continuity in learning and the learner is supported to align and realign, enroute utilising the benefits of constructive feedbacks that s/he receives because of continuous assessment. S/he will be empowered to enjoy the opportunities to explore, experiment and experience.

R-22 eliminates the melancholy of examinations. The expected severity of breakdown due to the anxiety of examination system is replaced by an affectionate assessment system, increasing the effectiveness in accomplishing the outcomes.

In brief, NEP-2020 compliant revised academic regulation of the University – the R-22, is VIGNAN's commitment to alleviate the acuteness in the present educational practices. It intends to provide a strategic solution to the critical observation made by Bharat ratna awardee, Professor. CNR Rao – "India has exam system, not education system. When will young people stop taking exams and do something worthwhile?" (Thought for the Day, Times of India 13.08.2022)

Here is R-22, which assures that the learners at VIGNAN are bound to do something worthwhile – very much worthwhile.

***

R22 B.Tech.
4 YEAR
DEGREE PROGRAMME

REGULATIONS - TABLE OF CONTENTS

			Page Numbers
	EXECUTIVE ABSTRACT		*vii*
1	**INTRODUCTION**		*vii*
	1.1	Definition	*viii*
	1.2	Academic Administration	*ix*
	1.3	Program Duration	*x*
	1.4	Courses and Credits	*x*
	1.4.1	Content Delivery of a Course	*x*
	1.5	B.Tech Degree	*x*
	1.5.1	B.Tech. with Honours (Specialization: XX)	*x*
	1.5.2	B.Tech. with Minor Specialization in XX	*xi*
	1.5.3	B.Tech. with Add-on Diploma	*xi*
	1.5.4	B.Tech. with Add-on Certification	*xi*
	1.5.5	Dual (B.Tech. + M.Tech./ MBA) degree programme of 5-years*	*xi*
	1.6	Composition of an Academic year	*xi*
	1.7	Semester wise provisions	*xii*
2.	**CURRICULUM**		*xii*
	2.1	Distribution of credits	*xii*
	2.2	Organization of course contents	*xiii*
3	**CHOICE BASED CREDIT SYSTEM**		*xiii*
	3.1	Common (Core) Courses for all branch disciplines	*xiii*
	3.1.1	Basic Sciences and Humanities	*xiii*
	3.1.2	Basic Engineering	*xiv*
	3.2	Professional Core	*xiv*
	3.3	Electives	*xiv*
	3.3.1	Department Electives	*xiv*
	3.3.2	Open Electives	*xiv*
	3.4	Inter Departmental Project	*xiv*
	3.5	Project	*xv*
	3.6	Internship	*xv*
	3.7	Binary Graded courses	*xv*
	3.7.1	Physical Fitness	*xv*
	3.7.2	Life Skills	*xv*
	3.7.3	Industry – Interface (I2) courses	*xv*
	3.7.4	Orientation course	*xvi*
	3.7.5	Courses with floating credits	*xvi*
4.	**ATTENDANCE**		*xvi*
5	**ASSESSMENT**		*xvi*

	5.1	Marks distribution	*xvii*
	5.2	Qualifying criteria	*xvii*
	5.3	L-based courses integrated with P/T	*xviii*
	5.3.1	Formative Assessment	*xviii*
	5.3.2	Summative Assessment	*xx*
	5.4	P-based Courses	*xx*
	5.4.1	Formative Assessment	*xx*
	5.4.2	Summative Assessment	*xxi*
	5.5	Assessment and Grading of MOOCs based electives	*xxi*
	5.6	Inter Departmental Project	*xxii*
	5.6.1	Formative Assessment	*xxii*
	5.6.2	Summative Assessment	*xxiii*
	5.7	Project	*xxiii*
	5.7.1	Formative Assessment	*xxiii*
	5.7.2	Summative Assessment	*xxiv*
	5.8	Semester-long Internship	*xxv*
	5.8.1	Formative assessment: Internal reviews at the place of internship	*xxv*
	5.8.2	Summative assessment – Internship	*xxvi*
6.	**SEMESTER-END ASSESSMENT ACTIVITES**		*xxvi*
7.	**COMPUTATION OF GRADING**		*xxvi*
8.	**SUPPLEMENTARY EXAMINATIONS**		*xxvii*
9.	**GRADE POINT AVERAGE**		*xxviii*
	9.1	SGPA	*xxviii*
	9.2	CGPA	*xxviii*
10.	**AWARD OF CLASS**		*xxviii*
11.	**AWARD OF DEGREE**		*xxix*
12.	**LATERAL ENTRY AND LATERAL EXIT OPTIONS**		*xxix*
	12.1	Lateral Entry into II year of B.Tech.	*xxix*
	12.2	Inter- Institutional Credit Transfer	*xxx*
	12.3	Honorable exit with B.Sc. or Engineering Diploma	*xxx*
	12.4	Volunteer 'Drop' with Sabbatical Semester option	*xxx*
	12.5	Volunteer 'Drop' of a Semester Drop option	*xxx*
13.	**CHANGE OF BRANCH DISCIPLINE**		*xxxi*
14.	**INTERPRETATION OF RULES**		*xxxi*
	ANNEXURES		
	Annexure–1: Supplement Regulation for Pursuing B.Tech. with Research Honours (Specilization : XX)		*xxxii*
	Annexure–2(a): Supplement Regulation for Pursuing Dual B.Tech. + M.Tech. Degree		*xxxiii*
	Annexure–2(b): Supplement Regulation for Pursuing Dual B.Tech. + MBA Degree		*xxxv*
	Annexure–3 – Inter – Institute Credit Transfer		*xxxvi*

* Programme will be offered based on satisfactory strength of students willing to register, after receiving the formal AICTE approval.

R22 B.Tech.
4 YEAR
DEGREE PROGRAMME

R22 B.Tech. 4 YEAR DEGREE PROGRAMME

Academic Regulations, Curriculum and Course Contents

EXECUTIVE ABSTRACT

R22 - Academic regulations, Curriculum and course contents, is an articulation of the VFSTR deemed to be University's commitment towards NEP-2020, with a view that it enables student(s) to maintain the spirit of continuous learning and continuous assessment to replace the normal tendency of preparing just before a test or an examination. The proposed framework accomplishes multi-disciplinary holistic education, continuous assessment along with multiple honorable exit options if a student falls short to complete the requirements to earn the degree within the stipulated period including the permissible spill over period.

R22 is oriented towards multi-disciplinary holistic education at the undergraduate level that includes integrated and rigorous exposure to science, humanities, management, and professional domains, as well as sufficient flexibility in curricular structures that allow students to choose electives from the same and other disciplines. Such holistic and diverse education will assist the candidate in transforming into all-rounded persons. Similarly, in line with NEP-2020, more weight will be given to continuous/ formative assessment, which is an Integrated learning model comprising Learning – Thinking – Understanding – Skilling – Applying – Creating. Emphasis on continuous formative assessment with a creative summative assessment will facilitate the candidate to "Move away from high stake examinations – towards more continuous and comprehensive evaluation".

The B.Tech. degree offered will be for four years' **(8 semesters) duration** with lateral entry and exit options within this period, with suitable certifications that will enable the candidate to have a professional career and as well as serve as a reminder to return and update his / her qualification in the future. Following three years of study and the completion of the required credits, an Engineering diploma or a Bachelor's degree will be conferred. However, the intention of the learners is not to join for the award of the diploma / degree with lateral exit, but to acquire a B.Tech. degree which offers add-on certifications such as Honours/ Minor/ Add-on Diploma.

Salient features of the regulation

- Multidisciplinary
- Continuous learning
- Continuous assessment
- Honours/ Research Honours / Minor / Add-on Diploma / Add-on Certification
- Lateral entry and lateral exit options
- Sabbatical Semester Drop option to pursue innovation, incubation, entrepreneurial and advanced exploratory activities and subsequent re-entry
- Dual B.Tech. + M. Tech. / MBA degree of 5 years
- Credit earning by credit transfer

1 INTRODUCTION

This document contains the academic regulations, scheme of assessments, curriculum, detailed syllabi, course contents with text / reference books recommended, course outcomes, skills acquired and the projects / assignments that are to be performed for each course for the conduct of 4-year B.Tech. degree programmes. The various B.Tech. degree programmes under different schools in VFSTR are as listed below. The 2-character codes indicated in parentheses are their branch discipline codes.

R22 B.Tech. 4 YEAR DEGREE PROGRAMME

I. **School of Agriculture and Food Technology**
- Agriculture Engineering (AG)
- Food Technology (FT)

II. **School of Biotechnology and Pharmaceutical Sciences**
- Bioinformatics (BI)
- Biotechnology (BT)

III. **School of Computing and Informatics**
- Computer Science and Engineering (CS)
- Computer Science and Engineering - Artificial Intelligence and Machine Learning (AI)
- Computer Science and Business Systems (CB)
- Computer Science and Engineering - Cyber Security (CY)
- Computer Science and Engineering – Data Science (DS)
- Information Technology (IT)

IV. **School of Conventional Engineering**
- Chemical Engineering (CH)
- Civil Engineering (CE)
- Mechanical Engineering (ME)
- Robotics and Automation (RA)
- Textile Technology (TT)

V. **School of Electrical, Electronics and Communication Engineering**
- Biomedical Engineering (BM)
- Electrical and Electronics Engineering (EE)
- Electronics and Communication Engineering (EC)

1.1 Definition

For the purpose of R22 regulation, definitions as follows shall apply:

- **"Degree"** shall refer to the B.Tech. Degree Program.
- **"Course"** shall refer to such Course(s) for which a student shall earn Credits after due assessment as per the laid provisions. Project is also treated as a Course.
- **"Academic activities"** shall refer to the activities like Lecture (Physical Lecture Session), Tutorial (Participatory discussion / Self-Study / Desk Work / Quiz / Seminar Presentation, *etc* activities that make the student absorb & assimilate, the delivered contents effectively) and Practical / Practice sessions (includes Hands on Experience / Lab experiments / Field Studies / Case Studies *etc* activities that enable the student to acquire the requisite skill).
- **"Continuous Assessment"** shall refer to the assessment of the student spread over the entire semester on the various constituent components of the prescribed course.
- **"Semester"** shall refer to a period covering the two assessment periods *viz* Formative and Summative Assessment period. A semester would generally be spread over twenty weeks.
- **"Course Drop"** shall refer to a student having to undertake a repeat of the Course(s) not being able to complete the Credit requirements of the Course(s), under the conditions stipulated in the regulation.
- **"Supplementary Examinations"** shall refer to the examination(s) conducted to allow the student to appear in the un-cleared Semester - End summative assessment component.
- **"Blank Semester"** shall refer to a Semester in which a student either does not register for any course at the beginning of the Semester OR chooses to DROP all courses OR is so compelled to DROP all the courses, as the case may be.
- **"Semester Drop"** shall refer to availing a blank semester. However, if drop is availed to pursue a creative extension activity, then it is defined as **semester sabbatical**.

- **"Spill Over Semester"** shall refer to the additional semester(s) beyond the completion of prescribed normal semesters.
- **"AAA Section"** shall refer to the Academics, Assessment and Award Section of the Institute.
- **"Attendance"** refers to the Physical personal presence in an academic activity session.
- **"Summer Semester"** refers to a Semester that is scheduled to be held during the intervening period of Even and Odd Semester (*i.e*. Summer Vacation period).
- **"Themes"** refer to the courses offered in a particular stream other than offered by the regular departments, for example NCC, Entrepreneurship, Fitness and Living, *etc*.
- **"School"** refers to a division of institute dealing with two or more specific areas of discipline / study comprising of the departments related with exclusive emphasis on trans-disciplinary research.
- **"Department"** refers to a division of institute dealing with a specific area of discipline / study.
- **"HoD"** refers to the Head of the respective Department, where the student is enrolled for his / her Branch of Study.
- **"Center"** refers to a structured unit within the school / department established with the purpose to carry out advanced research.
- **"Grade Point"** refers to the quantification of the performance of a candidate in a particular course as defined herein.
- **"SGPA"** refers to the Semester Grade Point Average and is calculated as detailed in the regulations subsequently.
- **"CGPA"** refers to the Cumulative Grade Point Average and is calculated as detailed in the regulations subsequently.
- **"Division"** refers to the Division awarded to the student as per the mechanism detailed in the regulations subsequently.
- **"Internship"** refers to onsite Practical Training offered by reputed companies / Institutions, in India or abroad. To be undertaken with (or seeking) prior approval of the respective HoD.
- **"Project"** refers to a course executed by a candidate on a specific research problem at VFSTR / any organization of repute. To be undertaken with (or seeking) prior approval of the respective HoD.
- **"Credit equivalence and credit transfer committee"** refers to the committee designated to look into for credit equivalence and credit transfer.
- **"Honorable Exit Option"** refers to the Exit Options available to students, when they are unable to complete the prescribed four-year B.Tech. Degree program in seven successive years.

1.2 Academic Administration

The academic programmes of VFSTR are governed by the rules and regulations approved by the Academic Council from time to time. The various academic activities are conducted following a fixed time schedule duly approved by the Academic Council in line with the AICTE / UGC regulations. The academic activities of VFSTR are followed meticulously as specified in the academic calendar as approved by the Academic Council. This academic calendar is shared with all the stake holders well before the beginning of the respective academic year. The curriculum and the course contents of all the programmes are discussed by the respective Board of Studies (BoS), analyzed and recommended for implementation. The Academic Council, being the highest statutory body, chaired by the Vice-Chancellor, meets at least twice or thrice a year and discusses, suggests and approves all the important academic matters related to curriculum and course contents in particular including the recommendations of BoS.

The intended revision in regulations (R22) was in principle accepted and recommended by the Academic Council in its 30th meeting on 07-05-2022. Subsequently respective Board of Studies brought necessary recommendations accordingly, which were duly placed before the Academic Council in its 31st meeting on 30-07-2022.

R22 B.Tech. 4 YEAR DEGREE PROGRAMME

1.3 Program Duration

For the branch disciplines listed in section (1), the regular courses including theory and practical are offered over a period of four years in eight semesters. The normal duration to complete the B.Tech. program is four years. However, a student can avail the benefit of spill over period for 3 years, that is the maximum duration of seven years can be availed by a candidate to complete the B.Tech. programme in a slower pace if he / she desires. The candidate failing to complete the requirements will be considered for the honorable exit as applicable.

1.4 Courses and Credits

The term course is used in a broader sense to refer to so called papers such as 'theory subject', 'laboratory', 'inter-departmental project', 'major-project' etc. A course can be of theoretical and/ or of practical nature, and certain number of credits are allotted to it depending on the number of hours of instruction per semester. For a course offered in a semester, one hour of lecture (L) instructions carried out in a week is considered equivalent to one credit, whereas two hours of practical (P) sessions done in a week are considered equivalent to one credit respectively. Depending on the course two hours of tutorial (T) sessions may be considered equivalent to one credit. A student earns these credits when he/she successfully completes the course. Add-on credits can also be obtained by successful completion of other recognized co-curricular and extra-curricular activities such as NCC, NSS, Yoga, Dance, Music, Painting, etc. The details of credits of such activities will be provided by the respective course coordinators and assessment of student performance in the activities will be carried out objectively by the constituted committees appointed by the Dean AAA. The criteria of assessment for these activities will include aspects like regular attendance in the programme and satisfactory completion of it through tests conducted at University level or by participation / performance at university level events, state level or national level participation etc. Add-on credits earned in a specified manner will lead to earning B.Tech., B.Tech. with Honours, B.Tech. with Research Honours, B.Tech. with Minor, B.Tech. with Add-on Diploma and B.Tech. with Add-on Certification. Provision is also created for a candidate to migrate from 4-years B.Tech. to 5 years of dual B.Tech. + M.Tech. / MBA degree.

1.4.1 Content Delivery of a Course

Content delivery of a Course in the B.Tech. Degree Program shall be through, either or all, of the following Methods:

i. **Lecture** - refers to Lecture Session(s) through classroom contact session wherein students will learn by listening. Denoted by "L".

ii. **Tutorial** - refers to transaction(s) consisting of Participatory discussion / Self-study / Desk work / Brief presentations by students along with such other novel methods that enable a student to efficiently & effectively absorb and assimilate the contents delivered in the lecture sessions. Denoted by "T".

iii. **Practice** - refers to Practice / Practical sessions and it consists of Hands-on- Experience / Laboratory Experiments / Field Studies / Case Studies / Minor / Major Project, that equip the students to acquire the much required skill component. Denoted by "P".

1.5 B.Tech. Degree

All students formally and conventionally enroll for B.Tech. degree programme. They have to earn **161** credits for the award of degree as specified in the Curriculum. However, additionally he/she can opt to earn up to 20 more credits as Add-on credits, to earn the academic benefits as specified below.

1.5.1 B.Tech. with Honours Specialization: XX

A candidate may earn additional 20 credits, cumulatively totaling to **181** credits in the respective discipline spread over fourth to eight semesters to become eligible for the award of B.Tech. with Honours in YY Engineering (Specialization: XX).

If the add-on credits are designed to take-up a research activity and to complete a research thesis under the scope of 181 credits, then he/she become eligible to receive the award of **B.Tech. with Research Honours in YY Engineering (Specialization: XX)**. However, admission stipulations are applicable for a candidate to pursue B.Tech. with Research Honours. **Annexure-1** provides the supplement regulations for the award of B.Tech. with Research Honours.

1.5.2 B.Tech. with Minor Specialization in XX

A candidate may earn additional 20 credits, cumulatively totaling to 181 credits in anyone of the other disciplines spread over fourth to eight semesters to become eligible for the award of B.Tech. in YY Engineering with Minor Specialization in XX.

1.5.3 B.Tech. with Add-on Diploma

If a candidate earns add-on 20 credits from assorted disciplines/ themes/ centers, then he/ she will be eligible for the award of B.Tech. with Add-on Diploma.

1.5.4 B.Tech. with Add-on Certification

If a candidate in accordance with the sections of 1.5.1, 1.5.2, and 1.5.3 earn additional credits, but will not satisfy the completion of 20 credits, then he / she will be eligible to receive B.Tech. with Add-on Certification. However, such a candidate may also avail a spill over semester to complete 20 credits.

1.5.5 Dual (B.Tech. + M.Tech. / MBA) degree programme of 5-years*

A provision is also created for a candidate who is enrolled for B.Tech. degree to switch over to Dual (B.Tech. + M.Tech. / MBA) degree. **Annexure-2a & 2b** provides the supplement regulations for Dual degree programme.

*Programme will be offered based on satisfactory strength of students willing to register, after receiving the formal AICTE approval.

Note: The consolidated transcript will contain the credits and grade details of all courses amounting to 161 + up to 20 credits.

Every candidate should express his/ her provisional intent at the time of B.Tech. admission for pursuing B.Tech. with Honours/ B.Tech. with Minor/ Dual (B.Tech. + M.Tech. / MBA) degree. However, he / she can exercise his option to change his intent and final confirmation should be submitted at 5th sem. for Research Honours and at 6th sem. for Dual degree.

YY refers to name of the Branch discipline (Eg. CSE / ECE / BT); XX refers to a particular specialization within a Branch discipline.

1.6 Composition of an Academic year

An academic year is composed of an Odd semester (20 – 22 weeks), an Even semester (20 – 22 weeks) and a Summer semester (6 – 8 weeks). The regular semester that begins in July / August is known as odd / first semester and the one that begins in December / January is known as even / second semester (Figure 1). The instructional days for a regular semester shall be a minimum of 90 working days exclusive of days earmarked for summative assessment.

YEAR OF 12 MONTHS											
1	**2**	**3**	**4**	**5**	6	7	8	9	**10**	**11**	**12**
July/ Aug.	Aug./ Sept.	Sept./ Oct.	Oct./ Nov.	Nov./ Dec.	Dec./ Jan.	Jan./ Feb.	Feb./ Mar.	Mar./ Apr.	Apr./ May	May/ June	June/ July
ODD SEM/ FIRST SEM					EVEN SEM/ SECOND SEM					SUMMER SEM	

Figure 1: Distribution of semesters during an Academic Year.

R22 B.Tech. 4 YEAR DEGREE PROGRAMME

1.6.1 Before the commencement of the semester, a candidate has to pay the stipulated tuition fee and submit an application detailing the courses he / she intended to register, valid for that respective Odd / Even semester. The maximum number of credits per semester will be 25 credits inclusive add-on credits. The intended semester wise coverage will be as presented in the curriculum.

1.6.2 **Summer semester** is a short duration semester program that will be generally conducted during the semester break between even semester and odd semester. The students having 'R' (Repeat grade) courses may register for the course work during this semester to get a chance for successfully completing the 'R' courses. In general, supplementary assessments are conducted in the later part of the summer semester. However, the courses offered in summer semester and the number of courses a student can register are subjected to academic and administrative convenience. A student may register up to a max. of 16 credits in a summer semester.

1.6.3 Exception to the routine practice of registering for 'R' courses in summer semester, a student can register in a course offered by a visiting expert during the summer vacation which may be equivalent to an open elective or a department elective or an Add-on-course. The candidates can register for such courses within the scope of 16 credits. Candidate may also avail summer semester for summer internship opportunities, which may be considered as Add-on credits.

1.6.4 Summer internship credits will also be appended to the credits of Honours / Minor / Add-on certification provided the theme of internship is in accordance with the specialization of Honours/ Minor.

1.7 Semester wise provisions

A student may register for a max of 25 credits per semester as prescribed or otherwise he/ she may include the Repeat courses in the event of having not successfully completed a course or courses in the earlier semester. However, a student may also opt to go in a slower pace to earn the credits less than the prescribed max of 25, including even 'Dropping' a semester for special reasons.

It should be clearly underscored that a candidate should on priority register for Repeat (R) credits if any, during a regular semester, within the said scope of 25 credits; in case he / she cannot be sure of completing the 'R' credits in Summer semester.

1.7.1 During the first four years from the date of admission to B.Tech., a candidate has to pay the semester / annual fees as prescribed irrespective of the less number of credits that he / she would register or even opt to Drop a semester.

1.7.2 If a candidate gets into spillover semester beyond four years up to a maximum of seven years he / she has to pay semester fee proportional to the credits that he/ she registered in that spill over semester as prescribed from time to time.

1.7.3 A candidate has to pay additional fee proportional to the number of credits for registering in a summer semester as prescribed from time to time.

2. CURRICULUM

Each School offers different B.Tech. degree programmes and the departments concerned prescribes semester-wise curriculum encompassing different courses. Every course offered will be designated in a L-T-P structure. The theory courses comprise of L (and / or T & P hours) whereas the practical courses include instructions (T) and practical sessions (P). Amalgamation of theory courses with practical sessions is predominantly seen in this curriculum. Specifically, during the first year, some of the courses may be offered in a swap mode, that is the same course is offered in both the semesters of I-year to accommodate the student strength and for operational convenience.

2.1 Distribution of credits

The overall distribution of credits for various categories of courses in the curriculum of B.Tech. programmes is represented in Table (1) as given below.

Table 1 : Credits Distribution for Various categories of courses.

Category of Courses	Number of Credits	Percentage of Credits	AICTE Recommendation (%)
Professional Core	53	32.92%	30 - 36%
Electives (Department & Open / I²C)	34	21.12%	19 - 23%
Basic Engineering	20	12.42%	10 - 18%
Humanities and Management	15	09.32%	08 - 09%
Basic Sciences	21	13.04%	12 - 16%
Life Skills & Physical Fitness	04	02.48%	--
Projects	14	08.70%	08 - 11%
Total	**161**	**100%**	

2.2 Organization of course contents

Courses offered in the program is composed of two modules covering all the course contents required for a candidate to obtain knowledge and skill. Content in each module is further distributed among two units; wherein Unit -1 contains 'Fundamentals and Broad perceptive' of the module. Unit-2 comprises of the extension / advanced topics of Unit-1 as well as necessary practice models for validation / applying the knowledge gained during L/T sessions. The modular period is about 8 weeks. The first unit in a module may be covered in 2 to 3 weeks and the second unit of the module maybe of 5 to 6 weeks (Figure 2). By the end of each module a candidate must be in a position to translate his/ her L-based knowledge into P-based skill as prescribed in the curriculum. Individual formative assessment shall be in place for each module and a single semester-end summative assessment for the course composed of both the modules.

YEAR OF 12 MONTHS											
1	**2**	**3**	**4**	**5**	6	7	8	9	**10**	**11**	**12**
July/ Aug.	Aug./ Sept.	Sept./ Oct.	Oct./ Nov.	Nov./ Dec.	Dec./ Jan.	Jan./ Feb.	Feb./ Mar.	Mar./ Apr.	Apr./ May	May/ June	June/ July
ODD SEM/ FIRST SEM					EVEN SEM/ SECOND SEM					SUMMER SEM	

Module- I		**Module- II**			**Module- I**		**Module- II**			
U1	**U2**	U1	U2		U1	U2	U1	U2		

Figure 2: Unit-wise distribution of course contents in a module and their mapping with Academic Calendar; U= Unit.

3. CHOICE BASED CREDIT SYSTEM

Each branch discipline of the B.Tech. programme comprises of a set of courses - basic sciences, humanities and management, basic engineering, professional core, electives, employability & life skill courses. VFSTR offers flexibility for students to choose courses of their choice and obtain the credits satisfying the minimum credits criterion in each category as given in Table (1).

3.1 Common (Core) Courses for all branch disciplines

Majority of the basic science courses, basic engineering courses, humanities and management courses and general life skill courses including a course on orientation are offered commonly to students of all disciplines of study.

3.1.1 Basic Sciences and Humanities

Basic science courses are included to offer the knowledge of scientific theories that form the foundation for all the engineering solutions. Courses of Management and Humanities are also

offered. The main purpose of offering Management courses is to impart Management skills to students so that they would pursue allied career opportunities. 'Humanities' courses are meant for gaining managerial and organisational skills. In this some aspects of civil services are also covered for the benefit of those interested students.

3.1.2 Basic Engineering

Basic engineering courses are those that every engineer, irrespective of his discipline, ought to understand to perform well in his/her discipline. Engineering drawing, IT workshop, computer programming are some of the courses that fall in this category.

3.2 Professional Core

Professional Core courses are individualized for each programme and they are mandatory for every student opting for that branch discipline. These are designed to offer the essential fundamental knowledge and skills required for that specific programme.

3.3 Electives

A candidate has a choice to choose the elective courses. A list of elective courses is pooled together, enabling a candidate to choose the electives from a pool so that he/she can focus to a specific theme. Otherwise also he/ she can exercise the choice to choose electives from across the pools. There may be courses which may not be listed under any pool, which are called 'Free elective courses'.

Elective courses offered for each programme are categorized as 'department electives' and 'open electives'. Some electives may have pre-requisites *i.e,* successful completion of a course / knowledge of a course / the department's consent.

3.3.1 Department Electives

'Department electives' are those courses that are aimed at offering the advanced/ additional knowledge in the chosen branch discipline.

Care should be exercised while opting for departmental elective courses and specialization courses that is a course opted as departmental elective should not be registered as a course under Specialization and vice versa.

3.3.2 Open Electives

The elective courses offered by the departments/ centers/ themes for other department students are called as open electives. These electives are offered to enable students understand interdisciplinary nature of courses. A student's enrollment for pursuing an open elective course(s) is based on his/her choice and order of merit and subject to availability of seats.

Elective courses are spread over four semesters from fourth to seventh semester, to enable students earn credits from a chosen pool or otherwise.

Care should be exercised by the candidate while opting for Open electives and minor courses, that is a course opted as Open elective should not be registered as a course under Minor and vice versa.

In 'electives' category, a student is required to secure a minimum of 33 credits. These credits can be obtained through a combination of courses from his / her own department and other departments. Of these 9 credits have to be earned through MOOCs offered via Swayam platform. A student may also be permitted to earn more elective credits through MOOCS. This will promote self-learning and drive students towards innovative learning approaches.

To facilitate the process of acquiring the elective credits through MOOCS, VFSTR has constituted "Online course committee" at Central level with 1-2 members represented from each of the department to guide students in selection of courses and to assist them with further steps if required until completion of the courses.

3.4 Inter Departmental Project

These projects are designed and executed by students during the third year of their program. By doing these projects, students will get an idea of how technologies or processes, prototype or working model can be developed by culmination of technologies from courses of different programs. The supervision of inter-departmental projects will be done by faculty from different departments who serve as supervisors. The minimum duration of inter-departmental project during each of the semester is 90 hours including writing of project report and submission for assessment. A batch of 2-5 students from across the departments can take part in each of the inter-departmental projects. Performance will also be assessed in the modular framework for formative and semester-end summative, however spread over 2 semesters with a weightage of 2 credits.

3.5 Project

Students may opt for Project work in lieu of internship. Such students may avail research -internship support from any institution well known for research and development (R&D). They may also take up project work in VFSTR itself. Each candidate has to submit interim reports and a final report which are mandatory requirements towards the partial fulfillment of project credits requirements. It bears a weightage of 12 credits with a duration of 90 working days. During the semester the student under the guidance of a faculty member(s) will involve in an innovative design / research through the application of his / her knowledge gained in various courses studied. He / she is therefore expected to present a survey of literature on the topic, work out a project plan and carry it out through experimentation / modelling / simulation / computation. Through such a project work, the student is expected to demonstrate system analysis, design, presentation and execution skills. Performance in the project will also be assessed in the modular framework for formative and semester-end summative.

3.6 Internship

A student can undertake internship in lieu of project work in industry for one complete semester during seventh/ eighth semester in lieu of major project work. If the Internship is under taken during 7th semester, then the regular course work of 7th semester should be taken up during the 8th semester (7th and 8th semester happen in a swapped mode). It bears a weightage of 12 credits. This is aimed at training students in solving / understanding real-life problems through application of engineering analysis, design, evaluation and creation, particularly in association with practitioners and experts in the industry. The procedures for obtaining the internship placements and allocation of the same to the students are as per University defined norms outlined in the 'internship programme operational guidelines' manual. Even during internship, a student is preferably expected to carry out a focused study on one topic/ problem in consultation with the interning institute. Internship progress report should be submitted periodically and finally a detailed internship report should be submitted duly certified by a mentor from the internship institute. Performance in the internship will also be assessed in the modular framework for formative and semester-end summative.

3.7 Binary Graded courses

3.7.1 Physical Fitness

As physical fitness contributes to physical, mental and social development, it is offered to students. The credits are spread across 2 semesters in I year with 1 credit per semester. Students are imparted training through physical exercises.

3.7.2 Life Skills

Realizing the VFSTR's Vision of preparing multifaceted personalities, VFSTR floats various credit based life skill activities for students such as yoga, dance, music etc. The credits are spread across 2 semesters in II year with 1 credit per semester.

3.7.3 Industry – Interface (I^2) courses

I^2 courses are offered with the support of industry personnel and are also included in department elective category. These courses are of one credit, and are offered by experts

from industry / academia. The I^2 course is offered to 3rd year B.Tech. students with 15 hours' duration. The duration of 15 hours can be spread across 3 days continuously or 3 different slots of 5 hours each on consecutive weeks. Students who successfully complete the course will be awarded one credit.

3.7.4 Orientation course

Course content covers cross-cutting issues of the society like Gender Equality, Human Values, Professional Ethics, Moral and Ethical Values. Orientation course is offered for I year B.Tech. students and on successful completion earns 3 credits. All the students have to register during first / second semester and mandatorily obtain successful completion grade.

3.7.5 Courses with floating credits

With a view to imbibe social responsibility and all round development students are encouraged to participate in NCC / NSS / Students Activity Council (SAC) / Entrepreneurship cell (E-cell) / Student Mentoring / Social activities / Publication with good impact factor (Only 2 students can claim 1 paper / patent). Such activities lead to awarding of 2 credits.

In case the candidate does not successfully complete the binary graded courses, they will be placed under 'R' category. However, such R should be completed in regular Odd / Even semester whenever these courses are offered. In special cases provision of summer semester may be provided.

4. ATTENDANCE

It is mandatory for the student to attend the course work in each semester as per the academic schedule of that semester. VFSTR expects 100% attendance. However, the attendance in each course shall not be less than 75 % of the aggregate of all L, T, P sessions conducted in that course.

a) The attendance calculations will be periodically reviewed at the end of every 4 weeks. The details of attendance status will be shared with the parents / guardian. The final status of attendance will be reported at end of 15th week granting the advantage of the attendance for the 16th week for the purpose of attendance shortage calculations.

b) The shortage of attendance may be condoned up to 10% on the ground of ill-health, social obligations, participating / representing in sports/cultural events, placement activities etc.

c) Documentary evidence like medical reports and certificates issued by concerned bodies is to be produced on time as support for the attendance shortage due to ill-health. These cases are subjected to the scrutiny of a committee constituted for this purpose by the Vice-Chancellor. The decision of the committee shall be final.

d) Prior approval has to be taken from the HoDs for the other types of leaves.

e) The courses where the student shortage of attendance was not condoned shall be considered as 'Repeat' category courses and will be under 'R' grade in the student's semester transcript. Student should re-register for these courses during the summer semester or whenever the course is offered next time during regular semesters. These re-registrations are subjected to the regulations at the time of re-registration. In case of core courses, the same core has got to be re-registered. However, in case of an elective a candidate may exercise a choice of choosing different elective in place of 'R' graded elective.

The students who are put into 'R' grade will not be allowed to take up the summative assessment in that semester. In case due to lack and/or delay in information, if he/she appears for the summative assessment in that course, office of AAA is empowered to cancel the attended exams. The scores obtained either in formative or summative assessment will not be considered for grading.

5. ASSESSMENT

Teaching-Learning and Assessment should go hand in hand and complement each other. Continuous assessment plays a vital role to enable the student to get synchronized with the teaching-learning process. Assessment mechanism adopted in the institute is aimed at testing the learning outcomes in tune with the outcome based model of education. The focus, is thus on assessing whether the outcomes are realized by the end of the course.

The performance of a student in each course is assessed on a continuous basis during the semester

through various in-semester and end-semester assessment models. The marks awarded through continuous assessment are referred to as Formative assessment marks. The marks awarded through end-semester tests are referred to as Summative assessment marks (Figure 3). Both the formative and summative assessment marks are considered for awarding the final marks and the grade point in a particular course.

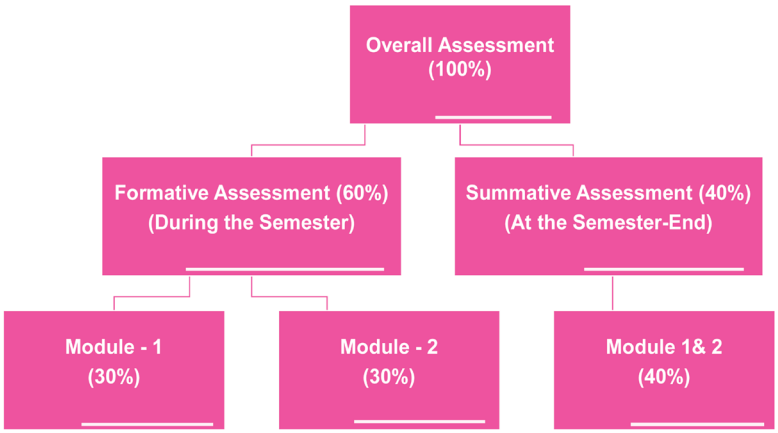


Figure 3: Categories of assessments in place for R22.

5.1 Marks distribution

For each course, the maximum sum of formative and summative assessment marks put together is 100, in the ratio of 60:40, respectively.

5.2 Qualifying criteria

To be declared successful in a course, a student must secure at least a grade 4.0 in a scale of 10 based on the total maximum marks which is inclusive of formative and summative assessment. The students should also get 35% from the maximum marks allotted for formative and summative assessments individually.

The hierarchy of qualifying criteria is as follows:

i. Attendance compliance should be 75% or within condonable range; else the candidate is put into 'R' grade.

ii. In formative assessment, a candidate should secure a minimum of 35% *i.e.* 21 marks out of 60; else the candidate is put into 'R' grade.

iii. In summative assessment, a candidate should secure a minimum of 35% *i.e.* 14 marks out of 40; else the candidate is put into 'I' (Incomplete) grade.

iv. Collectively the candidate should secure a min. grade of 4.0 in a scale of 10 after relative grading (section 7); else the candidate has to choose either 'R' or 'I' grade duly being counselled.

v. A candidate who has secured grade <5 in a course may be permitted (optional) to volunteer to improve his / her grade by opting suitably 'R' or 'I' grade in that course.

The candidates with 'R' grade should re-register for 'R' courses either in Summer semester or in a regular semester as and when the courses are offered. The candidates in 'I' grade are allowed to appear for supplementary summative assessment whenever the semester-end assessments are conducted.

To assess Binary graded courses / special projects / courses, not fitting into the categories described here, a suitable assessment procedure will be evolved in consultation with experts of that area and adjudicated by the committee constituted for that purpose. The decision given by the committee will be final. The appended assessment scheme shall be announced by the course coordinator during the commencement of course.

R22 B.Tech. 4 YEAR DEGREE PROGRAMME

YEAR OF 12 MONTHS											
1	**2**	**3**	**4**	**5**	6	7	8	9	**10**	**11**	**12**
July/ Aug.	Aug./ Sept.	Sept./ Oct.	Oct./ Nov.	Nov./ Dec.	Dec./ Jan.	Jan./ Feb.	Feb./ Mar.	Mar./ Apr.	Apr./ May	May/ June	June/ July
ODD SEM/ FIRST SEM					EVEN SEM/ SECOND SEM					SUMMER SEM	

Module- I		**Module- II**			**Module- I**		**Module- II**			
U1	**U2**	**U1**	**U2**		**U1**	**U2**	**U1**	**U2**		
Formative Assessment				SA	Formative Assessment				SA	

Figure 4: Schedules of formative and summative assessments in line with Academic calendar. SA = Summative assessment.

5.3 L-based courses integrated with P/T

5.3.1 Formative Assessment

The scheme of formative assessment is designed to promote the continuous learning. Scheme consists of assessments planned at institute level and assessment that may be scheduled by the course instructor (Figure 4). Institute level assessments shall be scheduled by the office of AAA. Respective Faculty Member(s) shall declare the schedule of Continuous Laboratory Assessments (CLA), Quiz, Tutorials, Assignments, Seminars, Discussions, etc. Some of the components may also however take place in an unscheduled manner like Surprise Tests. However, students shall be made aware of the assessment modalities that are going to be followed in a course by the faculty, under information to the HoD.

To monitor the progress of students, continuous assessment comprising of five targets (T1, T2, T3, T4 and T5) is advocated in each module for a maximum of 60 marks. For a class (or section) of 60 to 70 students, formative assessment commences by the announcement of module bank containing 10 problems for each module in a course. Nature of problems in the module bank shall be at the level of creative / exploratory / design / thought provoking covering the complete syllabus of a module at somewhat advanced / challenging level.

The purpose of creating module bank of 10 problems is to assign one problem each to 2 batches of 3 - 4 members. The batches are composed of randomly picked up candidates. These batches remain same for all courses and also for the P-sessions in the courses in that semester and are created in the beginning of that semester.

The purpose of assigning one problem to two batches is to create a healthy competitive spirit between the two batches.

The modality of evaluation of five targets is listed here under:

a) **T1**: During 5th or 6th week of each module a classroom test shall be conducted. T1 consists of two parts: A and B.

Part A consists of one random problem from the module bank and vary from batch to batch. All the questions in the module bank shall be distributed among students and students shall know the question to be answered only on the day of test in the examination hall.

Part B consists of one common problem at fairly application/ advanced level (**not at all prior notified**) from outside the module bank for all the students.

T1 shall be paper based and proctored test for a period of 60 min (maximum) which shall be assessed for 30 marks and downscaled to 10 marks.

For the students who for justifiable reasons could not attend the classroom test on the scheduled day, a re-test maybe conducted. However, Part-B will contain a new question and Part-B will have higher weightage than part-A or full weightage could even be allotted for Part-B in such an event.

b) **T2**: Immediately follows T1. Students in a specified batch who now have received the same question during T1 will work further on that problem for T2.

T2 is primarily an extension of problem received in T1 for carrying out validation study: Case studies / Simulations / Experimentation. Each batch shall interact with the course instructor to finalize the nature of validation and expected to complete the exercise within 10 to 15 days after T1.

Course instructor should ensure assigning a different case study / a different scope for validation study for each batch in case the same problem is assigned to two batches.

Course instructor shall assess every student in a batch for a max. of 10 marks based on his observation, interaction and/or reviewing (based on at least two reviews).

c) **T3**: T3 shall be conducted during the last week of each module. Student batches are expected to submit a report, clearly documenting the work executed during T2. The report should be in IEEE / APA format and additionally a voice in-built PPT should be prepared and submitted.

The report and presentation shall be assessed by the course instructor for 10 marks for every student. In certain cases, a course instructor can call for a physical presentation also by a batch.

d) **T4**: T4 is a comprehensive module test, conducted for 30 min. comprising of 20 multiple choice questions (MCQs) covering the holistic content of module. T4 shall be evaluated for a max. of 10 marks @ ½ mark for each question. T4 will be conducted in ON-LINE mode.

There shall be two tests in each course in a day and the best performance of the tests shall be considered for awarding the marks.

Two sets of question papers each containing 20 questions should be set. The theme of the questions could be similar across the sets. When the test is administered online, every student receives the questions in shuffled sequence and also the choices in shuffled sequence. Therefore, the choice like both 'a' & 'b' above. Neither 'a' nor 'b', all the three a, b, c will not be set.

e) **T5:** T5 assessment is based on Practice or Tutorial assignments. Implementation, Report presentation and Discussion shall happen in a continuous mode throughout the module period.

At least 4 such continuous lab practice assessments (CLPA) / assignments per module shall be conducted by course instructor. The marks will be @ 5 marks per assignment totaling up to 20 per module.

f) The scores of the targets are to be normally announced within three working days on completion of the assessment and the performance is to be discussed in the class.

g) The total marks per module is 60 - T1 (out of 10), T2 (out of 10), T3 (out of 10), T4 (out of 10) and T5 (out of 20).

h) Total marks for both the modules from formative assessment will be added up to 120, which will be ***suitably mapped down*** to a max. of 60 marks. The mapping policy should be decided by the lead instructor / instructors in consultation with the HoD. The mapping policy should be shared with Dean AAA for the purpose of documentation.

i) The marks scored in Module-1 for a max. of 60 should be entered / submitted latest by 9th week and of Module-2 latest by 17th week of the semester. Consolidated score of for a max. of 120 ***suitably mapped down*** to a max. of 60 marks should be submitted latest by 18th week of semester enabling the declaration of 'R'- grade before the commencement of summative assessment.

j) A candidate put under 'R' will not be permitted to take up the summative assessment.

R22 B.Tech. 4 YEAR DEGREE PROGRAMME

5.3.2 Summative Assessment

a) An instructor may choose one of the two formats for conducting summative assessment for L-based courses integrated with T/P.

i) 15 + 25 marks format or 20 + 20 marks format (following b, c, d below).

ii) 40 marks format (following c, d below).

b) **If summative assessment is in two parts format:**

i) Part-I will be the assessment of capstone project which is pre-assigned during the module-2 period or will be the exploratory review assessment of all lab practice assignments.

ii) Part-II will be based on a written examination for a max. marks of 80, as in c & d below, which is ***scaled down*** to 25 or 20 based on the selected pattern of format.

iii) A candidate should attend both the parts of summative assessments; else he will be put into I grade.

c) For each L-based course integrated with T/P, the summative assessment shall be conducted by the Institute for a duration of 150 min. and for a maximum of 80 marks. Contents for summative assessment shall cover the breadth and depth of the complete syllabus that is mentioned in the two modules of a course.

d) The question paper for end-semester theory examination consists of two parts as given in Table (2).

Table 2: L-based Summative Assessment Question Paper Pattern.

Part No.	No. of Questions	Marks for each Question	Marks	Choice
A	4	10	40	No
B	2	20	40	No
Total Marks			80	

e) The questions will be comprehensive covering the entire course syllabus and any single question should not necessarily be limited to any particular unit / module.

f) These marks are ***suitably mapped down*** to a score of 40.

g) Total marks of summative assessment will be for a max. of 40 irrespective of format of evaluation.

h) The award of 'I' grade is solely based on marks scored in summative assessment out of 40, if he/she does not score a min. 14 out of 40 (35%).

5.4 P-based Courses

The detailed information consisting of experiments, batch formations, experiment schedules, etc., will be displayed / informed to the student in the first week of the semester so that the student comes prepared for the lab sessions. Copies of the lab manual will be made available to the students along with the schedule. The lab manual will consist of the list of equipments, detailed procedure to conduct the experiment, format for record writing, outcomes for each experiment and possible set of short questions to help students gain critical understanding. The courses like workshop (IT), Engineering Graphics will also be treated as P-based courses.

5.4.1 Formative Assessment

During laboratory sessions, a brief viva-voce is conducted for each student on the experiment he/she is carrying out on that day. Some of the parameters that could be included in the Continuous Lab Assessment (CLA) are given in Table (3). The set of parameters may slightly differ from one laboratory to the other, and will be announced before the commencement of the lab session. These parameters are assessed for each laboratory session.

Table 3: Suggested parameters for Continuous Lab Assessment (CLA).

S. No	Component	Marks
1	Report of about 1 page on proposed experimental layout and background theory before the start of lab session	4
2	Viva and interaction to evaluate understanding of concepts	4
3	Experimentation and data collection	4
4	Analysis of experimental data and interpretation	4
5	Finalized report submitted in the next week	4
Total		**20**

This assessment is carried out for each practical session and the total marks of all practical sessions will be ***suitably mapped down*** to a max. of 60.

5.4.2 Summative Assessment

End semester examination for each practical course is conducted jointly by two examiners. The examiners are appointed by Dean, AAA from the panel of examiners suggested by the respective Heads of the Department. In some cases, one of the examiner may be from outside the institution and will be identified as external examiner. The scheme of assessment may vary depending on the nature of laboratory, which shall be shared with student by the laboratory in-charge. The summative assessment will be conducted for a max. marks of 40. The general scheme of assessment is given in Table (4).

Table 4: Suggested end-semester summative assessment pattern for P-based courses.

Component	Marks		
	Examiner 1	**Examiner 2**	**Total**
Objective & Procedure write up including outcomes	4	4	08
Experimentation and data collection	4	4	08
Computation of results	4	4	08
Analysis of results and Interpretation	4	4	08
Viva Voce	0	8	08
Total Marks	16	24	40

5.5 Assessment and Grading of MOOCs based elective

Whenever a candidate opts for a course through MOOCS offered via Swayam platform, he/ she has to learn and undergo assessment as per Swayam norms. Upon the declaration by Swayam, that the candidate has successfully completed the course, the candidate is said to have earned the credits under credit equivalence and credit transfer. The online course committee will also translate the score awarded by Swayam into an equivalent grade in a scale of 10 for incorporation by Dean AAA.

In case the candidate is unsuccessful, and then if the candidate has secured less than 20% in the final examination conducted by Swayam, the candidate will be placed into 'R' grade, and if it is ≥ 20%, then the candidate will be placed into 'I' grade. The online committee decides the equivalence score for formative assessment for a max. of 60 based on both the scores he/ she has earned in the assessments and the final examination conducted by Swayam in case of 'I' grade. Supplementary examination for 'I' grade will be conducted by VFSTR. The candidate has to re-register for the same Swayam course or an alternative Swayam course or may choose any other elective offered by the department in place of MOOCs course in case he/ she received a 'R' grade. However, a candidate should register and undergo MOOCs offered via Swayam platform course work for a min. of 9 credits.

R22 B.Tech. 4 YEAR DEGREE PROGRAMME

5.6 Inter Departmental Project

Inter departmental project work is undertaken in the 5th semester and continued till the end of 6th semester for earning 2 credits by each candidate. It is expected that the inter- disciplinary Projects result in publication of a technical paper in a peer-reviewed journal. For this purpose, Dean R&D and Heads of Department will identify faculty mentors who will guide the students by conducting workshops on drafting of research article, communication, process of peer-review, publication, etc. The faculty will support the students by all means to get the technical findings published in peer-reviewed journals.

5.6.1 Formative Assessment

The assessment will be carried in four reviews in a systematic way wherein; first two reviews are conducted in the 5th semester and remaining two reviews in the 6th semester. Reviews will be conducted jointly by the two supervisors allocated by the two departments representing the host and the collaborating department.

The detailed assessment guidelines and scheme are to be announced along with the assessment schedule as mentioned in the Table (5).

Table 5: Schedule and parameters followed for formative assessment.

Semester	No of Module	Schedule	No of reviews	Points to be considered	Formative assessment marks
5th semester	Module -1	7th – 8th week	Review -1	• Identification of specific area out of broad areas under the supervisor • Identification of outcomes in line with programme objectives. • Feasibility of contributing to the attainment of outcomes • Identification of tools / equipment / surveys / training needs / etc..	15
	Module – 2	15th – 16th week	Review -2	• Understanding by individual students on the overall aspect of the project. • Completion of literature survey • Design of equipment set up • Acquisition/learning of the tool required • Readiness of about 25% documentation	15
6th semester	Module – 1	7th – 8th week	Review -3	• Progress review as per mechanism/ schedule identified • Evaluation or validation techniques of the project • Structure of project report • Individual student contribution in above activities	15
	Module – 2	15th – 16th week	Review -4	• Presentation of results, analyses and conclusions • Meeting of objectives defined in first review • Preparation of report • Understanding by individual students on the overall project • Submission of technical article	15

5.6.2 Summative Assessment

Summative assessment will be done jointly by two examiners one is from VFSTR as internal examiner and the other may be from other institution as external examiner. These examiners will be appointed by Dean-AAA from the panel of examiners suggested by the respective Head of the Department. The scheme of assessment will be report (15 marks), presentation (10 marks) and demonstration (15 marks) respectively. Points to be considered during the review.

a) Presentation of results, analyses and conclusions

b) Meeting of objectives defined in first review

c) Preparation of report

d) Understanding by individual students on the overall project

e) Individual student contribution

f) Progress of project as per schedule

g) Submission of technical article/ prototype realization

5.7 Project

Those students who do not opt for the semester-long internship, carry out their major project at VFSTR and submit their report which is a mandatory requirement for the award of degree. These projects are usually done in batches (not exceeding five students in a batch), during the VIII semester, under the guidance of a faculty member. Every batch, in consultation with the guide, should define the project and also the probable procedure of carrying it out and submit the same to a committee consisting of 2 to 3 faculty members appointed by Head of the Department. This is to avoid the repetition and also to come up with a roadmap for completion of the project within the time stipulated. The students are encouraged to select topics related to ongoing research and consultancy projects. The students are expected to carry out and present a survey of literature on the topic, work out a project plan and its implementation through experimentation / modelling / simulation / computation. They are also expected to exhibit system analysis, design, and presentation and evaluation skills. The entire process of grouping of student batches, and identification of respective guides etc., is to be completed by the end of VII semester, so that students can start of their project work immediately after VII semester.

5.7.1 Formative Assessment

The progress of project is reviewed twice in a module by the Project Review Committee (PRC) and formative assessment marks are awarded based on these reviews. The Project review committee consists of

a) Head of Department or his/her nominee – Chairperson

b) A senior faculty member identified by the HoD – member

c) Project supervisor – member

Review schedules of PRC are to be announced by the department immediately after the commencement of class work. The review presentations are open to all the students of that section and attendance is compulsory. The first review should be of 15 minutes / batch; the remaining reviews should be around 30 minutes / batch. Before every review the batches should submit their PPT along with a brief report of not exceeding two pages. It is to be expected by the committee that student communicates/publishes research article based on the project work prior to graduation. The following aspects may be considered by the committee for assessment Table (6).

R22 B.Tech. 4 YEAR DEGREE PROGRAMME

Table 6: Schedule and suggested parameters to be considered for formative assessment.

Module	Schedule	Review	Points to be considered	Max. Marks
Module -1	4th week	First review	• Identification of specific area out of broad areas. • Identification of outcomes in line with programme objectives • Feasibility of contributing to the attainment of outcomes	15
	8th week	Second review	• Identification of tools/equipment/training needs/ etc.. • Understanding by individual students on the overall aspect of the project • Completion of literature survey • Design of project set up	15
Module – 2	12th week	Third review	• Acquisition/learning of the tool required • Readiness of the layout of the project report • Progress review as per mechanism/ schedule identified • Individual student contribution in above activities	15
	16th week	Fourth review	• Presentation of results and conclusions • Meeting of objectives defined in first review • Submission of draft report • Understanding by individual students on the overall project • Individual student contribution • Progress of project as per schedule	15

5.7.2 Summative Assessment

At the end of the semester, during 18th to 20th week of the semester the summative assessment will be conducted in two phases.

Phase–I (during 18th -19th week): This is an evaluation for a max of 20 marks. A committee of two members comprising of HoD's nominee and Guide will assess the project work which will involve going through the project report (6 marks), project presentation (7 marks) and demonstration of the project (7 marks).

Phase–II (during 20th week): A final presentation and defense assessment for a max. of 20 marks will be carried out by one-man committee composed of an external expert who is chosen by the Dean AAA from a panel of examiners suggested by the HoD. The format for evaluation will involve going through the project report's quality (6 marks), presentation (6 marks) and interaction and defense (8 marks).

The qualifying marks will be finalized considering the marks scored in both the phases (I & II) of summative assessment.

In case the candidate is placed in 'I' grade, he/she has to appear for both Phase-I and Phase-II assessments, which will be held within the 15 days after declaration of results. In the consecutive assessment also if the candidate fails to secure min. required score then he/ she will be placed in 'R' grade.

5.8 Semester-long Internship

Internship work is undertaken either in the VII or VIII semester by a student in an industry, under the joint supervision of industry personnel and an internal faculty member. Sixty percentage of the marks of Internship are allotted through continuous evaluation as formative assessment and the remaining 40% are based on end semester assessment.

Table 7: Assessment scheme for Internship reviews.

Module	Review	Schedule	Formative assessment marks
Module -1	First review	4th week	10
	Second review	8th week	15
Module – 2	Third review	12th week	15
	Fourth review	16th week	20
Total			**60**

a) The progress of internship work is reviewed twice in every module by the "Internship Review- Committee" and marks for formative assessment are awarded based on these reviews.

b) The Internship Review Committee (IRC) consists of Head of Department or his/her nominee (Chairperson), the internal and external (industry) supervisors.

c) The IRC may not be the same for all students; however, the same IRC should exist for entire duration of the internship program of any single student.

d) The schedule and the scheme of evaluation are to be announced with internship notification. The internship reviews may take place at the place of internship or at the university, as decided by the interning organization or may be conducted in the blended mode.

5.8.1 Formative assessment: Internal reviews at the place of internship

The internal supervisor will interact with the guide allotted at internship offering industry based on the schedule given to conduct the reviews. Scheduled reviews can be conducted by IRC on online mode for discussion/ presentation. The 15 marks obtained by students for each review will be scaled to allotted marks as given in Table (7).

a) Students should submit a report (not more than two pages) explaining about the progress of their work, mentioning clearly details like the machines or software handled / adopted, type of data collected and his/her understanding and contribution in the programme, and the same has to be presented before the supervisors.

b) The candidate should clearly present the completion of stipulated assignments set by the industry supervisor for that period.

c) The evaluation will be based on a & b above and also based on regularity and discipline maintained in the internship venue.

Table 8: Suggested scheme of assessment for every review

Component	Total
Regularity and interaction	3
Application of knowledge	2
Gaining of new knowledge /skills / literature survey	5
Internship progress	3
Report	2
Total marks	**15**

R22 B.Tech. 4 YEAR DEGREE PROGRAMME

5.8.2 Summative assessment – Internship

At the end of the semester, the student shall submit a comprehensive report of internship covering the work done and make a final presentation in two phases as follows:

Phase–I (during 18th -19th week): A committee of two members comprising of internal supervisor and HoD's nominee will assess the overall internship participation by the candidate and his final report through presentation made by the intern. The internship report (6 marks), presentation (7 marks) and overall impression (7 marks) during the internship will be evaluated respectively.

Phase–II (during 20th week): A final presentation and defense assessment for a max. of 20 marks will be carried out by one-man committee composed of an external expert who is chosen by the Dean AAA from a panel of examiners suggested by the HoD. The format for evaluation will involve going through the project report's quality (6 marks), presentation (6 marks) and interaction and defense (8 marks).

The qualifying marks will be finalized considering the marks scored in both the phases (I & II) of summative assessment.

In case the candidate is placed in 'I' grade, he / she has to appear for both Phase-I and Phase-II assessments, which will be held within the 15 days after declaration of results. In the consecutive assessment also if the candidate fails to secure min. required score then he/ she will be placed in 'R' grade.

6. SEMESTER-END ASSESSMENT ACTIVITIES

6.1 Setting of semester-end summative assessment question papers will be coordinated by the lead instructor assigned for a particular course. Two sets of question papers will be submitted latest by 12th week of the semester.

6.2 There shall be 'Summative Assessment Question Paper Scrutiny Committee' which would be constituted with external experts. Experts are empowered to modify / rephrase the questions to maintain a high standard of the semester-end assessment. The review should be completed by the 14th week of the semester. The review process will be coordinated by a committee of School Dean, HoDs and external experts.

6.3 The question wise marks scored in the summative assessment out of a total of 80 will be made available online within two weeks from the last date of examination and would be kept active for 24 hours. Latest by the end of 48 hours from the instant of notification any candidate can submit an appeal online providing question wise claim.

6.4 Claims for re-assessment on P-based courses are not allowed.

6.5 The appeals will be attended within next three working days. Fees for appeal, as decided from time to time, has to be remitted online along with the appeal.

6.6 Final results and grades will be computed as explained in the next section.

6.7 Final results and grades shall be announced within four weeks of completion of the last examination of the summative assessment (within two weeks from the last date of appeal). Grades are published on the University website, and also informed to the parents and students through SMS.

6.8 Provisional Grade cards will be issued within two weeks after the announcement of grades. Grade card will contain three parts. Part 1: Details of successfully completed courses. Part 2: Details of 'I' grade courses. Part 3: Details of 'R' grade courses.

7 COMPUTATION OF GRADING

7.1 Formative assessment decides the list of 'R'- candidates. Therefore, these candidates will not be considered for grading computation. Summative assessments decide the list of 'I' candidates. Therefore, these candidates will not be considered for grading computation.

7.2 The candidates who have successfully completed both formative and summative assessments will be considered for computation of relative grading.

7.3 Threshold value (**Th**) for relative grading in each course is arrived after studying the marks distribution in that course by a committee constituted by office of Dean AAA. The threshold value is decided by the upper bound marks of the major chunk of the class keeping the top outlier scores away from consideration (the least upper bound). The threshold value will be slightly greater than upper bound marks or may be equal to the upper bound marks.

7.4 The total marks (**m**) = marks scored in the formative assessment + marks scored in the summative assessment is transformed into relative grade expressed accurate to two decimal places as follows:

Relative grade point (P) = (m/Th) X 10 [and limited to 10]

7.5 If students require course wise percentage equivalence, then the calculation will be based on the following

Course wise percentage equivalence = (m/Th) X100
[truncated to two-digit integer and limited to 100]

7.6 After relative grading, a student is assigned a 'Letter Grade (G)' for each course as per Table (9). The grade and the corresponding letter grade represent the outcomes and assessments of a student's performance in a course.

Table 9: Grading information

Relative Grading Range (P)	Category	Grade (G)
≥ 9.50	Outstanding	O
≥ 8.50 to 9.49	Excellent	S
≥ 7.00 to 8.49	Very good	A
≥ 6.00 to 6.99	Good	B
≥ 5.00 to 5.99	Fair	C
≥ 4.00 to 4.99	Marginal	M
Transitional Grade	Repeat	R
Transitional Grade	Incomplete	I

8. SUPPLEMENTARY EXAMINATIONS

8.1 The supplementary examinations shall be conducted once in summer semester. Notifications will be released by the examination section informing the students about registration procedures, details of fee and timetables. Apart from these examinations the students who have courses with 'I'-grade can also write the supplementary examinations along with regular semester-end examinations of that academic (Odd / Even) semester.

8.2 Whenever a candidate clears courses with 'I' grade in a supplementary examination that are conducted during a regular semester, the Threshold value for computing his / her grade will be obtained from the same batch in which he / she had completed his/her formative assessment.

8.3 Whenever a candidate clears courses with 'R' / 'I' grade in a summer semester, the Threshold value for computing his / her grade will be carry forwarded from the preceding Odd / Even semester for the respective courses.

8.4 Whenever a candidate clears courses with a 'R' grade in a regular semester along with his/her junior batch then for this candidate the Threshold value will be corresponding to his/her junior batch for computing grade.

8.5 The results of summative assessment of Project / Internship will be announced only if the candidate successfully earn all the credits in courses registered during the program. If the candidate is with 'R' / 'I' graded courses the results will be kept under **'Announced Later (L)'** status and will be announced only after candidate clears these courses.

R22 B.Tech. 4 YEAR DEGREE PROGRAMME

9. GRADE POINT AVERAGE

The Academic Performance of a student in every semester is indicated by the Semester Grade Point Average (SGPA) and finally by Cumulative Grade Point Average (CGPA).

9.1 SGPA

The Semester Grade Point Average (SGPA) shall be computed using the formula given below:

$$SGPA = \frac{\sum_{i=1}^{n} C_i P_i}{\sum_{i=1}^{n} C_i}$$

Where

n = number of courses a student successfully completed in the semester under consideration

P_i = Grade points secured for the ith course registered in the semester under consideration.

C_i = the number of credits assigned to i^{th} course registered in the semester under consideration

9.2 CGPA

The Cumulative Grade Point Average (CGPA) shall be computed after successful completion of the programme. The CGPA shall be expressed in different flavours to reflect B.Tech. of 161 credits, B.Tech. with Research Honours of 181 credits, and different Add-on provisions up to 20 credits leading to Minor / Add-on Diploma/ Add-on certification.

Accordingly, the computations will be as below:

$$\text{CGPA} = \frac{\sum_{j=1}^{m} C_j P_j}{\sum_{j=1}^{m} C_j}$$

Where

m = total number of courses prescribed for the completion of the programme

P_j = grade points secured for the jth course.

C_j = the number of credits assigned to j^{th} course

and $\sum C_j = 161$

$\sum C_j = 181$ for CGPA calculations in case of B.Tech. with Research honours

$\sum C_j = 20$ for CGPA calculations of specialization part in case of B.Tech. with honours

$\sum C_j = 181$ for CGPA calculations in case of B.Tech. with honours

$\sum C_j = 20$ for CGPA calculations of minor part in case of B.Tech. with Minor

$\sum C_j = 181$ for CGPA calculations in case of B.Tech. with Minor

$\sum C_j = 20$ for CGPA calculations of Add-on Diploma part in case of B.Tech. with Add-on Diploma

$\sum C_j = 181$ for CGPA calculations in case of B.Tech. with Add-on Diploma

$\sum C_j < 20$ for CGPA calculations of Add-on certification part in case of B.Tech. with Add-on certification

$\sum C_j < 181$ for CGPA calculations in case of B.Tech. with Add-on certification

Percentage equivalence of SGPA & CGPA = (SGPA or CGPA) X10

10. AWARD OF CLASS

The students who have become eligible for award of degree shall be classified based on their CGPA secured, as per the Table (10) given below:

Table 10 : Class/ Division information.

Sl. No.	CGPA	Class / Division
1	8.0 and above	First Class with Distinction
2	6.5 and above but less than 8.0	First Class
3	6.0 and above but less than 6.5	Second Class
4	Less than 6.0	Pass Class

a) For the purpose of rewarding the accomplishers with ranks and awards, toppers in each branch discipline are identified, based on their academic performance (CGPA) in the following categories:

   i) Ranking in B.Tech.

   ii) Ranking in B.Tech. with Research Honours

   iii) Ranking in B.Tech. in Specialization

   iv) Ranking in B.Tech. in Minor

b) In addition, the 'Chairman's gold medal' and other 'Endowment Awards' are awarded to the 'outstanding students' based on the overall performance which includes academic, co-curricular and extra-curricular activities, campus placements and competitive examinations. A committee appointed by the Vice-Chancellor will recommend the eligible student for the award, selected from the nominations received from the departments.

c) In addition, the institution may recognize exceptional performance such as music, dance, sports etc. and display of exceptional bravery from time to time.

d) Only such candidates who complete 161 credits (+20 credits) in the first 8 successive semesters shall be eligible to receive awards/ ranks.

e) The candidates availing spill over semesters will not be eligible for the award of merit scholarships.

11. AWARD OF DEGREE

On successful completion of prescribed requirements of the programme, the degree shall be conferred during the convocation of the VFSTR.

For the conferment of degree, the student has to fulfill the following requirements:

a) a bonafide student and undergone the course work of not less than four academic years and not more than seven academic years from the date of joining.

b) successfully completed all the courses as prescribed in the respective curriculum.

c) acquired a minimum eligible credits i.e. 161 credits for the award of B.Tech. degree.

d) obtained no due certificates as prescribed by VFSTR.

e) no in-disciplinary proceedings pending against him / her.

Consequent upon being convinced, following an enquiry, the Academic council may resolve to withdraw the degree / diploma / any other certification provided by the institute. The aggrieved may however prefer for a review of such decision by the Academic Council, citing cogent reasons for review or go in for an appeal to the, BoM of the institute.

12. LATERAL ENTRY AND LATERAL EXIT OPTIONS

12.1 Lateral Entry into II year of B.Tech.

The students who have completed their diploma programme in recognized institutes can be admitted into the third semester of the undergraduate programme. The committee constituted by Vice-Chancellor, will establish the eligibility of admission into different branches offered by VFSTR. The committee may recommend additional courses to be studied by the candidate, which students will complete during their first year of study at VFTSR however subject to a maximum of 25 credits per semester. To earn B.Tech. degree the student has to earn the same number of credits specified for regular B.Tech. in their 2nd, 3rd and 4th year of study. The student is eligible to register for Add-on credits as presented in section (1.5).

R22 B.Tech. 4 YEAR DEGREE PROGRAMME

12.2 Inter- Institutional Credit Transfer

Students pursuing an undergraduate programme in other recognized Higher Education Institute (HEI) could be admitted into VFSTR without appearing for the entrance exam to continue their studies, subject to the approval by a committee constituted by Vice-Chancellor. In this case, the student shall furnish the transcripts or grade cards, syllabus copies, educational certificates, and other relevant documents while applying for admission. The committee constituted by Vice-Chancellor, will establish the equivalency based on the marks / credits obtained in the courses in the previous institute by following the Inter- Institutional Credit Transfer policy (Annexure-3).

VFSTR students also may earn credits by Inter- Institutional Credit Transfer.

12.3 Honorable exit with B.Sc. or Engineering Diploma

In line with NEP-2020, an optional exit is provided for a candidate who has earned a min. of 120 credits and has completed all the requirements up to the end of six semesters.

B.Sc. degree / Engineering Diploma will be awarded in respective branches of specialization. Candidates opted to exit with Engineering Diploma will also be awarded with B.Sc. equivalence certificate in case of disciplines existing in the University system.

In case the candidate fails to earn 120 credits, a suitable certification will be awarded during his / her exit from B.Tech. degree.

Semester-wise transcript and a consolidated transcript will be given to the candidates during their exit from the registered program.

Such a candidate who has exited can seek re-entry to complete B.Tech. by surrendering the B.Sc. Degree / Engineering Diploma. A committee constituted by Vice-Chancellor will scrutiny all such re-entry requests and recommend the plan of action. However, the max. duration of programme should be limited to seven years and further extension beyond the stipulated max. duration of study has to be approved by Academic Council, if the candidate appeals for an extension.

12.4 Volunteer 'Drop' with Sabbatical Semester option

A candidate may exercise his option to voluntarily exit from B.Tech. programme temporarily for a semester during the B.Tech. programme, by registering for a 'DROP option' in the beginning of the semester. The DROP can be exercised to take up special Internship / Innovation / Exploratory / Entrepreneurship / Advanced research / Start-up and such related activities. Under such circumstances a candidate can normally avail DROP over two successive semesters. Such 'Drop' semester will be identified as Sabbatical semesters.

Such a candidate has to pay the regular semester fee if such a Drop option is utilized during the first 8 semesters of B.Tech., and has to pay a nominal semester maintenance fee during the spillover period, if a candidate has not yet completed the credit requirements.

Upon returning from such a temporary exit, a candidate may continue his B.Tech. studies utilizing the provision of spillover period. A candidate may also submit a claim for Credit equivalence for the activities undertaken during the sabbatical period. The equivalence committee would evaluate and assess the academic equivalence of the work carried out and would recommend the credit equivalence and credit transfer to be granted together with the grades that could be attributed, if applicable. However, the max. duration of programme should be limited to seven years and further extension beyond the stipulated max. duration of study has to be approved by Academic Council, if the candidate appeals for an extension.

12.5 Volunteer 'Drop' with Semester Drop option

A candidate may exercise his option to voluntarily exit from B.Tech. programme temporarily for a semester during the B.Tech. programme, by registering for a 'DROP option' in the beginning of the semester to meet the family / personal exigencies. All the norms as mentioned in the section (12.4) shall be applicable for the candidates utilizing semester drop option.

13. CHANGE OF BRANCH DISCIPLINE

The students can request for change of branch after completing the first two semesters of study. One percent of the seats in each branch subject to a minimum of one seat is made available for accommodating such requests. These requests are considered subject to the following conditions:

a) Top one percent of the students in each branch based on CGPA at the end of the second semester subject to a minimum of one student in each branch are eligible for a change of branch.

b) If only one student is eligible from a branch and if he / she is not willing to change specialization, the opportunity can be availed by the second or third ranker of the branch in that order.

c) The seats must be available in the branch to which a student intends to change.

d) The student opting for a change should have completed all the credit requirements of the first two semesters.

e) The decision of the Institution is final in this regard.

All the above points are applicable for branch change cases based on the merit criterion. There may be branch change requests of another kind, from the students who are not able to cope up with the studies of the branch they are admitted into. All such requests are to be referred to a committee constituted for the purpose. The committee takes the decision based on the merit of each case, the availability of seats and various other factors. The decision of the committee in this regard is final. Because there is a difference of one or two courses in the first year curriculum among different specializations, when the student changes his specialization after completing first year, he / she is required to complete those first year courses of the new specialization which were not studied by him/her during the first year, which can be taken up in the immediate summer semester.

14. INTERPRETATION OF RULES

a) The academic rules and regulations should be read as a whole for the purpose of any interpretation.

b) For the matter(s) NOT covered herein above or for unforeseen circumstances, but arising during the course of the implementation of the above regulations. The Vice-Chancellor shall be authorized to remove the difficulties and decide upon the matters. The same shall be reported in the next meeting of Academic Council for ratification and subsequently informed to BoM.

c) The Institution may change or amend the academic rules and regulations or curriculum at any time, and the changes or amendments made shall be applicable to all the students with effect from the dates, notified by the Institution.

d) Procedure and explanation to any section can be floated by the office of Dean AAA as applicable from time to time with due approval by the chairman of Academic Council.

R22 B.Tech. 4 YEAR DEGREE PROGRAMME

ANNEXURE -1

SUPPLEMENT REGULATION FOR PURSUING B.TECH. WITH RESEARCH HONOURS (SPECIALIZATION: XX)

A candidate willing to transform his / her B.Tech. programme into research orientation should meet the following terms and conditions.

1. He / she should register add-on credits for Honours specializing in a particular stream XX in his discipline of study.
2. He / she should have maintained a CGPA of 8.0 up to the end of 5th semester (III-year Odd semester).
3. He / she should have successfully completed all the credit requirements up to the end of 5th semester.
4. He / she should indicate willingness to transfer his/her study from B.Tech. with Honours to B.Tech. with Research Honours during the Module-2 period of III year- Odd semester (5th Semester), when the applications are floated by the office of the Dean AAA for the purpose.
5. He / she has to appear for an aptitude test / interaction and the corresponding committee of experts constituted for the purpose has to recommend his / her name.
6. All recommended candidates may take up research project in a research lab / a research institution/ an institute of repute / a research organization or in the department of VFSTR.
7. All such candidates may compete for research internship support from the external research institutions or may be supported with research internship within VFSTR. This internship support will be for one full semester duration during his 8th semester extendable by one or two months during the Summer semester that falls in the sequel to 8th semester, in which case the extended period will be treated as 8th semester itself.
8. Such candidates are normally expected to pursue research in the area of specialization chosen for B.Tech. with Honours study.
9. Such candidates should register for a course on Research Methodology in III year- II semester as Open elective-3.
10. Such candidates should have identified a broad research problem by the end of 6th semester, as a part of his / her Module-2 requirement in the course on Research Methodology (Open elective-3).
11. In the 7th semester (IV year- I semester) he / she has to take up Research preliminaries for his / her Honours- 4 credit requirements. This will be a research project work by itself in the broad area of the proposed research problem. The candidate shall be able to complete a good literature survey, execute basic / preliminary implementations and by the end of module-2 of Honours – 4, he / she should have identified the specific research problem on which he / she will take up research during 8th semester. The entire course (Honours - 4) will be evaluated as per the requirements of a project work.
12. The candidate takes up the intensive research work during IV year – II semester for 16 credits (which is combined project work of 12 credits and Honours-5 of 4 credits). He / she should spend a minimum duration of 5 months on the research work. If required, he / she will be considered for research internship period in the extended period also. In fact, he/ she may conserve time by starting the research work immediately after the completion of 7th semester.

Thus a candidate should have successfully earned 161 + 20 = 181 credits following the aforementioned procedure to qualify for B.Tech. in YY Engineering with Research Honours (Specialization: XX).

R22 B.Tech. 4 YEAR DEGREE PROGRAMME

ANNEXURE – 2(a)

SUPPLEMENT REGULATION FOR PURSUING DUAL B.TECH. + M.TECH. DEGREE

The proposal to institute Dual B.Tech. + M.Tech., in line with the practices in Institutions of National Importance, is to elevate the Gross Qualification Index (GQI) of India, and specifically to enhance the technological/ engineering competency of the Technocrats.

1. In order to attract the committed learners towards earning M.Tech. immediately after B.Tech. the following scheme is proposed:
2. Every candidate should express his / her provisional intent at the time of B.Tech. admission for pursuing Dual B.Tech. + M.Tech. degree. However, he / she can exercise his option to change his intent and final confirmation should be submitted by the end of III-year B.Tech. programme (VI semester) for Dual degree.
3. A candidate needs to be well informed about the advantages that he / she will reap by completing a Master's degree in one stretch as an immediate follow up to B.Tech. degree program. Such M.Tech. program could enhance his orientation to research studies and could elevate as a researcher.
4. To make the proposal very attractive, the scheme incorporated to effectively reduce the study period from 4+2 = 6 years to around five years only by exploiting the continuity of the study sustaining the tempo and through making use of two Summer semesters between III & IV year and between IV & V year, and still technically maintaining the requirement to complete 8 semesters of B.Tech. + 4 semesters of M.Tech. = 12 Semesters intact.

A candidate willing to continue Dual B.Tech. + M.Tech. instead of regular B.Tech. programme should follow the following terms, conditions and procedures:

1. He / she should be a candidate maintaining his studentship through proper registration process of his B.Tech. programme.
2. He / she can exercise his / her option to earn Add-on credits up to 20 along with his / her regular B.Tech. to receive B.Tech. with Honours / with Minor / with Add-on Diploma.
3. He / she should have completed all the credit requirements up to the end of 5th semester successfully with a CGPA of at least 7.5. the candidate is expected to likely to complete 6th semester successfully and to maintain the CGPA of at least 7.
4. He / she should indicate his willingness to continue his study in Dual B.Tech. + M.Tech. during the Module-2 period of III-year – Even semester (6th semester). When the call is given by the office of the Dean AAA for the purpose.
5. He / she has to appear for an aptitude test / interaction and the corresponding committee of experts constituted for the purpose has to recommend his / her name.
6. The candidate formally enrolls for Dual B.Tech. + M.Tech. by the end of VI semester before the semester-end summative assessment period of VI semester.
7. The study pattern will be as follows:

Period	M.Tech. semester count	Cumulative semester count	Expected study coverage
Sandwiched summer semester between III & IV year	I semester	7th semester	14 credits of M.Tech. aligning with M.Tech. Odd semester
Regular B.Tech. VII semester		8th semester	As per B.Tech VII semester curriculum

R22 B.Tech. 4 YEAR DEGREE PROGRAMME

Regular VIII semester	II semester	• Internship / Project 12 credits of B.Tech. is suspended. • 4 credits of Honours / Minor would continue in 9th semester	16 credits of M.Tech. aligned with M.Tech. even semester
Sandwiched summer semester between IV & V year	III semester	10th semester	Remaining core / electives required in M.Tech. - 14 credits
Regular odd semester in V year	Suspended B.Tech. VIII semester credits as a preamble to M.Tech. project work	11th semester	• Internship / project work 12 credits corresponding to B. Tech. • 4 credits of IV semester of M. Tech. on Research methodology • 4 credits of IV semester of M. Tech. on preliminary Research work with a project
Regular even semester in V year	IV semester	• 12th semester • If required IV semester may be extended by 1 or 2 months to complete the research work.	Research project of M. Tech. 16 credits.

Thus the credit requirements specified for both B.Tech. and M.Tech. are satisfied along with the 12 semester requirement for B.Tech. and M.Tech. put together

- B.Tech. degree = 161 + 20 Add-on credits (opted)
- M. Tech degree = 68 (14+ 16 +14 +08 +16 credits)

A higher research orientation is created.

8. In the convocation after V year the candidate receives both the degrees B. Tech. & M. Tech. under dual mode.
9. The candidate should not opt for campus placements during III year / IV year beginning. He / she will be permitted to pursue for campus placement during 4th year end / 5th year.
10. Such a candidate may try for fellowship support during V year if he / she gets qualified in GATE; such candidates may also be eligible for support consideration by VFSTR during V year.
11. Such a candidate during his / her V year could be considered as Teaching cum Research Assistant and suitable compensation may be provided for his / her association as a Teaching Assistant in the VFSTR campus.

The proposed Dual B.Tech. + M.Tech. may be offered as a customized Dual B.Tech. + M.Tech. program jointly with an Industry for the candidates freshly placed in that industry. Then the orientation would be towards on-Job training, Advanced Industry skills and Job basis internship & project work.

R22 B.Tech. 4 YEAR DEGREE PROGRAMME

ANNEXURE – 2(b)

SUPPLEMENT REGULATION FOR PURSUING DUAL B.Tech. + MBA DEGREE

The proposal to institute Dual B.Tech. + MBA, in line with the practices in Institutions of National Importance, is to elevate the Gross Qualification Index (GQI) of India, and specifically to enhance the technological / management competency of the Technocrats. In order to attract the committed learners towards earning MBA immediately after B.Tech. the following scheme is proposed:

1. Every candidate should express his / her provisional intent at the time of B.Tech. admission for pursuing Dual B.Tech. + MBA degree. However, he / she can exercise his option to change his intent and final confirmation should be submitted during the III semester of B.Tech. programme for Dual degree.
2. A candidate needs to be well informed about the advantages that he / she will reap by completing a Master's degree in one stretch as an immediate follow up to B.Tech. degree program. Such MBA program could enhance his orientation to management studies and could elevate as a Manager / Entrepreneur.
3. To make the proposal very attractive, the scheme incorporated to effectively reduce the study period from 4+2 = 6 years to around five years only by exploiting the continuity of the study sustaining the tempo and through making use of two Summer semesters between III & IV year and between IV & V year, and still technically maintaining the requirement to complete 8 semesters of B.Tech. + 4 semesters of MBA = 12 Semesters intact.

A candidate willing to continue Dual B.Tech. + MBA instead of regular B.Tech. programme should follow the following terms, conditions and procedures:

1. He / she should be a candidate maintaining his studentship through proper registration process of his B.Tech. programme.
2. He / she should opt Engineering Management stream as Minor during regular B.Tech and the 20 credits earned in Minor will be considered for extension into dual B.Tech. + MBA.
3. He / she should have completed all the credit requirements up to the end of 3rd semester successfully and the candidate is expected to likely to complete 4th semester successfully.
4. He / she should indicate his willingness to continue his study in Dual B.Tech. + MBA during the Module-2 period of II-year – odd semester (3th semester). When the call is given by the office of the Dean AAA for the purpose.
5. The study pattern will be as follows:

Period	MBA semester count	Cumulative semester count	Expected study coverage
Regular B.Tech. IV Semester	-	4th semester	4 credits of Management Minor
Regular B.Tech. V Semester	-	5th semester	4 credits of Management Minor
Regular B.Tech. VI Semester	-	6th semester	4 credits of Management Minor
Sandwiched summer semester between III & IV year	I Year I semester	7th semester	16 credits of MBA
Regular B.Tech. VII semester		8th semester	4 credits of Management Minor
Regular B.Tech. VIII semester		9th semester	4 credits of Management Minor
Sandwiched summer semester between IV & V year	I Year II Semester	10th semester	16 credits of MBA

R22 B.Tech. 4 YEAR DEGREE PROGRAMME

Regular odd semester in V year	II Year I semester	11th semester	20 credits of MBA
Regular even semester in V year	II Year II semseter	12th semester	Internship of MBA 12 credits.

Thus the credit required of both B.Tech. and MBA are satisfied along with the 12 semester requirement for B.Tech. and MBA put together

- B.Tech. degree = 161
- MBA degree = 64 (16+ 16 +20 +12 credits) +20 Add-on credits (opted in B.Tech. for Management minor)

Total credits = 161 + 20 + 64 = 245

6. In the convocation after V year the candidate receives both the degrees B. Tech. & MBA under dual mode.
7. The candidate should not opt for campus placements during III year / IV year beginning. He / she will be permitted to pursue for campus placement during 4th year end / 5th year.

Annexure – 3

INTER – INSTITUTE CREDIT TRANSFER

Students pursuing an undergraduate programme in other recognized Higher Education Institute (HEI) could be admitted into VFSTR without appearing for the entrance exam to continue their studies, subject to the approval by a committee constituted by Vice-Chancellor. In this case, the student shall furnish the transcripts or grade cards, syllabus copies, educational certificates, and other relevant documents while applying for admission. The committee constituted by Vice-Chancellor, will establish the equivalency based on the marks/credits obtained in the courses in the previous institute by following the admission policy as mentioned below.

PROCEDURE

- Admissions are only at the beginning of an academic year and depends on availability of vacancies in the respective program.
- The courses studied by the candidates elsewhere will be mapped with the courses offered by VFTSR and the committee may suggest for study of additional courses for maintaining credit equivalency or for satisfying pre-requisites. The committee may also recommend the plan of study for the courses identified.
- After admission, the student has to submit the following documents
  - Transfer Certificate
  - Grade card / Marks Memo
  - “No objection certificate (NOC)” from the concerned Institute / University where the candidate completed part of the programme.
- Provisional admission of the candidates will be confirmed only after verification of the authenticity of the certificates from the concerned authorities.
- Min. 50% of the credits have to be earned at VFSTR.
- Courses which are completed by the candidate in the previous institute will be transferred as recommended by the equivalence committee under credit cum grade transfer. Inclusive these credits the candidate has to complete the credit requirements as per VFSTR.
- Consolidated grade card / transcript will clearly show the credits under credit transfer and corresponding CGPA and credits earned in VFSTR and corresponding CGPA. Over all CGPA may also be indicated.

EXTENSION:

VFSTR candidates may also be permitted to earn credits up to max of 20% by the way of credit transfer from taking up the courses from other institutes of repute.

B.Tech. COMPUTER SCIENCE AND ENGINEERING

TABLE OF CONTENTS

		Page Numbers
Foreword		03
VFSTR - Vision & Mission		04
CSE - Vision & Mission		04
Program Educational Objectives, Program Specific Outcomes, Program Outcomes		05
Course Structure		06
I YEAR I SEMESTER		
22MT103	Linear Algebra and Ordinary Differential Equations	13
22PY105	Semiconductor Physics and Electromagnetics	15
22EE101	Basics of Electrical and Electronics Engineering	17
22CS103	IT Workshop and Tools	19
22TP105	Problem Solving through Programming - I	22
22EN102	English Proficiency and Communication Skills	28
22TP101	Constitution of India	30
22SA101	Physical Fitness, Sports and Games – I	
I YEAR II SEMESTER		
22MT106	Algebra	32
22MT107	Discrete Mathematical Structures	34
22ME101	Engineering Graphics	36
22TP106	Problem Solving through Programming - II	38
22EN104	Technical English Communication	43
22CS104	Python Programming	46
22SA102	Orientation Session	
22SA103	Physical Fitness, Sports and Games – II	
II YEAR I SEMESTER		
22ST202	Probability and Statistics	53
22TP201	Data Structures	55
22MS201	Management Science	60
22CS201	Database Management Systems	62
22CS202	Digital Logic Design	65
22CS203	Object-Oriented Programming through JAVA	67
22CT201	Environmental Studies	71
22SA201	Life Skills – I	
	NCC/ NSS/ SAC/ E-cell/ Student Mentoring/ Social activities/ Publication	
II YEAR II SEMESTER		
22TP203	Advanced Coding Competency	73
22TP204	Professional Communication	76
22CS205	Computer Organization and Architecture	78
22CS206	Design and Analysis of Algorithms	81
22CS207	Operating Systems	84
22CS208	Theory of Computation	87
22SA202	Life Skills - II	

III YEAR I SEMESTER		
22TP301	Soft Skills Laboratory	93
22CS301	Introduction to Artificial Intelligence	95
22CS302	Compiler Design	98
22CS303	Web Technologies	101
22CS304	Inter-Disciplinary Project - Phase I	
22CS305	Industry Interface Course	
	Department Elective - 1	
	NCC/ NSS/ SAC/ E-cell/ Student Mentoring/ Social activities/ Publication	
III YEAR II SEMESTER		
22TP302	Quantitative Aptitude and Logical Reasoning	104
22CS204	Computer Networks	106
22CS306	Data Mining Techniques	108
22CS307	Software Engineering	110
22CS308	Inter-Disciplinary Project - Phase II	
	Department Elective - 2	
IV YEAR I SEMESTER		
22CS401	Cryptography and Network Security	115
22CS402	Big Data Analytics	117
22CS403	Cloud Computing	119
	Department Elective - 3	
	Department Elective - 4	
IV YEAR II SEMESTER		
22CS404	Internship / Project Work	
DEPARTMENT ELECTIVES		
22CS801	Advanced Data Structures	123
22CS802	Advanced JAVA Programming	127
22CS803	Computer Graphics	129
22CS804	Deep Learning	133
22CS805	Digital Image Processing	136
22CS806	Machine Learning	138
22CS807	Mobile Ad-hoc Networks	142
22CS808	Mobile Application Development	144
22CS809	Text Mining	149
22CS810	Numerical Algorithms	151
22CS811	Operating System Design	153
22CS812	Optimization Techniques	157
22CS813	Simulation and Modeling	161
HONOURS FOR CSE		
22CS951	Advanced Graph Algorithms	167
22CS952	Biometrics	169
22CS953	Parallel and Distributed Computing	171
22CS954	Internet of Things	174
22CS955	Wireless Sensor Networks	176
22CS956	Capstone Project	

FOREWORD

The vital aspect of Computer Science & Engineering program is problem solving through programming. It is an essential skill required to carry out profession of Software Engineer. CSE graduates practice the activities (analysis, design, development, testing and maintenance) of software development life cycle to solve complex and challenging real world problems.

R22 Curriculum enables our students to start with the basic science, basic engineering and introductory programming courses. Further, they learn the mathematical foundations of computing; get hands-on experience in building software solutions using various technologies for various real-world problems and pursue advanced courses such as Artificial Intelligence, Machine Learning, Cloud Computing, Big Data & Analytics, Internet of Things and much more.

Salient Features of R 22 Curriculum:

- *Multidisciplinary holistic education with continuous learning and continuous assessment.*
- *Lateral entry and lateral exit options.*
- *Credit Earning by credit transfer.*
- *Honors/ Research Honors/ Minor/ Add-on Diploma/ Add-on Certification and Dual B.Tech.+ M.Tech./ MBA Degree of 5 Years.*
- *Semester drop option to pursue innovation, incubation, entrepreneurial and advanced exploratory activities and subsequent re-entry.*

Emphasis on continuous formative assessment with a creative summative assessment will facilitate the student to "Move away from high stake examinations – towards more continuous and comprehensive evaluation".

The Board of Studies of B.Tech. CSE Programme consists of a right mix of eminent personalities from Academic, Research and Industry Organizations, besides experienced faculty members of the University.

External BoS Members:

1. Prof. C. R. Rao, Professor SCIS, University of Hyderabad.
2. Prof. R. B. V. Subramanyam, Professor, Department of CSE, NIT Warangal.
3. Dr. B. Venkata Ramana, Assoc. Professor & HoD, Department of CSE, IIT Tirupathi.
4. Dr. Nagesh Bhattu Sristy, Asst. Professor, Department of CSE, NIT AP.

I thank all the BOS members, Academic Council Members and University authorities for their continuous support and encouragement towards design of this innovative curriculum for CSE.

Dr. K.V. Krishna Kishore
Head, Department of CSE
VFSTR Deemed to be University

VIGNAN'S
Foundation for Science, Technology & Research
(Deemed to be UNIVERSITY)
-Estd. u/s 3 of UGC Act 1956

VISION

To evolve into a Centre of Excellence in Science & Technology through creative and innovative practices in teaching – learning, towards promoting academic achievement and research excellence to produce internationally accepted, competitive and world class professionals who are psychologically strong & emotionally balanced, imbued with social consciousness & ethical values.

MISSION

To provide high quality academic programmes, training activities, research facilities and opportunities supported by continuous industry - institute interaction aimed at promoting employability, entrepreneurship, leadership and research aptitude among students and contribute to the economic and technological development of the region, state and nation.

Department of COMPUTER SCIENCE & ENGINEERING

VISION of the department

To evolve as a centre of high repute in Computer Science & Engineering and create computer software professionals trained on problem solving skills imbued with ethics to serve the ever evolving and emerging requirements of IT Industry and society at large.

MISSION of the department

M_1: Imparting quality education through well designed curriculum, innovative teaching and learning methodologies integrated with professional skill development activities to meet the challenges in the career.

M_2: Nurture research and consultancy activities amongst students and faculty by providing State-of-art facilities and Industry-Institute Interaction.

M_3: Developing capacity to learn new technologies and apply to solve social and industrial problems to become an entrepreneur.

B.Tech in Computer Science and Engineering

Program Educational Objectives (PEOs)

Program Educational Objectives (PEOs) are established through a consultation process. PEOs are broad statements that describe the career and professional accomplishments that the graduates should achieve within three to five years after their graduation.

Graduates of the UG-CSE program will be able to

PEO1: Graduates acquire extensive technical knowledge and related skills required to demonstrate themselves as professionals or pursue higher education.

PEO2: Graduates adapt to upskilling and excel in their careers despite future technological changes, and demonstrate research aptitude to generate innovative engineering solutions.

PEO3: Graduates acquire the potential to contribute for the field of computing as well as for the societal development by demonstrating professional, social and ethical practices.

Program Specific Outcomes (PSOs)

The students will be able to

PSO1: **Application Development Skills:** Design and development of web applications using various technologies such as HTML, JSP, PHP, ASP and ASP.NET to cater the needs of the society.

PSO2: **Enrich Research Skills:** Offer solutions which impact geo-socio-economic and environmental scenarios by using Machine Learning, Artificial Intelligence and IoT.

Program Outcomes (POs)

Program Outcomes (POs), are attributes acquired by the student at the time of graduation. The POs given in below, ensure that the POs are aligned to the Graduate Attributes (GAs) specified by National Board of Accreditation (NBA). These attributes are measured at the time of Graduation.

PO1: **Engineering knowledge:** Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

PO2: **Problem analysis:** Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

PO3: **Design/development of solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

PO4: **Conduct investigations of complex problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

PO5: **Modern tool usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modelling to complex engineering activities with an understanding of the limitations.

PO6: **The engineer and society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

PO7: **Environment and sustainability:** Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

PO8: **Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

PO9: **Individual and team-work:** Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

PO10: **Communication:** Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO11: **Project management and finance:** Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PO12: **Life-long learning:** Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

R22 B.Tech.
4 YEAR
DEGREE PROGRAMME

COURSE STRUCTURE - R22

I Year I Semester

Course Code	Course Title	L	T	P	C
22MT103	Linear Algebra and Ordinary Differential Equations	3	2	0	4
22EE101	Basics of Electrical and Electronics Engineering	2	0	2	3
22ME101	Engineering Graphics	2	0	2	3
22CS103	IT Workshop and Tools	2	0	2	3
22TP103	Programming in C	2	0	4	4
22EN102	English Proficiency and Communication Skills	0	0	2	1
22TP101	Constitution of India	0	2	0	1
22SA101	Physical Fitness, Sports and Games-I	0	0	3	1
Total		**11**	**4**	**15**	**20**
		30 Hrs			**20**

I Year II Semester

Course Code	Course Title	L	T	P	C
22MT106	Algebra	3	2	0	4
22MT107	Discrete Mathematical Structures	2	2	0	3
22PY105	Semiconductor Physics and Electromagnetics	2	0	2	3
22TP104	Basic Coding Competency	0	1	3	2
22EN104	Technical English Communication	2	0	2	3
22CS104	Python Programming	2	0	2	3
22SA102	Orientation Session	0	0	6	3
	Physical Fitness, Sports and Games – II	0	0	3	1
Total		**11**	**5**	**18**	**22**
		34 Hrs			**22**

COURSE STRUCTURE - R22

R22 B.Tech. 4 YEAR DEGREE PROGRAMME

II Year I Semester

Course Code	Course Title	L	T	P	C
22ST202	Probability and Statistics	3	0	2	4
22TP201	Data Structures	2	2	2	4
22MS201	Management Science	2	2	0	3
22CS201	Database Management Systems	2	2	2	4
22CS202	Digital Logic Design	2	0	2	3
22CS203	Object-Oriented Programming through JAVA	2	0	4	4
22CT201	Environmental Studies	1	1	0	1
22SA201	Life Skills-I	0	0	2	1
Total		**14**	**7**	**14**	**24**
		35 Hrs			**24**

II Year II Semester

Course Code	Course Title	L	T	P	C
22TP205	Front-End Application Development	0	0	2	1
22TP204	Professional Communication	0	0	2	1
22CS205	Computer Organization and Architecture	2	2	0	3
22CS206	Design and Analysis of Algorithms	2	2	2	4
22CS207	Operating Systems	2	0	2	3
22CS208	Theory of Computation	3	2	0	4
22SA202	Life Skills - II	0	0	2	1
	Open Elective – 1	3	0	0	3
	Sub-Total	**12**	**6**	**10**	**20**
	Minor / Honours - 1	3	0	2	4
Sub-Total		**15**	**6**	**12**	**24**
Total		**33 Hrs**			**24**

R22 B.Tech.
4 YEAR
DEGREE PROGRAMME

COURSE STRUCTURE - R22

III Year I Semester

Course Code	Course Title	L	T	P	C
22TP301	Soft Skills Laboratory	0	0	2	1
22CS301	Introduction to Artificial Intelligence	2	2	0	3
22CS312	Compiler Design	2	2	0	3
22CS313	Web Technologies	0	0	2	1
22CS314	Computer Networks	2	0	2	3
22CS306	Data Mining Techniques	2	0	2	3
22CS304	Inter-Disciplinary Project – Phase I	0	0	2	0
	Department Elective – 1	3	0	2	4
	Open Elective – 2	3	0	0	3
Total		**14**	**4**	**12**	**21**
	Minor / Honours - 2	3	0	2	4
Total		**17**	**4**	**14**	**25**
		35 Hrs			**25**

III Year II Semester

Course Code	Course Title	L	T	P	C
22TP302	Quantitative Aptitude and Logical Reasoning	0	4	0	2
22CS407	Cryptography and Network Security	2	0	2	3
22CS311	Parallel and Distributed Computing	2	2	0	3
22CS307	Software Engineering	2	0	2	3
22CS305	Industry Interface Course	1	0	0	1
22CS308	Inter-Disciplinary Project – Phase II	0	0	2	2
	Department Elective - 2	3	0	2	4
	Open Elective – 3	3	0	0	3
Sub-Total		**13**	**6**	**8**	**21**
	Minor / Honours - 3	3	0	2	4
Sub-Total		**16**	**6**	**10**	**25**
Total		**17**	**4**	**14**	**25**
		32 Hrs			**25**

COURSE STRUCTURE - R22

R22 B.Tech. 4 YEAR DEGREE PROGRAMME

IV Year I Semester

Course Code	Course Title	L	T	P	C
22CS406	Privacy and Intrusion Detection	2	2	0	3
22CS402	Big Data Analytics	3	0	2	4
22CS403	Cloud Computing	3	0	2	4
22CS310	Computing Ethics	1	2	0	2
	Department Elective – 3	3	0	2	4
	Department Elective – 4	3	0	2	4
	Sub-Total	**15**	**2**	**10**	**21**
	Minor / Honours – 4	3	0	2	4
	Sub-Total	**18**	**2**	**12**	**25**
		32 Hrs			**25**

IV Year II Semester

Course Code	Course Title	L	T	P	C
22CS404	Internship / Project Work	0	2	22	12
	Total	**0**	**2**	**22**	**12**
	Minor / Honors – 5	0	2	6	**4**
	Total	**0**	**4**	**28**	**16**
		32 Hrs			**16**

R22 B.Tech. 4 YEAR DEGREE PROGRAMME

COURSE STRUCTURE - R22

Department Electives

Course Code	Course Title	L	T	P	C
22CS801	Advanced Data Structures	2	2	2	4
22CS802	Advanced JAVA Programming	2	2	2	4
22CS803	Computer Graphics	2	2	2	4
22CS804	Deep Learning	3	0	2	4
22CS805	Digital Forensics	3	0	2	4
22CS806	Digital Image Processing	2	2	2	4
22CS807	Web and Database Security	3	0	2	4
22CS808	Machine Learning	3	0	2	4
22CS809	Mobile Ad-hoc Networks	3	0	2	4
22CS810	Mobile Application Development	2	0	4	4
22CS811	Text Mining	3	0	2	4
22CS812	Numerical Algorithms	3	2	0	4
22CS813	Operating System Design	3	0	2	4
22CS814	Optimization Techniques	3	2	0	4
22CS816	Simulation and Modeling	3	0	2	4

Honours for CSE

Course Code	Course Title	L	T	P	C
22CS951	Advanced Graph Algorithms	3	0	2	4
22CS952	Biometrics	3	0	2	4
22CS953	Parallel and Distributed Computing	3	2	0	4
22CS954	Internet of Things	3	0	2	4
22CS955	Wireless Sensor Networks	3	0	2	4
22CS956	Capstone Project	0	2	6	4

Open Electives for Non-CSE

Course Code	Course Title	L	T	P	C
22CS851	Database Systems	2	0	2	3
22CS852	Mobile Application Design and Development	3	0	2	4
22CS853	Java Programming	2	0	2	3
22CS854	Python Programming	2	0	2	3
22CS855	Design and Development of Internet Applications	2	0	2	3

I

YEAR

B.Tech.

COMPUTER SCIENCE AND ENGINEERING

I SEMESTER

22MT103	-	Linear Algebra and Ordinary Differential Equations
22EE101	-	Basics of Electrical and Electronics Engineering
22ME101	-	Engineering Graphics
22CS103	-	IT Workshop and Tools
22TP103	-	Programming In C
22EN102	-	English Proficiency and Communication Skills
22TP101	-	Constitution of India
22SA101	-	Physical Fitness, Sports and Games-I

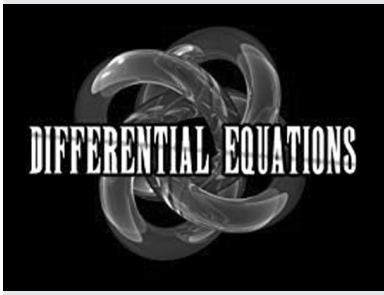
II SEMESTER

22MT106	-	Algebra
22MT107	-	Discrete Mathematical Structures
22PY105	-	Semiconductor Physics and Electromagnetics
22TP104	-	Basic Coding Competency
22EN104	-	Technical English Communication
22CS104	-	Python Programming
22SA102	-	Orientation Session
22SA103	-	Physical Fitness, Sports and Games-II

COURSE CONTENTS

I SEM & II SEM

22MT103 LINEAR ALGEBRA AND ORDINARY DIFFERENTIAL EQUATIONS



Source: https://www.amazon.com/Differential-Equations/dp/B01H30X2JA

Hours Per Week :

L	T	P	C
3	2	0	4

PREREQUISITE KNOWLEDGE: Basics of matrices, Differentiation and Integration.

COURSE DESCRIPTION AND OBJECTIVES:

The goal of this course is to build a grasp of the principles of mathematics through matrices, differential equations and applications that serves as an essential tool in several engineering applications.

MODULE-1

UNIT-1 **12L+8T+0P=20 Hours**

MATRICES

Definition of matrix; Types of matrices; Algebra of matrices, adjoint of a matrix, inverse of a matrix through adjoint and elementary row operations, Rank of a matrix, Echelon form, Normal form. Eigen values and Eigen vectors (up to 3 x 3 matrices only) and properties (without proofs).

UNIT-2 **12L+8T+0P=20 Hours**

APPLICATIONS OF MATRICES

Consistency of system of linear equations, Solution of system of linear equations having unique solution and involving not more than three variables by Gauss elimination method and Gauss Jordan method. Cayley-Hamilton theorem (without proof), Power of a matrix, Inverse of a matrix. Strength of materials and strength of beams using Eigen value and Eigen vectors.

PRACTICES:

- Compute inverse of a matrix if exists.
- Explain with suitable examples how rank of matrix is independent of the elementary operations.
- Explain with suitable examples how rank of matrix is unique.
- Discuss with suitable examples when eigen values and eigen vectors are possible for a matrix.
- Discuss the possibility of solution of a system of equations.
- Discuss when inverse and power of a matrix exist using Cayley-Hamilton theorem.

MODULE-2

UNIT-1 **12L+8T+0P=20 Hours**

ORDINARY DIFFERENTIAL EQUATIONS (ODE)

First Order Differential Equations: Introduction to ODE, variable separable method, homogenous and non-homogenous differential equations, linear differential equations, Bernoulli's equations.

Second Order Differential Equations: Linear differential equations with constant coefficients with RHS of the form eax, xn, sin(ax) or cos(ax).

SKILLS:

- ✓ *Find rank of a matrix using different methods.*
- ✓ *Compute the eigen values and eigen vectors of a matrix.*
- ✓ *Find analytical solution of a differential equation using appropriate method.*
- ✓ *Demonstrate any one numerical method to solve differential equation*

UNIT-2 **12L+8T+0P=20 Hours**

APPLICATIONS OF ODE

Applications of ODE: Newton's law of cooling, Law of natural growth and decay, LR Circuit.

PRACTICES:

- Check the order and degree of an ODE.
- Find solution for any four ordinary differential equations by applying suitable method.
- Find numerical solution for any four ordinary differential equations by applying suitable method.
- Discuss some applications of ODE.

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Module No.	Mapping with POs
1	Apply the concepts of rank, eigen values and eigenvectors of a matrix and finding inverse of a matrix and powers of a matrix.	Apply	1	1, 2, 9, 10, 12
2	Apply differential equations in real life problems.	Apply	2	1, 2, 9, 10, 12
3	Analyse the solution of a system of linear equations and find it.	Analyze	1	1, 2, 9, 10, 12
4	Inspect the analytical method for solving differential equations and applications.	Analyze	2	1, 2, 9, 10, 12

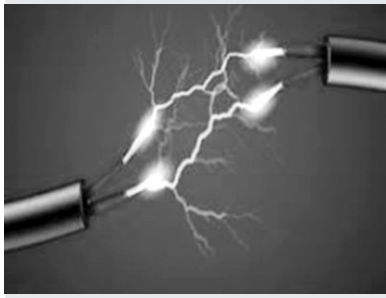
TEXT BOOKS:

1. N. P. Bali, K. L. Sai Prasad, "A Textbook of Engineering Mathematics I, II, III", 2nd edition Universal Science Press, New Delhi, 2018.
2. B. S. Grewal,"Higher Engineering Mathematics", 44th edition, Khanna Publishers, 2018.

REFERENCE BOOKS:

1. Erwin Kreyszig, "Advanced Engineering Mathematics", 10th Edition, John Wiley & Sons, Inc, 2015.
2. H. K. Dass and Er. RajanishVerma, "Higher Engineering Mathematics", 3rd revised edition, S. Chand & Co., 2015.
3. B. V. Ramana, "Advanced Engineering Mathematics", TMH Publishers, 2020.
4. T. K.V. Iyengar et al, "Engineering Mathematics, I, II, III", S. Chand & Co., New Delhi.

22EE101 BASICS OF ELECTRICAL AND ELECTRONICS ENGINEERING



Source: https://vita.vision.org.in/emerging-technologies-in-electrical-engineering/

Hours Per Week :

L	T	P	C
2	0	2	3

PREREQUISITE KNOWLEDGE: Electrostatics and Electromagnetism.

COURSE DESCRIPTION AND OBJECTIVES:

This course provides an insight into the functioning of basic electrical components like resistor, inductor and capacitor. It deals with the constructional and operational details of AC machines. It also deals with the basic electronic components like P-N junction diode, Zener diode, Transistor and their characteristics.

MODULE–1

UNIT-1 **8L+0T+8P=16 Hours**

FUNDAMENTALS OF ELECTRIC CIRCUITS

DC Circuits: Concept of network, Active and passive elements, Voltage and current sources, Concept of linearity and linear network, Unilateral and bilateral elements, R, L and C as linear elements, Ohm's Law, Kirchhoff's Laws, Application to simple series, Parallel circuits, Mesh and nodal analysis of resistive circuits with DC source.

AC circuits: Generation of AC voltage, Frequency, Average value, R.M.S. value, Form factor, Peak factor for sinusoidal only.

UNIT-2 **8L+0T+8P=16 Hours**

SEMICONDUCTOR DEVICES

Classification of semiconductors, P-N junction diode -operation and its characteristics, Half wave rectifier - operation, efficiency; Full wave rectifiers -types, operation, efficiency; Zener diode and its characteristics, Zener diode as Voltage regulator.

Bi polar junction transistor- operation, types (NPN & PNP)

PRACTICES:

- Verification of Ohm's law.
- Verification of Kirchhoff's current law.
- Verification of Kirchhoff's voltage law.
- Determination of R.M.S. Values of sinusoidal waveform.
- Verification of PN junction diode characteristics under both forward and reverse bias.
- Verification of Zener diode characteristics under reverse bias.

MODULE–2

UNIT-1 **8L+0T+8P=16 Hours**

ANALYSIS OF AC CIRCUITS

Analysis of single- phase ac circuits consisting of R, L, C, RL, RC (series and parallel) (simple numerical problems). Introduction to three phase system, Relation between phase and line quantities of voltages and currents in star and delta connected systems (Elementary treatment only).

SKILLS:

- ✓ *Distinguish between linear and nonlinear elements by looking at VI characteristics.*
- ✓ *Develop a simple loop generator.*
- ✓ *Design a voltage regulator using Zener diode.*
- ✓ *Design a half wave rectifier using PN junction diode.*
- ✓ *Design a full wave rectifier using PN junction diodes.*

UNIT-2 **8L+0T+8P=16 Hours**

AC MACHINES

Electromagnetism: Concepts of Magneto motive force, Reluctance, Flux and flux density, Concept of self-inductance and mutual inductance, Coefficient of coupling .

Static AC Machine: Principle of operation of single phase transformer, Constructional features, EMF equation (simple numerical problems).

Rotating AC Machine Principle of operation of three phase induction motor, Slip ring and squirrel cage motors, Torque equation; Constructional details of synchronous machine.

PRACTICES:

- Transformation ratio of a single phase transformer at different loads.
- Measurement of Energy in single phase resistive load circuit.
- Measurement of Power in single phase resistive load circuit.
- Determination of impedance in complex AC circuits.
- Verification of line and phase quantities in a balanced three phase system.

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Module No.	Mapping with POs
1	Analyze the resistive circuits with independent sources and find its solution.	Analyze	1,2	1,2,6,9
2	Solve the AC (single and three phase) and DC circuits using different methods.	Apply	1,2	1,2,9,12
3	Apply the concepts of electromagnetism for its applications.	Apply	2	1,2,3,9,12
4	Examine the different electrical equipment.	Evaluate	2	1,2,9,12
5	Acquire the knowledge of semiconductor devices to create circuits.	Create	1	1,2,3,9,12

TEXT BOOKS:

1. V. K. Mehta, “Principles of Electrical Engineering and Electronics”, 1st edition, S.Chand& Co., Publications, New Delhi, 2019.
2. D.P. Kothari, “Basic Electrical and Electronics Engineering”, 2nd edition, TMH, New Delhi, 2017.

REFERENCE BOOKS:

1. Millman and Halkias, “Electronic Devices and Circuits”, Mc Graw Hill, 2006.
2. A.K. Thereja and B.L.Thereja, “Electrical Technology”, Vol.–II, S. Chand & Co., Publications, 2020.
3. U. Bakshi and A. Bakshi, “Basic Electrical Engineering”, 1st edition, Technical Publications, Pune, Nov 2020.

22ME101 ENGINEERING GRAPHICS



Source: https://depositphotos.com/5087383/stock-photo-the-engineering-drawing.html

Hours Per Week :

L	T	P	C
2	0	2	3

PREREQUISITE KNOWLEDGE: Basics of Geometry.

COURSE DESCRIPTION AND OBJECTIVES:

Engineering graphics is the language of engineers and is the most effective way of communicating and sharing technical ideas in the form of pictures/drawings. The objective of this course is to familiarize the students with the conventional concepts of engineering drawing and computer aided drawing.

MODULE-1

UNIT-1 **6L+0T+6P=12 Hours**

ENGINEERING CURVES

Types of lines; Lettering, Dimensioning, Geometric constructions - lines, polygons (Angle, ARC, General and Inscribe in circle method), Conical curves (General method), Ellipse by Oblong method.

UNIT-2 **10L+0T+10P=20 Hours**

ORTHOGRAPHIC PROJECTIONS OF POINTS, LINES & PLANES

Principles of projection; Projections of points; Projection of straight lines - Inclined to one plane, inclined to both planes; Projection of planes - Inclined to one plane.

PRACTICES:

- Construction of polygons using different methods (i.e. ARC, Angle, General).
- Inscribe a regular hexagon & pentagon in a circle of the given diameter.
- Tracing of conical curves (Ellipse, Parabola, Hyperbola) by using General Method.
- Draw the projections of the points situated in all the 4 quadrants.
- Draw the projections of a line when it is inclined to one plane (HP or VP).
- Draw the projections of a line when it is inclined to both the planes (HP &VP).
- Draw the projections of a plane when it is inclined to one plane (HP or VP).

MODULE-2

UNIT-1 **6L+0T+6P=12 Hours**

PROJECTIONS OF SOLIDS

Projection of solids axis inclined to one reference plane - Prisms, pyramids, Cylinder and cone.

DEVELOPMENT OF SURFACES

Development of lateral surfaces of simple solids - Prisms, Pyramids, Cylinder and cone.

UNIT-2 **10L+0T+10P=20 Hours**

ORTHOGRAPHIC VIEWS

Conversion of pictorial views into orthographic views.

Drafting Using Computer Package: Introduction to 2D modelling software - AutoCAD; Conversion of Isometric view into Orthographic views of simple castings; Conversion of Orthographic views into Isometric view of simple solids - Prisms, Pyramids, Cylinders and cones.

SKILLS:

- ✓ *Convert isometric views of objects into orthographic views and vice versa.*
- ✓ *Visualize the shape of the 3D components.*
- ✓ *Create pictorial views by using AutoCAD.*
- ✓ *Know projections by visualization.*

PRACTICES:

- Draw the projections of Prisms, when they are inclined to one reference plane (HP or VP).
- Draw the projections of Pyramids, when they are inclined to one reference plane (HP or VP).
- Draw the projections of cylinder & cone, when they are inclined to one reference plane (HP or VP).
- Draw the complete surface development of prisms & pyramids with the given dimensions.
- Draw the complete surface development of cylinder & cone with the given dimensions.
- Draw the orthographic view's (i. e. front view, top view, and side view) of the given pictorial view of the sketches by using AutoCAD.
- Draw the Isometric view of simple solids (Prisms & Pyramids) by using AutoCAD.
- Draw the Isometric view of simple solids (Cylinder & Cone) by using AutoCAD.

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Module No.	Mapping with POs
1	Communicate the technical ideas in the form of drawings.	Apply	1	1,2,3,5
2	Apply the drawing skills in representing various geometrical features.	Apply	1	1,2,3,5
3	Develop orthographic projections and isometric views of various objects.	Apply	1	1,2,3,5
4	Estimate the lateral surface area of regular geometrical solids.	Analyze	2	1,2,3,5
5	Sketch simple objects and their pictorial views using AutoCAD.	Analyze	2	1,2,3,5

TEXT BOOKS:

1. J Hole, "Engineering Drawing", Tata McGraw-Hill, 2nd Edition, 2019.
2. N D Bhatt, "Engineering Drawing", Charotar Publication, 53rd Edition, 2014.

REFERENCE BOOKS:

1. Basant Agrawal and C.M. Agrawal "Engineering Drawing", , Tata Mc Graw- Hill, 2nd Edition 2018.
2. K L Narayana, "Engineering drawing", SciTech Publications, 3rd Edition, 2011.
3. Colin H. Simmons, Dennis E. Maguire, Manual of Engineering Drawing, 2nd Edition, 2003.

22CS103 IT WORKSHOP AND TOOLS



Source: https://www.facebook.com/TheITWorkshopWA

Hours Per Week :

L	T	P	C
2	0	2	3

PREREQUISITE KNOWLEDGE: Basics of Computer knowledge, Applications of Computers.

COURSE DESCRIPTION AND OBJECTIVES:

This course enables the students to learn various components of a computer system, assembly and dis-assembly of various components, troubleshooting, installation of OS and other applications. Also practicing of the usage of software tools such as word, excel, ppt and LaTex, text and image editors.

MODULE-1

UNIT-1 **0L+8T+8P=16 Hours**

PRACTICING EXERCISES ON HARDWARE DEVICES:

- Demo of various physical components of a computer system.
- Integration of various components of a computer system and dismantling.
- Installation of OS in a computer system through various storage devices.
- Installation of OS in a computer system through cloning.
- Demonstration of booting process of a computer system
- Detection of faulty components such as hard disk, RAM, SMPS, network interface in a computer system.
- Demonstration of program execution environment.
- Demo of Windows/Linux file system.
- Demo of location OS files in the file system (Windows/Linux).
- Configuration of network interface in a computer system and troubleshooting of network connectivity issues.
- Demo of shell scripts for maintenance and administration of a computer system.
- Usage of editor tools.
- Installation of software tools such as C compiler / interpreter, Java IDE, Python IDLE, Pycharm etc.
- Installation of antivirus software, web browsers and application of servers such as Apache server etc.

ACTIVITIES:

- Assemble and dis-assemble of various components of a computer system.
- Connect devices to various interfaces-(a) Serial Port, (b) Parallel Port, (c) USB Port, (d) Fire wire, (e) RJ45 connector, (f) VGA connector, (g) Audio plugs (Line-In, Line-Out and microphone), (h) PS/2 Port, and (h) SCSI Port.
- Install Linux/windows OS in your computer.
- Identification of different Computer hardware problems and troubleshooting of the same.
- Editing of image, audio and video files using different editor tools.
- Build Ethernet and Wi-Fi LANs.

SKILLS:

- ✓ *Integration of various components of a computer system.*
- ✓ *Trouble shooting of components of a computer system.*
- ✓ *Installation of OS and its various tools/ applications.*
- ✓ *Usage of IT tools such as MS-Word, LaTex etc.*
- ✓ *Creating the documents using MS-Word and LaTex.*
- ✓ *Analysing and visualizing data with excel.*
- ✓ *Developing various power point presentations*

MODULE-2

UNIT-1 **0L+8T+8P=16 Hours**

PRACTICING EXERCISES USING SOFTWARE TOOLS:

- Prepare your resume using MS-word.
- Design a “Birthday Invitation” card.
- Design a Timetable given to you at the beginning of the semester without grid lines.
- Using Draw Table feature, insert a 7-column, 6-row table to create a calendar for the current month.
  a. Enter the names of the days of the week in the first row of the table.
  b. Centre the day names horizontally and vertically.
  c. Change the font and font size as desired.
  d. Insert a row at the top of the table.
  e. Merge the cells in the row and enter the current month and year using a large font size.
  f. Shade the row.
  g. Enter and right-align the dates for the month in the appropriate cells of the table.
  h. Change the outside border to a more decorative border. Identify two important dates in the calendar and shade them.
- Prepare mark sheet using MS-Excel.
- Create a pivot table to analyse your worksheet data.
- Prepare a presentation on your university using MS-PowerPoint.
- Design a Magazine cover. Use the following:
  1) Select a theme for the page,
  2) Insert either a picture or clipart, and
  3) Use WordArt.
- Design a poster inviting all students of your university to the Computer Festival.
- Installation and demonstration of LaTeX.
- Prepare professional pdf documents using LaTeX.
- Prepare LaTex document containing mathematical equations.

ACTIVITIES:

- Create a 5-page document. Use hyperlinks, insert bookmarks in the same document.
- Design a worksheet using the following functions-MODE, STDDEV, VARIANCE, MEDIAN, SIN, COS, TAN, COUNT, MAX, MIN, ABS, MOD, SUM, SUMIF, POWER.
- Create bar graphs, pie charts and line charts in excel.
- Create a 5-slide presentation on any topic. Use Images, Graphs, Chart, Tables, Animation, Time, Bullets, Transition, Sound, Hyperlink, Background template, Header and Footer.
- Create a following numbered list using LaTex.

INTRODUCTION:

This is a display of numbered list

- Abstract
- Introduction.
- Section 1.
  a) Section 1.1.
  b) Sedtion 1.2.
- Section 2.

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Module No.	Mapping with POs
1	Ability to assemble and disassemble the computer system components and trouble shooting.	Apply	1	1
2	Installing Operating Systems and understanding the system booting process.	Under-stand	1	1
3	Ability to develop system maintenance using shell scripts.	Apply	1	1
4	Create word documents, presentations and spread sheets by applying various tools.	Create	2	2,5

TEXT BOOKS:

1. Fundamentals of Computers by Reema Thareja, Oxford University Press 2nd edition 2019, India.
2. Stefan Kottwitz, “LaTeX Beginner’s Guide: Create visually appealing texts, articles, and books for business and science using LaTeX”, 2nd Edition, Kindle, 2021.

REFERENCE BOOKS:

1. Priti Sinha and Pradeep K. Sinha, “Computer Fundamentals: Concepts, Systems and Applications”, 8th edition, BPB Publications, 2004.
2. John Walkenbach, Herb Tyson, Michael R.Groh and FaitheWempen, “Microsoft Office 2010 Bible”, Wiley.



Source: Techgig.com

22TP103 PROGRAMMING IN C

Hours Per Week :

L	T	P	C
2	0	4	4

PREREQUISITE KNOWLEDGE: Fundamentals of Problem Solving.

COURSE DESCRIPTION AND OBJECTIVES:

This course is aimed to impart knowledge on basic concepts of C programming language and problem solving through programming. It covers basic structure of C program, data types, operators, decision making statements, loops, functions, strings, pointers, and also file manipulations. At the end of this course, students will be able to design, implement, test and debug complex problems using features of C.

MODULE-1

UNIT-1 **8L+8T+8P=24 Hours**

INTRODUCTION TO ALGORITHMS AND PROGRAMMING LANGUAGES

Introduction to Algorithms: Basics of algorithms; Flow charts; Generations of programming languages. Introduction to C: Structure of a C program - pre-processor statement, inline comments, variable declaration statements, executable statements; C Tokens - C character set, identifiers and keywords, type qualifiers, type modifiers, variables, constants, punctuations and operators.

Data Types and Operators: Basic data types; Storage classes; Scope of a variable; Formatted I/O; Reading and writing characters; Operators - assignment, arithmetic, relational, logical, bitwise, ternary, address, indirection, sizeof, dot, arrow, parentheses operators; Expressions - operator precedence, associative rules.

UNIT-2 **8L+8T+8P=24 Hours**

CONTROL STATEMENTS

Control Statements: Introduction to category of control statements; Conditional branching statements - if, if - else, nested-if, if - else ladder, switch case; Iterative statements - for, while, do - while, nested loops; Jump statements - break, continue, goto, and return.

PRACTICES:

QUESTIONS ON DATA HANDLING – LEVEL 1:

- Write a program to accept a character as input from the user and print it.
- Write a program to accept a number as input from the user and print it.
- Write a program to accept a float value from the user and print it.
- Write a program to accept a message as input from the user and print it.
- Write a program to accept a message from the user as input and print it in 3 different lines.
- Write a program to accept 2 numbers from the user as input and print their sum.
- Write a program to accept 2 numbers from the user as input and print their product.
- Write a program to accept a number as input from the user which denotes the temperature in Celsius, convert it to Fahrenheit reading and print it.
- Write a program to accept a number as input from the user which denotes the radius and print the area of the circle.
- Write a program to accept a character as input from the user and print it's corresponding ASCII value.

QUESTIONS ON CONTROL STATEMENTS - LOOPING – LEVEL 1:

- Write a C program to print all the characters from a to z once.
- Write a C program to print all the characters from Z to A once.
- Write a C program to print all the characters from A to Z 3 times.
- Write a C program to print the first N natural numbers, where N is given as input by the user.
- Write a C program to print the first N natural numbers and their sum, where N is given as input by the user.
- Write a C program to print all the odd numbers between 1 and N where N is given as input by the user.
- Write a C program to print all the even numbers between I and N where N is given as input by the user.
- Write a C program to print the squares of the first N natural numbers between 1 and N, where N is given as input by the user.
- Write a C program to print the cubes of the first N natural numbers between 1 and N, where N is given as input by the user.
- Write a C program to print the squares of every 5th number starting from 1 to N, where N is given as input by the user.

QUESTIONS ON CONTROL STATEMENTS – DECISION MAKING – LEVEL 1:

- Write a program to accept two numbers as input check if they are equal.
- Write a program to accept two characters as input and check if they are equal.
- Write a program to accept two numbers as input and print the greater of the 2 numbers.
- Write a program to accept two numbers as input and print the lesser of the 2 numbers.
- Write a program to accept 3 numbers as input and print the maximum of the 3.
- Write a program to accept 3 numbers as input and print the minimum of the 3.
- Write a program to accept a number as input and print EVEN if it is an even number and ODD if it is an odd number.
- Write a program to accept a number as input and check if it is divisible by 3. If it is divisible by 3 print YES else print NO.
- Write a program to accept a number as input and check if it is divisible by both 3 & 5. If it is divisible print YES else print NO.
- Write a program to accept a number as input and check if it is positive, negative or zero.

QUESTIONS ON PATTERNS – LEVEL 1:

1. Write a program to accept a number N as input from the user and print the following pattern. Sample N = 5.

```
*****
*****
*****
*****
*****
```

2. Write a program to accept a number N as input from the user and print the following pattern. Sample N = 5.

```
*****
*   *
*   *
*   *
*****
```

3. Write a program to accept a number N as input from the user and print the following pattern. Sample N = 5.

```
*
**
```

SKILLS:

- ✓ *Analysis of the problem to be solved.*
- ✓ *Select static or dynamic data structures for a given problem and manipulation of data items.*
- ✓ *Application of various file operations effectively in solving real world problems.*
- ✓ *Develop C programs that are understandable, debuggable, maintainable and more likely to work correctly in the first attempt.*

***
****
*****

4. Write a program to accept a number N as input from the user and print the following pattern. Sample N = 5.

*
**
***
****
*****

5. Write a program to accept a number N as input from the user and print the following pattern. Sample N = 5.

1
12
123
1234
12345

6. Write a program to accept a number N as input from the user and print the following pattern. Sample N = 5.

1
22
333
4444
55555

7. Write a program to accept a number N as input from the user and print the following pattern. Sample N = 5.

54321
4321
321
21
1

8. Write a program to accept a number N as input from the user and print the following pattern. Sample N = 5.

12345
2345
345
45
5

9. Write a program to accept a number N as input from the user and print the following pattern. Sample N = 5.

A
AB
ABC
ABCD
ABCDE

10. Write a program to accept a number N as input from the user and print the following pattern. Sample N = 5.

A
BC

DEF
GHIJ
KLMNO

QUESTIONS ON NUMBER CRUNCHING – LEVEL 1:

- Write a program to accept a number as input and print the number of digits in the number.
- Write a program to accept a number as input print the sum of its digits.
- Write a program to accept a number as input, reverse the number and print it.
- Write a program to accept a number and digit as input and find the number of occurrences of the digit in the number.
- Write a program to accept a number as input and check if it is an Armstrong number.
- Write a program to accept a number as input and check if it is an Adam number.
- Write a program to accept a number as input and check if is a prime number.
- Write a program to accept 2 numbers as input and check if they are amicable or not.
- Write a program to accept a number as input and check if it is a power of 2.
- Write a program to accept 2 numbers as input and find their LCM.

MODULE-2

UNIT-1 **8L+8T+8P=24 Hours**

ARRAYS & STRINGS

Arrays: Introduction; Types of arrays; Single dimensional array - declaration, initialization, usage, reading, writing, accessing, memory representation, operations; Multidimensional arrays.

Strings: Character array, Reading string from the standard input device, Displaying strings on the standard output device, Importance of terminating a string, Standard string library functions.

UNIT- 2 **8L+8T+8P=24 Hours**

FUNCTIONS

Functions: User-defined functions; Function declaration-definition, header of a function, body of a function, function invocation, Call by value, Call by address, Passing arrays to functions; Recursion; Library functions.

PRACTICES:

QUESTIONS ON STRINGS – LEVEL 1:

- Write a program to accept a string as input and print it.
- Write a program to accept a string as input and count the number of vowels in it.
- Write a program to accept a string as input and count the number of consonants in it.
- Write a program to accept a string as input and print its length.
- Write a program to accept a string as input and print the reversed string.
- Write a program to accept 2 strings as input and check if they are the same.
- Write a program to accept a string as input and copy the contents into a second string and print the second string.
- Write a program to accept 2 strings as input and concatenate them into a third string and print the third string.
- Write a program to accept a string as input and check if it is a palindrome.
- Write a program to accept two strings as input and check if the second string is a substring of the first.

QUESTIONS ON STRINGS – LEVEL 2:

- Implement the string length function.
- Implement the string copy function.
- Implement the string concatenate function.

- Implement the string compare function.
- Implement the vowel count function.
- Implement the consonant count function.
- Implement the count words function.
- Implement the string reverse function.
- Implement the strstr function.
- Complete the code snippet to implement the is Palindrome function that checks if a given string is a palindrome. You will need to use the 3 functions string Copy, str Reverse and string Compare functions provided to accomplish this.

QUESTIONS ON STRINGS – LEVEL 3:

- Write a program to swap two given strings and print the swapped strings.
- Write a program to swap two given words of the given sentence and print the altered string.
- Return the maximum occurring character in the string.
- Write a program to print the character in the string with the count where count is the occurrence of the character.
- Write a program to print the duplicate characters in the given string.
- Write a program to remove the duplicate characters in the given string.
- Write a program to remove the vowels from a given string.
- Write a program to rotate a given string N number of times.
- Write a program to check if 2 strings are rotations of each other.
- Write a program to remove the characters from the first string that are present in the second string.

QUESTIONS ON 2D ARRAYS – LEVEL 1:

- Print the contents of a 2D array row-wise.
- Print the contents of a 2D array column-wise.
- Print the contents of a 2D array in a zig-zag order.
- Print the contents of a 2D array diagonal-wise.
- Print the contents of a 2D array right-diagonal order.
- Print the contents of a 2D array left-diagonal order.
- Print the contents of a 2D array in the upper triangular order – left top to right bottom.
- Print the contents of a 2D array in the lower triangular order.
- Find and print the maximum element along with its position in a matrix.
- Find and print the minimum element along with its position in a matrix.

QUESTIONS ON 2D ARRAYS – LEVEL 2:

- Find and print the maximum element of each row of a matrix.
- Find and print the minimum elements of each row of a matrix.
- Find and print the maximum element of each column of a matrix.
- Find and print the minimum element of each column of a matrix.
- Find the lowest value in the upper triangle area and the largest value in the lower triangular area of a matrix and print their product.
- Find the sum of the elements of each row and each column of a matrix and print the minimum row sum and maximum sum column.
- Write a program to find the row with the maximum number of 1's in a matrix consisting of only 0's and 1's.
- Write a program to print the quotient and remainder on dividing sum of left-top to right-bottom diagonal by sum of right-top to left-bottom diagonal.
- Write a program to print the absolute difference of the sum of major diagonal elements and the sum of minor diagonals of the given matrix.
- Write a program to search a given element in a row-wise and column-wise sorted 2D array.

QUESTIONS ON 2D ARRAYS – LEVEL 3:

- Write a program to find the Kth smallest element in the given matrix.
- Write a program to find the Kth largest element in the given matrix.
- Write a program to check whether the given two two-dimensional array of same dimensions are equal or not.
- Write a program to add the given two two-dimensional array of same dimensions.
- Write a program to subtract the given two two-dimensional array of same dimensions.
- Write a program to multiply the given two two-dimensional array of same dimensions.
- Write a program to sort each row of a matrix.
- Write a program to find the sum of the elements in 'Z' sequence of the given 2D array.
- Write a program to print the unique rows of the given two-dimensional array consisting of only 0's and 1's.
- Write a program to print the unique columns of the given two-dimensional array consisting of only 0's and 1's.

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

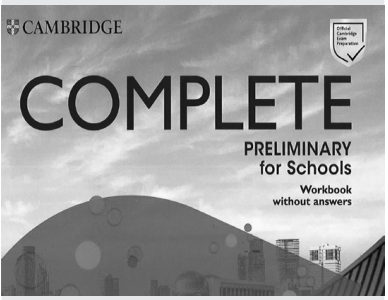
CO No.	Course Outcomes	Blooms Level	Module No.	Mapping with POs
1	Identify suitable data type for operands and design of expressions having right precedence.	Apply	1,2	1
2	Apply decision making and iterative features of C Programming language effectively.	Apply	1,2	1
3	Select problem specific data structures and suitable accessing methods	Analyze	1,2	1,2
4	Design and develop non- recursive and recursive functions and their usage to build large modular programs and also able to design string manipulation functions.	Create	1,2	3
5	Develop C programs that are understandable, debuggable, maintainable and more likely to work correctly in the first attempt.	Evaluate	1,2	3,4

TEXT BOOKS:

1. Behrouz A. Forouzan, Richard F.Gilberg, "Programming for Problem Solving", 1st edition, Cengage publications, 2019.
2. Ajay Mittal, "Programming in C - A Practical Approach", 1st edition, Pearson Education, India, 2010.

REFERENCE BOOKS:

1. Reema Thareja, "Computer Fundamentals and Programming in C", 1st edition, Oxford University Press, India, 2013.
2. Herbert Schildt, "C: The Complete Reference", 4th edition, Tata McGraw-Hill, 2017.
3. R G Dromey and Pearson, "How to solve it by Computer", 2nd Edition, 1998.



Source: https://www.scribd.com/document/502301821/Cambridge-Complete-B1-Preliminary-for-Schools-Workbook-2020-Edition

22EN102 ENGLISH PROFICIENCY AND COMMUNICATION SKILLS

Hours Per Week :

L	T	P	C
0	0	2	1

PREREQUISITE KNOWLEDGE: Basics of grammar, Read and understand for global context, Cultural sensitivity and Basic writing skills.

COURSE DESCRIPTION AND OBJECTIVES:

English Proficiency and Communication Skills seeks to develop the students' abilities in grammar, speaking, reading, writing and overall comprehension skills. The course will provide students an exposure on a wide range of language use in everyday situations. It will make the students to equip with functional English and make them use it confidently in their professional and social contexts. Finally, students will strengthen their reading, writing, listening and speaking skills in English

MODULE-1

UNIT-1 **0L+0T+8P=8 Hours**

MY LIFE AND HOME - MAKING CHOICES - HAVING FUN

Reading: Understanding main message, factual information global meaning, specific information and paraphrasing.

Writing: Developing hints based mail, Writing short messages/paragraphs.

Listening: Understanding short monologues or dialogues and choose the correct visual.

Speaking: Express simple opinions /cultural matters in a limited way.

Vocabulary: Discerning use of right word suiting the context, B1 Preliminary word list.

Grammar: Frequency Adverbs, State Verbs, AFV and Prepositions.

UNIT-2 **0L+0T+8P=8 Hours**

ON HOLIDAY - DIFFERENT FEELINGS – THAT'S ENTERTAINMENT!

Reading: Longer text for detailed comprehension, gist and inference.

Writing: Developing notes and responding to penfriends or 'e-pals'.

Listening: Understand straightforward instructions or public announcements.

Speaking: Describing people, things and places in a photograph.

Vocabulary/Grammar: Comparatives and Superlatives, Gradable and non-gradable adjectives, Cloze tests.

PRACTICES:

- Developing hints based mail.
- Writing short message.
- Writing paragraphs.
- Expressing opinions and cultural matters.
- Understanding short monologues.
- Understanding straightforward instructions and public announcements.
- Describing people, things and places in a photograph.

MODULE-2

UNIT-1 **0L+0T+8P=8 Hours**

GETTING AROUND – INFLUENCES - STAY FIT AND HEALTHY

Reading: Reading for understanding coherence of the text and drawing inferences.

Writing: Reading an announcement from a magazine or website for preparing an article.

Listening: Discussion activities and listening to understand the gist of each short dialogue.

Speaking: Snap Talks, Make and respond to suggestions, discuss alternatives and negotiate agreement.

Vocabulary / Grammar: Punctuation, Prepositions, Phrasal Verbs, B1 Preliminary word list.

UNIT-2 **0L+0T+8P=8 Hours**

LOOKS AMAZING! - THE NATURAL WORLD - EXPRESS YOURSELF!

Reading: Content, Communicative Achievement, Organisation and Language.

Writing: Developing a story with clear links to the given opening sentence.

Listening: An interview for a detailed understanding of meaning and to identify attitudes and opinions.

Speaking: Discuss likes, dislikes, experiences, opinions, habits, etc.

Vocabulary/Grammar: Modals, Conditionals, Verb forms (Time and Tense).

PRACTICES:

- Listening to understand the gist of each short dialogue.
- Listening to an interview for a detailed understanding of meaning and to identify attitudes and opinions.
- Preparing an article.
- Discuss for alternatives and negotiate agreement.
- Discussion on likes, dislikes, experiences, opinions, habits, etc.

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Module No.	Mapping with POs
1	Apply to read and grasp content on a range of topics/texts related to their everyday life like notifications, advertisements, travel brochures, news reports, articles.	Apply	1	7, 8, 9, 10, 12
2	Apply suitable strategies to achieve comprehension, like listening for main points and checking comprehension using contextual clues etc.	Apply	1	7, 8, 9, 10, 12
3	Use functional English to communicate and interact effectively in everyday situations.	Apply	1, 2	7, 8, 9, 10, 12
4	Demonstrate vocabulary beyond that of the familiar subjects.	Analyze	1, 2	7, 8, 9, 10, 12
5	Show sufficient control of English grammar and sentence variety to coherently organise information at sentence and discourse levels.	Evaluate	2	7, 8, 9, 10, 12

TEXT BOOK:

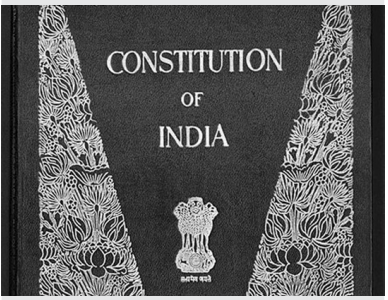
1. Emma Heyderman and Peter May, “Complete Preliminary”, Student’s Book with Answers, 2nd edition, Cambridge University Press, 2019.

REFERENCE BOOKS:

1. Annette Capel and Rosemary Nixon, “Introduction to PET”, Oxford University Press, 2009.
2. Adrian Doff and Craig Thaine, “Empower Pre intermediate”, Cambridge University Press, 2015.
3. Louise Hashemi and Barbara Thomas, “Objective PET”, Cambridge University Press, 2010.

SKILLS:

- ✓ *Use of appropriate grammar and vocabulary with syntactic patterns in short texts.*
- ✓ *Read and extract the main message, global meaning, specific information, detailed comprehension, understanding of attitude, opinion and writer purpose and inference.*
- ✓ *Listen to understand key information, specific information, gist and detailed meaning and to interpret meaning.*
- ✓ *Understand questions and make appropriate responses and talk freely on everyday topics*



Source: https://commons.wikimedia.org/wiki/File:Constitution_india.jpg

22TP101 CONSTITUTION OF INDIA

Hours Per Week :

L	T	P	C
0	2	0	1

PREREQUISITE KNOWLEDGE: High School-level Civics and Social Studies.

COURSE DESCRIPTION AND OBJECTIVES:

To provide students with a basic understanding of Indian Polity and Constitution and make students understand the functioning of government at the center and state level besides local self-government. This course also equips students with knowledge pertaining to fundamental rights and fundamental duties of a citizen in a democracy such as India.

MODULE-1

UNIT-1 **0L+8T+0P=8 Hours**

HISTORICAL BACKGROUND TO THE INDIAN CONSTITUTION

Meaning of the constitution law and constitutionalism; Historical perspective of the Constitution of India; Salient features and characteristics of the Constitution of India.

UNIT-2 **0L+8T+0P=8 Hours**

FUNDAMENTAL RIGHTS, DUTIES, DIRECTIVE PRINCIPLES, AND AMENDMENT

Scheme of the fundamental rights - scheme of the Fundamental Right to Equality; scheme of the Fundamental Right to certain Freedom under Article 19; scope of the Right to Life and Personal Liberty under Article 21; Scheme of the Fundamental Duties and its legal status; Directive Principles of State Policy – its importance and implementation; Amendment of the Constitution - Powers and Procedure.

PRACTICES:

- Enactment of Constituent Assembly debates to further understand the rationale for the provisions of the constitution.
- Fundamental Rights in our popular culture - discussion in the movie Jai Bhim.

MODULE-2

UNIT-1 **0L+8T+0P=8 H Hours**

STRUCTURE AND FORM OF GOVERNMENT

Federal structure and distribution of legislative and financial powers between the Union and the States; Parliamentary Form of Government in India – The constitution powers and status of the President of India; Emergency Provisions: National Emergency, President Rule, Financial Emergency.

UNIT-2 **0L+8T+0P=8 Hours**

LOCAL SELF GOVERNMENT

Local Self Government – Constitutional Scheme in India – 73rd and 74th Amendments.

PRACTICES:

- Debate on federalism in India.
- Collect news published in the local papers about panchayats in the nearby areas.

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Module No.	Mapping with POs
1	Analyse major articles and provisions of the Indian constitution.	Analyze	1	6
2	Appreciation for the constitution and safeguarding individual rights.	Apply	1	6
3	Evaluating functions of various organs of the State in a democracy.	Evaluate	2	6

TEXT BOOK:

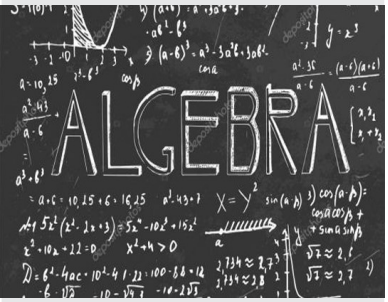
1. PM Bhakshi, “Constitution of India”, 15th edition, Universal Law Publishing, 2018.

REFERENCE BOOKS:

1. B. R. Ambedkar, “The Constitution of India” Educreation Publishing, India, 2020.
2. Subhash Kashyap, “Our Constitution” 2nd edition, National Book Trust, India, 2011.
3. Arun K. Thiruvengadam, “The Constitution of India: A Contextual Analysis”, Hart Publishing India, 2017.

SKILLS:

- ✓ *Knowledge the basics of the Indian constitution.*
- ✓ *Know the fundamental rights, fundamental duties, and Directive Principles of State Policy.*
- ✓ *Fair knowledge about the functioning of various institutions in a democracy*



Source: https://pll.harvard.edu/course/college-algebra.

22MT106 ALGEBRA

Hours Per Week :

L	T	P	C
3	2	0	4

PREREQUISITE KNOWLEDGE: Basics of sets, Relations and Functions.

COURSE DESCRIPTION AND OBJECTIVES:

This course emphasizes on motivation and justification for the algorithmic usage of group theory in different domains. The objective of this course is to introduce the concepts of Groups, Rings, Integral domains and Fields. Develop the ability to form and evaluate group theory and its actions. Understand the fundamental concepts of algebra. The fundamental notions viz. linear dependence, basis and dimension and linear transformations on these spaces have to be studied thoroughly.

MODULE-1

UNIT-1 **12L+8T+0P=20 Hours**

GROUP THEORY

Algebraic structures with binary operations, Semigroup, Monoid, Group, Subgroup, Cosets, Lagrange's theorem, Normal subgroup, Quotient group.

UNIT-2 **12L+8T+0P=20 Hours**

APPLICATIONS OF GROUP THEORY

Introduction to Rings, Integral Domains, Fields with examples.

Properties of groups, order of an element in a group, homomorphism, isomorphism.

PRACTICES:

- List all the properties for group.
- Give examples for groups and other binary structures.
- In a group of even order there is an element a e such that a2 = e.
- For any two subgroups discuss the possibility of their intersection and union being a subgroup.
- Any two groups of order 6 are isomorphic, Verify.

MODULE-2

UNIT-1 **12L+8T+0P=20 Hours**

VECTOR SPACES

Vector space, Subspace, linear span, linearly independent and dependent vectors, Bases, Dimension, Linear transformations, Inner product spaces.

UNIT-2 **12L+8T+0P=20 Hours**

APPLICATIONS OF VECTOR SPACES

Matrix of Linear Transformation, Change of Coordinates, Rank and Nullity, Orthogonality, Cauchy's Schwartz Inequality, Gram Schmidt Orthogonalization.

PRACTICES:

- Examine whether or not a given algebraic structure is a vectorspace.
- Verify whether a given set forms a basis or not of R3.
- Testing orthogonality of given set of vectors.
- Finding Rank and Nullity of linear transformation.

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Module No.	Mapping with POs
1	Apply the concepts of cosets to study properties of subgroups.	Apply	1	1, 2, 9, 10, 12
2	Outline the various properties and apply group actions critically.	Apply	1	1, 2, 9, 10, 12
3	Understand and apply the concepts of vector spaces, subspaces, bases, dimension and their properties.	Apply	2	1, 2, 9, 10, 12
4	Analyse inner product spaces for their orthogonality.	Analyse	2	1, 2, 9, 10, 12

TEXT BOOKS:

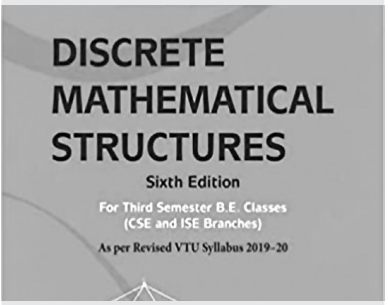
1. Tremblay, J.P. and Manohar. R, "Discrete Mathematical Structures with Applications to Computer Science", 30th Reprint, Tata McGraw Hill Pub. Co. Ltd, New Delhi, 2017.
2. Rosen, K.H., "Discrete Mathematics and its Applications", 7th Edition, Tata McGraw Hill Pub. Co. Ltd., New Delhi, 2018.

REFERENCE BOOKS:

1. R.P. Grimaldi, "Discrete and Combinatorial Mathematics: An Applied Introduction", 4th Edition, Pearson Education Asia, Delhi, 2017.
2. S. Lipschutz and Mark Lipson., "Discrete Mathematics", Schaum's Outlines, Tata McGraw Hill Pub. Co. Ltd., New Delhi, 2021.
3. T. Koshy, "Discrete Mathematics with Applications", Elsevier Publications, 2015.
4. S B Singh. "Discrete Structures", Khanna Book Publishers Co-Pvt. Ltd. 2019

SKILLS:

- ✓ *Identifying identity elements of an Algebraic structure and inverses of elements.*
- ✓ *Evaluate the rank and nullity of a Linear Transformation.*



Source: https://www.amazon.in/Discrete-Mathematical-Structures-Dr-D-S-C/dp/9388478398

22MT107 DISCRETE MATHEMATICAL STRUCTURES

Hours Per Week :

L	T	P	C
2	2	0	3

PREREQUISITE KNOWLEDGE: Basic of set theory, Algebra.

COURSE DESCRIPTION AND OBJECTIVES:

The course objective is to provide students with an overview of discrete mathematics. Students will learn about of group and expected to demonstrate analytical and combinatorial methods such as propositional logic , Mathematical Induction, Boolean functions, combinatorial ,recurrence relation, generating function and graph theory

MODULE-1

UNIT-1 **8L+8T+0P=16 Hours**

BOOLEAN ALGEBRA AND LOGIC

Group: Group, Subgroup, Lagrange's theorem, Introduction to rings and fields.

Boolean algebra: Boolean algebra, Truth table, Basic logic gates, Postulates of Boolean algebra, Principle of duality, Propositions, Connectives, Equivalence and Normal form.

UNIT-2 **8L+8T+0P=16 Hours**

APPLICATIONS OF LOGIC & BOOLEAN ALGEBRA

Normal form: CNF, DNF, CDNF, CCNF, Conversion of CNF to DNF and vice versa. Minimization of Boolean function (Karnaugh Maps), Generalized Pigeonhole Principle.

PRACTICES:

- Determine order of an element of group.
- Determine the order of a subgroup of a finite group.
- Determine whether algebraic structure is a Ring or a Field.
- Construct Truth table of propositions.
- Check whether propositions are equivalence.
- Obtain CNF, DNF of expression.
- Draw 2 – variables, 3-variables K-map.
- Minimize the Boolean function by K-map.
- Simplify the Boolean expression using Boolean algebra laws.
- Determine the homogeneous solution and particular solution for recurrence relation.
- Switching Circuit in Boolean algebra, Combination of two switches in a Circuit

MODULE-2

UNIT-1 **8L+8T+0P=16 Hours**

COMBINATORICS AND GRAPHS

Combinatorics: The basics of counting, Permutations and combinations, Discrete Numeric Functions. Recurrence relations and Generating functions.

Graph theory: Graph terminology, Special types of graphs, Connected graph, Weighed graph, Graph Isomorphism, Euler and Hamiltonian paths and circuits, Planar graphs, Bipartite graph, Tree.

UNIT-2 **8L+8T+0P=16 Hours**

APPLICATIONS OF COMBINATORICS AND GRAPHS

Graph Coloring, Chromatic number, Matrix representation of graph, DFS, BFS algorithms, Minimum spanning tree.

PRACTICES:

- Represent sequence by Generating function.
- Solve Generation function by recurrence relation.
- Find degree of digraph and undirected graph.
- Identify Isomorphic graph, Euler circuit, Hamiltonian circuit.
- Determine matrix representation of graph.
- Determine the regions of planer connected graph.
- Draw Kn, Km,n.
- Determine the chromatic number of graphs.
- Find the weight of minimum spanning tree.
- Find number of arrangements that are possible.

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Module No.	Mapping with POs
1	Apply the concepts of mathematical logic and Boolean algebra.	Apply	1	1, 2, 9, 10, 12
2	Apply Karnaugh map to minimize.	Apply	1	1, 2, 9, 10, 12
3	Solve generating function by recurrence relations.	Apply	2	1, 2, 9, 10, 12
4	Model and solve real world problems using graphs and trees.	Apply	2	1, 2, 9, 10, 12

TEXT BOOKS:

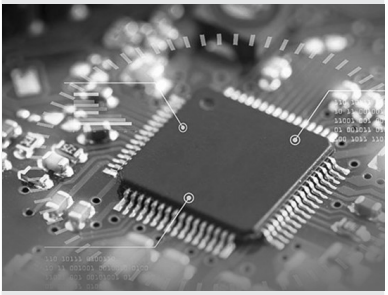
1. Tremblay, J.P. and Manohar. R, “Discrete Mathematical Structures with Applications to Computer Science”, 30th Reprint, Tata McGraw Hill Pub. Co. Ltd, New Delhi, 2017.
2. Rosen, K.H., “Discrete Mathematics and its Applications”, 7th Edition, Tata McGraw Hill Pub. Co. Ltd., New Delhi, 2018.

REFERENCE BOOKS:

1. R.P. Grimaldi, “Discrete and Combinatorial Mathematics: An Applied Introduction”, 4th Edition, Pearson Education Asia, Delhi, 2017.
2. S. Lipschutz and Mark Lipson., “Discrete Mathematics”, Schaum‘s Outlines, Tata McGraw Hill Pub. Co. Ltd., New Delhi, 2021.
3. T. Koshy, “Discrete Mathematics with Applications”, Elsevier Publications, 2015.
4. S B Singh. “Discrete Structures”, Khanna Book Publishers Co-Pvt. Ltd. 2019.

SKILLS:

- ✓ *Familiarity of concepts of statements, logic and truth tables.*
- ✓ *Analyze closed form of discrete numeric function.*
- ✓ *Know some basic properties of graphs, trees and related discrete structures.*



Source: https://www.scitusacademics.com/product/semiconductor-devices-and-circuits/

22PY105 SEMICONDUCTOR PHYSICS AND ELECTROMAGNETICS

Hours Per Week :

L	T	P	C
2	0	2	3

PREREQUISITE KNOWLEDGE: Basics of vectors and semiconductors.

COURSE DESCRIPTION AND OBJECTIVES:

This course ensures commensurable understanding of electrostatics and magnetostatics. It enunciates the electron dynamics in solids through the conceptual grasp of principles of quantum mechanics. This embark perspective outlook on optoelectronic devices and optical fibres in the backdrop of semiconductor physics.

MODULE-1

UNIT-1 **10L+0T+10P=20 Hours**

ELECTROSTASTICS AND MAGNETOSTASTICS

Electrostatics: Introduction to Vector analysis, Computation of electric field and potential due to Point charge, linear charge density, surface charge density, bulk charge density, Coulomb's law, Electric field due to line of charges, Gauss law, Differential Form of Gauss law, Applications, Electric field due to a charged sphere – inside, on the surface, and outside, Electric field due to a spherical shell- inside and outside.

Magnetostatics: Introduction to magnetic force – Lorentz force, Biot-Savart's law, Magnetic field due to a linear conductor – magnetic field due to a circular loop – Ampere's law, Faraday's law in integral form; Lenz's law, Maxwell's equations – correction to Ampere's law.

UNIT-2 **6L+0T+6P=12 Hours**

QUANTUM MECAHNICS AND FREE ELECTRON THEROY

Quantum mechanics: Introduction to Quantum mechanics; Concepts of wave and particle duality of radiation; de Broglie's concepts of matter waves, Schrödinger's time-independent wave equation – Eigen values and Eigen functions; Particle confined in a one-dimensional infinite Potential square well.

Free electron theory of solids: Classical and Quantum free electron theory of metals; Fermi- Dirac distribution; Density of states – derivation -Bloch's Theorem (Qualitative); Classification of solids based on energy bands.

PRACTICES:

- Photoelectric effect-Determination of plancks constant.
- Stewart & Gee's Experiment- Study of magnetic field along the axis of a current carrying coil.
- Melde's Experiment - determination of the frequency of tuning fork.
- Sonometer- Determination of AC frequency.

MODULE-2

UNIT-1 **8L+0T+8P=16 Hours**

SEMICONDUCTOR PHYSICS AND OPTOELECTRONICS

Introduction, Classification of Semiconductors, Direct and indirect band gap semiconductors, Intrinsic semiconductors; Variation of Intrinsic carrier concentration with temperature, Fermi level, and conductivity; Extrinsic semiconductor, the effect of temperature on carrier concentration in extrinsic semiconductors, Band diagrams of extrinsic semiconductors; Hall effect, Classification of optoelectronic devices; Photo voltaic cell, LED.

UNIT-2 **8L+0T+8P=16 Hours**

SKILLS:

- ✓ *Able to compute the electric and magnetic field and potentials in different applications*
- ✓ *Apply the quantum laws to understand the electron dynamics of solids*
- ✓ *Realizing the importance of optoelectronic devices*

LASERS AND OPTICAL FIBERS

Introduction to lasers, Population inversion & pumping processes, Semiconductor diode laser, Applications of lasers. Optical fiber-Numerical Aperture, types of optical fibres, Fiber optic communication system.

PRACTICES:

- Laser - Determination of wavelength.
- Optical fiber - Determination of Numerical aperture – Acceptance angle.
- Determination of Energy Band gap of p-n junction diode.
- Hall Effect - Determination of Hall coefficient.
- Solar cell – Determination of Fill factor & efficiency.

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Module No.	Mapping with POs
1	Apply Maxwell's equations to unravel electron dynamics in amidst of electric and magnetic fields.	Apply	1	1, 2, 4, 5, 9, 10
2	Discriminate solids based on principles of quantum mechanics.	Analyse	1	1, 2, 3, 4, 9, 10
3	Assessment of semiconductors in the perspective of optoelectronic devices.	Evaluate	2	1, 3, 4, 5, 6, 9, 10
4	Comprehend the knowledge of Lasers and optical fibers to conceive their applications in vivid domains.	Apply	2	1, 2, 3, 5, 9, 10

TEXT BOOKS:

1. S.O. Pillai, "Solid State Physics", New age International publishers, 8th edition, 2018.
2. H.C. Varma, "Classical Electromagnetism", Bharathi Bhavan Publication, 2022.

REFERENCE BOOKS:

1. D. Halliday, R. Resnick and J. Walker, "Fundamentals of Physics", 6th edition, John Wiley and Sons, New York, 2001.
2. M.N. Avadhanulu, "Engineering Physics", S. Chand publications 2010.
3. Charles Kittel, "Introduction to Solid State Physics", 7th edition, Wiley, Delhi, 2007.
4. Donald A. Neamen, "Semiconductor Physics and Devices: Basic Principle", 4th edition, McGraw-Hill, New York, 2012.
5. David J. Griffiths, "Introduction to Electrodynamics", 3rd edition, Prentice Hall of India, New Delhi, 2012.
6. N.W. Ashcroft and N.D. Mermin, "Solid State Physics", International student edition, Brooks Cole, 2008.



Source: www.geeksforgeeks.org/best-way-to-start-with-competitive-programming-geeksforgeeks-cp-live-course/

22TP104 BASIC CODING COMPETENCY

Hours Per Week :

L	T	P	C
0	1	3	2

PREREQUISITE KNOWLEDGE: Programming in C.

COURSE DESCRIPTION AND OBJECTIVES:

This course is aimed to impart knowledge on advanced concepts of C programming language and problem solving. At the end of this course, students will be able to design, implement, test and debug complex problems using features of C.

MODULE-1

UNIT-1 **0L+4T+12P=16 Hours**

NUMBER CRUNCHING

PRACTICES:

Problems On Number Crunching

- Write a program to check if a given number is perfect or not.
- Write a program to check if a given number is deficient or not.
- Write a program to check if 2 given numbers are amicable or not.
- Write a program to check if 2 given numbers are betrothed or not.
- Write a program to check whether a given number is an Armstrong number or not.
- Write a program to print the series of prime numbers in the given range.
- Write a program to print all the perfect numbers in a given range.
- Write a program to generate all deficient numbers in a given range.
- Write a program to generate all the amicable numbers in a given range.
- Write a program to generate all the betrothed numbers in a given range.
- Write a program to find the largest prime factor of a given number.
- Write a program to check whether the given number is a palindrome or not.
- Write a program to calculate sum of the individual digits for the given number.
- Write a program to find the first number that has more than ‘n’ factors, excluding 1 and that number.
- Write a program to accept a number as input and print its factorial.
- Write a program to accept a number n, print first N Fibonacci numbers.
- Write a program to check if an input number is Armstrong number or not.
- Write a program that takes input a,b. Print a power b.
- Write a program that takes input a number n, check if it a perfect square or not.
- Print array in spiral format.
- Print sum of each row in a matrix.
- Print sum of each column in matrix.
- Print left->right and right->left diagonals in a matrix.
- Initially you are at (0,0) find the shortest path count to reach the (n, n) block in matrix.
- Remove all the elements present in row and column of unsafe elements. An element is called unsafe if it is equal to smallest or largest value. Count number of remaining elements.
- Write a program to check if the string contains all the letters of alphabet.
- Check if a string is matching password requirements.

- Check if String A contains String B (String searching).
- Check if a number is harshad number or not.
- Write a program to get 3 numbers as input. The first is the number num1 and second is the digit that needs to be replaced. The third is the digit that is to replace the 2nd digit. Print the number after performing this operation.
- Write a program to accept a number and swap its alternate digits. Print the number generated.
- Write a program to accept a number and choice as input. If the choice is 0 rearrange the number such that the odd digits are ordered first followed by the even digits. If the choice is 1 rearrange the number such that the even digits are ordered first followed by the odd digits. Print the rearranged number. The order of occurrence of the digits is to be preserved.
- Write a program to determine that whether the given quadrilateral is cyclic or not. You are given the sizes of angles of a simple quadrilateral (in degrees) A, B, C and D, in some order along its perimeter.

  Note: A quadrilateral is cyclic if and only if the sum of opposite angles is 180180^0.
- Chef is a very lazy person. Whatever work is supposed to be finished in x units of time, he finishes it in max units of time. But there is always a limit to laziness, so he delays the work by at max d units of time. Given x,m,d, find the maximum time taken by Chef to complete the work.
- Suppose Chef is stuck on an island and currently he has x units of food supply and y units of water supply in total that he could collect from the island. He needs xr units of food supply and yr units of water supply per day at the minimal to have sufficient energy to build a boat from the woods and also to live for another day. Assuming it takes exactly D days to build the boat and reach the shore, tell whether Chef has the sufficient amount of supplies to be able to reach the shore by building the boat? Read five integers x,y,xr,yr,D from the user and display “YES” if Chef can reach the shore by building the boat and “NO” if not (without quotes).
- There are 3 problems in a contest namely A,B,C respectively. Alice bets Bob that problem C is the hardest while Bob says that problem B will be the hardest.

  You are given three integers SA,SB,SC which denotes the number of successful submissions of the problems A,B,C respectively. It is guaranteed that each problem has a different number of submissions. Determine who wins the bet.

  1) If Alice wins the bet (i.e. problem C is the hardest), then output Alice.

  2) If Bob wins the bet (i.e. problem B is the hardest), then output Bob.

  3) If no one wins the bet (i.e. problem A is the hardest), then output Draw.

  Note: The hardest problem is the problem with the least number of successful submissions.

  Input Format
- The first line of input contains a single integer T denoting the number of test cases. The description of T test cases follows.
- The first and only line of each test case contains three space-separated integers SA,SB,SC, denoting the number of successful submissions of problems A,B,C respectively.

  Output Format

  For each test case, output the winner of the bet or print Draw in case no one wins the bet.

  Sample Input 1

  3

  1 4 2

  16 8 10

  14 15 9

  Sample Output 1

  Draw

  Bob

  Alice
- In a season, each player has three statistics: runs, wickets, and catches. Given the season stats of two players A and B, denoted by R, W, and C respectively, the person who is better

SKILLS:

- ✓ *Analysis of the problem to be solved.*
- ✓ *Application of various file operations effectively in solving real world problems.*
- ✓ *Develop C programs that are understandable, debuggable, maintainable and more likely to work correctly in the first attempt.*

- than the other in the most statistics is regarded as the better overall player. Tell who is better amongst A and B. It is known that in each statistic, the players have different values.

  Input

  The first line contains an integer T, the number of test cases. Then the test cases follow.

  Each test case contains two lines of input.

  The first line contains three integers R1, W1, C1, the stats for player A.

  The second line contains three integers R2, W2, C2, the stats for player B.

  Output

  For each test case, output in a single line “A” (without quotes) if player A is better than player B and “B” (without quotes) otherwise.

- Write a program to find the direction.

  Chef is currently facing the north direction. Each second he rotates exactly 90 degrees in clockwise direction. Find the direction in which Chef is facing after exactly X seconds.

  Note: There are only 4 directions: North, East, South, West (in clockwise order). Initially chef is at 0th second i.e., facing North direction.

  Input Format

- First line will contain T, number of testcases. Then the testcases follow.
- Each testcase contains of a single integer X.

  Output Format

  For each testcase, output the direction in which Chef is facing after exactly X seconds.

  Sample Input 1

  3

  1

  3

  6

  Sample Output 1

  East

  West

  South

- Chef is playing in a T20 cricket match. In a match, Team A plays for 20 overs. In a single over, the team gets to play 6 times, and in each of these 6 tries, they can score a maximum of 6 runs. After Team A’s 20 overs are finished, Team B similarly plays for 20 overs and tries to get a higher total score than the first team. The team with the higher total score at the end wins the match.

  Chef is in Team B. Team A has already played their 20 overs, and have gotten a score of R. Chef’s Team B has started playing, and have already scored C runs in the first O overs. In the remaining 20−O overs, find whether it is possible for Chef’s Team B to get a score high enough to win the game. That is, can their final score be strictly larger than R?

  Input: There is a single line of input, with three integers, R, O, C.

  Output: Output in a single line, the answer, which should be “YES” if it’s possible for Chef’s Team B to win the match and “NO” if not.

- Make Array Zeros using pointers

  You are given an array A of length N (size should be created using Dynamic memory allocation) and can perform the following operation on the array:

  Select a subarray from array A having the same value of elements and decrease the value of all the elements in that subarray by any positive integer x.

  Find the least possible number of operations required to make all the elements of array A equal to zero.

  The first line contains an integer N denoting the number of elements in the array.

  The next line contains space-separated integers denoting the elements of array A.

Print the least possible number of operations required to make all the elements of array A equal to zero.

Sample Test case

Input:

5

2 2 1 3 1

Output:

4

UNIT-2 **0L+4T+12P=16 Hours**

PATTERNS

PRACTICES:

Problems on Number Patterns

- Write a program to generate Floyd triangle. Sample input N= 4.
  1
  2 3
  4 5 6
  7 8 9 10
- Write a program to generate the following pattern. Sample input N=5.
  13579
  3579
  579
  79
  9
- Write a program to generate the following pattern. Sample input N=4.
  1111111
  222222
  33333
  4444
  333
  22
  1
- Write a program to generate the following pattern. Sample input N=5.
  5432*
  543*1
  54*21
  5*321
  *4321
- Write a program to generate the following pattern. Sample input N=5.
  12 21
  123 321
  1234 4321
  123454321
- Write a program to generate the following pattern. Sample input N=5.
  1
  2*2

3*3*3
4*4*4*4
4*4*4*4
3*3*3
2*2
1

- Write a program to generate the following pattern. Sample input N=4.
  1
  212
  32123
  4321234
- Write a program to generate the following pattern. Sample input N=5.
  *
  * *
  * * * *
  * *
  *
- Write a program to print Pascal triangle for the given number of rows. Sample input N=5.
  1
  1 1
  1 2 1
  1 3 3 1
  1 4 6 4 1
- Write a program to generate the following pattern. Sample input N=4.
  1234
  2341
  3421
  4321
- Print Hollow Diamond pattern.
- Print pascals triangle.
- Print Floyds triangle.
- Print Butterfly Pattern.
- Print palindromic pattern.
- Print full inverted number triangle.
- Check if a number is prime or not (Efficient Approach).
- Find sum of all the digits of the number.
- Print transpose of given matrix.
- Rotate a two dimensional matrix by 90, 180, 270 degrees.

MODULE-2

UNIT-1 **0L+4T+12P=16 Hours**

ARRAYS

PRACTICES:

Problems On Arrays

- Given an unsorted array of size N, and the array elements are in the range of 1 to N. There are no duplicates, and the array is not sorted. One of the integers is missing. Write a program to find the missing number.

- Given an array consisting of only 0s and 1s in random order rearrange the array such that all the 0s are to the left of the array and 1s to the right.
- Give an array consisting of odd and even numbers in random order, rearrange the array such that all the odd numbers are to the left of the array and even numbers are to the right of the array.
- Write a program to find all the unique elements in an array.
- Write a program to merge two arrays of the same size sorted in descending order.
- Write a program to count the frequency of each element in an array of integers.
- Write a program to find the second largest element in an array.
- Write a program to find the second smallest element in an array.
- Write a program to find that one element in array that occurs odd number of times, where every other element appears even number of times.
- Create a jagged array (adjacency list representation of a graph) with no of rows and no of columns in each row as specified by the user.

  Hint: Use Dynamic memory allocation (malloc() or calloc())

  Input:

  Enter no of rows: 3

  Enter no of columns Row in 1: 3

  Enter no of columns Row in 2: 5

  Enter no of columns Row in 3: 2

  Enter the elements row wise:

  8 6 5

  8 4 6 9 7

  9 2

  Output:

  8 6 5

  8 4 6 9 7

  9 2
- Write a program to find second largest number in the array.
- Write a program to find first repeating element in the array.
- Write a program to left rotate the array.
- Write a program to right rotate the array.
- Write a program to find the largest continuous sum.
- Write a program to print the sum of 2nd largest and 2nd smallest elements.
- Write a program to find the maximum product of two numbers multiplies in array (same index should not be used twice).
- Rearrange an array consisting of 1s and 0s such that they are alternatively arranged. Print minimum number of moves required.
- In a given array, find two numbers whose sum equal k.
- Find the difference between positive and negative elements in the array.
- Implement sorting algorithms (Insertion, selection, bubble).

UNIT-2 **0L+4T+12P=16 Hours**

STRINGS

PRACTICES:

Problems on Strings:

- Write a program to reverse a given string word by word.
- Write a program to find the first occurrence of non-repeating character in the given string.
- Write a program to compress the string as provided in the example.
- Write a program to expand a string as provided in the example.

- Write a program to reverse those words of a string whose length is odd.
- Write a program to check if a given matrix is symmetric or not.
- Write a program to convert all the cases of letter (Lower case -> Upper Case, Upper Case-> Lower Case).
- Write a program to reverse all the words (Not the entire sentence but individual words).
- Find the longest palindrome in a given string.
- Check if two strings are anagrams or not.
- Find minimum number of changes to be done to make a string palindrome.
- Convert Excel sheet name to number (A-1, B-2, Z-26, AA-27).
- Find number of possible palindromes present in a string.
- Write a C program to read a string s, and determine the number of words in s.
  Example: s=oneTwoThree

  There are 3 words in the string: 'one', 'Two', 'Three'.
- Write a C program that reads a string S and remove all duplicates characters from the given string S.
  NOTE: 1) Order of characters in output string should be same as given in input string.

  2) String S contains only lowercase characters ['a'-'z'].

  Example: S = Vignanuniversity

  The program should generate the output as: Vignauersty
- Today Ron is reading the book. Due to some reason, he started hating the word 'are' (without quotes). So he decided to replace the substring 'are' with 'R'. Write a C program that reads a line of message 's' and replace the substring 'are' with 'R'. Example: s= Howareyou.
  The program should generate the output as: HowRyou
- Write a program to concatenate the characters of the two given strings alternatively.
- Given a string S consisting of uppercase and lowercase letters, change the case of each alphabet in this string. That is, all the uppercase letters should be converted to lowercase and all the lowercase letters should be converted to uppercase.
  Input: Vignan University

  Output: vIGNAN uNIVERSITY
- Write a program to insert a given character at the beginning and end of the given string.
- Given two Strings A and B. They are said to be friends if ASCII sum of the each individual string is divisible by 4 else they are not friends. You need to find whether given two strings are friends or not.
  Sample Test case:

  Input:

  man nam

  vignan university

  Output:

  YES

  NO
- Write a program to find the frequency of each digit in the given string.
  Input Format

  The first line contains a string, which is the given number.

  Output Format

  Print ten space-separated integers in a single line denoting the frequency of each digit, indicate that the integers are from 0 to 9.

  Sample Input 0

  a11472o5t6

  Sample Output 0

0 2 1 0 1 1 1 1 0 0

Explanation 0

In the given string:

1 occurs two times.

2,4,5,6 and 7 occur one time each.

The remaining digits and don't occur at all.

- Sherlock considers a string to be valid if all characters in the given string appear the same number of times. It is also valid if he can remove just 1 character at 1 index in the string, and the remaining characters will occur the same number of times.

  Write a C program that reads a string s and determine whether it is valid or not. If valid, return YES, otherwise return NO.

  Example: S=abc

  This is a valid string because frequencies are {a:1,b:1,c:1}

  S=abcc

  This is a valid string because we can remove one c and have 1 of each character in the remaining string.

  S=abccc

  This string is not valid as we can only remove 1 occurrence of c. That leaves character frequencies of {a:1,b:1,c:2}

- Read a string containing characters A and B only. Your task is to change it into a string such that there are no matching adjacent characters. To do this, you are allowed to delete zero or more characters in the string.

  Write a C program that finds the minimum number of deletions required.

  Example: S=AABAAB

  Remove A at positions 0 and 3 to make S=ABABA in 2 deletions.

  Input Format

  The first line contains an integer (the number of queries).

  The next q lines each contain a string s to analyze.

  Sample Input:

  5

  AAAA

  BBBBB

  ABABABAB

  BABABA

  AAABBB

  Sample Output:

  3

  4

  0

  0

  4

- Write a C program that reads a string 's' and it is said to be complete if it contains all the characters from a to z.

  Input Format

  First line of the input contains the number of strings N. It is followed by N lines each contains a single string.

  Output Format

  For each test case print "YES" if the string is complete, else print "NO"

Constraints 1 <= N <= 10

The length of the string is at max 100 & the string contains only the characters a to z.

- Write a C program that reads two strings and determine whether they share a common substring or not. A substring may be as small as one character.

  Example;

  S1=and

  S2=art

  The common substring in these two strings: a.

  Sample Input

  2

  hello

  world

  hi

  world

  Sample Output

  YES

  NO

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Module No.	Mapping with POs
1	Write simple, but complete, C programs.	Apply	1,2	1
2	Identify suitable data type for operands and design of expressions having right precedence.	Apply	1,2	1
3	Apply decision making and iterative features of C Programming language effectively.	Apply	1,2	1
4	Select problem specific data structures and suitable accessing methods.	Analyse	1,2	1,2
5	Design and develop non- recursive and recursive functions and their usage to build large modular programs and also able to design string manipulation functions.	Create	1,2	3
6	Develop C programs that are understandable, debuggable, maintainable and more likely to work correctly in the first attempt.	Create	1,2	3,4

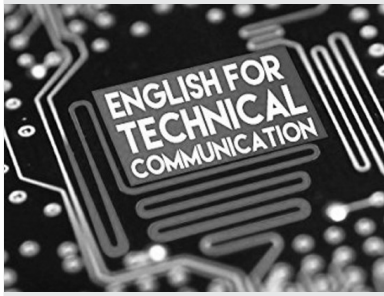
TEXT BOOKS:

1. Behrouz A. Forouzan, Richard F.Gilberg, “Programming for Problem Solving”, 1st edition, Cengage publications, 2019.
2. Ajay Mittal, “Programming in C - A Practical Approach”, 1st edition, Pearson Education, India, 2010.

REFERENCE BOOKS:

1. Reema Thareja, “Computer Fundamentals and Programming in C”, 1st edition, Oxford University Press, India, 2013.
2. Herbert Schildt, “C: The Complete Reference”, 4th edition, Tata McGraw-Hill, 2017.
3. Byron S Gottfried, “Programming with C”, 4th edition, Tata McGraw-Hill, 2018.

22EN104 TECHNICAL ENGLISH COMMUNICATION



Source: https://www.abebooks.com/9781316640081/English-Technical-Communication-Students-Book-1316640086/plp

Hours Per Week :

L	T	P	C
2	0	2	3

PREREQUISITE KNOWLEDGE: Basic sentence formation, understanding contextual meanings, basic writing skills and moderate fluency in English.

COURSE DESCRIPTION AND OBJECTIVES:

In this course students will read, analyze, and interpret material from technical and general fields, and practice reading, writing, listening and speaking skills to gain exposure and functional English on a variety of contemporary topics. The overall course objective is to provide English for Specific Purposes(ESP) instruction to enhance students' reading, writing, listening and speaking skills through a practice in the language. It will aim to build students' confidence and motivation through exposure to academic skills like Note making/taking, Paraphrasing, Summarizing, Report Writing, Making Presentations etc., so as to generate interest in the language from an ESP perspective. Finally, students are expected through the course to gain key strategies and expression for communicating with professionals and non-specialists.

MODULE-1

UNIT-1 **8L+0T+8P=16 Hours**

GENETICS

Reading: Reading for Note Making Sub skills: Reading for global understanding (skimming), specific information (scanning), understanding main ideas and supporting ideas, guessing contextual meanings from the text. -Vocabulary building: commonly used roots, prefixes, and suffixes.

Writing: Note making, organising main points and sub points, numbering and sequencing, suggesting titles, paraphrasing and summarising.

Functional grammar: Common Errors in Articles and Prepositions (Handout).

Listening: Listening for Note Taking: top down and bottom up approach, listening for main ideas and supporting points.

Speaking: Presentation in teams - ideas on the topic summarised, making a PPT, effective introductions and conclusions, logical organisation of content, using appropriate structure and cohesive devices.

UNIT-2 **8L+0T+8P=16 Hours**

ALIENS

Reading: Reading: predicting, skimming, scanning, reading for inference, extrapolative reading.

Vocabulary building: Academic vocabulary from the text: synonyms, antonyms, Words often confused.

Writing: Paragraph writing; writing a topic sentence, supporting sentences, effective introductions and conclusions, use of cohesive devices. Types of Paragraphs: Descriptive, narrative, argumentative and expository.

Functional grammar: Common Errors inVerb forms and Conditional sentences (Handout).

Listening: Listening for identifying parts from a description, listening to and sorting information, listening for specific information.

Speaking: Narrating/Retelling an incident, using suitable cohesive devices/discourse markers Speaking of past and present habits/activities/events - Speaking of future plans.

SKILLS:

- ✓ *Apply different sub skills like skimming, scanning, reading for information, reading for inference etc. to understand different kinds of text.*
- ✓ *Apply different sub skills like top down, bottom up approaches to listening.*
- ✓ *Use functional vocabulary relevant to engineering and technology to express ideas lucidly.*
- ✓ *Use appropriate sentence structure, cohesive devices to construct simple text in regular correspondence like e-mails and letters.*

PRACTICES:

- Note making.
- Summarizing.
- Paragraph Writing.
- Error correction and Restructuring.
- Vocabulary building.
- Listening comprehension.
- Note taking.

MODULE-2

UNIT-1 **8L+0T+8P=16 Hours**

SOCIAL MEDIA – HEALTH AND NUTRITION

Reading: Reading for factual information researching for supporting evidence - skimming, scanning.

Vocabulary building: One-word substitutes.

Writing: Letter Writing- E-mail writing – New age communication – Format, protocol, and style- WhatsApp, Facebook and Twitter Functional grammar: Common Errors in Sub-Verb Agreement and Modals.

Listening: Listening to a Business Presentation: Listening for deducing information, for abstract details and specific details, listening for taking a message.

Speaking: Making a presentation with a PPT on a topic assigned- organising the presentation using appropriate discourse markers - presenting a point of view - Extempore.

UNIT-2 **8L+0T+8P=16 Hours**

FASHION

Reading: Reading for data interpretation and information transfer from graphical aids to text reports (pictograms. tables, graphs, pie charts, flow charts), deducing specific information and general information.

Vocabulary building: business vocabulary, collocations, idioms and phrasal verbs.

Writing: Writing a Report: Drafting general and factual reports - writing an overview - an effective introduction - organising information into paragraphs (Stages of writing: planning /organising /writing / editing /rewriting).

Functional grammar: transformations and miscellaneous common errors.

Listening: Listening to a Ted talk and sorting information - taking notes from a discussion.

Speaking: Group Discussion - prerequisites -generating content - initiating a discussion - expressing one's opinion - leading a discussion - agreeing/ disagreeing to someone's view - cutting into a speech - body language and voice modulation.

PRACTICES:

- E-mail writing.
- Letter writing.
- Report writing.
- Messaging in Social media.
- Extempore.
- Making PPTs.

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Module No.	Mapping with POs
1	Apply a variety of strategies to interpret and comprehend spoken texts/ discourse using contextual clues.	Apply	1	6, 7, 8, 9, 10, 12
2	Apply appropriate reading strategies to interpret content / material related to engineering and technology domain.	Apply	1	6, 7, 8, 9, 10, 12
3	Participate in discussions and make short presentations on general and technical topics.	Apply	1, 2	6, 7, 8, 9, 10, 12
4	Possess an ability to write clearly on topics related to technology and workplace communication.	Analyze	2	6, 7, 8, 9, 10, 12
5	Choose functional language, grammar structures, cohesive devices and skills of organisation to express clearly in speaking.	Evaluate	2	6, 7, 8, 9, 10, 12

LANGUAGE LAB ACTIVITIES

Session - 1: Dictionary Skills.

Session - 2: Introduction to Phonetics and Identifying Phonemes.

Session - 3: Pronunciation Practice (Commonly mispronounced words).

Session - 4: Rosetta Stone (Exercises on LSRW).

Session - 5: Listening Comprehension (Summarising exercise on a Ted Talk).

Session - 6: Technical Presentations (Individual).

Session - 7: Technical Presentations (Team).

Session - 8: TOEFL Mastery.

TEXT BOOK:

1. N P Sudharshana & C Savitha, “English For Technical Communication”, Cambridge University Press, 2016.

REFERENCE BOOKS:

1. Balasubramanian T,“A Text book of Phonetics for Indian Students”, Orient Longman, New Delhi, 1989.
2. Krishnaswamy, N and Sriraman, T, “Current English for Colleges”, Trinity publications, 2016.
3. Mohan Krishna and Meera Banerjee, “Developing Communication Skills”, Macmillan India Ltd. New Delhi, 1990.
4. Ashraf Rizvi M, “Effective Technical Communication”, 2ndEdition, McGraw Hill Education, 2017.
5. Narayana Swamy V R, “Strengthen your Writing”, Third Edition Orient Black Swan, New Delhi, 2005.



Source: https://www.hitalent.co/blog/2019/12/tech-jobs-python-programming-language-and-aws-skills-demand-has-exploded

22CS104 PYTHON PROGRAMMING

Hours Per Week :

L	T	P	C
2	0	2	3

PREREQUISITE KNOWLEDGE: Prior knowledge of any programming language and object-oriented concepts is helpful but not mandatory.

COURSE DESCRIPTION AND OBJECTIVES:

This course offers sufficient knowledge required to understand the fundamental concepts of Python programming language. This course enables students to choose appropriate data structures (lists, dictionaries, tuples, sets, strings) for the given problem. In addition, the students will be able to create reliable, modular and reusable applications using Object- Oriented Programming approaches. At the end they will get an idea of how to access database using python programming, develop web applications, and using web Services using python Programming.

MODULE-1

UNIT-1 **6L+0T+6P=12 Hours**

INTRODUCTION

Introduction to python, Variables, Assignment, Keywords, Built-in functions, Indentation, Comments, Basic data types - integers, float, complex, Boolean, strings; Python program development, running python using REPL shell, Python scripts.

Operators and Expressions: Operators- arithmetic operators, comparison (relational) operators, assignment operators, logical operators, bitwise operators, membership operators, identity operators; Expressions and order of evaluations.

Control Structures: Conditional control structures - if, elif, else; Loop control structures - for, while, for... else, while…else, nested loops, break, continue, pass.

UNIT-2 **10L+0T+10P=20 Hours**

PYTHON DATA STRUCTURES AND FUNCTIONS

Data Structures: Lists, Tuples, Sets, Strings, Dictionary - creation, accessing, basic operators and methods; List comprehensions.

Functions: Defining functions, calling functions, Passing arguments - keyword arguments, default arguments, positional arguments, variable-length arguments; Types of functions- anonymous functions, fruitful function, recursive functions; Scope of the variables- global and local variables, Development of recursive and non-recursive functions.

PRACTICES:

- A. Given an integer N, write a program to find its 1's complement.
- B. Given two integers N1 and N2, write a program to find their product without using multiplication ('*') operator and loops.
- C. Given two integers N1 and N2 having same value, write a program to check whether N1 and N2 points to the same object or not.
- A. Given an Integer N, write a program to check whether given number is even or odd without using modulus operator.
- Given a number N, number of bits K and starting position P, write a program to extract K bits from a position P (from right) in the binary representation of N. Convert the extracted bits in decimal number.

- Given coordinates of centre of a circle, radius and a point coordinate, write a program to check whether the given point lies inside or on the circle, or outside the circle.
- Write a program to find the sum of digits in a given integer.

  Given an integer N as an input, decides the geometrical figure for which the area has to be calculated, for example N=1 for circle, N=2 for rectangle, and N=3 for triangle. Write a program to display the area of the respective figure.
- A semi prime number is an integer which can be expressed as a product of two distinct primes. For example, 15 = 3*5 is a semi prime number but 9 = 3*3 is not.

  For a Given an integer number N, write a program to find whether it can be expressed as a sum of two semi-primes or not (not necessarily distinct).
- Given an integer amount X, write a program to find the minimum number of currency notes $ (500, 100, 50, 20, 10, 5, 2, 1) required for the given amount.

  Input:

  575

  Where input is the amount for which we have to calculate the number of currency notes.

  Output: 4

  Explanation: Total amount = 1(500 dollar note) + 1(50 dollar note) + 1(20 dollar note) + 1(5 dollar note) = 575, hence the minimum number of notes required is 4.
- For a given a string S and width W, write a program to wrap the string S into a paragraph of width W.

  Example:

  Input:

  ABCDEFGHIJKLIMNOQRSTUVWXYZ

  4

  Output:

  ABCD EFGH IJKL IMNO QRST UVWX YZ
- Write a program to Measure the required time to access the first element, nth element and n/2 element stored in list and tuple data structure.

  Given a list L of N numbers (integers), Write a program to find the sum of the elements of given list L with the corresponding elements of the reverse of list L. If list L has elements [1,2,3], then reverse of the list L will be [3,2,1] and the resultant list should be [4,4,4].
- Given a positive integer number n. Write a program to generates a dictionary d which contains (i, i*i*i) such that i is the key and i*i*i is its value, where i is from 1 to n (both included). Print the content of the dictionary d.
- Write a program to create a data structure to store student information such as regd no, name, percentage of marks, phone number and display the student details based on the order of percentage of marks.
- For a Given a string, design and implement functions to perform the following:

  a) remove vowels in the given string.

  b) count number of uppercase and lowercase letters.

  c) remove all special characters.

  d) check whether it is a palindrome or not.

  e) swap case of each letter.
- Create a function that receives 3 numbers and returns the median, i.e. the number that is not the min and not the max, but the one in between.
- Given two lists of integer numbers, write a function to perform the following operations.

  a. print elements that are common in both the lists. (Print without duplicates).

  b. print elements that are present in the first list and not present in second list.

  c. print elements that contain the first element of the first list and last element of the second list.

  d. print elements that contain sum of elements of first list and sum of elements of second list.

SKILLS:

- ✓ *AInstallation and usage of python libraries.*
- ✓ *Working with varieties of data structures.*
- ✓ *Improved analytical and problem-solving abilities.*
- ✓ *Developing structured modular and Object-oriented programming solutions.*

e. print largest number of both the lists.

f. print smallest number of both the lists.

MODULE-2

UNIT-1 **8L+0T+8P=16 Hours**

MODULES

Creating modules, Import Statement, From...Import Statement, Name Spacing, Creating user defined modules.

Standard Modules: sys, math, date, time, os, random and itertools modules.

Packages: Numpy, Pandas, Matplotlib, Requests, Nltk.

File Processing: Reading and writing files -creating a new file, writing to a file, reading text files, opening and closing files, reading, writing, tell (), seek (), rename ().

UNIT-2 **8L+0T+8P=16 Hours**

ERRORS AND EXCEPTIONS

Introduction to Exceptions, Handling Exception, Try Except Else and Finally Block, Raising Exceptions.

Simple Graphics and Image Processing: Overview of Turtle Graphics, Two Dimensional Shapes, Colours and RBG System and Image Processing

PRACTICES:

- Given a string 'S', find all possible permutations of the string S in lexicographic sorted order. Each Permutation size is "2" or "3".

  Sample Input:

  HACK

  Expected Output: AC AH AK CA CH CK HA HC HK KA KC KH
- Write a program that finds area of the pentagon when length from center of a pentagon to vertex are given, the formula for computing the area of pentagon is $\sqrt[3]{3}/2\ s2$, where s is the length of the side, the side can be computed using formula $s= 2r \sin[\pi/5]$, where r is the length from the center of a pentagon to vertex.

  Given X as a date. Write a program to find what the day is on that date.

  Sample Input: 08 05 2015

  Expected Output: Wednesday
- Arun is working in an office which is N blocks away from his house. He wants to minimize the time it takes him to go from his house to the office. He can either take the office cab or he can walk to the office. Arun's velocity is V1 m/s when he is walking. The cab moves with velocity V2 m/s but whenever he calls for the cab, it always starts from the office, covers N blocks, collects Arun and goes back to the office. The cab crosses a total distance of N meters when going from office to Arun's house and vice versa, whereas Arun covers a distance of $(2–\sqrt{}*N)(2*N)$ while walking. Help Arun to find whether he should walk or take a cab to minimize the time.

  Input Format:

  A single line containing three integer numbers N, V1, and V2 separated by a space.

  Example-1:

  Input:

  5 10 15

  Output:

  Cab
- Create a binary NumPy array (containing only 0s and 1s) and convert a binary NumPy array in to to a Boolean NumPy array Convert the first column of a Data Frame as a Series by using suitable packages.

Sample Input:

Original Data Frame

col1 col2 col3

0 1 4 7

1 2 5 5

2 3 6 8

3 4 9 12

4 7 5 1

5 11 0 11

Sample Output:

0 1

1 2

2 3

3 4

4 7

11

- Create two text files and read data from two text files. Display a line from first file followed by the corresponding line from the second file.
- Define the following functions that are more robust to erroneous input data

  a) To divide two numbers (To handle Zero Division Error).

  b) To read two integer numbers and display them (To handle Value Error).

  c) To display elements of a list (To handle Index Error).

  d) To open a file and display file contents (To handle File Not Found Error).
- Write a python program to handle multiple exceptions using raise keyword.
- Draw the spiral hexagon, where we use turtle to create a spiral structure. The final shape is a hexagon and there are various colors used in producing the sides of the hexagon.
- Implement a program to print it in a counterclockwise spiral form for a given square matrix.

  Sample Input:

  4

  25 1 29 7

  24 20 4 32

  16 38 29 1

  48 25 21 19

  Sample Output:

  25 24 16 48 25 21 19 1 32 7 29 1 20 38 29 4
- Write a function that finds the nearest prime number of a given number.

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Module No.	Mapping with POs
1	Analyze various features of programming language and their application in problem solving in computer programming to write, compile, and debug programs in python language.	Analyze	1	1, 2
2	Make use of different data types to design programs involving decisions, loops, and functions.	Apply	1	1, 2, 5
3	Analyze the usage of different data structures for practical and contemporary applications for a given problem.	Analyze	1	1, 2, 3, 5
4	Develop functional, reliable and User-friendly Python programs for given problem statement and constraints.	Apply	2	1, 2, 3,5
5	Installing the python environment and related packages that are required for practical and contemporary applications.	Apply	2	1, 2,3,5

TEXT BOOKS:

1. Kenneth A. Lambert, “The Fundamentals of Python: First Programs”, Cengage Learning, 2011.
2. Mark Lutz, “Learning Python”, 5th edition, Orielly Publishers, 2013.

REFERENCE BOOKS:

1. Introduction to Computation and Programming Using Python. John V. Guttag, The MIT Press.
2. James Payne, “Beginning Python using Python 2.6 and Python 3”, Wrox publishing.
3. Paul Gries, “Practical Programming: An Introduction to Computer Science using Python 3”, The Pragmatic Bookshelf, 2nd edition, 4 Oct. 2013.
4. Allen B. Downey, “Think Python”, 1st edition, Orielly publishing.

II

YEAR

B.Tech.

COMPUTER SCIENCE AND ENGINEERING

I SEMESTER

22ST202	-	Probability and Statistics
22TP201	-	Data Structures
22MS201	-	Management Science
22CS201	-	Database Management Systems
22CS202	-	Digital Logic Design
22CS203	-	Object Oriented Programming Through Java
22CT201	-	Environmental Studies
22SA201	-	Life Skills - I
	-	NCC/NSS/SAC/E-Cell/Student Mentoring/Social activities/Publication

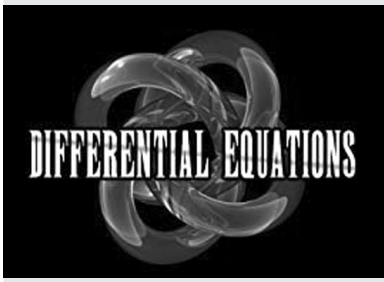
II SEMESTER

22TP205	-	Front-End Application Development
22TP204	-	Professional Communication
22CS205	-	Computer Organization and Architecture
22CS206	-	Design And Analysis of Algorithms
22CS207	-	Operating Systems
22CS208	-	Theory of Computation
22SA202	-	Life Skills - II
	-	Open Elective - 1
	-	Minor / Honours - 1

COURSE CONTENTS

I SEM & II SEM

22ST202 PROBABILITY AND STATISTICS



Source: https://www.amazon.com/Differential-Equations/dp/B01H30X2JA

Hours Per Week :

L	T	P	C
3	0	2	4

PREREQUISITE KNOWLEDGE: Basic knowledge in statistics and mathematics.

COURSE DESCRIPTION AND OBJECTIVES:

To provide students with foundation in elementary topics of statistics and probability such as descriptive statistics, correlation, probability, random variables, correlation, regression, and testing of hypothesis. The course emphasizes statistics to solve engineering and management problems.

MODULE-1

UNIT-1 **12L+0T+8P=20 Hours**

DESCRIPTIVE STATISTICS

Basic Definitions, Frequencies, Graphical Representation, Histogram, Ogive curves; Measures of Central tendency, Arithmetic mean, Median, Mode, Mean deviation, Standard deviation; Symmetry and Skewness, Karl Pearson's Coefficient of skewness.

UNIT-2 **12L+0T+8P=20 Hours**

PROBABILITY AND RANDOM VARIABLES

Probability: Introduction, Definition (Classical and Axiomatic approach), Addition theorem, Conditional probability, Multiplication theorem and Bayes theorem.

Random Variables: Random variables, Discrete and Continuous variables and distribution function.

PRACTICES:

- Various graphical presentation techniques.
- Measures of central tendency.
- Skewness.
- Karl Pearson's coefficient of skewness.
- Applications of addition theorem.
- Applications of multiplication theorem.

MODULE-2

UNIT-1 **12L+0T+8P=20 Hours**

REGRESSION ANALYSIS AND DISTRIBUTIIONS

Correlation and Regression: Correlation, types, Pearson's coefficient of correlation, regression, regression lines.

Distributions: Introduction to distributions: Binomial, Poisson and Normal distributions with properties and applications.

UNIT-2 **12L+0T+8P = 20 Hours**

TESTING OF HYPOTHESIS

Testing large samples-one mean, two means, one proportion and two proportions. Testing small

SKILLS:

- ✓ *Collect the data from various data sources and evaluate mean, median, mode mean deviation and standard deviation.*
- ✓ *Identify the areas which we can apply the probability theory.*

samples- one mean, two means (independent and paired samples), Chi square tests-goodness of fit and independence of attributes.

PRACTICES:

- Correlation.
- Karl Pearson's coefficient of correlation.
- Regression and regression lines.
- Applications of statistical distributions.
- Testing the large sample tests-one mean and two sample means.
- One proportion and two proportion tests.
- Testing small samples-one, two samples and paired tests.
- Chi-square test for goodness of fit.
- Chi-square test for independence of attributes.

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Module No.	Mapping with POs
1	Apply measures of central tendency, skewness, and Karl Pearson's coefficient of skewness to study the statistical data sets.	Apply	1	1,2
2	Apply the probability theory and their applications to measure the uncertainty.	Apply	1	1,2
3	Study the relations between statistical variables and can fit the mathematical models for association.	Analyze	2	1,2,3
4	Test the statistical significances for various samples.	Evaluate	2	1,2,4
5	Identify the distribution type to measure the occurrences of chance.	Evaluate	2	1,4,5

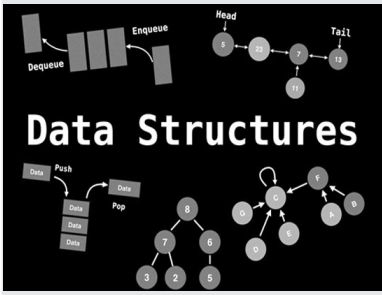
TEXT BOOKS:

1. Sheldon M. Ross, An Introduction to Probability and Statistics for Engineers and Scientists, 3rd Edition, Academic Press, Elsevier.
2. S. C. Gupta and V. K. Kapoor, "Fundamentals of Mathematical Statistics", Sultan Chand & Sons, 2012.

REFERENCE BOOKS:

1. P. R. Vittal, "Mathematical Statistics", Margham Publications, Chennai, 2018.
2. Kishore S. Trivedi, "Probability and Statistics with Realiability, Queueing and Computer Science Applications", 2nd edition, Wiley Student edition, 2008.
3. A. Singaravelu, "Probability and Statistics", 22nd edition, Meenakshi Agency, 2015.

22TP201 DATA STRUCTURES



Source: https://www.youtube.com/watch?v=Qmt0QwzEmh0

Hours Per Week :

L	T	P	C
2	2	2	4

PREREQUISITE KNOWLEDGE: Programming in C.

COURSE DESCRIPTION AND OBJECTIVES:

This course is aimed at offering fundamentals concepts of data structures and explains how to implement them. It begins with the basic concepts of data, data structures and then introduces the primitive and non-primitive data structures in detail. It forms the basis for understanding various ways of representing data and its usage in different computing applications.

MODULE-1

UNIT-1 **6L+6T+6P=18 Hours**

DATA STRUCTURES BASICS

Basic Terminology – data, information, datatype; Data Structures – Introduction, storage structures-sequential and linked storage representations; classification of data structures; Applications of data structures.

Sorting: Selection Sort, Bubble Sort, Insertion Sort, Quick Sort and Merge Sort.

Searching: Linear Search and Binary Search.

UNIT-2 **10L+10T+10P=30 Hours**

LINKED LISTS AND STACKS, QUEUES

Linked List: Introduction, Types of linked list – Singly linked list, doubly linked list and circular linked list, representation of linked list, Operations of linked list: Traverse forward/ reverse order, searching, insertion and deletion; Applications of linked lists.

Stack – Introduction, array and linked representations, implementation and their applications; Queue – Introduction, array and linked representations, implementation; Types – Linear, circular and doubly ended queues – operations; Applications of Queues.

PRACTICES:

PROBLEMS ON RECURSION – LEVEL 1

- Find the product of 2 numbers using recursion.
- Find the sum of natural numbers using recursion.
- Find the factorial of a number using recursion.
- Find the Nth term of Fibonacci series using recursion.
- Calculate the power using recursion.
- Write a recursive program for checking if a given number is a prime number.
- Given two integers write a function to sum the numbers without using any arithmetic operators.
- Convert a decimal to binary using recursion.
- Print all factors using recursion.
- Find the maximum product of digits among numbers less than or equal to N.

SKILLS:

- ✓ *Experienced to Store data and various types of data to handle.*
- ✓ *Ordering and sorting of data.*
- ✓ *Indexing and Searching of required data from large data sequences.*
- ✓ *Exposed to various characteristics such as Linear or non-linear, Homogeneous or heterogeneous and Static and Dynamic.*

PROBLEMS RECURSION – LEVEL 2

- Implement insertion sort recursively.
- Write a program to find the numbers less than N that are product of exactly 2 distinct prime numbers - using recursion.
- Implement selection sort recursively.
- Find the middle of a singly linked list using recursion.
- Find the sum of even numbers of an array using recursion.
- Check if a given array is in sorted order using recursion.
- Print alternate nodes of a linked list using recursion.
- Reverse a doubly linked list using recursion.
- Write a recursive function that returns all permutations of a given list.
- Implement bubble sort recursively.

PROBLEMS ON SORTING AND SEARCHING – LEVEL 1

- Implement the insertion sort function.
- Implement the bubble sort function.
- Implement the quick sort function.
- Implement the merge sort function.
- Implement the selection sort function.
- Implement linear search function.
- Implement binary search function.

PROBLEMS ON SLL – LEVEL 1

- Implement the insert function to insert nodes into a singly linked list (ascending order).
- Implement the insert function to insert nodes into a singly linked list (descending order).
- Implement the search node function.
- Implement the delete node function.
- Display forwards function.
- Display backwards function.
- Count the number of nodes in a singly linked list.
- Swap alternate nodes of a singly linked list.
- Move last node to the front of the linked list.
- Move first node to the last of the linked list.

PROBLEMS ON STACKS – LEVEL 1

- Implement two stacks using a single array.
- Given an array replace every element with nearest greater element on the right.
- Given a stack reverse the elements using only push and pop functions.
- Postfix evaluation using stack.
- Balance symbols.
- Find middle element in a stack.
- Remove middle element from a stack.
- Implement push and pop using linked list.
- Given an array of characters with the middle marked by X, check if the string is a palindrome.
- Maximum sum in sliding window.

PROBLEMS ON QUEUES – LEVEL 1

- Write a program to accept two numbers as input check if they are equal.
- Write a program to accept two characters as input and check if they are equal.
- Write a program to accept two numbers as input and print the greater of the 2 numbers.
- Write a program to accept two numbers as input and print the lesser of the 2 numbers.
- Write a program to accept 3 numbers as input and print the maximum of the 3.

- Write a program to accept 3 numbers as input and print the minimum of the 3.
- Write a program to accept a number as input and print EVEN if it is an even number and ODD if it is an odd number.
- Write a program to accept a number as input and check if it is divisible by 3. If it is divisible by 3 print YES else print NO.
- Write a program to accept a number as input and check if it is divisible by both 3 & 5. If it is divisible print YES else print NO.
- Write a program to accept a number as input and check if it is positive, negative or zero.

PROBLEMS ON DLL – LEVEL 1

- Implement insert function.
- Implement display forward function.
- Implement display backward function.
- Implement search function.
- Implement delete function.
- Reverse a doubly linked list from M to N.
- Find the sum of the odd and even nodes.
- Count odd keys of the linked list.
- Merge two sorted lists.
- Delete adjacent duplicate nodes.

PROBLEMS ON CLL – LEVEL 1

- Insert function (circular doubly linked list).
- Search function.
- Display forward.
- Display backward.
- Delete node (circular doubly linked list).
- Print the middle N nodes of a circular singly linked list.
- Move the last node of a circular singly linked list to the beginning.
- Delete adjacent duplicate nodes of a circular singly linked list.
- Delete nodes greater than a value from a circular doubly linked list.
- Find the sum of the nodes of a circular linked list.

PROBLEMS ON LINKED LIST – LEVEL 2

- Given 2 sorted linked lists, print the common elements.
- Reverse a list (using Stack).
- Given a pointer to a node (not the last node), delete the node.
- Reverse a list (Recursive).
- Reverse a list (Iterative).
- Reverse a singly linked list in pairs (recursive).
- Reverse a singly linked list in pairs (iterative).
- Check if a singly linked list is a palindrome or not.
- Remove the loop if exists.
- Given 2 linked lists with data in the ascending order, merge them into a single list.

MODULE-2

UNIT-1 **8L+8T+8P=24 Hours**

TREES

Trees: Basic Terminology, Types of Trees, Binary Tree – Introduction, properties, array and linked representations; Tree traversals and their implementation; Expression trees; BST – definition and operations, AVL trees – definition and construction; Applications of binary trees.

UNIT-2 **8L+8T+8P=24 Hours**

GRAPHS & HASHING

Graphs: Basic Terminology, Types of Graphs, Graphs representations – adjacency matric, adjacency list; Traversals - breath first search and depth first search; Applications of graphs.

Hashing: Introduction, Different hash functions, collision: avoidance and handling methods.

PRACTICES:

PROBLEMS ON BST – LEVEL 1

- Insert function.
- Insert function (recursive).
- Search function.
- Pre order traversal.
- Post order traversal.
- In order traversal.
- Level order traversal.
- Delete child node.
- Delete parent node.
- Delete nodes greater than a value from a circular doubly linked list.

PROBLEMS ON PRIORITY QUEUES – LEVEL 1

- Meeting rooms problem.
- Ugly number.
- Find median from data stream.
- Find the top K frequent elements.
- Find K Pairs with smallest sums.
- Find the Kth smallest element in a sorted matrix.
- Trapping Rain Water.
- Rearrange String k distance apart.
- Sort characters by frequency.
- Solve the maze problem.

PROBLEMS ON GRAPHS – LEVEL 1

- Implement Graph data structure.
- Implement BFS - iterative solution.
- Implement BFS - recursive solution.
- Implement DFS - iterative solution.
- Implement DFS - recursive solution.
- Check if given graph is strongly connected or not.
- Check if given graph is strongly connected or not - using DFS.
- Given a graph find the arrival and departure time of its vertices in DFS. Arrival time is the time when the vertex was explored for the first time, and departure time is the time at which all the neighbours are explored and are ready to backtrack.
- Given a directed acyclic graph and a source vertex, find the cost of the shortest path from source vertex to all other vertices present in the graph. If a vertex cannot be reached from given source vertex that distance may be printed as infinite.
- Given an undirected graph, check if the graph is 2 edge connected or not.

PROBLEMS ON HASHING – LEVEL 1

- Print a binary tree in vertical order.
- Find whether an array is subset of another array.
- Given an array A [ ] and a number x, check for pair in A [ ] with sum as x.

- Minimum operation to make all elements equal in array.
- Maximum distance between two occurrences of same element in array.
- Check if a given array contains duplicate elements within k distance from each other.
- Find duplicates in a given array when elements are not limited to a range.
- Most frequent element in an array.
- Smallest subarray with all occurrences of a most frequent element.
- First element occurring k times in an array.

PROBLEMS ON GRAPHS – LEVEL 2

- Find the shortest graph distances between every pair vertex in a given path. Assume that the graph does not have any negative edges.
- Find the shortest graph distances between every pair of vertices in a given path. The graph can have negative edges.
- Detect cycle in DFS.
- Count the number of connected components of a graph represented in the adjacent matrix.
- Count the number of connected components of a graph represented in the adjacent matrix - using DFS.
- Find a spanning tree - not necessarily a minimum spanning tree.
- Detect cycle in an undirected graph.
- Given an undirected graph, find its depth.
- Determine if a directed graph has a unique topological ordering.
- Given a directed acyclic graph and two vertices v and w, find the lowest common ancestor.

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Module No.	Mapping with POs
1	Explore the organization of several ADTs and the manipulation (searching, insertion, deletion, traversing) of data stored in various data structures.	Apply	1,2	1
2	Apply different data structures to solve a given problem.	Apply	1,2	1
3	Analyze the efficiency of using different data structures and choose the efficient data structure for solving a given problem.	Analyze	1,2	2
4	Develop new algorithms to solve various problems.	Create	1,2	3,4

TEXT BOOKS:

1. Reema Thareja, “Data Structures Using C”, 2nd Edition, Oxford University Press, 2014.
2. Seymour Lipschutz, “Data Structures with C”, 1st Edition, McGraw Hill Education, 2017.

REFERENCE BOOKS:

1. Ellis Horowitz and Sartaj Sahni, “Fundamentals of Data Structures”, illustrated edition, Computer Science Press, 2006.
2. Richard F. Gilberg and Behrouz A. Forouzan, “Data Structures: A Pseudocode Approach with C”, 2nd Edition, CENAGE Learning, 2005.
3. R G Dromey and Pearson, “How to solve it by Computer”, 2nd edition, Impression edition, 1998.



Source: https://xueqi326.wordpress.com/semester-3/management-science/

22MS201 MANAGEMENT SCIENCE

Hours Per Week :

L	T	P	C
2	2	0	3

PREREQUISITE KNOWLEDGE: Basic knowledge on management .

COURSE DESCRIPTION AND OBJECTIVES:

The goal of this course is to analyse the importance of management, significance of operation management and carry out production operations through work-study. Students will be able to analyse the markets, customers, competitors, and then plan HR function effectively.

MODULE-1

UNIT-1 **6L+6T+0P=12 Hours**

INTRODUCTION TO MANAGEMENT

Concepts of Management and organization- nature, importance and Functions of Management, Systems approach to Management - Taylor's Scientific Management Theory, Fayol's Principles of Management, Mayo's Hawthorne Experiments, Maslow's Theory of Human Needs, Douglas McGregor's Theory X and Theory Y, Herzberg's Two-Factor Theory of Motivation, Leadership Styles, Social responsibilities of Management.

UNIT-2 **10L+10T+0P=20 Hours**

OPERATIONS MANAGEMENT

Principles and Types of Plant Layout-Methods of production (Job, batch and Mass Production), Work Study -Basic procedure involved in Method Study and Work Measurement, Objectives, Need for Inventory control, EOQ, ABC Analysis, Purchase Procedure, Stores Management and Stores Records. Statistical Quality Control: control charts for variables and attributes (simple Problems), Acceptance Sampling.

PRACTICES:

- Collect some examples with videos for types of production.
- Carry out production operations through work-study.
- Practice problems with Inventory control methods and Quality Control charts.

MODULE-2

UNIT-1 **8L+8T+0P=16 Hours**

HUMAN RESOURCES MANAGEMENT

Concepts of Human Resource Management, Basic functions of HR Manager: Manpower planning, Recruitment, Selection, Training and Development, Placement, Wage and Salary Administration, Promotion, Transfer, Separation, Performance Appraisal, Grievance Handling and Welfare Administration, Job Evaluation and Merit Rating.

UNIT-2 **8L+8T+0P =16 Hours**

MARKETING MANAGEMENT

Evolution of Marketing, Functions of Marketing Selling Vs Marketing, 4 P's of Marketing - Product Mix - Product Life Cycle – Place Mix - Channels of Distribution - Price Mix - Pricing Methods - Promotion Mix - Tools of Promotions.

PRACTICES:

- Select any Designation in an organization and try to describe its job description and job specifications.
- How do you deal with grievances at your work.
- Analyze marketing mix in various situations.

SKILLS:

- ✓ *Expert in managerial skills.*
- ✓ *Maintain social relations.*
- ✓ *Evaluate pricing strategies.*

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

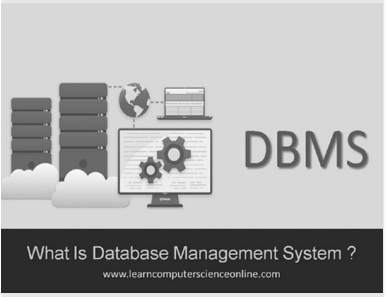
CO No.	Course Outcomes	Blooms Level	Module No.	Mapping with POs
1	Analyze the nature and importance of management	Analyze	1	1,2,4,6
2	Significance of Operations Management.	Analyze	1, 2	1,2,5
3	Carry out production operations through work-study	Apply	1, 2	1, 2, 3, 5
4	Analyze the markets, customers, and competition	Analyze	2	1,2,4,5,6
5	Plan and control the HR function effectively	Evaluate	1, 2	1,2,3,4,5,6

TEXT BOOKS:

1. Stoner, Freeman, Gilbert, Management, 6th Ed, Pearson Education, New Delhi, 2004.
2. P. Vijay Kumar, N. Appa Rao and Ashnab, Chnalill, Cengage Learning India, 2012.

REFERENCES :

1. Kotler Philip & Keller Kevin Lane: Marketing Mangement 12/e, PHI, 2005.
2. Koontz & Weihrich: Essentials of Management, 6/e, TMH, 2005.
3. Thomas N. Duening & John M .Ivancevich Management - Principles and Guidelines, Biztantra, 2003.
4. Aryasri: Management Science, TMH, 2004.



Source: https://www.youtube.com/watch?v=lDpB9zF8LBw

22CS201 DATABASE MANAGEMENT SYSTEMS

Hours Per Week :

L	T	P	C
2	2	2	4

PREREQUISITE KNOWLEDGE: Discrete Mathematical Structures.

COURSE DESCRIPTION AND OBJECTIVES:

This course presents an introduction to database management systems with an emphasis on how to organize, maintain and retrieve data efficiently from a relational database. It also focuses on requirements gathering and conceptual, logical, physical database design. The objective of the course is to enable the student to understand database design, expressing queries using SQL, query optimization and transaction processing.

MODULE-1

UNIT–1 **10L+6T+4P=20 Hours**

DATABASE SYSTEM CONCEPTS

Databases And Database Users: Introduction; Characteristics of the database approach; Actors on the scene; Advantages of using DBMS approach.

Database System Concepts and Architecture: Data models, Schemas and instances; Three-Schema architecture and data Independence; Database languages and interfaces; The database system environment; Centralized and Client-Server architectures for DBMS.

Conceptual Data Modeling and Database Design: Entity types, Entity sets, Attributes and keys; Relationship types, Relationship sets, Roles and structural constraints; Weak entity types; Relationship types.

UNIT–2 **6L+10T+12P=28 Hours**

RELATIONAL DATABASE DESIGN

Relational Database Design by ER–To-Relational Mapping: Relational Database design using ER-to-Relational mapping.

The Relational Data Model and Relational Database Constraints: Relational model concepts; Relational model constraints and Relational database schemas.

Relational Algebra: Unary relational operations - SELECT and PROJECT; Relational algebra operations from set theory; Binary relational operations - JOIN and DIVISION.

SQL: SQL data definition and data types; specifying constraints in SQL, Basic retrieval queries in SQL; INSERT, DELETE, and UPDATE statements in SQL.

PRACTICES:

- Design ER Model for various real time database applications.
- Development of Relational Database schemas for Company/Student/Sailors/ using DDL constructs of SQL.
- Apply various DML Commands such as select, insert, update etc. of SQL on Relational Database.
- Design of Relational Database schemas by specifying different types of Constraints.
- Apply various Relational Database operators (Arithmetic, Logical &comparison) and string-matching constructs of SQL.
- Expressing queries using Aggregate Functions of SQL on Relational Database.
- Queries on Relational Database using GROUP BY, HAVING and ORDER BY clauses of SQL.

MODULE-2

UNIT–1 **8L+8T+12P=28 Hours**

NORMALIZATION

Complex Queries, Triggers, Views: More complex SQL retrieval queries; Specifying constraints as assertions and actions as triggers; Views (virtual tables) in PI/SQL.

Basics of Functional Dependencies and Normalization for Relational Databases: Informal design guidelines for relation schemas; Functional dependencies-inference rules, equivalence and minimal cover; Normal forms based on primary keys; Boyce-Codd normal form; Properties of relational decompositions, multi valued dependency, join dependencies.

UNIT–2 **8L+8T+4P=20 Hours**

TRANSACTION PROCESSING

Introduction To Transaction Processing Concepts and Theory: Introduction to transaction processing; Transaction and system concepts; Desirable properties of transactions; Characterizing schedules based on serializability.

Concurrency Control Techniques: Two-phase locking techniques for concurrency control, concurrency control based on timestamp ordering.

Database Recovery Techniques: Recovery concepts; Shadow paging; The ARIES recovery algorithm.

Indexing Structures for Files and Physical Database Design: Single level and multi-Level indexing; Dynamic multi-level indexing using B-trees and B+ trees.

PRACTICES:

- Design and Development of company database and expressing Nested queries using SQL.
- Design and Development of student database and specifying queries using set operations.
- Design and Development of sailor's database and specifying queries using different types of JOINs.
- Implementation of PL/SQL programs with Control Structures.
- Implementation of PL/SQL programs with Procedures.
- Implementation of PL/SQL programs with Function.
- Implementation of PL/SQL programs with Triggers.
- Creation and dropping of VIEWS.
- Relation R has eight attributes ABCDEFGH. Fields of R contain only atomic values. F = {CH -> G, A -> BC, B -> CFH, E -> A, F -> EG} is a set of functional dependencies (FDs) so that F+ is exactly the set of FDs that hold for R. How many candidate keys does the relation R have?
- Apply various DCL and TCL constructs of SQL on Relational Database.

SKILLS:

- ✓ *Develop E-R model for real life applications.*
- ✓ *Design of relational databases for real world applications.*
- ✓ *Devise queries using relational algebra and SQL.*
- ✓ *Analyze transaction processing, concurrency control and recovery techniques*

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Module No.	Mapping with POs
1	Develop an E-R model for real life applications.	Apply	1	1,10
2	Design and normalize databases for real time applications.	Create	1	1,3
3	Devise queries using Relational Algebra and SQL.	Analyze	2	2
4	Express queries using database tools like Oracle, DB2, MYSQL.	Apply	2	5,10

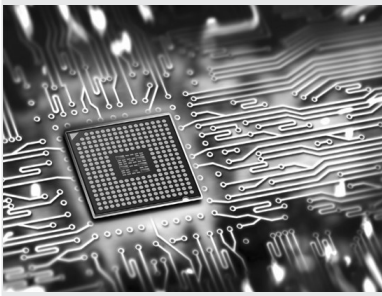
TEXT BOOKS:

1. Ramez, Elmasri and Shamkant B. Navathe, “Fundamentals of Database Systems”, 7th Edition, Pearson Education, 2016.
2. Raghu Rama Krishnan and Johannes Gehrke, “Database Management Systems”, 3rd Edition, Tata McGraw Hill, 2013.

REFERENCE BOOKS:

1. Abraham Silberschatz, Henry F.Korth and S.Sudarshan, “Database System Concepts”, 7th edition, Tata Mc Graw Hill,2019.
2. Allen G. Taylor “Database Development for Dummies” 1st Edition, 2011
3. C. J. Date “Introduction to Database Systems” 7th Edition, Addison Wesley, 2003.

22CS202 DIGITAL LOGIC DESIGN



Source: https://collegeacademy.in/images/ex-blogs/deld.jpg

Hours Per Week :

L	T	P	C
2	0	2	3

PREREQUISITE KNOWLEDGE: Basics of Computers.

COURSE DESCRIPTION AND OBJECTIVES:

This course introduces the basic knowledge on number systems, analysis and design of combinational and sequential circuits. The course mainly focuses on designing digital circuits in optimized manner by using components like decoders, encodes, multiplexers. It also deals with design of sequential circuits and Programmable logic devices.

MODULE-1

UNIT-1 **8L+0T+8P=16 Hours**

INTRODUCTION

Number Systems: Binary Numbers, Number base Conversions, Complements, Binary codes.

Boolean Algebra: Fundamental concepts of Boolean algebra basic theorems and properties.

Gate-Level Minimization: Canonical and standard forms - SOP and POS forms, Logic gates, Algebraic simplification and realization with basic gates and universal gates, The map method – two, three, four variable K map; POS and SOP simplification; Don't care conditions; NAND and NOR implementation.

UNIT-2 **8L+0T+8P=16 Hours**

COMBINATIONAL LOGIC CIRCUITS

Combinational circuits analysis, design procedure; Half adder, Full adder, Half subtractor, Full subtractor, Binary adder/subtractor; BCD adder; Binary multiplier; Magnitude comparator; Decoders; Encoders; Multiplexers; De-Multiplexer.

PRACTICES:

- Design a combinational circuit with three inputs and one output. The output is 1 when the binary value of the inputs is less than 3. The output is 0 otherwise.
- Design a combinational circuit with three inputs x, y, z and three outputs A, B, C. When the binary inputs is 0, 1, 2 or 3, the binary output is one grater than the input. When the binary input is 4, 5, 6, or 7 then the binary output is one less than the input.
- Design a code converter that converts a decimal digit from the 8, 4, -2, -1 code to BCD.
- Implement a Full – Adder using 4 X 1 multiplexer.
- Design a 16 X 1 Multiplexer with five 4 X 1 multiplexer.
- Design a 5-to-32 line decoder with four 3-to-8-line decoders with enable and 2-to-4-line decoder.
- Design and realize 4bit parallel adder and subtractor.

MODULE-2

UNIT-1 **10L+0T+10P=20 Hours**

CIRCUITS, REGISTERS AND COUNTERS

Sequential Logic Circuits: Latches, Flip-Flops - SR, JK, D, T; Flip-flop conversion; Analysis of sequential circuits; Design procedure.

Registers and Counters: Shift registers; Ripple counters; Synchronous counters.

SKILLS:

- ✓ *Learn different data and number representations.*
- ✓ *Design of logical circuits using all types of gates.*
- ✓ *Minimizing of Boolean functions.*
- ✓ *Design of simple logical circuits*

UNIT-2 **6L+0T+6P=12 Hours**

MEMORY AND PROGRAMMABLE LOGIC DEVICES

Random access memory; Read only memory; Programmable logic array; Programmable array logic.

PRACTICES:

- Design a JK flip-flop using a D flip-flop.
- Design a sequential circuit with two D flip-flops A and B and, one input x. When x=0, the state of the circuit remains same. When x = 1, the circuit goes through the state transitions from 00 to 01 to 11 to 10 back to 00 and repeats.
- Design a 4-bit binary synchronous counter with D flip-flop.
- Design and realize a decimal counter.
- A sequential circuit has two J-K fiip-flops A and B. Two inputs x and y, and one output z. The flip-flop input equations and circuit output equation are:
  JA=Bx+B'y'
  KA=B'xy'
  JB=A'x
  KB=A+xy'
  Z=Ax'y'+Bx'y'
  a) Tabulate the state table. b) Derive the state equations.
- Realize the given two Boolean functions with a PLA:
  F1(A, B, C) = ∑(0, 1, 2, 4)
  F2(A, B, C) = ∑ (0, 5, 6, 7)
  F3(A, B, C) = ∑ (1, 3, 4, 5, 7)
- Tabulate the PAL programming table for the four Boolean functions listed below. Minimize the numbers of product terms.
  A(x, y, z) = ∑ (1, 3, 5, 6) B(x, y, z) = ∑ (0, 1, 6, 7)
  C(x, y, z) = ∑ (3, 5) D(x, y, z) = ∑ (1, 2, 4, 5, 7).

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Module No.	Mapping with POs
1	Apply the knowledge of digital logic concepts to optimize digital circuits.	Apply	1	1
2	Apply Boolean algebra rules and karnaugh map method to reduce the Boolean functions.	Apply	1	1, 2
3	Design Combinational digital circuits for the given problem statement by applying the digital techniques.	Analyze	1	3
4	Design and analyze sequential digital circuits for the given problem statement and improve the performance by reducing the complexities.	Analyze	2	3
5	Categorize various types of Programmable Logic Devices.	Analyze	2	2

TEXT BOOK:

1. M Morris Mano and Michael D. Ciletti, "Digital Design", 5th Edition, Pearson Education, 2013.

REFERENCE BOOKS:

1. John F.Wakerly, "Digital Design Principles and Practices", 3rd Edition, Pearson/PHI, 2015
2. Charles H.Roth. "Fundamentals of Logic Design", 6th Edition, Thomson Learning, 2013.
3. John. M Yarbrough, "Digital Logic Applications and Design", Thomson Learning, 2006.

22CS203 OBJECT-ORIENTED PROGRAMMING THROUGH JAVA



Source: https://www.datasciencecentral.com/wp-content/uploads/2021/10/866750 7462.jpeg

Hours Per Week :

L	T	P	C
2	0	4	4

PREREQUISITE KNOWLEDGE: Classes, Objects and Inheritance.

COURSE DESCRIPTION AND OBJECTIVES:

This course is about the fundamentals of Object-Oriented Programming (OOP) Concept and OOP-based software development methodology. Java as a class-based and pure OOP language is used to demonstrate and implement appropriate concepts and techniques. The students are exposed to the concepts, fundamental syntax, and the thought processes behind object- oriented programming. By end of the course, students will acquire the basic knowledge and skills necessary to implement object-oriented programming techniques in software development using Java.

MODULE-1

UNIT-1 **6L+0T+12P=18 Hours**

INTRODUCTION

History of Java, Byte code, JVM, Java buzzwords, OOP principles, Data types, Variables, Scope of variables, Operators, Control statements, Type conversion and casting, Arrays.

Concepts Of Classes and Objects: Introduction to methods, Method over loading, Constructors, Constructor overloading, Usage of static with data and method, Access control, this keyword, Garbage collection, String class, String Tokenizer.

UNIT-2 **10L+0T+20P=30 Hours**

INHERITANCE AND EXCEPTIONS

Inheritance: Types of inheritance, Member access rules, Usage of super keyword, Method overriding, Usage of final keyword, Abstract classes, Interfaces - differences between abstract classes and interfaces, defining an interface, implementing interface, applying interfaces, variables in interface and extending interfaces. Packages-defining, creating and accessing a package, importing packages, access control in packages.

Exception Handling: Concepts of exception handling, Types of exceptions, Usage of try, catch, throw, throws and finally keywords, Built-in exceptions, User defined exception.

PRACTICES:

- There is a telecommunication company called “Powered Air” who have approached you to build their Interactive Voice Response (IVR) system. write a Java program and be able to provide the following menu (given below):
  Note: User should provide an input for each menu display. Welcome to Powered Air service. What would you like to do?
  a. Know my balance. b. Know my validity date
  c. Know number of free calls available. d. More
  1. Prepaid Bill Request 2. Customer Preferences 3. GPRS activation
  4. Special Message Offers 5. Special GPRS Offers 6. 3G Activation
  7. Go back to Previous menu
  You are free to display your own messages in this IVR.
- Create a class Rectangle. The class has attributes length and width. It should have methods that calculate the perimeter and area of the rectangle. It should have read Attributes method to read length and width from user.

SKILLS:

- ✓ *To analyse and develop algorithm for real life problems using Java.*
- ✓ *Experience with developing and debugging programs in different IDEs.*
- ✓ *Develop multi-threaded applications.*
- ✓ *Creating web applications*

Hint: Area of rectangle = length * width, Perimeter of rectangle = 2*(length+width).

- Implement a Java Program that reads a line of integers, and then displays each integer, and the sum of all the integers (use StringTokenizer class).
- Implement a java program to print all tokens of a string on the bases of multiple separators (use StringTokenizer class).
- Using inheritance, one class can acquire the properties of others. Consider a class Animal that has only one method “walk”. Next, create a Bird class that also has a fly method. Finally, create a bird object that can both fly and walk.
- Using inheritance, Write the following code in your editor :

  1. A class named Arithmetic with a method named “add” that takes integers as parameters and returns an integer denoting their sum.

  2. A class named Adder that inherits from a superclass named Arithmetic.

  Note: Your classes should not be Public.
- When a subclass inherits from a superclass, it also inherits its methods; however, it can also override the superclass methods (as well as declare and implement new ones). Consider the Sports class having methods getName()[which returns name of sport] and getNumberOf TeamMembers()[which returns noof team members] create a Soccer class that inherits from the Sports class. We can override the get Name method and return a different subclass-specific string and override getNumberOfTeamMembers method and return noof team members
- Implement a java program to create an abstract class named Shape that contains an empty method named number Of Sides ( ).Provide three classes named Trapezoid, Triangle and Hexagon such that each one of the classes extends the class Shape. Each one of the classes contains only the method number Of Sides ( ) that shows the number of sides in the given geometrical figures.
- You are given an interface Advanced Arithmetic which contains a method signature int divisor_ sum(int n). You need to write a class called My Calculator which implements the interface. divisor_sum function just takes an integer as input and return the sum of all its divisors. For example divisors of 6 are 1, 2, 3 and 6, so divisor_sum should return 12. The value of n will be at most 1000.
- Implement a Java program for the following

  a) Creation of simple package.

  b) Accessing a package.
- Implement a Java program to read two numbers a,b from user and perform division a/b,if the user passes b value as zero, handle the exception using try and catch otherwise display the result.
- Create a class called Customer with data members account_number, balance (initialize with 10000), and member functions print(), deposit(), and withdraw(). Print method display account number and balance. If withdraw amount is less than current balance while withdrawing, throw an exception “In Sufficient Funds”. If the input is 1 do print. If the input is 2 withdraw (). If the input is 3 deposit. If the input is 4 terminate program.
- Implement a Java program which accepts age as input from the user and throws an exception “Not Eligible to Vote” when age is <=18 otherwise print “Eligible to Vote”.

MODULE-2

UNIT-1 **8L+0T+16P=24 Hours**

MULTI THREDING AND FRAMEWORK

Multithreading: Concepts of multi threading, Differences between process and thread, Thread life cycle, Creating multiple threads using Thread class and Runnable interface, Synchronization, Thread priorities, Inter thread communication.

Collection Framework: Collections Overview, Collection Interfaces - List, Set, Map, List – Array List, Linked List, Vector, Set - HashSet, TreeSet, Map - HashTable, HashMap, accessing a collection via an Iterator, comparator, comparable.

UNIT-2 **8L+0T+16P=24 Hours**

SWINGS

GUI Programming With Swing: Delegation event model-Events, Event sources, Event Listeners, Event classes, handling mouse and keyboard events.

Exploring Swing Controls: JLabel and Image Icon, JText Field, JButton, JCheckBox, JRadioButton, JTabbed Pane, JList, JCombo Box.

PRACTICES:

- Print in Order

  Suppose we have a class:

  public class Foo {

  public void first() { print("first"); }

  public void second() { print("second"); }

  public void third() { print("third"); }

  }

  The same instance of Foo will be passed to three different threads. Thread A will call first(), thread B will call second(), and thread C will call third(). Design a mechanism and modify the program to ensure that second() is executed after first(), and third() is executed after second().

  Note: We do not know how the threads will be scheduled in the operating system, even though the numbers in the input seem to imply the ordering. The input format you see is mainly to ensure our tests' comprehensiveness.

  Example 1:

  Input: nums = [1,2,3] Output: "firstsecondthird"

  Explanation: There are three threads being fired asynchronously. The input [1,2,3] means thread A calls first(), thread B calls second(), and thread C calls third(). "firstsecondthird" is the correct output.

  Example 2:

  Input: nums = [1,3,2] Output: "firstsecondthird"

  Explanation: The input [1,3,2] means thread A calls first(), thread B calls third(), and thread C calls second(). "firstsecondthird" is the correct output.

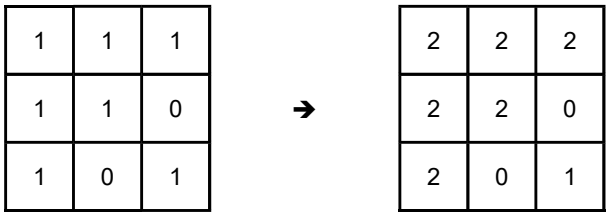
- **Flood Fill:**

  An image is represented by an m x n integer grid image where image[i][j] represents the pixel value of the image.

  You are also given three integers sr, sc, and color. You should perform a flood fill on the image starting from the pixel image[sr][sc].

  To perform a flood fill, consider the starting pixel, plus any pixels connected 4-directionally to the starting pixel of the same color as the starting pixel, plus any pixels connected 4-directionally to those pixels (also with the same color), and so on. Replace the color of all of the aforementioned pixels with color.

  Return the modified image after performing the flood fill.



  Example 1:

  Input: image = [[1,1,1],[1,1,0],[1,0,1]], sr = 1, sc = 1, color = 2

  Output: [[2,2,2],[2,2,0],[2,0,1]]

  Explanation: From the centre of the image with position (sr, sc) = (1, 1) (i.e., the red pixel), all pixels connected by a path of the same color as the starting pixel (i.e., the blue pixels) are colored with the new color.

Note the bottom corner is not coloured 2, because it is not 4-directionally connected to the starting pixel.

Example 2:

Input: image = [[0,0,0],[0,0,0]], sr = 0, sc = 0, color = 0

Output: [[0,0,0],[0,0,0]]

Explanation: The starting pixel is already colored 0, so no changes are made to the image.

- Count words in a given string

  The input parameter is a list of strings representing lines of text.

  Count how often the word occurs in the text.

  If the word "kitten" occurred in a text 23 times, then its entry would be "kitten - 23\n". Return statistics as a String containing all the entries.

  Omit all words which contain less than 4 letters and appear less than 10 (the words which are too small or to rare) The entries in the resulting String should be also sorted by their amount and then in alphabetical order if it is needed.
- Implement a Java program for handling mouse events when the mouse entered, exited, clicked, pressed, released, dragged and moved in the client area.
- Implement a Java program for handling key events when the key board is pressed, released, typed.
- Implement a Java swing program that reads two numbers from two separate text fields and display sum of two numbers in third text field when button "add" is pressed.
- Implement a Java program to design student registration form using Swing Controls. The form which having the following fields and button "save". Form Fields are: Name, RNO, Mail id, Gender, Branch, and Address.
- Implement a java program using swings to design a multiple choice question having three options (use radio button) ,display the message using dialog box "Your answer is wrong" if the user selects wrong option otherwise display ,"Your answer is correct."

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Module No.	Mapping with POs
1	Apply object oriented concepts on real time scenarios.	Apply	1	1,2
2	Apply the concepts of Multithreading and Exception handling to develop efficient and error free codes	Apply	1, 2	1,2
3	Design and develop Java applications to solve real world problems by using modern tools and collection framework	Create	2	3,5
4	Design and develop GUI based applications using swings for internet and system based applications.	Create	2	3,5

TEXT BOOKS:

1. Herbert Schildt, "Java the complete reference", 12th Edition, McGraw Hill, Education, 2021.
2. M.T. Somashekara, D. S. Guru, K.S. Manjunatha, "Object-Oriented Programming with Java", 1st Edition, PHI Learning, 2017.

REFERENCE BOOKS:

1. E. Balagurusamy, "Programming with Java", 6th Edition, McGraw Hill, 2019.
2. Mark Lassoff," Java Programming for Beginners: Learn the fundamentals of programming with Java", 1st Edition, Packt Publishing Limited,2017.
3. Philip Conrod, Lou Tylee," Learn Java GUI Applications : A JFC Swing Tutorial", 11th Edition, Kidware Software, 2019.

22CT201 ENVIRONMENTAL STUDIES



Source: Biogas plant at VFSTR

Hours Per Week :

L	T	P	C
1	1	0	1

PREREQUISITE KNOWLEDGE: General awareness regarding environmental problems and importance of environmental protection.

COURSE DESCRIPTION AND OBJECTIVES:

It is a multidisciplinary subject where different aspects of society and environment are dealt using a holistic approach. It is evolving to be the education for sustainable and ethical development both at a local and global level. It helps to prepare the next generation for planning appropriate strategies to address environmental issues. It identifies and creates solutions that conserve to manage ecosystem and biodiversity and helps to eliminate pollutants, toxicants, preserve air, water and soil quality. Environmental education recognizes impacts of global issues, enhances the public awareness and helps to take decisions towards environmentally responsible actions.

MODULE-1

UNIT-1 **4L+4T+0P=8 Hours**

INTRODUCTION TO ENVIRONMENT

NATURAL RESOURCES, ECOSYSTEMS AND BIODIVERSITY

Environment and sustainable development; Natural resources- forest, water, energy and land resources; Ecosystem – basic structural components, function and interactions in ecosystem, ecological succession.

UNIT-2 **4L+4T+0P=8 Hours**

BIODIVERSITY AND CONSERVATION

Introduction to biodiversity, types of biodiversity- species, genetic and ecosystem diversity; Threats to biodiversity - natural and anthropogenic, species extinctions, man wildlife conflicts; Biodiversity conservation - principles and strategies; in-situ and ex-situ conservation.

PRACTICES:

- Visit to a Biogas plant, Solar Power plant.
- Visit to a local area: river / pond / lake / forest / grassland / hill / mountain and study of different types of ecosystems, biodiversity study and documentation (herbarium sheet preparation).
- Set up an aquarium.
- Case study: Renewable energy use.

MODULE-2

UNIT-1 **4L+4T+0P=8 Hours**

ENVIRONMENTAL POLLUTION AND CLIMATE CHANGE

Air, water, soil, radioactive and noise pollution; Study of different pollutants (SOx, NOx, PAN, PAH etc.); Toxicity study; Climate change - greenhouse effect, acid rain, ozone layer depletion.

UNIT-2 **4L+4T+0P=8 Hours**

POLLUTION CONTROL DEVICES AND WASTEWATER TREATMENT TECHNOLOGIES

Air pollution control devices - Gravitational settling chambers, cyclonic separators, electrostatic precipitators, fabric filters and bio filters, Wastewater management.

SKILLS:

- ✓ *Create a biodiversity map of any habitat/ ecosystem.*
- ✓ *Strategize different ways of using renewable energy resources.*
- ✓ *Design novel strategies and approaches for pollution control and waste management*

PRACTICES:

- Visit to a sewage treatment plant and wastewater analysis.
- Case study: Recycling Technologies.
- Case study: Effects of contaminants on microorganisms.
- Report writing: 12 principles of green chemistry for environmental sustainability.
- Report writing: Environmental Impact Analysis, Local Disaster Management Plan.

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Module No.	Mapping with POs
1	Apply the basic concepts of sustainable development, natural resource utilization and ecology for the purpose of environmental protection.	Apply	1	1,6,7, 9, 10, 11, 12
2	Design remediation technologies for their abatement	Apply	2	1, 3,6,7, 9, 10, 11, 12
3	Analyze the biodiversity of different ecosystems and formulate various conservation approaches	Analyze	1	1, 7, 8, 9, 10, 11, 12
4	Analyze the presence of various environmental pollutants.	Analyze	2	1, 6,7,9, 10, 11, 12
5	Recommend various waste management approaches and their implementation strategies.	Evaluate	2	1,2, 7,8,9,10,11, 12

TEXT BOOKS:

1. A. Kaushik and C. P. Kaushik, “Perspectives in Environmental Studies”, New Age International Publishers, 5th Edition, 2016.
2. Y. Anjaneyulu, “Introduction to Environmental Science”, B. S. Publications, 2015.

REFERENCE BOOKS:

1. B. Joseph, “Environmental Studies”, Mc Graw Hill Education, 2nd Edition, 2015.
2. S. Subash Chandra, “Environmental Science”, New Central Book Agency, 2011.
3. M. Basu and S. Xavier, “Fundamentals of Environmental Studies”, Cambridge University Press, 2016.
4. K. Mukkanti, “A Textbook of Environmental Studies”, S. Chand Company Ltd., 2009.
5. M. Anji Reddy, “A Textbook of Environmental Science and Technology”, B. S. Publications, 2008.

22TP205 FRONT-END APPLICATIOIN DEVELOPMENT



Source: https://www.geeksforgeeks.org/best-way-to-start-with-competitive-programming-geeksforgeeks-cp-live-course/

Hours Per Week :

L	T	P	C
0	0	2	1

PREREQUISITE KNOWLEDGE: OOPs through JAVA. No Previous Development Experience Required. Access to various Open -Source (Free)Software.

COURSE DESCRIPTION AND OBJECTIVES:

This course covers web development principles such as static webpage design, building a simple HTML text site, styling web pages with CSS, and programming websites using Javascript.

MODULE-1

UNIT-1 **0L+0T+8P=8 Hours**

WEB PAGE DESIGNING:

Web technologies, WWW, website, web browser, web server, global and local impact of web technologies on modern society.

HTML: Creating structured documents, Links and navigation, Tables, Forms, and Frames.

HTML 5: Introduction to HTML5, The HTML5 Canvas, HTML5 audio and Video.

UNIT-2 **0L+0T+8P=8 Hours**

CSS: INTRODUCTION

CSS Syntax, CSS selectors: (ID, Class, Tags, Attributes)

CSS Properties: Align-Content, Align-items, Align-self, Animation.

PRACTICES:

- Design a webpage having four frames named a) Top, b)Center, c)Bottom, and d) Left. The top frame should contain the company logo and title. The bottom frame should contain copyright information. The left frame should contain various links like Home, Products, Services, Branches, About, etc., When clicked on respective links, the content should display on the center frame.
- Design a catalog page that should contain the details of all the books available on the website in a table. The details should contain the following: a) Snapshot of Cover Page b) Author Name c) Publisher. d) Price. e) Add to cart button.
- Design a timetable schedule for your current semester using the Table tag.
- Design a HTML page for Student Registration Form using Form Elements that includes Form, input upload.
- Write a HTML program with at least two <h1>, two images, two buttons and appropriate CSS to display,
  - All <h1> with font-size 12pt, and italic in Verdana font using Inline CSS.
  - All <img> with border color red, thickness 15px using Document Level CSS
  - All <input type='button'> should change background color to red on mouse over them using External CSS.
- Design a web page to display the videos on-page, on user selection using frames and HTML5 tags.
- Design a web page to display different types of objects using HTML5 Canvas.
-

SKILLS:

✓ .

MODULE-2

UNIT-1 **0L+0T+8P=8 Hours**

JAVASCRIPT

JAVASCRIPT: Learning JavaScript-how to add scripts to your page, DOM, Variables, Operators, JavaScript Data Types, JavaScript Special Characters.

UNIT-2 **0L+0T+8P=8 Hours**

JavaScript Arrays, JavaScript Boolean, Loops, For-In, For, Do-While, While Loops, JavaScript Events. Built in Objects, Form and regular expression, Validation.

PRACTICES:

- Write Java script programs to demonstrate the following objects with atleast five methods: a) Math b) String c) Array c) Date
- Write a Java script program to display message on OnBlur and OnFocus events
- Design a HTML page having a textbox and four buttons viz Factorial Fibonacci Prime and Palindrome. When a button is pressed an appropriate java script function should be called to display the following
  - Factorial of that number.
  - Fibonacci series up to that number.
  - Prime numbers up to that number.
  - Is it Palindrome or not?
- Design a web page that contains a color pallet, when the user moves the mouse to the particular area, then it changes the background color of the web page.
- Write JavaScript Function to modify Style of the Paragraph Text.
- Design a registration page to validate the following fields using Java Script.Make sure the user name starts with an upper case letter.
  - The user name must have at least one digit.
  - Ensure that Email is valid.
  - Ensure that the password length is between 8 to 20 characters.
  - Make sure the password contains at least one upper case letter, one lower case, and one special character exclude [. dot, , comma, ; semi colon, :colon].

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

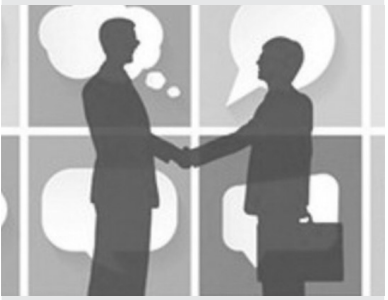
CO No.	Course Outcomes	Blooms Level	Module No.	Mapping with POs
1	Usage of HTML, HTML5, CSS, Java Script in web application development.	Apply	1,2	1
2	Create sand style webpages with html and css.	Under-standing	1	2
3	Analyze JavaScript frameworks and how they can be utilized to save time when writing custom Javascript code.	Apply	2	4
4	Design and develop Script Forms,Validation.	create	2	1

TEXT BOOKS:

1. Learning Web Design:A beginners guide to HTML,CSS,Javascript and Web Graphics,By Jennifer Niederst Robbins.
2. Eloquent JavaScript, by Marijn Haverbeke.
3. HTML And CSS: Design And Build Website by Jon Duckett.
4. JavaScript And JQuery: Interactive Front-End Web Development by Jon Duckett.
5. Front-End Development: The Big Nerd Ranch Guide, by Chris Aquino and Todd Gandee.

REFERENCE BOOKS:

1. JavaScript: The Good Parts, Book by Douglas Crockford
2. JavaScript and JQuery: Interactive Front-End Web Development, Book by Jon Duckett
3. JavaScript for dummies.
4. Head First JavaScript (English, Morrison Michael).



Source: https://www.coursera.org/specializations/improve-english

22TP204 PROFESSIONAL COMMUNICATION

Hours Per Week :

L	T	P	C
0	0	2	1

PREREQUISITE KNOWLEDGE: High School-level English.

COURSE DESCRIPTION AND OBJECTIVES:

To improve the overall professional communication skills (LSRW) of students and prepare them for their profession as engineers and managers. To provide them exposure to conventions of corporate communication and training them on how to function in the business world.

MODULE-1

UNIT-1 **0L+0T+8P=8 Hours**

BASICS OF BUSINESS WRITING SKILLS, PRACTICING BUSINESS CORRESPONDENCE AND REPORT WRITING

Business English Vocabulary: Glossary of most commonly used words (formal and informal usage).

Elements of Technical Writing: Sentence structure, reducing verbosity, arranging ideas logically, building coherence, cohesive devices and transitional words.

Mechanics of Writing: elementary rules of grammar, choice of diction, elementary principles of composition, matters of form, punctuation, conventions of business communication, language and professional tone, code of conduct (not sending illegal, offensive, disparaging personal remarks or comments) in written business communication.

Business Correspondence: E-mail: nature and scope, e-mail etiquette, clear call for action, common errors in composing e-mails, office communication such as meeting agenda and notice, circular and memo.

Letter-Writing: Formal and informal letters, structure of formal letters, expressions of salutations, different types of letters [such as sales letter, complaint letter, response to the complaint letter (dispute resolution), letter of permission, letter of enquiring, claim letter – letter of apology etc], introductory and concluding paragraphs and clear call for action.

Professional Proposal/Report: Differentiating proposals and reports, Drafting formal business proposals, types of reports such as factual reports, feasibility reports and survey reports, parts of a report (such as title page, declaration, acknowledgements, table of contents, abstract, introduction, findings, conclusion and recommendations).

New Age Corporate Communication Media: Importance of social media communication and Etiquettes, form and structure, sharing texts through Twitter, Whatsapp, instgram etc.

UNIT-2 **0L+0T+8P=8 Hours**

PRACTICING COMMUNICATIVE LANGUAGE IN VARIOUS PROFESSIONAL CONTEXTS

Speaking: Speaking in business context, assertiveness, politeness, making requests, queries and questions, negotiations, asking for information, offering suggestions, conflict resolution, contacting clients, initiating, addressing delegates (in public), delivering the presentation effectively, telephone etiquettes, delivering seminar/proposal/report effectively, team meeting etiquettes (face to face and conference call), making effective one minute presentations(JAM) and participating in Group Discussions.

PRACTICES:

- Basic grammar practice, framing paragraphs on topics allocated, paraphrasing an article or a video in your own words, finding topic sentences in newspaper articles, finding out new words

from a professional viewpoint and understanding the meaning and its usage.

- Perusing samples of well-prepared business emails, memo, letter writing and short proposals and reports, students will draft business correspondence writing tasks and different proposals/ reports on topics assigned.
- Watching videos/listening to audios of business presentations, classroom activities of team and individual presentations, using PPTs, mock exercises for BEC speaking, agreeing, disagreeing politely, developing content, extended speaking in Group Discussion(s).

MODULE-2

UNIT-1 **0L+0T+8P=8 Hours**

READING AND COMPREHENDING BUSINESS DOCUMENTS

Reading: Reading and comprehending business documents, learning business register, regularizing the habit of reading business news, suitable vocabulary, skimming and scanning a text for effective and speedy reading and dealing with ideas from different sectors of corporate world in different business contexts.

UNIT-2 **0L+0T+8P=8 Hours**

IMPARTING AND PRACTICING LISTENING SKILLS

Listening: Specific information in business context, listening to telephonic conversations / messages and understanding the correct intended meaning, understanding the questions asked in interviews or in professional settings, summarizing speaker's opinion or suggestion, enable active listening.

PRACTICES:

- Hand-outs; matching the statements with texts, finding missing appropriate sentence in the text from multiple choices, using right vocabulary as per the given context and editing a paragraph.
- Working out BEC/TOEFL/IELTS listening exercises with hand-outs; matching the statements with texts, finding missing appropriate sentence in the text from multiple choice- multiple choices, using right vocabulary in context-editing a paragraph, listening to a long conversation such as an interview and answer MCQ s based upon listening.

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Module No.	Mapping with POs
1	Possess comprehensive skills in listening and reading business texts in formal context.	Apply	2	7
2	Communicate effectively both in their academic as well as professional environment.	Apply	1, 2	10
3	Clear grasp on the register of business language.	Analyze	1	8
4	Possess the ability to write business reports and proposals clearly and precisely to succeed in their future.	Create	1	12
5	Make effective presentations and participate in formal context.	Create	2	10

TEXT BOOK:

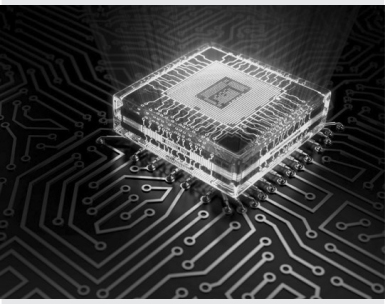
1. S. Schnurr, "Exploring Professional Communication: Language in Action", London: Routledge, 2013.

REFERENCE BOOKS:

1. Brook Hart Guy, "Cambridge English Business Bench Mark: Upper Intermediate", 2nd Edition: CUP, 2014.
2. Cambridge University Publication, "Cambridge: BEC VANTAGE Practice Papers", CUP, 2002.
3. J. Seely, "The Oxford Guide to Effective Writing and Speaking", Oxford University Press, 2005.

SKILLS:

- ✓ *To enhance listening and spoken abilities of students needed for professional and social success in interpersonal situations, group interactions, and personal and professional presentations.*
- ✓ *Know and practice specific functions and vocabulary in a business context.*
- ✓ *Produce short business reports, proposals and correspondence.*
- ✓ *Write various business documents through reading techniques.*



Source: https://machinelearningmedium.com/assets/images/computer-architecture.jpg

22CS205 COMPUTER ORGANIZATION AND ARCHITECTURE

Hours Per Week :

L	T	P	C
2	2	0	3

PREREQUISITE KNOWLEDGE: Digital logic design.

COURSE DESCRIPTION AND OBJECTIVES:

This course covers the basics of modern Computer Organization and Architecture. The emphasis is on understanding the design of computer and its components. The student will learn the concepts of data representation, micro-operations, memory organizations and input output organization.

MODULE-1

UNIT-1 **8L+8T+0P=16 Hours**

INTRODUCTION, RTL, DATA REPRESENTATION AND COMPUTER ARITHMETIC

Introduction, Register Transfer language & Data Representation: Organization and Architecture, Register Transfer, Bus and Memory Transfers, Data Representation-Fixed Point Representation, Floating Point Representation.

Computer Arithmetic: Fixed point arithmetic operations such as Addition and Subtraction, Multiplication Algorithms, Division Algorithms.

UNIT-2 **8L+8T+0P=16 Hours**

MICRO OPERATIONS AND BASIC COMPUTER ORGANIZATION AND DESIGN

Micro operations: Arithmetic Micro operations, Logic Micro Operations, Shift Micro Operations, Arithmetic Logic Shift Unit.

Basic Computer Organization and Design: Instruction Codes, Computer Register, Computer Instructions, Instruction Cycle, Memory – Reference Instructions. Register Reference Instructions, Input –Output and Interrupt.

PRACTICES:

- Design a Common bus system for eight registers with eight bits each using multiplexers.
- Design a Common bus system for four registers with four bits each using three state gate buffers.
- A digital computer has a common bus system for 16 registers of 32 bits each. The bus is constructed with multiplexers.
  How many selection inputs are there in each multiplexer?
  What size of the multiplexers are needed?
  How many multiplexers are there in the bus?
- Perform arithmetic operations (+42) + (-13) and (-42) – (-13) in binary using signed 2's complement representation for negative numbers.
- Find the product using Booth Multiplication Algorithm.
- a. (9) X (13) b. (9) X (-13) c. (-9) X (13) d. (-9) X (-13)
- Perform the division of 27 and 4 using Division algorithm.
- Design a 4- bit combinational circuit decrementer using 4 full adder circuits.
- Register A holds the 8-bit binary 11011001. Determine the B operand and the logic micro operation to be performed in order to change the value in A to :
  a) 01101101 b) 11111101

- An 8-bit register contains the binary value 10011100. What is the register value after an arithmetic shift right? Starting from the initial number 10011 100, determine the register value after an arithmetic shift left, and state whether there is an overflow.
- Starting from an initial value of R =11011101, determine the sequence of binary values in R after a logical shift-left, followed by a circular shift-right, followed by a logical shift-right and a circular shift-left.
- Design arithmetic logic shift unit that performs different operations on 4 bits.

SKILLS:

- ✓ *Learn different data representations.*
- ✓ *Design digital circuitry for implementing different operations.*
- ✓ *Identify the types of memories and their uses.*
- ✓ *Study various data transfer mechanisms in digital computer and I/O.*

MODULE-2

UNIT-1 **8L+8T+0P=16 Hours**

CPU AND MEMORY ORGANIZATION

Central Processing Unit: General Register Organization, STACK Organization. Instruction Formats, Addressing Modes, Data Transfer and Manipulation, Program Control.

Memory Organization: Memory Hierarchy, Main Memory, Auxiliary Memory, Associative Memory, Cache Memory.

UNIT-2 **8L+8T+0P=16 Hours**

I/O ORGANIZATION

Input-Output Organization: Peripheral Devices, Input-Output Interface, Asynchronous data transfer, Modes of Transfer, Priority Interrupts, Direct Memory Access.

PRACTICES:

- The content of PC in the basic computer is 3AF (all numbers are in hexadecimal). The content of AC is 7EC3. The content of memory at address 3AF is 932E. The content of memory at address 32E is 09AC. The content of memory at address 9AC is 8B9F.

  What is the instruction that will be fetched and executed next?

  Show the binary operation that will be performed in the AC when the instruction is executed.

  Give the contents of registers PC, AR, DR, AC, and IR in hexadecimal and the values of E, I, and the sequence counter SC in binary at the end of the instruction cycle.
- Implement the given expressions into different addressing architectures.
  Y=(A-B)/(C*D + E) b. Y=A-B+C*(D *E+F)
- How many 128 x 8RAM chips are needed to provide a memory capacity of 2048 byte?
- How many lines of the address bus must be used to address 2048 bytes of memory? How many of these lines will be common to all chips?
- How many lines must be decoded for chip select and design the size of the decoders.
- A computer uses RAM chips or 1024 x 1 capacity.

  How many chips are needed, and show the connection of memory capacity 1024 bytes?

  How many chips are needed to provide a memory capacity or 16K bytes? Explain in words how the chips are to be connected to the address bus.
- How many characters per second can be transmitted over a 1200-baud line in each of the following modes? (Assume a character code of eight bits.)

  Synchronous serial transmission.

  Asynchronous serial transmission with two stop bits.

  Asynchronous serial transmission with one stop bit.
- Information is inserted into a FIFO buffer at a rate of m bytes per second. The information is deleted at a rate of n byte per second. The maximum capacity of the buffer is k bytes.

  How long does it take for an empty buffer to fill up when m >n?

  How long does it take for a full buffer to empty when m <n?

  Is the FIFO buffer needed if m = n?

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Module No.	Mapping with POs
1	Analyze Computer Organization and Computer Architecture, different arithmetic operations.	Analyze	1	1, 2, 12
2	Design different digital circuits required to perform the micro operations.	Apply	1	1, 2, 3, 12
3	Design interface circuits for memory and peripheral, DMA and communication devices. Compare various modes of transfer.	Analyze	2	1, 2, 3, 4, 12
4	Evaluate the performance of a processor and memory in terms of speed, size and cost.	Evaluate	2	1, 2, 12

TEXT BOOKS:

1. M. Morris Mano, “Computer System Architecture”, 3rd Edition update, Pearson, 2017.
2. William Stallings, “Computer Organization & Architecture: Designing for Performance”, 11th Edition, Pearson, 2019.

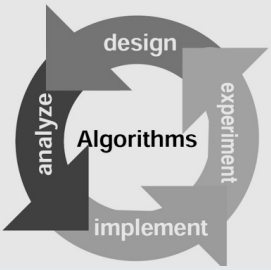
REFERENCE BOOKS:

1. Carl Hamacher, ZvonkoVranesic, SafwatZaky, “Computer Organization”, 5th Edition, McGraw Hill, 2002.
2. Vincent P. Heuring and Harry F Jordan, “ Computer Systems Design and Architecture “, 2nd edition , Pearson/ Prentice Hall India 2004.
3. David A. Patterson and John L. Hennessy, “Computer Organization and Design-The Hardware/ Software Interface”, ARM Edition, 5th Edition, Elsevier, 2009.

22CS206 DESIGN AND ANALYSIS OF ALGORITHMS

Hours Per Week :

L	T	P	C
2	2	2	4



Source: https://www.facebook.com/Design-and-Analysis-of-Algorithms-153902878155564/

PREREQUISITE KNOWLEDGE: Programming for problem solving, Discrete Mathematical Structures, Data Structures.

COURSE DESCRIPTION AND OBJECTIVES:

This course offers the basic knowledge required to analyze the asymptotic performance of algorithms. In addition, this course provides the knowledge required to solve different problems using suitable design strategies such as the greedy method, divide and conquer, dynamic programming, backtracking and branch & bound. This course helps to understand the impact of the choice of data structures and algorithm design strategies on the performance. This course also provides the understanding of advanced graph applications and throws light on tractable and intractable problems.

MODULE-1

UNIT-1 **6L+6T+6P=18 Hours**

INTRODUCTION

Algorithm, Pseudo-code for expressing algorithms, Performance analysis – space and time complexity; Asymptotic notation - big oh notation, Omega notation, Theta notation and little oh notation; Analysis of recursive algorithms through recurrence relations- substitution method, Recursion tree method, Masters Theorem.

Disjoint sets: Disjoint set operations, Union and find algorithms.

UNIT-2 **10L+10T+10P=30 Hours**

DIVIDE & CONQUER AND GREEDY METHOD

Divide and Conquer: General method, Applications - Binary search, Quick sort, Merge sort and Strassen's matrix multiplication.

Greedy Method: Applications - job sequencing with deadlines, Knapsack problem, Minimum cost spanning trees.

PRACTICES:

- Sort a given set of elements using the following methods and determine the time required to sort the elements. Repeat the experiment for different values of n, the number of elements in the list to be sorted and plot a graph of the time taken versus n inputs. The elements can be read from a file or can be generated using the random number generator.
  a. Quick sort b. Merge sort
- Search for a given set of elements using the following methods and determine the time required to search the given element. Repeat the experiment for different values of n, the number of elements in the list to be sorted and plot a graph of the time taken versus no. of elements. The elements can be read from a file or can be generated using the random number generator.
  a. Linear Search b. Binary Search
- Implement the following using divide and conquer approach.
  To multiply two given square matrices.
  To multiply two given square matrices using Strassen's matrix multiplication.
- Design the Algorithm to solve Job sequencing with deadlines problem and Analyze its time complexity. Implement the above algorithm using Greedy method.

SKILLS:

- ✓ *Analyze the given algorithm concerning space and time complexities and compare it with other algorithms.*
- ✓ *Develop algorithms for solving problems using divide and conquer, greedy, dynamic programming, backtracking and branch & bound techniques.*
- ✓ *Application of existing design strategies to solve real-world problems.*

- Design the Algorithm to solve fractional Knapsack problem using Greedy method. Analyze the time complexity and implement the above algorithm.
- Design the Algorithm to find minimum spanning tree and its cost for an undirected graph. Analyze the time complexity and implement the above algorithm.

MODULE-2

UNIT-1 **10L+10T+10P=30 Hours**

DYNAMIC PROGRAMMING AND BACKTRACKING

Dynamic Programming: General method, Applications - optimal binary search trees, Matrix chain multiplication, 0/1knapsackproblem, All pairs shortest path problem, Travelling sales person problem.

Backtracking: General method, Applications - N-Queen problem, Sum of subsets problem, Graph colouring and Hamiltonian cycles.

UNIT-2 **6L+6T+6P=18 Hours**

BRANCH & BOUND AND P, NP, NP - HARD AND NP-COMPLETE

Branch and Bound: General method, Applications- Travelling sales person problem, 0/1 knapsack problem using LC branch and bound solution and FIFO branch and bound solution.

P, NP, NP - HARD and NP-Complete: Basic Concepts - Non-Deterministic Algorithms - The Classes NP-Hard and NP Complete- NP Hard Problems- Clique Decision Problem-Cook's Theorem.

PRACTICES:

- Design the Algorithm to find all pairs shortest path problem by using dynamic programming approach. Analyze its time complexity and implement the above algorithm.
- Design the Algorithm to find optimal binary search tree and its cost by using dynamic programming approach. Analyze its time complexity and implement the above algorithm.
- Design the Algorithm to find optimal order of matrix chain multiplication and its cost using dynamic programming approach. Analyze its time complexity and implement the above algorithm.
- Design the Algorithm to find optimal route for travelling sales person problem and its cost by using dynamic Programming approach. Analyze its time complexity and implement the above algorithm.
- Design the Algorithm to solve N-queens problem by using backtracking approach and Analyze its time complexity. Implement the above algorithm.
- Design the Algorithm to solve sum of subsets problem using backtracking approach and Analyze its time complexity. Implement the above algorithm.
- Design the Algorithm to solve 0/1 Knapsack problem using Branch and Bound method. Analyze the time complexity and Implement the above algorithm.

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Module No.	Mapping with POs
1	Analyze the efficiency of a given algorithm using time and space complexity theory. Understanding algorithmic design strategy like divide and conquer approach.	Analyze	1	1, 2, 12
2	Apply greedy algorithm Strategy for suit able problems and argue the correctness of such algorithms with respect to the global optimization.	Apply	1	1, 2,3, 5, 12
3	Apply the dynamic programming paradigm and identify the kind of problem best suited to solve using dynamic programming.	Apply	2	1, 2, 3, 5, 12
4	Compare and contrast the design principles of branch and bound with backtracking strategy.	Apply	2	1, 2,3,5, 12
5	Investigate computational complexity of different class of problems.	Analyze	2	1, 2,4,12

TEXT BOOKS:

1. Ellis Horowitz, SatrajSahni and Rajasekharan, “Fundamentals of Computer Algorithms”, 2nd Edition, Galgotia publications, 2006.
2. Thomas H. Coremen, Charles E. Leiserson and Ronald L. Rivest, “Introduction to Algorithm”, 2nd Edition, MIT press Ltd., 2014.

REFERENCE BOOKS:

1. Anony Levitin, “Introduction to Design and Analysis of Algorithms”, 3rd Edition, Pearson Education, 2016.
2. Donald E. Knuth, “The Art of Computer Programming”, 2nd Edition, Addison Wesley Publishing Company, 1998.
3. Ronald L. Graham, Donald E. Knuth and Oren Patashnik, “Concrete Mathematics”, 2nd Edition, Addison wesley Publishing Company,1998.
4. Dasgupta, Papadimitriou and Vazirani,“Algorithms”, 1st Edition, McGraw-Hill publishers, 2008.
5. Weiss, “Data Structures and Algorithm Analysis”, 1st Edition, Addison-Wesley Publishing Company, 2016.



Source: https://www.123rf.com/stock-photo/operating_system.html

22CS207 OPERATING SYSTEMS

Hours Per Week :

L	T	P	C
2	0	2	3

PREREQUISITE KNOWLEDGE: Knowledge of computers fundamentals, Computer organization & Digital logic and its design.

COURSE DESCRIPTION AND OBJECTIVES:

This course aims at concepts and principles of Operating Systems, its overall responsibility inacting as an interface between the system's hardware components and the user. Further, it also helps students to understand the different scheduling policies, process synchronization mechanisms, deadlock handling mechanisms and memory management techniques.

MODULE-1

UNIT-1 **10L+0T+10P=20 Hours**

LINUX FILE SYSTEM & PROCESS SCHEDULING

Introduction to LINUX File System: The LINUX file System, File System Hierarchy, File system Commands, File Attributes, File Permissions.

Filters: cmp, comm, diff, head, tail, find, cut, paste, sort, uniq.

Regular Expressions: grep, egrep, fgrep, Sed- line addressing, context addressing, text editing, substitution.

Introduction to Operating System: What Operating System do; Operating System Structure; Process concept-overview, Process Scheduling, Operations on Process; Inter Process Communication; Threads;

Process (CPU) Scheduling-Scheduling Criteria, Scheduling Algorithms; Multiple-Processor scheduling;

UNIT-2 **6L+0T+6P=12 Hours**

PROCESS SYNCHRONIZATION AND DEADLOCKS

Process Synchronization: The critical-section problem; Peterson's solution; Synchronization hardware; Semaphores; Classical problems of synchronization; Monitors.

Deadlocks: Deadlock characterization; Methods of handling deadlocks; Deadlock prevention; Deadlock avoidance; Deadlock detection and recovery.

PRACTICES:

- Use the cat command to create a file containing the following data. Call it mytable.txt usetabsto separate the fields.

  1425 ravi 15.65
  4320 ramu 26.27
  6830 sita 36.15
  1450 raju 21.86

  a. Use the cat command to display the file, mytable.txt.
  b. Use the vicomm and to correct any errors in the file, mytable.txt.
  c. Use the sort command to sort the file mytable.txt according to the first field.
  d. Call thesortedfilemytable.txt (same name)
  e. Printthefilemytable.txt.
  f. Use the cut &paste commands to swap fields 2and 3mytable.Call itmytable.txt (same name)

g. Print the new file, mytable.txt.

- Write a shell script that takes a command–line argument and reports on whether it is directory, a file, or something else.
- Write a shell script that accepts one or more file name as arguments and convertsall of them to uppercase, provided they exist in the current directory.
- Write a shell script that displays a list of all the files in the current directory to which the user has read, write and execute permissions.
- Write a shell script that computes the total and average marks of a student according to the following;
- Ifaveragemarks≥69thenresultis-Distinction‖.
- Ifaveragemarks≥59and≤70thenresultis-FirstClass‖.
- Ifaveragemarks≥49and≤60thenresultis-SecondClass‖If average marks ≤50 then result is -Pass‖.
- Note that any subject marks ≤ 40then result is-Fail‖.
- Accept student name and six subject marks through the keyboard.
- Write an interactive file-handling shell program. Let it offer the user the choice of copying, removing, renaming, or linking files. Once the user has made a choice, have the program ask the user for the necessary information, such as the file name, new name and so on.
- Write a shell script, which receives two file names as arguments. It should check whether the two file contents are same or not. If they are same then second file should be deleted.
- Write a shell script that accepts a file name starting and ending line numbers as arguments and displays all the lines between the given line numbers.
- Write a shell script that deletes all lines containing a specified word in one or more files supplied as arguments to it.
- Implementation of new process creation and its communications.
- Implement of thread creation and deletion.
- Implementation of FCFS scheduling.
- Implementation of SJF and RR Scheduling.
- Implementation of producer consumer problem.
- Implementation of Banker's algorithm for Dead lock avoidance.

SKILLS:

- ✓ *Manage open-source operating systems like Ubuntu, Fedora etc.*
- ✓ *Know the concepts of Processes scheduling and File Systems.*
- ✓ *Identification of different disk scheduling methodologies.*

MODULE-2

UNIT-1 **8L+0T+8P=16 Hours**

MEMORY MANAGEMENT

Memory Management: Basic concept tofmemory management, Swapping, Contiguous Memory Allocation, Paging, Structure of the Page Table, Segmentation.

Virtual Memory Management: Demand Paging, Page Replacement: Optimal, First in First Out (FIFO), Second Chance (SC), Not recently used (NRU), Least Recently used (LRU), Allocation of Frames.

UNIT-2 **8L+0T+8P=16 Hours**

SECONDARY STORAGE STRUCTURE

Secondary Storage Structure: Over view of mass-storage structure, disk structure, disk scheduling;

File System Interface - File concept, Access Methods, Directory & Disk Structure, File-System Mounting, File Sharing, Protection; File-system structure.

File System Implementation- Directory implementation, Allocation Methods, Free Space Management.

PRACTICES:

- Assume that you have a page-reference string for a process with m frames (initially all empty). The page-reference string has length p, and n distinct page numbers occur in it.
  a) What is a lower bound on the number of page faults?
  b) What is an upper bound on the number of page faults?
- Consider the following page-replacement algorithms. Rank these algorithms on a five-point scale from "bad" to "perfect" according to their page-fault rate. Separate those algorithms that

suffer from Belady's anomaly from those that do not.

a) LRU replacement. b) FIFO replacement.

c) Optimal replacement. d) Second-chance replacement.

- Consider the page reference string: 1, 2, 3, 4,2, 1, 5, 6, 2, 1, 2, 3,7, 6, 3, 2, 1, 2, 3, 6.
- How many page faults would occur for the following replacement algorithms, assuming one, two, three, four, five, six, and seven frames? Remember that all frames are initially empty, so your first unique pages will cost one fault each.
- LRU replacement.
- FIFO replacement.
- Optimal replacement.
- How many page fault soccur for your algorithm for the following reference string with four page frames? 1, 2, 3,4, 5, 3, 4, 1, 6,7, 8, 7, 8, 9, 7, 8,9, 5, 4, 5,4, 2.
- What is the minimum number of page faults for an optimal page replacement strategy for thereference string above with four page frames?
- Consider a demand-paged computer system where the degree of multiprogramming is currently fixed at four. The system was recently measured to determine utilization of the CPU and the paging disk. Three alternative results are shown below. For each case, what is happening?

  a) Can the degree of multiprogramming be increased to increase the CPU utilization? Is the paging helping?

  b) CPU utilization 13 percent; disk utilization 97 percent.

  c) CPU utilization 87percent; disk utilization 3 percent.

  d) CPU utilization 13 percent; disk utilization 3 percent.
- Implementation of Disk scheduling algorithm–FCFS.
- Implementation of Disk scheduling algorithm–SSTF and SCAN.

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Module No.	Mapping with POs
1	Classify the basic concepts of operating system and explore Linux ecosystem.	Analyze	1	1
2	Apply the concepts of process scheduling algorithms and process synchronization techniques to derive the efficiency of resource utilization.	Apply	1	1, 2, 3, 5, 12
3	Analyze the requirements for attempting Operating systems principles.	Analyze	1,2	1,2,12
4	Design the various memory management schemes For a given scenario.	Create	2	3,5
5	Apply the concepts of file system interface and implementation.	Apply	1,2	2,5

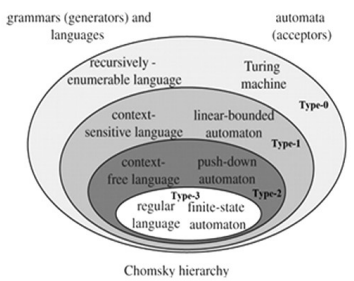
TEXT BOOKS:

1. Sumitabha Das, Unix concepts and applications‖, TMH Publications, 4th Edition, July 2017.
2. Abraham Silberschatz, Peter Baer Galvin and Greg Gagne, "Operating System Concepts", 9th Edition, John Wiley & SonsInc, 2013.

REFERENCE BOOKS:

1. Richard. Stevens and Stephen A Rago, "Advanced Programming in the Unix Environment", 3rd Edition, Addison-Wesley, 2013.
2. William Stallings, "Operating Systems-Internals and Design principles" PHI, 7th Edition, 2012.
3. Gary J. Nutt. Addison-Wesley, "Operating Systems: A Modern Perspective", 2nd Edition, 2001.
4. B.A. Forouzan & R.F.Giberg, - Unix and shell Programming‖, Thomson, 1st Edition, New Delhi, 2003.

22CS208 THEORY OF COMPUTATION



Source: https://sameer9247.wordpress.com/2016/11/15/theory-of-computation/

Hours Per Week :

L	T	P	C
3	2	0	4

PREREQUISITE KNOWLEDGE: Knowledge of graphs, trees and logic.

COURSE DESCRIPTION AND OBJECTIVES:

This course aims to teach the student to identify different formal language classes and their relationships, strong theoretical foundation for designing compilers. In addition to this the student will be able to learn the techniques for information processing, design different grammars, automata and recognizers for different formal languages.

MODULE-1

UNIT-1 **12L+8T+0P=20 Hours**

INTRODUCTION

Alphabets, Strings and languages, Automata and Grammars, Regular languages, Chomsky hierarchy of languages, Deterministic finite automata (DFA)-Formal definition, Simplified notation, State transition graph, Transition table, Language of DFA; Nondeterministic finite automata (NFA), NFA with epsilon transition, Language of NFA, Equivalence of NFA and DFA, Minimization of finite automata, FA with output - Moore and Mealy machine, Equivalence of Moore and Mealy machine, Applications and Limitation of FA.

UNIT-2 **12L+8T+0P=20 Hours**

REGULAR EXPRESSIONS

Regular Expression (RE): Definition, Operators of regular expression and their precedence, Algebraic laws for Regular Expressions, Kleen's Theorem, Regular Expression to FA, DFA to regular expression, Arden theorem, Non regular languages, pumping lemma for regular languages (proofs not Required), Application of pumping lemma, Closure properties of regular languages, Decision properties of regular languages.

Grammar Formalism: Regular Grammars-Right linear and left linear grammars, Equivalence between regular linear grammar and FA;

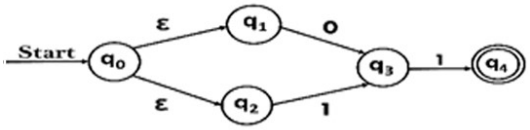
PRACTICES:

- Design DFA and NFA which accepts the following languages over the alphabet {0,1}. And also covert NFA to DFA. Give separate Automata for each and also write RE for the obtained automata.
  a) The set of all strings ends with 00.
  b) With three consecutive 0's.
  c) With 011 as a substring.
  d) Either begin or ends with 01.
  e) Strings whose fourth symbol from the right end is 1.
  f) Even number of 0's.
  g) number of 1 's are divisible by three.
- Design NFA to recognize the following set of strings.
  a) abc, abd, and aacd: Assume the alphabet is {a,b,c,d}.
  b) 0101,101 and 011: Assume the alphabet is {0,1}.
  c) ab,bc and ca: Assume the alphabet is {a,b,c}.

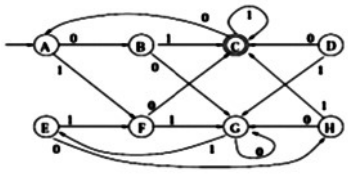
SKILLS:

- ✓ *Investigate syntax and semantics of a regular and context free languages.*
- ✓ *Develop the problem understanding solving ability.*
- ✓ *Design optimized solutions for a language.*

- Convert epsilon NFA to DFA.



- Minimize the following DFA.



- Construct Mealy and Moore Machines and equivalent them for the residue (remainder) mod 3 of binary input.
- Construct Finite Automata for the following Regular Expressions.
  a) RE=ab(a+b)*
  b) RE=(a+ab)(ab+ab)*
- Prove that the following languages are nor Regular.
  a) L={a^p | p is a prime number}
  b) L={b^n | n=i^2 and i>1}
  c) L={ WW^R | W is (a,b)*}
  d) L={a^nb^n+1 |n≥1}

MODULE-2

UNIT-1 **12L+8T+0P=20 Hours**

CONTEXT FREE GRAMMAR

Definition, Examples, Derivation, Derivation trees, Ambiguity in grammar, Inherent ambiguity, Ambiguous to unambiguous CFG, Useless symbols, Simplification of CFGs; Normal forms for CFGs - CNF and GNF, CFLs; Closure properties of Decision properties of CFLs-Emptiness, Finiteness and membership, pumping lemma for CFLs (proofs not Required), Application of pumping lemma.

UNIT-2 **12L+8T+0P=20 Hours**

PDA AND TM

Push Down PDA AND TM Automata (PDA): Description and definition, Instantaneous description, Language of PDA, Acceptance by final state, Acceptance by empty stack, Deterministic PDA, Equivalence of PDA and CFG, CFG to PDA and PDA to CFG, Two stack PDA.

Turing Machines (TM): Basic model, Definition and representation, Instantaneous Description, Language acceptance by TM, Computable functions, Types of Turing Machines, Universal TM, Recursive and Recursively Enumerable Languages, undecidability, Church Turing Thesis, Universal Turing Machine, The universal and diagonalization languages, Reduction between languages and Rice's Theorem.

PRACTICES:

- Construct CFG for the following:
  a) L={a^nb^n|n>1}
  b) L={WW^R | W is (a,b)*}
  c) L={a^p | p is a prime}

- Derive the strings 10001 using left most derivation and right most derivation and parse tree by using the following grammar. And show that grammar is ambiguous.
  S-> T000T
  T->0T|1T|∈
- Convert the following CFG to CNF.
  S->ABC|Aa
  A->a
  B->d|∈
  C-> Aab|a
- Convert the following CFG to GNF.
  S->AA | 0
  A->SS | 1
- Prove that the following are not CFL.
  e) L={a^p | p is a prime number}
  f) L={b^n | n=i^2 and i>1}
  g) L={ WW^R | W is (a,b)*}
  h) L={a^nb^{n+1} |n≥1}
- Convert the following language or PDA to CFG.
  a) L={a^nb^n|n≥1} and
  b) δ(q,0,z)={(q,xz)}
  c) δ(q,0,x)={(q,xx)}
  d) δ(q,1,x)={(q,x)}
  e) δ(q, ∈, x)={(p, ∈)}
  f) δ(p,1,x)={(p,xx)}
  g) δ(p,∈, x)={(p, ∈)}
  h) δ(p, 1,z)={(p, ∈)}
- Construct PDA for the following Languages.
  a) L={0^n1m | n≥m}
  b) L={a^nb^n|n≥1}
  c) L= L={w\w ∈ {a, b}*
  d) L={w| $n_a(w) > n_b(w)$}
  e) L={0^n1^{2n} | n>0}
  f) L=Where $w^R$is reverse of w
  g) L= Where $w^R$is reverse of w
  h) L= $\{wcw^R \mid w \in \{a,b\}^*\}$ Where $w^R$is reverse of w
- Construct PDA for the following Languages.
  a) L= {$a^nb^nc^n$ | n >1}
  b) L={$a^nb^ma^mb^n$ | n,m≥1}
- Construct Turing Machine for the following Languages.
  a) L={a^nb^n | n ≥ 1}
  b) L={$0^{2n}1^n$ |n >0 }
  c) L={ww^r | w is (0+1)*}
  d) L={$a^nb^nc^n$|n≥1}.
  e) Well balanced Parenthesis for example: ()()

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Module No.	Mapping with POs
1	Design abstract models of computing, including Deterministic Finite Automata (DFA), non-deterministic Finite. Automata (NFA), Push Down Automata (PDA) and Turing Machine (TM) models and their power to recognize the languages.	Analyze	1,2	1,2,3
2	Design different finite state machines to perform various operations.	Apply	1,2	1,2,3
3	Analyze the given language is regular or not regular, CFL or not, Ambiguous unambiguous, Recursive and Recursive Enumerable.	Analyze	1,2	2
4	Design Regular grammar and context free grammars for a language.	Apply	1,2	1,3

TEXT BOOK:

1. Hopcroft and Ullman, “Introduction to Automata Theory, Languages and Computation”, 2nd Edition, Pearson/ Prentice Hall India, 2007.

REFERENCE BOOKS:

1. Zed A Shaw, Learn C the Hard Way: Practical Exercises on the Computational Subjects You Keep Avoiding (Like C), Addison Wesley, 2015.
2. Christoph Dürr, Sorbonne University, Jill-Jênn Vie, Inria, Competitive programming in Python, Cambridge University Press, 2020.
3. Michael Sipser, “Introduction to Theory of Computation”, 3rd Edition, Course Technology, 2012.

III

YEAR

COMPUTER SCIENCE AND ENGINEERING

B.Tech.

COURSE CONTENTS

I SEM & II SEM

I SEMESTER

22TP301	-	Soft Skills Laboratory
22CS301	-	Introduction to Artificial Intelligence
22CS312	-	Compiler Design
22CS313	-	Web Technologies
22CS314	-	Computer Networks
22CS306	-	Data Mining Techniques
22CS304	-	Inter-Disciplinary Project - Phase - I
	-	Department Elective - 1
	-	Open Elective - 2
	-	Minor / Honours - 2
	-	NCC/NSS/SAC/E-Cell/Student Mentoring/Social Activities/Publication

II SEMESTER

22TP302	-	Quantitative Aptitude and Logical Reasoning
22CS407	-	Cryptography and Network Security
22CS311	-	Parallel and Distributed Computing
22CS307	-	Software Engineering
22CS305	-	Industry Interface Course
22CS308	-	Inter-Disciplinary Project - Phase - II
	-	Department Elective - 2
	-	Open Elective - 3
	-	Minor / Honours - 3

22TP301 SOFT SKILLS LABORATORY



Source: https://choosework.ssa.gov/blog/2019-07-23-soft-skills-an-intro-to-effective-communication

Hours Per Week :

L	T	P	C
0	0	2	1

PREREQUISITE KNOWLEDGE: Grasp on their own academic achievements.

COURSE DESCRIPTION AND OBJECTIVES:

To impart employability skills like resume preparation and facing interviews. To enable trainees to develop interpersonal and leadership skills and to train them on work place skills like making presentations, participating in group discussions etc.

MODULE-1

UNIT-1 **0L+0T+8P=8 Hours**

PERSONALITY DEVELOPMENT

Soft Skills: Need for soft skills, professionalism, employability skills; Communication: Need for effective communication - the process of communication, levels of communication, flow of communication, choice of diction and style with reference to setting (formal, semi-formal or informal); communication networks, barriers to communication, miscommunication, noise and ways to overcome the barriers; Career Planning: Job vs. career, SWOT analysis.

UNIT-2 **0L+0T+8P=8 Hours**

LANGUAGE AND VOCABULARY

Vocabulary Building: Word etymology, roots, prefixes & suffixes, synonyms & antonyms, collocations, one-word substitutes, analogies, idioms and phrases, contextual guessing of unfamiliar words, task-oriented learning; Reflection of language on Personality, Gender sensitive language in MNCs, Mind your language, Seven essential skills for a team player; attentive listening, intelligent questioning, gently persuading, respecting other's views, assisting others, sharing, participating actively.

PRACTICES:

- Self-Introduction.
- Personal and Academic SWOC.
- Johari Window.
- Giving and taking opinions of Self Vs others and assessing oneself.
- Goal setting.
- Short, Mid and Long Term goals planning the semester.
- Time management: four quadrant system.
- Stephen Covey Time Management Matrix planning a semester.
- Stress-management.
- Questionnaire to assess level of stress.
- 50 words towards resume preparation and interviews.
- Newly coined words.
- Gender sensitive words and Words acceptable in Indian context and objectionable international context.

MODULE-2

UNIT-1 **0L+0T+8P=8 Hours**

LANGUAGE IN ACTION

Functional English: Situational dialogues, Role plays (including small talk); Group Discussion: Articulation and flow of oral presentation, dynamics of group discussion, intervention, summarizing and conclusion, voice modulation, content generation, Key Word Approach (KWA), Social, Political, Economic,

SKILLS:

- ✓ *Balance social and emotional intelligence quotients though SWOC, JOHARI etc. activities.*
- ✓ *Prepare tailor made resume and face various job interviews with enriched personality traits.*
- ✓ *Career planning with clear personal and professional goals.*
- ✓ *Solve personal and professional life hiccups with confidence and maturity*

Legal and Technical Approach (SPELT), View Point of Affected Part (VAP), language relevance, fluency and coherence – 11th and 12th weeks; Resume preparation: Structure and presentation, defining career objective, projecting one's strengths and skill-sets, summarizing, formats and styles and covering letter-Statement of Purpose.

UNIT-2 **0L+0T+8P=8 Hours**

PREPARING FOR PRESENTATIONS AND INTERVIEWS

Facing Interviews: Interview process, understanding employer expectations, pre-interview planning, opening strategies, impressive self-introduction, answering strategies, other critical aspects such as body language, grooming, other types of interviews such as stress-based interviews, tele- interviews, video interviews, frequently asked questions (FAQs) including behavioral and HR questions and the aspect looked at by corporate during interviews; Presentation Skills: Selection of a topic, preparing an abstract, gathering information, organizing the information, drafting the paper, citing reference sources – writing striking introductions, discussing the methodology used, developing the argument, presentation style, language, presenting the paper and spontaneously answering audience questions.

PRACTICES:

- Opening and closing a telephonic conversation.
- Making an appointment.
- Making a query.
- Offering/Passing on information.
- Communicating with superiors.
- Expressing agreement/objection.
- Opening bank account (combination of prepared and impromptu situations given to each student).
- Group Discussions on various topics.
- Preparing SoP and Resume.
- Mock interviews on the FAQs including feedback.
- Oral presentation with the help of technology (Preparing PPT and presenting).

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Module No.	Mapping with POs
1	Have the ability to introspect on individual strengths and weaknesses, and emerge as a balanced personality with improved self-awareness and self-worth.	Apply	1	12
2	Observe gender sensitive language and workplace etiquette in his professional life.	Analyze	1	9
3	Be able to prepare a resume and gain the confidence to face an interview.	Create	1, 2	10
4	Possess the interpersonal skills to conduct himself/herself effectively in everyday professional and social contexts.	Apply	2	8
5	Bring professionalism into his/her daily activities.	Create	2	8

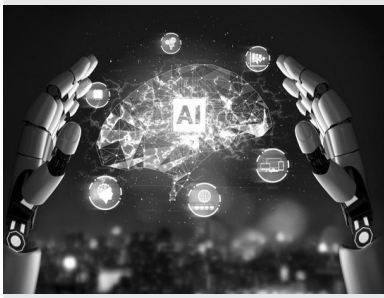
TEXT BOOKS:

1. Adrian Furnham, "Personality and intelligence at work", Psychology Press, 2008.
2. S. P. Dhanvel, "English and Soft skills", Orient Blackswan, 2011.

REFERENCE BOOKS:

1. Edward Holffman, "Ace the corporate personality", McGraw Hill, 2001.
2. John Adair Kegan Page, "Leadership for innovation", Kogan, 2007.
3. Krishna Mohan & NP Singh, "Speaking English effectively", Macmillan, 2008.
4. Rajiv K. Mishra, "Personality Development", Rupa & Co. 2004.

22CS301 INTRODUCTION TO ARTIFICIAL INTELLIGENCE



Source: https://www.forbes.com/sites/bernardmarr/2020/08/03/3-important-ways-artificial-intelligence-will-transform-your-business-and-turbocharge-success/

Hours Per Week :

L	T	P	C
2	2	0	3

PREREQUISITE KNOWLEDGE: Probability and statistics.

COURSE DESCRIPTION AND OBJECTIVES:

The primary objective of this course is to introduce the basic principles, techniques, and applications of Artificial Intelligence. In addition to this, student will understand the building blocks of AI such as search, knowledge representation, inference, logic and learning. This course enables the students to develop a small AI system for real time problems.

MODULE-1

UNIT-1 **6L+6T+0P=12 Hours**

INTELLIGENT SYSTEMS

Introduction, what is AI, Examples of AI systems, Brief history of AI Agent, Agents and environments, Structure of agents, the concept of rationality, the nature of environments, Types of agents, problem solving approaches to typical AI problem.

UNIT-2 **10L+10T+0P=20 Hours**

PROBLEM SOLVING

State Space Problem; Searching: Uniform search, Informed Search: Solving problems by searching: Heuristic functions, Hill climbing, Best First Search, A* algorithm, AO* algorithm, Searching game trees: Min Max Search, Alpha Beta pruning.

PRACTICES:

- In the classical vacuum cleaner problem, we have two rooms and one vacuum cleaner. There is dirt in both the rooms and it is to be cleaned. The vacuum cleaner is present in any one of these rooms. Find the solution, how we can reach to reach a state in which both the rooms are clean and are dust free.
- In this problem, three missionaries and three cannibals must cross a river using a boat which can carry at most two people, under the constraint that, for both banks, that the missionaries present on the bank cannot be outnumbered by cannibals. The boat cannot cross the river by itself with no people on board. Find the solution, how to solve the given problem.
- You are given two jugs, a 4-gallon one and a 3-gallon one, a pump which has unlimited water which you can use to fill the jug, and the ground on which water may be poured. Neither jug has any measuring markings on it. Find the solution, how can you get exactly 2 gallons of water in the 4-gallon jug?
- There is a farmer who wishes to cross a river but he is not alone. He also has a goat, a wolf, and a cabbage along with him. There is only one boat available which can support the farmer and either of the goat, wolf or the cabbage. So at a time, the boat can have only two objects (farmer and one other). But the problem is, if the goat and wolf are left alone (either in the boat or onshore), the wolf will eat the goat. Similarly, if the Goat and cabbage are left alone, then goat will eat the cabbage. The farmer wants to cross the river with all three of his belongings: goat, wolf, and cabbage. What strategy he should use to do so?
- Either place a block that doesn't have other blocks stacked on top of it on another block with the same behaviour, or on the table. The initial and the goal state are described by the exact position of each block. Find the solution, how to solve the given problem.

SKILLS:

- ✓ *Analyze Intelligent systems.*
- ✓ *Apply problem solving techniques.*
- ✓ *Interface various knowledge representation.*
- ✓ *Create a dynamic planning.*

- Given a 3×3 board with 8 tiles (every tile has one number from 1 to 8) and one empty space. The objective is to place the numbers on tiles to match the final configuration using the empty space. We can slide four adjacent (left, right, above, and below) tiles into the empty space. Find the solution, how to solve the given problem by using using A* search algorithm.
- The rules of tic-tac-toe on the 3 × 3 field are as follows. Before the first turn all the field cells are empty. The two players take turns placing their signs into empty cells (the first player places Xs, the second player places Os). The player who places Xs goes first, the another one goes second. Find the solution, how to solve the given problem where the winner is the player who first gets three of his signs in a row next to each other (horizontal, vertical or diagonal).
- In crypt arithmetic problem, the digits (0-9) get substituted by some possible alphabets or symbols. The task in crypt arithmetic problem is to substitute each digit with an alphabet to get the result arithmetically correct. Find the solution, how to solve the given problem, where we can perform all the arithmetic operations on a given crypt arithmetic problem.

MODULE-2

UNIT-1 **10L+10T+0P=20 hours**

KNOWLEDGE REPRESENTATION AND PLANNING

Propositional logic: Inference in propositional logic, Resolution, Forward chaining, Backward chaining, First order logic: Reasoning patterns in First order logic, Resolution, Forward chaining, Backward chaining, The planning problem: Planning with state space search, Partial order planning, Planning graphs.

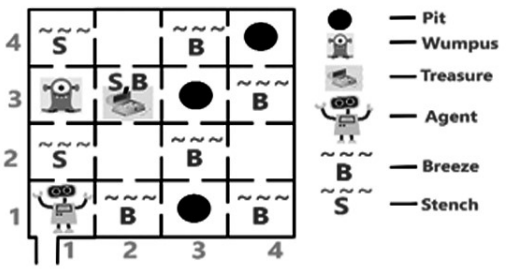
UNIT-2 **6L+6T+0P=12 hours**

LEARNING

Forms of learning: Supervised Learning, Unsupervised learning, Reinforcement learning, Learning Decision Trees, Ensemble Learning, Expert system.

PRACTICES:

- With logic programming, compare expressions and find out unknown values.
- The Wumpus world is a cave with 16 rooms (4×4). Each room is connected to others through walkways (no rooms are connected diagonally). The knowledge-based agent starts from Room [1, 1]. The cave has – some pits, a treasure and a beast named Wumpus. The Wumpus cannot move but eats the one who enters its room. If the agent enters the pit, it gets stuck there. The goal of the agent is to take the treasure and come out of the cave. The agent is rewarded, when the goal conditions are met. The agent is penalized, when it falls into a pit or being eaten by the Wumpus. Some elements support the agent to explore the cave, like -The Wumpus's adjacent rooms are stench. -The agent is given one arrow which it can use to kill the Wumpus when facing it (Wumpus screams when it is killed). – The adjacent rooms of the room with pits are filled with breeze. -The treasure room is always glittery. Find the Wumpus presented room.



- you are on one side of a river with a wolf, a goat, and a cabbage. You want to transport all three to the other side of the river, but you can only transport one object at a time. You cannot leave the wolf and the goat alone, or the cabbage and the goat alone; you are the only thing keeping them from eating each other. How can you transport everything from one side of the river to the other? Formulate it in terms of a Planning Domain Definition Language (PDDL).

- Impliment the working of the decision tree based ID3 algorithm. Use an appropriate data set for building the decision tree and apply this knowledge to classify a new sample.
- Build an Artificial Neural Network by implementing the Back propagation algorithm and test the same using appropriate data sets.
- Implement k-nearest neighbors classification using python.
- Implement linear regression using python.
- Implement the naïve Bayesian classifier for a sample training dataset. Compute the accuracy of the classifier, considering few test data sets.

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Module No.	Mapping with POs
1	Apply AI search Models and Generic Search strategies for problem solving.	Apply	1	1,3
2	Inspect and analyze Logic for representing Knowledge and Reasoning of AI systems and Conduct investigation and implement project using AI learning techniques.	Analyze	1	2
3	Apply and evaluate the searching strategies to achieve the goal for a given situation..	Apply	2	6
4	Design different learning algorithms for improving the performance of AI systems.	Apply	2	4

TEXT BOOK:

1. S. Russel and P. Norvig, “Artificial Intelligence – A Modern Approach”, 4th Edition, Pearson Education, 2010.

REFERENCE BOOKS:

1. David Poole, Alan Mackworth, Randy Goebel, “Computational Intelligence : a logical approach”, Oxford University Press,2013.
2. G. Luger, “Artificial Intelligence: Structures and Strategies for complex problem solving”, 4th Edition, Pearson Education, 2008.
3. J. Nilsson, “Artificial Intelligence: A new Synthesis”, Elsevier Publishers. 2012.



Source: https://www.javatpoint.com/compiler-tutorial

22CS312 COMPILER DESIGN

Hours Per Week :

L	T	P	C
2	2	0	3

PREREQUISITE KNOWLEDGE: Programming for problem solving- I & II and Formal languages and automata theory.

COURSE DESCRIPTION AND OBJECTIVES:

This course introduces the foundation for understanding the theory and practice of compilers and compiler design concepts; symbol table management, compiler parsing techniques, semantic analysis and optimized code generation. This course introduced the concepts of lexical analyzer, parser, code generation and code optimization techniques. The objective of this course is to enable the student to acquire the knowledge of various phases of compiler such as lexical analyzer, parser, code optimization and code generation.

MODULE-1

UNIT-1 **10L+6T+0P=16 Hours**

INTRODUCTION

The evolution of programming languages and basic language processing system; The structure of a compiler; Bootstrapping; Lexical analyser and its Role; Input buffering; Specifications and recognition of tokens; LEX.

UNIT-2 **14L+10T+0P=24 Hours**

SYNTAX ANALYSIS

The role of the parser; Context-free grammars; Types of parsers with examples, YACC.

Semantic Analysis: Type checking; Syntax directed definition (SDD) and translation schemes (TS); Application of SDD and TS; Translation of expressions and control flow statements.

PRACTICES:

- Implement various phases of compiler in detail. Write down the output of each phase for expression Total = (b + c) + (b + c) * 50.
- Construct the symbol table for any input file with the help of LEX tool.
- Consider the context free grammar.
  S→SS+, S→SS*, S→a and the string aa+a*.
  i) Give the leftmost derivation for the string.
  ii) Give the rightmost derivation of the string.
  iii) Is the grammar ambiguous or not.
- Check whether the following grammar is a LL (1) grammar.
  S →iEtS | iEtSeS | a, E → b.
- Construct the FIRST and FOLLOW procedures for the following grammar.
  S → Aa | bAC | dc | bda, A → d.
- Consider the grammar,
  E→TE', E'→+TE' | €, T → FT', T'→ *FT' | €, F → (E) | id.
  Construct a predictive parsing table for the grammar given above. Verify whetherthe input string id + id * id is accepted by the grammar or not.

MODULE-2

UNIT-1 **10L+8T+0P=18 hours**

INTERMEDIATE REPRESENTATIONS

Three address code; Syntax tree; DAG.

Run-Time Environment: Storage organization; Stack allocation - Activation Trees, Activation Records.

UNIT-2 **14L+8T+0P=22 Hours**

OPTIMIZATION AND CODE GENERATION

The principal sources of optimization; Basic blocks and flow graphs; Local optimization; Global optimization and loop optimization.

Code Generation: Issues in the design of code generator; Code-generation algorithm – register allocation and assignment and peephole optimization.

PRACTICES:

- Translate the executable statements of the following C-code segment into three address code.

```
int i:
int a[10]
i = 0;
While (I <= 10) {
a[i] = i + 1; i + + ;
}
```

- Compute the DAG for the following three address statements. Considering this DAG as an example, explain the process of code generation from DAG.

  t1 = a + b t2 = c + d t3 = e – t2 t4 = t1 - t3

- What is Data flow equation? Represent the Data flow information for the following

  a = b + c; d = c * d; e = a – c; f = d + e.

- Draw a flow graph for the below code. Show the basic blocks clearly in your control flow graph?

```
If ( i&gt;=0){
sum = B[0];
i = 0;
L1: if (A[i]&lt; B[i]){
j=i;
L2:
if( B[i]&gt;=0){
sum = sum +B[j];
}
j = j+1
if ( j&lt;N) goto L2;
}
i = i+1
if ( i&lt;N) goto L1;
}
```

SKILLS:

- ✓ *Design parsers using top-down and bottom-up approaches.*
- ✓ *Usage of tools like LEX and YACC.*
- ✓ *Design a simple code generation*

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Module No.	Mapping with POs
1	Apply the different phases of compiler with various examples.	Apply	1	1, 12
2	Design different parsing and optimization techniques in the design of compile.	Design	1	1, 2, 12
3	Analyze the code optimization techniques.	Analyze	2	1, 2, 3, 12
4	Analyze the algorithm for compiler segments and evaluate the algorithm for optimized code generation.	Analyze	2	1, 2, 3,12

TEXT BOOKS:

1. Alfred V. Aho, Monica S. Lam, Ravi Sethi and Jeffrey D. Ulman, “Compilers: Principles, Techniques and Tools”, 3rd Edition, Pearson Education, 2019.
2. Thomson, “Introduction to Theory of Computation”, 2nd Edition, Sipser, 2016.

REFERENCE BOOKS:

1. V. Raghavan, “Principles of Compiler Design”, 2nd Edition, Mc Graw Hill, 2016.
2. John R.Levin, Tony Mason and Doug Brown, “Lex & YAAC”, 2nd Edition, O Reilly, 2012.
3. Ms. Manisha Bharambe, “Compiler Construction”, 2nd Edition, Nirali Prakashan, 2017.

22CS313 WEB TECHNOLOGIES



Source: https://www.dreamstime.com/web-development-coding-programming-internet-technology-business-concept-web-development-coding-programming-internet-technology-image121903546

Hours Per Week :

L	T	P	C
0	0	2	1

PREREQUISITE KNOWLEDGE: OOPs through JAVA. No Previous Development Experience Required. Access to various Open -Source (Free)Software.

COURSE DESCRIPTION AND OBJECTIVES:

This course covers web development principles such as static webpage design, building a simple HTML text site, styling web pages with CSS, and programming websites using Javascript.

MODULE - 1

UNIT-1 **0L+0T+8P=8 Hours**

INTRODUCTION

HTML Basics: Introduction, Elements & Tags, Attributes, Forms.

HTML Advanced: Semantic Elements, Multimedia, Tables

CSS Basics: Introduction to CSS, Selectors, Properties & Values, Box Model

CSS Layout: Positioning, Flexbox, Grid

CSS Advanced: Media Queries, Transitions & Animations, CSS Variables, Preprocessors.

UNIT-2 **0L+0T+8P=8 Hours**

JAVASCRIPT BASICS

Javascript Basics: Introduction, Syntax & Variables, Operators, Functions, Control Structures

DOM Manipulation: Introduction to DOM, Selecting Elements, Events, Modifying Elements

JavaScript Advanced: ES6 Features, Asynchronous JavaScript, Fetch API & AJAX, Error Handling

Object-Oriented JS: Objects & Prototypes, Classes

JavaScript in Web Development: JSON, Local Storage & Session Storage, Libraries & Frameworks, Testing & Debugging**.**

PRACTICES:

- Build a RESTful API using Express that allows users to perform CRUD operations on a single resource (e.g., a todo list, a blog post). Include routes for GET , POST , PUT , and DELETE operations.
- Create and use middleware functions to log request details, authenticate users, and handle errors in your Express.js application.
- Implement user authentication using JSON Web Tokens (JWT) in a Node.js and Express application. Create endpoints for user registration, login, and authentication middleware to protect routes.
- Create a basic server with the help of HTTP that listens on port 3000. Set up routes for a simple to-do list application where users can add, view, update, and delete to-do items.
- Demonstrate the use of callbacks, promises, and async/await in Node.js by fetching weather data from an external API, processing the data, and storing it in a MongoDB collection.
- Write a script to create a database named store and a collection named products
  a. Insert multiple documents into the products collection, each document representing a

SKILLS:

- ✓ *Perform client-side validation using Java Script and Angular JS.*
- ✓ *Store and retrieve data using Node JS.*
- ✓ *Generate dynamic web pages using JSP and PHP.*
- ✓ *Develop a web application or website for any real-time requirements.*

product with fields such as name, price, category, and stock.

b. Write a script to update the stock field of all products in the electronics category by reducing their stock by 5.

c. Create an index on the price field of the products collection to improve query performance.

d. Define a schema for the products collection using MongoDB's validation feature to ensure each product has a name, price, category, and stock.

e. Write a query to find all products in the products collection that have a price greater than $50.

- Create a global error handling middleware in Express (app.use) that catches unhandled errors and sends appropriate error responses (500 Internal Server Error) to clients. Log detailed error messages (console.error).
- Create a global error handling middleware in Express (app.use) that catches unhandled errors and sends appropriate error responses (500 Internal Server Error) to clients. Log detailed error messages (console.error).
- Write a Node.js script that :
  a. Reads the contents of a text file (data.txt) synchronously and logs its content to the console.
  b. read the contents of data.txt asynchronously. Ensure that the file content is logged to the console once it's fully loaded.
- Implement JWT-based authentication in your Express.js application. Create routes for user registration, login, and profile management. Protect certain routes to ensure only authenticated users can access them.

MODULE-2

UNIT–1 **0L+0T+8P=8 Hours**

INTRODUCTION

MERN Stack: Understand the components and architecture of the MERN stack

MongoDB Basics: Learn the basics of MongoDB, its data model, and CRUD operations

Advanced MongoDB: Deepen knowledge of MongoDB features like indexing, aggregation, and transactions

Node.js Fundamentals: Learn the basics of Node.js and asynchronous programming

Express.js Basics: Understand the basics of Express.js and building RESTful APIs

Advanced Express.js: Learn advanced concepts of Express.js like error handling and security.

UNIT–2 **0L+0T+8P=8 Hours**

React Basics: Understand the basics of React, JSX, and component-based architecture

Advanced React: Learn advanced React concepts like hooks, context API, and lifecycle methods

Integration: Frontend & Backend: Integrate React with Express and MongoDB to create a full-stack application

Authentication & Authorization: Implement user authentication and authorization in your application

State Management: Learn state management in React using Redux or Context API

Testing & Debugging: Understand the basics of testing and debugging in MERN stack applications

Deployment: Learn how to deploy your MERN stack application to cloud platforms like Heroku or AWS

Capstone Project (Virtual Support): Apply all the learned concepts to build a comprehensive MERN stack application.

PRACTICES:

- Set up a new React project using Create React App. Create a 'ProductList' component that displays a list of products, each with a name, price, and image, passed as props. Render this

component in the main 'App' component.

- Build a traffic light where the lights switch from green to yellow to red after predetermined intervals and loop indefinitely. Each light should be lit for the following durations:
  1. Red light: 4000ms
  2. Yellow light: 500ms
  3. Green light: 3000ms
- Develop a small (test) application to use Redux (or Context API for state management). Implement actions, reducers, and connect components to the store to manage application-wide state.
- Create a component that fetches a list of users from a backend API and displays them in a table.
- Create a 'Weather' component that uses the 'useEffect' hook to fetch weather data from an API and the 'useState' hook to manage the fetched data. Display the current weather and forecast for a given city.
- Build a React application that displays a list of books fetched from the Express API. Implement a form to add new books to the collection and ensure the list updates in real-time after a new book is added.
- Set up Redux in a React application to manage the state of a blog platform. Implement actions and reducers to handle fetching, adding, updating, and deleting blog posts.
- Write a Higher-Order Component (HOC) called withAuthentication that wraps a component and provides an isAuthenticated prop indicating whether the user is authenticated or not. Assume there's an authenticate function that returns a boolean value.
- Build a simple mortgage calculator widget that takes in a loan amount, interest rate, loan term, and calculates the monthly mortgage payment, total payment amount, and total interest paid.

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

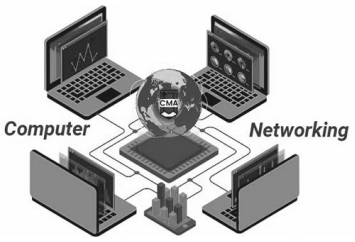
CO No.	Course Outcomes	Blooms Level	Module No.	Mapping with POs
1	Usage of HTML, HTML5, CSS, Java Script in web application development.	Apply	1, 2	1
2	Create sand style webpages with html and css.	Under-standing	1	1
3	Analyze JavaScript frameworks and how they can be utilized to save time when writing custom Javascript code.	Apply	2	2
4	Design and develop Script Forms,Validation.	Creating	2	3

TEXT BOOKS:

1. Learning Web Design:A beginners guide to HTML,CSS,Javascript and Web Graphics,By Jennifer Niederst Robbins.
2. Eloquent JavaScript, by Marijn Haverbeke.
3. HTML And CSS: Design And Build Website by Jon Duckett.
4. JavaScript And JQuery: Interactive Front-End Web Development by Jon Duckett.
5. Front-End Development: The Big Nerd Ranch Guide, by Chris Aquino and Todd Gandee.

REFERENCEBOOKS:

1. JavaScript: The Good Parts, Book by Douglas Crockford.
2. JavaScript and JQuery: Interactive Front-End Web Development, Book by Jon Duckett.
3. JavaScript for dummies8.
4. Head First JavaScript (English, Morrison Michael).



Source: https://snabaynetworking.com/what-is-computer-network-and-its-types/

22CS314 COMPUTER NETWORKS

Hours Per Week :

L	T	P	C
2	0	2	3

PREREQUISITE KNOWLEDGE: JAVA programming and UNIX commands.

COURSE DESCRIPTION AND OBJECTIVES:

This course focuses on imparting knowledge about various protocols involved in LANs and WANs. In addition, it gives a good foundation on different protocols such as data link protocols, internet protocols, and transport protocols present in the respective layers of the data communication system.

MODULE-1

UNIT-1 **8L+0T+6P=14 Hours**

INTRODUCTION TO COMPUTER NETWORKS AND INTERNET

Understanding of network and Internet, the network edge, the network core, Understanding of Delay, Loss and Throughput in the packet switching network, protocols layers and their service model, History of the computer network.

UNIT-2 **16L+0T+10P=26 Hours**

APPLICATION LAYER & TRANSPORT LAYER

Principles of computer applications, Web and HTTP, E-mail, DNS, Socket programming with TCP and UDP.

Introduction and transport layer services, Multiplexing and Demultiplexing, Connectionless transport (UDP), Principles of reliable data transfer, Connection-oriented transport (TCP), Congestion control.

PRACTICES:

- Install Network Simulator 2/3. Create a wired network using dumbbell topology. Attach agents, generate both FTP and CBR traffic, and transmit the traffic. Vary the data rates and evaluate the performance using metric throughput, delay, jitter and packet loss.
- Create a static wireless network. Attach agents, generate both FTP and CBR traffic, and transmit the traffic. Vary the data rates and evaluate the performance using metric throughput, delay, jitter and packet loss.
- Create a mobile wireless network. Attach agents, generate both FTP and CBR traffic, and transmit the traffic. Vary the data rates and evaluate the performance using metric throughput, delay, jitter and packet loss.
- Implementation of one-way and two-way communication using TCP / UDP.
- Hello command is used to know whether the machine at the other end is working or not. Echo command is used to measure the round trip time to the neighbour. Implement Hello and Echo commands using JAVA.

MODULE-2

UNIT-1 **12L+0T+8P = 20 Hours**

NETWORK LAYER

Introduction to forwarding and routing, Network Service models, Virtual and Datagram networks, study of router, IP protocol and addressing in the Internet, Routing algorithms, Broadcast and Multicast routing.

UNIT-2 **12L+0T+8P=20 Hours**

THE LINK LAYER AND LOCAL AREA NETWORKS

Introduction to link layer services, error detection, and correction techniques, Multiple access protocols, addressing, Ethernet, switches, and VLANs.

PRACTICES:

- Find all the IP addresses on your network using Unicast, Multicast, and Broadcast on your network.
- Use Packet tracer software to build network topology and configure using Distance vector routing and Link State routing protocols.
- Study different types of Network cables (Copper and Fiber) and prepare cables (Straight and Cross) to connect two or more systems.
  a. Use a crimping tool to connect jacks.
  b. Use a LAN tester to connect the cables.
  c. Install and configure Network Devices: HUB, Switch and Routers (Consider both manageable and non-manageable switches. Perform logical configuration of the system and set the bandwidth of different ports).
  d. Install and Configure Wired and Wireless NIC and transfer files between systems in Wired LAN and Wireless LAN. Consider both ad-hoc and infrastructure modes of operation.
- Apply the commands such as Ping, Tracert, Ipconfig, pathping, telnet, FTP, getmac, ARP, Hostname, Nbtstat, netdiag, and Nslookup to solve various problems.

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Module No.	Mapping with POs
1	Build the basic concepts of Network hardware, software and reference models.	Apply	1	1, 2, 12
2	Evaluate different physical layer media and switching methods.	Evaluation	1	1, 2, 5, 12
3	Implement various protocols with modern tools.	Apply	1	1, 2, 3, 5, 12
4	Apply different protocols to perform end-to-end delivery and interaction with users.	Analyze	2	1, 2, 12
5	Analyze various design issues, protocols and functionalities of network layer.	Analyze	2	1, 2, 12
6	Demonstrate various protocols involved in data link layer operations.	Apply	2	1,2, 5

TEXT BOOKS:

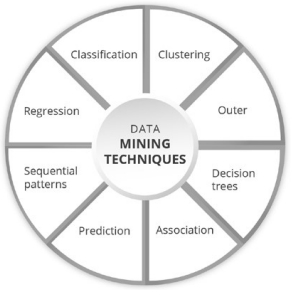
1. Kurose and Ross, “Computer Networking- A Top-Down approach”, 6th Edition, Pearson, 2017.
2. Behrouz Forouzan, “Computer Networks- A Top-Down approach”, McGraw Hill, 2014.

REFERENCE BOOKS:

1. Andrew S. Tanenbaum, “Computer Networks”, 5th edition. Pearson Education, 2014.
2. Behrouz A. Forouzan, “Data communications and Networking”, 5th edition, TMH, 2017.
3. William Stallings, “Data and Computer Communications”, 10th edition, Pearson Education, 2017.
4. Fred Halsall, “Computer Networking and the Internet”, 5th edition, Addison Wesley, 2005.

SKILLS:

- ✓ *Establish local area networks with different topologies.*
- ✓ *Design of new routing protocols.*
- ✓ *Network troubleshooting such as installing network interface card drivers, setting IP addresses, subnet masking, etc.*



Source: https://alternative-spaces.com/blog/8-data-mining-techniques-you-must-learn-to-succeed-in-business/

22CS306 DATA MINING TECHNIQUES

Hours Per Week :

L	T	P	C
2	0	2	3

PREREQUISITE KNOWLEDGE: Probability and statistics, Python programming.

COURSE DESCRIPTION AND OBJECTIVES:

This course introduces the basic concepts, principles, methods, implementation techniques, and applications of data mining, with a focus on three major data mining functions: (1) Association rule mining (2) Classification and (3) cluster Analysis. It also focuses on issues relating to the feasibility, usefulness, effectiveness and scalability of techniques for the discovery of patterns hidden in large data sets.

MODULE-1

UNIT-1 **8L+0T+8P=16 Hours**

INTRODUCTION

What is Data Mining? Why data mining?; What kinds of data can be mined?;What kinds of patterns can be mined?; Which technologies are used?; What kinds of applications are targeted? Major issues in data mining; Data objects and attribute types; Basic statistical descriptions of data, Data matrix versus dissimilarity matrix.

Data Pre-processing: Overview - data quality, major tasks in data preprocessing; Data cleaning - missing values, noisy data; Data Integration - entity identification problem, redundancy and correlation analysis tuple duplication; Data value conflict detection and resolution; Data reduction - PCA, attribute subset selection, regression and log linear models; Histogram; Data transformation - data transformation by normalization; Discretization by binning;

UNIT-2 **8L+0T+8P=16 Hours**

ASSOCIATION ANALYSIS

Market basket analysis; Frequent Item sets; Closed item sets and association rules; Frequent Item set Mining Methods-apriori algorithm, generating association rules, improving apriori, FP growth method, vertical format method; Which patterns are interesting? Pattern evaluation method; Pattern Mining in multilevel multidimensional space, Pattern Mining in Multilevel, Multidimensional Space.

PRACTICES:

- Apply the following data pre-processing techniques on dataset (download from n UCI/ Kaggle/
- NCBI data repository) to illustrate the need of the pre-processing in data mining
  a) Data Cleaning
  b) Data Normalization
  c) Data Discretization
  d) Computation of correlation coefficient to analyze the data behavior
  e) Dimensionality reduction using PCA and Wavelets
- Construct Heat Map Table to understand the Correlation among the attributes in a given dataset.
- Extract the interesting association rules from a given dataset using A priori algorithm.
- Extract the interesting association rules from a given dataset using Frequent Pattern growth algorithm.

MODULE-2

UNIT-1 **8L+0T+8P=16 Hours**

CLASSIFICATION

What is classification?, General approach to classification, Decision tree induction - attribute selection measures; Tree pruning; Bayes Classification methods - Bayes theorem; Naïve Bayesian classification; Classification by back propagation - a multilayer feed forward neural network; Defining a network topology; Back propagation; K nearest neighbor classifier; Support vector machine, Linearly separable and inseparable cases, Model evaluation and selection; Techniques to improve classification accuracy.

UNIT-2 **8L+0T+8P=16 Hours**

CLUSTER ANALYSIS

Partition methods - K means and K medoid; Hierarchical methods; Agglomerative and divisive method; Density based methods - DBSCAN; Optics; Grid based methods-STING; Cluster evaluation methods; Clustering high dimensional data; Problems, Challenges and major methodologies

PRACTICES:

- Apply the following classifiers on a given dataset and analyze their performance.
  a) J48 and visualize the decision tree.
  b) Naive Bayes.
  c) Support Vector Machine.
  d) Multi-Layer Perceptron.
  e) K-Nearest Neighbor.
- Illustrate the performance of Ensemble Classification algorithms such as Bagging and Boosting Methods.
- Evaluate the performance of partitioning clustering algorithms on a given dataset.
- Evaluate the performance of hierarchical clustering algorithms on a given dataset.

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Module No.	Mapping with POs
1	Investigate various patterns that can be extracted from different types of data.	Analyze	1,2	1, 2,
2	Apply various pre-processing techniques and classification algorithms on different domains of data.	Apply	1,2	1, 2, 5, 6
3	Build decision making systems using data mining algorithms for a given real time data set.	Apply Create	1,2	3, 5, 8
4	Construct models using modern tools such as WEKA, R and Python etc.	Apply Create	1,2	1, 2,5,9

TEXT BOOKS:

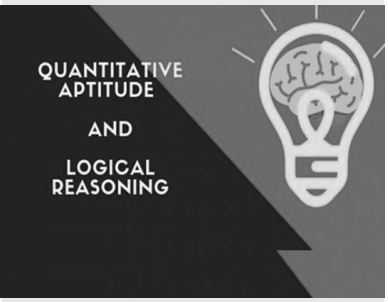
1. Jiawei Han, Micheline Kamber and Jian Pei, “Data mining Concepts and Techniques”, 3rd Edition, Morgan Kaufmann. 2012.
2. Pang-Ning Tan, Michael Steinbach and Vipin Kumar, “Introduction to Data Mining”, 2nd Edition, Pearson, 2018.

REFERENCE BOOKS:

1. Jure Leskovec, Anand Raja raman and Jeffrey D Ullman, “Mining of Massive Datasets”, 5th Edition, Stanford University, 2014.
2. GK Gupta, Introduction to Data Mining with Case Studies, Prentice Hall. 3rd Edition,2014.
3. Margaret H Dunham, “Data Mining: Introductory and Advanced Topics”, PEA, 2008.

SKILLS:

- ✓ *Handle various types of Data and able to explore the characteristics of data*
- ✓ *Perform various Data Visualisation tasks over the data and present the data with ease of access*
- ✓ *Perform descriptive and predictive mining tasks over the data to carry out decision making.*



Source: https://images.app.goo.gl/kvtVgA8TkvDCqLhj7

22TP302 QUANTITATIVE APTITUDE AND LOGICAL REASONING

Hours Per Week :

L	T	P	C
0	4	0	2

PREREQUISITE KNOWLEDGE: Basic logical thinking and Problem solving ability.

COURSE DESCRIPTION AND OBJECTIVES:

The Students will be introduced to various Arithmetic and Reasoning Problems. The students will have acquaintance with various problems like Time & Work, Time & distance, Percentages, Profit & Loss etc. besides solving puzzles and Critical Reasoning.

MODULE-1

UNIT-1 **0L+4T+12P=16 Hours**

Number system, LCM & HCF of numbers, Percentage, Ratio and proportion, Profit, loss and discount, Average & Mixtures, Simple Interest & Compound interest.

UNIT-2 **0L+4T+12P=16 Hours**

Time and work, Time & distance, Problems on trains, Problems on ages, Permutation & Combinations, Probability.

PRACTICES:

- Each concept would be taught in detail in the class followed by 10 problems solved in the class.
- Students would have to solve 10 additional problems as a homework assignment in each concept.

MODULE-2

UNIT-1 **0L+4T+12P=16 Hours**

Number series, Letter series, Analogy, Odd man out, Coding and decoding, Syllogisms- Statement & Conclusions, Puzzle test.

UNIT-2 **0L+4T+12P=16 Hours**

Blood relations, Direction sense test, Order & Ranking, Seating Arrangements, Calendar & Clocks.

PRACTICES:

- Each concept would be taught in detail in the class followed by 10 problems solved in the class. Students would have to solve 10 additional problems as homework assignments in each concept.

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Module No.	Mapping with POs
1	Meet the demands of current job market besides equipping them higher studies like CAT, GMAT etc.	Apply	1	2, 5
2	Solve Arithmetic and Reasoning Problems within shortest possible time without paper work.	Apply	1	2, 5
3	Exhibit better analytical skills and aptitude skills.	Analyse	2	2, 4
4	Develop interpretational skills.	Evaluation	2	2, 4

SKILLS:

- ✓ *Helps in developing and improving problem-solving skills.*
- ✓ *Flexing and honing logical abilities.*
- ✓ *Allow students to develop critical thinking skills.*

TEXT BOOKS:

1. R. S. Aggarwal, "Quantitative Aptitude for Competitive Examinations", S. CHAND Publications- Revised Edition, 2017.
2. ARIHANT, "A New Approach to Verbal & Non-Verbal Reasoning", Arihant Publication- Revised Edition, 2021.

REFERENCE BOOKS:

1. Trishna Knowledge Systems, "Quantitative Aptitude for Competitive Examinations", Pearson Publication, 2013.
2. R. S. Aggarwal, "A Modern Approach to Verbal & Non-Verbal Reasoning", Revised Edition, S. CHAND Publications, 2018.



Source: https://www.brainkart.com/subject/CRYPTOGRAPHY-AND-NETWORK-SECURITY-PRINCIPLES-AND-PRACTICE_136/

22CS401 CRYPTOGRAPHY AND NETWORK SECURITY

Hours Per Week :

L	T	P	C
2	0	2	3

PREREQUISITE KNOWLEDGE: Computer networks.

COURSE DESCRIPTION AND OBJECTIVES:

This course focuses on the modern concepts of network security using various cryptographic algorithms and underlying network security applications. It enables to understand various symmetric and asymmetric cryptographic techniques. It focuses on providing security services in applications such as e-mail functioning, web security and secure electronic transactions protocol and system security.

MODULE-1

UNIT-1 **12L+0T+8P=20 Hours**

INTRODUCTION

Introduction To Computer and Network Security Concepts: Computer Security Concepts, Security attacks, Security services, Security mechanisms, Fundamental Security Design Principles, Attack Surfaces and Attack trees, A model for network security.

Classical Encryption Techniques: Symmetric cipher model, Substitution techniques, Transposition techniques

UNIT-2 **12L+0T+8P=20 Hours**

SYMMETRIC AND ASYMMETRIC CRYPTOGRAPHY

Symmetric Ciphers: Block cipher principles, Data encryption standard, Strength of DES, Blockcipher design principles, AES cipher, Multiple encryption and triple DES, Block cipher modes ofoperation, RC4.

Asymmetric Ciphers and Cryptographic Hash Functions: Principles of public keycryptosystems, RSA algorithm, Diffie-Hellman Key Exchange, Message Authentication requirements,Authentication functions, Message authentication Codes, Hash functions, Security of hash functionsand MACs, Digital signature standard.

PRACTICES:

- Implement Substitution and Transposition Ciphers
  - Ceaser cipher
  - Playfair cipher
  - Hill cipher
  - Rail fence cipher
- Implement Symmetric Cipher
  - S-DES
  - RC4
- Implement Asymmetric Cipher
  - RSA
  - Diffie-Hellman
  - Hash Function

MODULE-2

UNIT-1 **12L+0T+8P=20 Hours**

SECURITY APPLICATIONS

Network Security Applications: Kerberos, X.509 authentication service, Public key infrastructure,

E-Mail Security: Pretty good privacy, S/MIME.

IP Security Overview: IP security architecture, Authentication header, Encapsulating security payload, Combining security associations, key management.

UNIT-2 **12L+0T+8P=20 Hours**

WEB AND SYSTEM SECURITY

Web Security: Secure socket layer and transport layer security, HTTPS, Secure Shell (SSH)

System Security: Intruders, Intrusion detection, Malicious software, Firewalls

PRACTICES:

- Configure IP Address in a system in LAN (TCP/IP Configuration)
- Configure DNS to establish interconnection between systems
- Configuring Windows Firewall
- Adding users, setting permissions
- Configure Mail server
- Demonstrate the usage of Wireshark to identify abnormal activity in network communication.
- Demonstrate usage of NMAP (Zenmap) Tool in Network Scanning.

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Module No.	Mapping with POs
1	Apply cryptographic techniques in various security service solutions effectively in everyday professional and social contexts.	Apply	1,2	1,2
2	Analyze the usage of secure protocols to safeguard sensitive data using internet.	Analyze	1,2	1,2
3	Usage of tools to Identify abnormal activity in network communication to take appropriate action.	Apply	2	5
4	Apply various security protocols to safe guard the data internet using SSL/TCL.	Apply	2	1,2

TEXT BOOK:

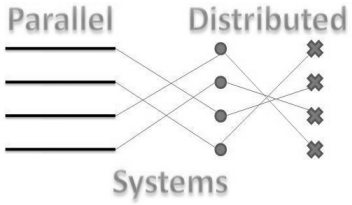
1. William Stallings, “Cryptography and Network security”, 7th Edition, Pearson Education, 2017.

REFERENCE BOOKS:

1. William Stallings “Network Security Essentials Applications and Standards”, 2nd Edition, Pearson Education, 2009.
2. Eric Malwald, “Fundamentals of Network Security”, 4th Edition, Pearson Education, 2010.
3. Buchmann, “Introduction to Cryptography”, 2nd Edition, Pearson Education, 2009.
4. Charlie Kaufman, “Radis Perlman and Mike Speciner, Network Security - Private Communication in a Public World”, 1st Edition, Pearson Education, 2009.

SKILLS:

- ✓ *Design various security services for appropriate applications.*
- ✓ *Identifying the appropriate firewall, password management and anti-virus models for specific applications.*
- ✓ *Test and resolve threats and malfunctions in network.*
- ✓ *Apply different security mechanisms for web applications.*
- ✓ *Build authentication system for security protocols.*



Source: http://uceou.edu/PDS/About%20PDS.html

22CS311 PARALLEL AND DISTRIBUTED COMPUTING

Hours Per Week :

L	T	P	C
2	2	0	3

PREREQUISITE KNOWLEDGE: Fundamental of programming and operating system, computer networking, and computer Organization..

COURSE DESCRIPTION AND OBJECTIVES:

This course covers the challenges faced in constructing parallel and distributed applications, Various implementation techniques, paradigms, architectures and parallel algorithms. And current trends in parallel and distributed computing like Open MP, POSIX Threads, Apache Hadoop (DFS).

MODULE-1

UNIT-1 **12L+8T+0P=20 Hours**

INTRODUCTION TO PARALLEL COMPUTING

The idea of Parallelism, Power and potential of parallelism, examining sequential and parallel programs, Scope and issues of parallel and distributed computing, Goals of parallelism, Parallelism and concurrency using multiple instructions streams.

Parallel Architecture: Pipeline architecture, Array processor, Multi-processor architecture, Systolic architecture, Dataflow architecture, Architectural classification schemes, Memory access classification, Memory Issues: Shared vs. distributed, Symmetric multiprocessing (SMP), SIMD, Vector processing, GPU co-processing, Flynn's Taxonomy, Instruction Level support for parallel programming, Multiprocessor caches and Cache Coherence, Non-Uniform Memory Access (NUMA).

UNIT-2 **12L+8T+0P=20 Hours**

PARALLEL ALGORITHM DESIGN PRINCIPLES AND PROGRAMMING

Need for communication and coordination/synchronization, Scheduling and contention, Independence and partitioning, Task- Based Decomposition, Data Parallel Decomposition, Characteristics of task and interaction, Load balancing, Data Management, parallel algorithm models, Sources of overhead in parallel programs, Performance metrics for parallel algorithm implementations, Parallel algorithmic patterns like divide and conquer, Map and Reduce, Specific algorithms like parallel Merge Sort, Parallel graph Algorithms.

PRACTICES:

- Identify Multiple Instruction Single Data, or MISD. How would an MISD system work? Give an example.
- Suppose a shared-memory system uses snooping cache coherence and write-back caches. Also suppose that core 0 has the variable x in its cache, and it executes the assignment x = 5. Finally suppose that core 1 doesn't have x in its cache, and after core 0's update to x, core 1 tries to execute y = x. What value will be assigned to y? Why?
- Consider a simplified version of bucket-sort. You are given an array A of n random integers in the range [1...r] as input. The output data consist of r buckets, such that at the end of the algorithm, Bucket i contains indices of all the elements in A that are equal to i.
    - Describe a decomposition based on partitioning the input data (i.e., the array A) and an appropriate mapping onto p processes. Describe briefly how the resulting parallel algorithm would work.
    - Describe a decomposition based on partitioning the output data (i.e., the set of r buckets) and an appropriate mapping onto p processes. Describe briefly how the resulting parallel algorithm would work.

- Consider seven tasks with running times of 1, 2, 3, 4, 5, 5, and 10 units, respectively. Assuming that it does not take any time to assign work to a process, compute the best- and worst-case speedup for a centralized scheme for dynamic mapping with two processes.

MODULE-2

UNIT-1 **12L+8T+0P=20 Hours**

INTRODUCTION TO DISTRIBUTED SYSTEMS

Goals of the Distributed Systems, Relation to parallel systems, synchronous versus asynchronous execution, design issues and challenges, Types of Distributed Systems, Distributed System Models, Hardware and software concepts related to distributed systems, middleware models.

Distributed Computing and Communication design principles: A Model of distributed executions, Models of communication networks, Global state of distributed system, Models of process communication. Communication and Coordination: Shared Memory, Consistency, Atomicity, Message- 08 Passing, Consensus, Conditional Actions, Critical Paths, Scalability, and cache coherence in multiprocessor systems, synchronization mechanism.

UNIT-2 **12L+8T+0P=20 Hours**

PARALLEL AND DISTRIBUTED PROGRAMMING FRAMEWORKS

Overview of CUDA, Open MP, POSIX Threads, Apache Hadoop (DFS), and current trends in parallel and distributed computing.

PRACTICES:

- Give five types of hardware resource and five types of data or software resource that can usefully be shared. Give examples of their sharing as it occurs in practice in distributed systems.
- The host computers used in peer-to-peer systems are often simply desktop computers in users' offices or homes. What are the implications of this for the availability and security of any shared data objects that they hold and to what extent can any weaknesses be overcome through the use of replication?
- Consider two communication services for use in asynchronous distributed systems. In service A, messages may be lost, duplicated or delayed and checksums apply only to headers. In service B, messages may be lost, delayed or delivered too fast for the recipient to handle them, but those that are delivered arrive with the correct contents. Describe the classes of failure exhibited by each service. Classify their failures according to their effects on the properties of validity and integrity. Can service B be described as a reliable communication service?
- Illustrate distributed design through a substantial case study, examining in detail the design of the Google infrastructure, a platform and associated middleware that supports both Google search and a set of associated web services and applications including Google Apps.
- Implementation of the parallel algorithms (on a PC-cluster under Linux platform). The programs will be based on POSIX Thread, MPI programming, Hadoop, Apache Spark etc.

SKILLS:

- ✓ *Recognize parallelism in computational problems.*
- ✓ *Know different parallel systems and their classification.*
- ✓ *Design parallel algorithms for different applications.*
- ✓ *Compare replication schemes with respect to performance, availability and consistency concerns.*
- ✓ *Design, implement, and debug distributed systems.*
- ✓ *Implement parallel algorithms using MPI and OpenMP environments.*
- ✓ *Element parallel algorithms using MPI and OpenMP environments 3.0 3.5*
- ✓ *AVERAGE.*

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Module No.	Mapping with POs
1	Apply the fundamentals of parallel and parallel computing including architectures and paradigms.	Apply	1	1
2	Analyse the various design principles of parallel algorithms.	Analyse	1	2
3	Learn the intricacies of distributed programming.	Under-stand	2	1
4	Develop and execute basic parallel and distributed applications using basic programming models and tools.	create	2	5

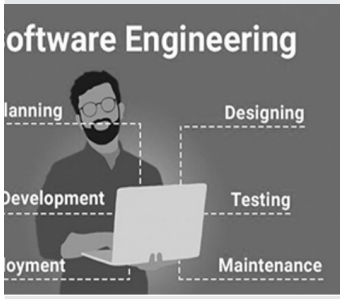
TEXT BOOKS:

1. DISTRIBUTED SYSTEMS Concepts and Design Fifth Edition. George Coulouris. Cambridge University. Jean Dollimore formerly of Queen Mary, University of London.
2. Distributed Systems Principles and Paradigms Andrew S. Tanenbaum Maarten Van Steen, 3rd Edition, 2017.

REFERENCE LINKS:

1. Introduction To Parallel Programming, Peter S. Pacheco University of San Francisco.
2. Introduction To Parallel Processing, M.Sasikumar, Dinesh Shikhare and P. Ravi Prakash, Randy Chow, T. Johnson, Distributed Operating Systems and Algorithms, Addison Wesley.
3. Ian Foster: Designing and Building Parallel Programs – Concepts and tools for Parallel Software Engineering, Pearson Publisher, 1st Edition, 2019.
4. Parallel Programming in C with MPI and Open MP Michael J.Quinn, McGrawHill Higher Education.
5. https://hpc.llnl.gov/training/tutorials/introduction-parallel-computing-tutorial.
6. https://www.geeksforgeeks.org/introduction-to-parallel-computing/.
7. https://nptel.ac.in/.
8. https://www.coursera.org/.

22CS307 SOFTWARE ENGINEERING



Source: https://artoftesting.com/software-engineering

Hours Per Week :

L	T	P	C
2	0	2	3

PREREQUISITE KNOWLEDGE: Data base management systems, OOPs through Java.

COURSE DESCRIPTION AND OBJECTIVES:

This course focuses on the concepts of software life cycle, role of process models and methods to prepare software requirement specification document. In addition to that, it also imparts knowledge of design, development and testing of software. The objective of this course is to enable the student to develop efficient, cost effective, feasible software as per user requirements.

MODULE - 1

UNIT-1 **8L+0T+8P=16 Hours**

INTRODUCTION

Introduction to Software Engineering: Introduction to Software and Software engineering, Software characteristics, Software project, Software myths, Project Planning, Scheduling and Management.

Generic View of Process: Software Engineering - A layered technology, A process framework, Software Development Life Cycle (SDLC), The Capability Maturity Model Integration (CMMI).

Process Models: Conventional Model, Agile process models - Unified process model, Extreme Programming, Scrum.

UNIT-2 **8L+0T+8P=16 Hours**

REQUIREMENTS ENGINEERING

Requirements Engineering: Functional and Non-functional requirements, User requirements, System requirements, Requirement engineering tasks, formal requirements specification and verification, Feasibility Study.

Building the Analysis Model: Data modeling - Data objects, Attributes, Relationships, Cardinality and modality. Class based modeling - Identify analysis classes, specify attributes and Define operations.

Design Engineering: Design model, Design concepts. Creating an Architectural Design-Architectural styles and patterns.

Performing User Interface Design: Golden rules; User interface analysis and design.

PRACTICES:

Laboratory session of this course is designed in such a way that the student should complete three projects of the given type by performing the below experiments.

- Development of software requirements specification using Mind-Map tool.
- Project planning using Gantt charts.
- Project estimation using metrics.
- Capture Use Case Scenarios and model UML Use Case Diagrams.
- Model the UML state chart and Activity diagrams.
- Model the UML Class and Sequence diagrams.

MODULE - 2

UNIT-1 **8L+0T+8P=16 Hours**

TESTING

Testing Strategies: A strategic approach to software testing, Unit testing, Integration testing, Validation testing, System testing,

SKILLS:

- ✓ *Know the software requirements and find out various ways to gather and specify them.*
- ✓ *Choose a process model for developing software solutions without schedule/ effort overruns and good quality.*
- ✓ *Analyse and model (diagrammatical representations) a software product.*

Testing Tactics: Black-Box and White-Box testing techniques, Art of debugging.

Product Metrics: Metrics for analysis model; Metrics for design model, Metrics for source code; Metrics for testing; Metrics for maintenance.

UNIT–2 **8L+0T+8P=16 Hours**

RISK AND QUALITY MANAGEMENT

Risk Management: Software risks, Risk identification; Risk projection; Risk refinement, Reactive vs Proactive risk strategies, RMMM.

Quality Management: Quality concepts, Formal technical reviews, Statistical Software Quality Assurance.

Computer-Aided Software Engineering (CASE): Use of appropriate CASE tools- Requirement engineering tools, Project planning tools, Testing tools.

PRACTICES:

Laboratory session of this course is designed in such a way that the student should complete three projects of the given type by performing the below experiments.

- Estimate the test coverage and Structural complexity of product using metrics.
- Develop the test cases for all the functional requirements of projects selected.
- Perform the functional testing using Selenium tool.

LIST OF PROJECTS:

Project-1: A Point-Of-Sale (PoS) System: A POS system is a computerized application used to record sales and handle payments; it is typically used in a retail store, it includes hardware components such as a computer and bar code scanner, and software to run the system. It interfaces to various service applications, such as a third-party tax calculator and inventory control. These systems must be relatively fault tolerant; that is, even if remote services are temporarily unavailable they must still be of capturing sales and handling at least cash payments. A POS system must support multiple and varied client-side terminals and interfaces such as browser, PDAs, touch-screens.

Project-2: Online Bookshop Example: Following the model of amazon.com or bn.com, design and implement an online bookstore.

Project-3: A Simulated Company: Simulate a small manufacturing company. The resulting application will enable the user to take out a loan, purchase a machine, and over a series of monthly production runs, follow the performance of their company.

Project-4: A Multi-Threaded Airport Simulation: Simulate the operations in an airport. Your application should support multiple aircrafts using several runways and gates avoiding collisions/ conflicts. Landing: an aircraft uses the runway, lands, and then taxis over to the terminal. Take-Off: an aircraft taxies to the runway and then takes off.

Project-5: An Automated Community Portal: Business in the 21st Century is above all BUSY. Distractions are everywhere. The current crop of “enterprise intranet portals” is often high noise and low value, despite the large capital expenditures it takes to stand them up. Email takes up 30 - 70% of an employee’s time. Chat and Instant Messaging are either in the enterprise or just around the corner. Meanwhile, management is tasked with unforeseen and unfunded leadership and change-agent roles as well as leadership development and succession management. What is needed is a simplified, repeatable process that enhances communications within an enterprise, while allowing management and peers to self-select future leaders and easily recognize high performance team members in a dynamic way. Additionally, the system should function as a general-purpose content management, business intelligence and peer-review application. Glasscode’s goal is to build that system.

Project-6: Content Management System: The goal is to enable non-technical end users to easily publish, access, and share information over the web, while giving administrators and managers complete control over the presentation, style, security, and permissions. Features: Robust Permissions System, Templates for easy custom site designs, Total control over the content, Search engine friendly URL’s, Role based publishing system, Versioning control, Visitor profiling.

Project-7: An Auction Application: Several commerce models exist and are the basis for several companies like eBay.com, pricellne.com etc. Design and implement an auction application that provides auctioning services. It should clearly model the various auctioneers, the bidding process, auctioning etc.

Project-8: A Notes And File Management System: During one's student years and professional career one produces a 1 lot of personal notes, documents. All these documents are usually kept 1 on papers or individual files on the computer. Either way the bulk of the information is often erased corrupted and eventually lost. The goal of this 1 project is to build a distrib- VFSTR 106 III Year I Semester uted software application that addresses this problem. The system will provide an interface to create, organize and manage personal notes through the Internet for multiple users. The system will also allow users to collaborate by assigning permissions for multiple users to view and edit notes.

Project-9: Library Management System(LMS): The goal is to enable students and librarians to easily access and manage the library and run it smoothly. Each physical library item - book, tape cassette, CD, DVD, etc. could have its own item number. To support it, the items may be barcoded. The purpose of barcoding is to provide a unique and scannable identifier that links the barcoded physical item to the electronic record in the catalog. Barcode must be physically attached to the item, and barcode number is entered into the corresponding field in the electronic item record. Barcodes on library items could be replaced by RFID tags. The RFID tag can contain item's identifier, title, material type, etc. It is read by an RFID reader, without the need to open a book cover or CD/DVD case to scan it with barcode reader.

Project-10: Hospital Management System: Simulate to show and explain hospital structure, staff, and relationships with patients, and patient treatment terminology

Project-11: Draft Software Requirement Analysis for the following Problem Statement: Fuel Delivery System: An unattended petrol (gas) pump system that includes a credit card reader. The customer swipes the card through the reader and then specifies the amount of fuel required. The fuel is delivered, and the customer's account debited.

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Module No.	Mapping with POs
1	Use basic concepts of software engineering for designing software product.	Usage	1	1, 11
2	Compare different process models and identify appropriate process model based on project requirements.	Evaluation	1	2, 4
3	Build Software Requirement Specification (SRS) document for any software product.	Design	1	3, 5
4	Design of solutions using UML diagrams like Use case, Sequence diagrams etc.	Design	1	3, 4, 5
5	Create an appropriate architecture for a given project that meets all quality constraints.	Create	2	5
6	Apply different testing techniques to ensure bug free software and metrics to measure the software size, complexity, and budget etc.	Apply	2	4, 5, 11

TEXT BOOKS:

1. Roger S. Pressman, "Software Engineering, A practitioner's Approach", 6th Edition, McGrawHill International Edition, 2008.
2. Booch G., Rumbaugh J. and Jacobsons I, "The Unified Modeling Language User Guide", 2nd Edition, Addison Wesley, 2005.

REFERENCE BOOKS:

1. Simon Sennet, Steve McRobb and Ray Farmer, "Object Oriented Systems Analysis and Design, 2nd edition, 2004.
2. Dr. Pankaj Jalote "Software Engineering: A Precise Approach"-edition 2010.

IV

YEAR

B.Tech.

COMPUTER SCIENCE AND ENGINEERING

I SEMESTER

22CS406	-	Privacy and Intrusion Detection
22CS402	-	Big Data and Analytics
22CS403	-	Cloud Computing
22CS310	-	Computing Ethics
	-	Department Elective - 3
	-	Department Elective - 4

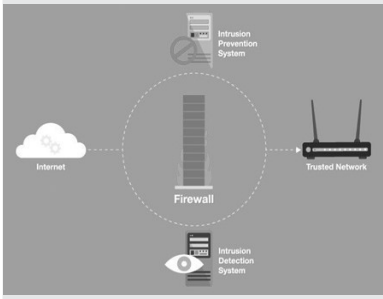
II SEMESTER

22CS404	-	Internship / Project Work

COURSE CONTENTS

I SEM & II SEM

22CS406 PRIVACY PRESERVING AND INTRUSION DETECTION



Source: https://www.kirkpatrickprice.com/blog/idps-techniques

Hours Per Week :

L	T	P	C
2	2	0	3

PREREQUISITE KNOWLEDGE: Cryptography and Network Security.

COURSE DESCRIPTION AND OBJECTIVES:

This course aims to understand modern concepts related to Intrusion Detection System. The course compares alternative tools and approaches for Intrusion Detection through quantitative analysis to determine the best tool or approach to reduce risk from intrusion Students undergoing this course can identify and describe the parts of all intrusion detection systems and characterize new and emerging IDS technologies according to the basic capabilities all intrusion detection systems share.

MODULE-1

UNIT-1 **8L+8T+0P=16 Hours**

INTRODUCTION

Basic Concepts of Security, Introduction to Intrusions, Need of Intrusion Detection, Classification of Intrusion Detection Systems, Sources of Vulnerabilities, Attacks against various security objectives, countermeasures of attacks.

UNIT-2 **8L+8T+0P=16 Hours**

INTRUSION DETECTION AND PREVENTION TECHNOLOGIES

Host-based intrusion detection system (HIDS), Network-based IDS, Information Sources for IDS, Host and Network Vulnerabilities and Countermeasures. Intrusion detection techniques, misuse detection: pattern matching, rule-based and state-based anomaly detection: statistical based, machine learning based, data mining-based hybrid detection.

IDS and IPS Architecture: Tiered architectures, Single-tiered, Multi-tiered, Peer-to-Peer. Sensor: sensor functions, sensor deployment and security.

PRACTICES:

- Installing Snort into the Operating System
- Configuring and Starting the Snort IDS
- Writing and Adding a Snort Rule
- Triggering an Alert for the New Rule

MODULE- 2

UNIT-1 **8L+8T+0P=16 Hours**

AGENTS

Agent functions, agent deployment and security. Manager component: manager functions, manager deployment and security. Information flow in IDS and IPS, defending IDS/IPS, Case study on commercial and open-source IDS.

UNIT-2 **8L+8T+0P=16 Hours**

ALERT MANAGEMENT AND CORRELATION DATA FUSION

Alert correlation, Pre-process, Correlation Techniques, Post-process, Alert Correlation architectures. Cooperative Intrusion Detection, Cooperative Discovery of Intrusion chain, Abstraction-based Intrusion

SKILLS:

- ✓ *Installation and configuration of IDS.*
- ✓ *Compare various anomaly detection techniques.*
- ✓ *Evaluate security of intrusion detection tool.*

Detection, Interest-based communication and cooperation, agent-based cooperation.

PRACTICES:

- Demo of Eavesdropping attack and its Prevention using SSH.
- Demonstrate how to provide secure data storage, secure data transmission and for creating digital signatures (GnuPG)
- Setup a honey pot and monitor the honeypot on network (KF Sensor)
- Installation of rootkits and study about the variety of option.
- Perform wireless audit on an access point or a router and decrypt WEP and WPA. (NetStumbler)
- Install any open-source IDS and study the logs

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Module No.	Mapping with POs
1	Apply Intrusion Detection tools and techniques in order to improve their security posture.	Apply	2	1,2
2	Apply the knowledge to the architecture, configuration, and analysis of specific intrusion detection systems.	Apply	2	1, 2
3	Analyse appropriate situations and scenarios where intrusion detection may be applied to achieve an increased level of situational awareness and information assurance.	Analyse	1	1,2
4	Evaluate the security of an organization for better performance.	Evaluate	1	1,2

TEXT BOOKS:

1. C. Endorf, E. Schultz and J. Mellander, Intrusion Detection & Prevention, McGraw-Hill/Osborne , 2006.

REFERENCE BOOKS:

1. Ali A. Ghorbani, Network intrusion detection and prevention concepts and techniques, Springer, 2010.
2. J. M. Kizza, Computer Network Security, Springer, 2005.
3. Chris Sanders and Jason Smith, Applied Network Security Monitoring: Collection, Detection, and Analysis, Syngress, 2013.

22CS402 BIG DATA AND ANALYTICS



Source: https://miro.medium.com/max/844/0*ARAg3FnAzy2e02Wy.png

Hours Per Week :

L	T	P	C
3	0	2	4

PREREQUISITE KNOWLEDGE: Basics of databases, Data mining.

COURSE DESCRIPTION AND OBJECTIVES:

This course serves as an introductory course to gain knowledge on analyzing Big Data. Expecting to face Big Data storage, processing, analysis, visualization, and application issues on both workplaces and research environments. Get insight on what tools, algorithms, and platforms to use on which types of real world use cases.

MODULE-1

UNIT-1 **12L+0T+8P=20 Hours**

INTRODUCTION TO BIG DATA

Data, Characteristics of data and Types of digital data, Sources of data, Working with unstructured data, Evolution and Definition of big data, Characteristics and Need of big data, Challenges of big data.

Big data analytics: Overview of business intelligence, Data science and Analytics, Meaning and Characteristics of big data analytics, Need of big data analytics, Classification of analytics, Challenges to big data analytics, Importance of big data analytics, Basic terminologies in big data environment.

UNIT-2 **12L+0T+8P=20 Hours**

INTRODUCTION TO HADOOP

Introducing Hadoop, need of Hadoop, limitations of RDBMS, RDBMS versus Hadoop, Distributed Computing Challenges, History of Hadoop, Hadoop Overview, Use Case of Hadoop, Hadoop Distributors, HDFS (Hadoop Distributed File System), Processing Data with Hadoop, Managing Resources and Applications with Hadoop YARN (Yet another Resource Negotiator), Interacting with Hadoop Ecosystem.

PRACTICES:

- Hadoop installation in standalone machine.
- Pig installation.
- Setup of Hadoop cluster.
- HDFS basic command-line file operations.
- HDFS monitoring User Interface.

MODULE-2

UNIT-1 **12L+0T+8P=20 Hours**

MAPREDUCE PROGRAMMING

Introduction, Mapper, Reducer, Combiner, Partitioner, Searching, Sorting, Compression, Real time applications using Map Reduce, combiner, Partitioner, matrix multiplication using Map Reduce and page rank algorithm using Map Reduce.

SKILLS:

- ✓ *Build and maintain reliable, scalable, distributed systems with Apache Hadoop.*
- ✓ *Develop Map Reduce based applications for Big data.*
- ✓ *Design and build applications using Hive and pig based Big data applications.*
- ✓ *Learn tips and tricks for big data use cases and solutions.*

UNIT-2 **12L+0T+8P=20 Hours**

PIG

Introduction to Pig, The Anatomy of Pig, Pig on Hadoop, Pig Philosophy, Use Case for Pig: ETL Processing, Pig Latin Overview, Data Types in Pig, Running Pig, Execution Modes of Pig, HDFS Commands, Relational Operators, Piggy Bank, Word Count Example using Pig, Pig at Yahoo!.

Hive: Introduction to Hive, Hive Architecture, Hive Data Types, Hive File Format, Hive Query Language (HQL), Partitions and bucketing, working with XML files, User-Defined Function (UDF) in Hive, Pig versus Hive.

Spark Programming: Introduction, features of Spark, components of Spark, Programming with Resilient Distributed datasets (RDDS).

PRACTICES:

- Word Count Map Reduce program using Hadoop.
- Implementation of word count with combiner Map Reduce program.
- Practice on Map Reduce Monitoring User Interface.
- Implementation of Sort operation using Map Reduce.
- Map Reduce program to count the occurrence of similar words in a file by using partitioner.
- Design Map Reduce solution to find the years whose average sales is greater than 30.
  - input file format has year, sales of all months and average sales.
  - Year Jan Feb Mar April May Jun July Aug Sep Oct Nov Dec Average.
- Map Reduce program to find Dept wise salary.
  - Empno Emp Name Dept Salary.
- Designing of Pig Latin scripts to sort, group, join, project and filter the data.
- Implementation of Word count using Pig.
- Creation of Database and tables using Hive query language.
- Implementation of partitions and buckets using Hive query language.
- Implementation of word count using spark RDD.

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Module No.	Mapping with POs
1	Use of Big data frameworks like Hadoop and NOSQL to efficiently store and process Big data to generate analytics.	Apply	1	1, 2, 5,9,10,12
2	Design a solution for data intensive problems using Map Reduce paradigm.	Apply	1	1, 2, 5, 9,10,12
3	Design and analyze the solutions of Big data using Pig and Hive to solve data intensive and to generate analytics.	Apply	2	1, 2, 3, 5, 9,10,12
4	Analyze Big data using Spark programming.	Analyze	2	1, 2, 3, 5, 9,10,12

TEXT BOOKS:

1. Seema Acharya, Subhashini Chellappan, “Big Data Analytics”, Wiley, 2015.
2. Holden Karau, Andy Konwinski, Patrick Wendell, MateiZaharia, “Learning Spark: Lightning-Fast Data Analysis”, O’Reilly Media, Inc., 2015.

REFERENCE BOOKS:

1. Boris Lublinsky, KevinT. Smith, Alexey Yakubovich, “Professional Hadoop Solutions”, Wiley, 2015.
2. Chris Eaton,Dirk deRooset al. , “Understanding Big data ”, McGraw Hill, 2012.
3. Tom White, “HADOOP: The definitive Guide”, O Reilly 2012.

22CS403 CLOUD COMPUTING



Source: https://tse4.mm.bing.net/th/id/OIP.vN86lZCAdr3RDSXc0cuHcAHaE8?pid=ImgDet&rs=1

Hours Per Week :

L	T	P	C
3	0	2	4

PREREQUISITE KNOWLEDGE: Operating systems and Computer networks.

COURSE DESCRIPTION AND OBJECTIVES:

This course deals with the computing model, which enables information, software, and shared resources to be provisioned over the network as services in an on-demand manner. The main objective of this course is to enable the student to understand the evolution of cloud computing through its supporting technologies virtualization and the architectures of top cloud platforms.

MODULE-1

UNIT-1 **12L+0T+8P=20 Hours**

INTRODUCTION

Introduction: Definition, Historical developments, Computing platforms, and technologies.

Principles Of Parallel and Distributed Computing: Parallel versus distributed computing, Elements of parallel computing, Elements of distributed computing, Technologies for distributed computing.

UNIT-2 **12L+0T+8P=20 Hours**

VIRTUALIZATION

Introduction, Characteristics, Virtualization techniques, Virtualization and cloud computing, Pros. and cons. of virtualization, Technology examples.

Cloud Computing Architecture: Introduction, Cloud reference model, Types of clouds, Economics of clouds, Open challenges.

Cloud Platforms in Industry: Amazon web Services, Google app engine, Microsoft Azure.

PRACTICES:

- Probability and statistical analysis of performing hardware virtualization using VMware workstation.
- Launch Amazon Linux EC2 Instance and connect the windows client to it.
- Launch Windows EC2 instance in AWS and connect windows client to it.
- Configure Web Server on Amazon Linux instance with Elastic IP.
- Manage Elastic Block Storage(EBS).
- Configure Amazon Simple Storage Service (Amazon s3).
- Configure Amazon S3 Glacier.
- Configure Amazon EFS.

MODULE-2

UNIT-1 **12L+0T+8P=20 Hours**

ANEKA

Cloud application platform, Framework overview, Anatomy of the Aneka container, Building Aneka clouds, Cloud programming, and management.

High Throughput Computing- Task Programming: Task computing, Task-based application models,

SKILLS:

- ✓ *Gain knowledge of different types of Cloud Service Providers.*
- ✓ *Explore basic design issues of Cloud Applications.*
- ✓ *Compare & evaluate the optimum costs in the data transmissions.*

Aneka task-based programming.

UNIT-2 **12L+0T+8P=20 Hours**

CLOUD APPLICATIONS & GLOBAL AND LOCAL IMPACT OF CLOUD COMPUTING ON SOCIETY

Scientific applications in healthcare, biology, geo science; Business applications in– CRM and ERP, productivity, social networking, media applications, multiplayer online gaming.

ETHICAL CONSIDERATION FOR CLOUD COMPUTING: Cloud Security Risks, Security: The Top Concern for Cloud Users, Privacy and Privacy Impact Assessment, Trust, Operating System Security, Virtual Machine Security,

PRACTICES:

- Configure Amazon Virtual Private Cloud (VPC).
  a) Create your own VPC.
  b) Create a public subnet.
  c) Create a private subnet.
  d) Create an Internet gateway and attach to your VPC.
  e) Create Pubic Routing Table, associate subnet and add routing rules.
  f) Create Private Routing Table, associate subnet and add routing Rules.
  g) To launch Windows instance in Public subnet.
- Configure Amazon Elastic Load Balancer.
- Configure Relational Database Service (RDS).

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Module No.	Mapping with POs
1	Deploying a VM Image to understand the evolution of cloud computing in contrast to the traditional approach.	Apply	1	1,5
2	Evaluate the concepts of various virtualization technologies.	Evaluate	1	2,5
3	Analyze the trade-offs, security, and privacy issues among application deployment in the various cloud and the local infrastructure.	Analyze	2	2,5
4	Deploy applications over commercial cloud computing infrastructures.	Apply	2	1,5

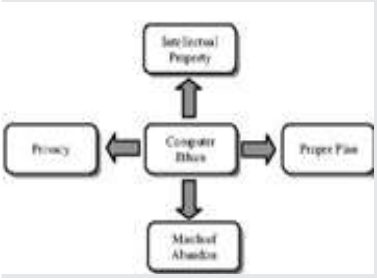
TEXT BOOKS:

1. Raj Kumar Buyya, C Vecchiola and S TSelvi , “Mastering Cloud Computing”, 1st Edition, Tata McGraw Hill Education (India), 2013.
2. RajKumarBuyya, Broberg J and GoscinskiA, “Cloud Computing - Principles and Paradigms”, 1st Edition, Wiley, 2011.

REFERENCE BOOKS:

1. David S. Linthicum, Cloud Computing and SOA Convergence in Your Enterprise A Step-by-Step Guide, Pearson 2010.
2. Dr. Kumar Saurabh, Cloud Computing, 2nd Edition, Wiley India 2012.
3. Rittinghouse J W, and Ransome J F, “Cloud Computing - Implementation, Management, and Security”, 1st Edition, CRC Press, 2009.
4. Michael Wittig and Andreas Wittig, “Amazon Web Services in Action”, 2nd Edition, Manning Publications, 2015.
5. Tim Mather, Subra Kumaraswamy, Shahed Latif, “Cloud Security and Privacy: An Enterprise Perspective on Risks and Compliance”, O’ReillyMediaInc, 2009.

22CS310 COMPUTING ETHICS



Source: Figure 2 from Computer and Information Security Ethics -- Models | Semantic Scholar

Hours Per Week :

L	T	P	C
1	2	0	2

PREREQUISITE KNOWLEDGE: General awareness on ethical guidelines.

COURSE DESCRIPTION AND OBJECTIVES:

The course will provide students with an understanding on Engineering Ethics and the nature of moral issues and dilemmas faced by Computer Science and engineers in their professional lives. It will give them an awareness on professional rights and responsibilities of an engineer and acquaint them on the Code of Conduct and Ethics prescribed by professional bodies like IEEE, ACM etc. for its members..

MODULE-1

UNIT-1 **0L+8T+8P=16 Hours**

INTRODUCTION

Introduction to Sociotechnical Computer Ethics.

Introduction: Why Computer Ethics? The Sociotechnical Systems Perspective, Sociotechnical Computer Ethics, Micro- and Macro-Level Analysis.

Ethics and Information Technology

Introduction: Doing Ethics, Ethical Theories and Concepts.

UNIT-2 **12L+0T+8P=20 Hours**

ETHICS IN IT CONFIGURED SOCIETIES

Technology as the Instrumentation of Human Action, Three Features of IT-Configured Activities, IT Configured Domains of Life, Democracy and the Internet.

Information Flow, Privacy and Surveillance

Introduction: Information Flow with and without Information Technology, why care about Privacy, Is Privacy over? Strategies for shaping Personal Information Flow.

PRACTICES:

- Surprises about Social Networking
- RFID and caring for the Elderly People
- Turing does not need to know
- Turnitin Dot Com
- Email Privacy and Advertising
- Workplace Spying
- Data Mining and e-Business

MODULE-2

UNIT-1 **0L+8T+8P=16 Hours**

DIGITAL INTELLECTUAL PROPERTY

The complexities of Digital Property, Protecting Property rights in Software, The Philosophical Basis of Property, Free and Open Source Software, PS Versus FOSS.

Introduction: Law and Order in the Internet, Sociotechnical Order, Online Crime, Hackers and Hacker

SKILLS:

✓ .

Ethic, Sociotechnical Security.

High Throughput Computing- Task Programming: Task computing, Task-based application models, Aneka task-based programming.

UNIT-2 **0L+8T+8P=16 Hours**

PROFESSIONAL ETHICS

Introduction: Why Professional Ethics, The Paradigm of Professions, Characteristics of Professions, Sorting out Computing and its Status as a Profession, Codes of Ethics, Professional Relationships, A Legal perspective on professionalism in Computing.

PRACTICES:

Laboratory session of this course is designed in such a way that the student should complete three projects of the given type by performing the below experiments.

- Obtaining pirated Software Abroad
- Free Software that follows Proprietary Software
- Using Public Domain Software in Proprietary Software
- Wiki Warfare
- Yahoo and Nazi Memorabilia
- Bot Roast
- Software Safety
- Conflicts of Interest
- Security in a Custom Database.

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Module No.	Mapping with POs
1	Engage in an informed critical reflection on the nature of professionalism and ethical challenges inherent in engineering profession	Under-standing	1	8
2	Understand the role of professional bodies, and the code of ethics and industrial standards pre-scribed for engineers	Under-standing	2	8
3	Apply awareness of professional rights and re-sponsibilities of an engineer to conduct themselves ethically within an organization	Applying	1	8
4	Apply understanding of safety norms to highlight ethical issues in risky situation	Applying	2	8

TEXT BOOKS:

1. Deborah G. Johnson, Keith W. Miller and Prentice Hall, “Computer Ethics”, 4th Edition, Pearson, 2009.

REFERENCE BOOKS:

1. Charles E. Harris, Michael S. Pritchard and Michael J. Rabins, “Engineering Ethics: Concepts and Cases”, 4th edition, Wadsworth Thompson Learning, 2009.
2. M. Govindarajan, S.Natarajan and V. S. Senthil Kumar, “Engineering Ethics”, Prentice Hall of India, Reprint 2013.

DEPT. ELECTIVES

B.Tech.

COMPUTER SCIENCE AND ENGINEERING

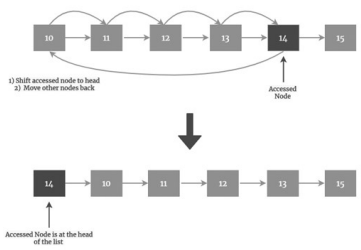
DEPARTMENT ELECTIVES

- 22CS801 - Advanced Data Structures
- 22CS802 - Advanced JAVA Programming
- 22CS803 - Computer Graphics
- 22CS804 - Deep Learning
- 22CS805 - Digital Forensics
- 22CS806 - Digital Image Processing
- 22CS807 - Web and Database Security
- 22CS808 - Machine Learning
- 22CS809 - Mobile Ad-hoc Networks
- 22CS810 - Mobile Application Development
- 22CS811 - Text Mining
- 22CS812 - Numerical Algorithms
- 22CS813 - Operating System Design
- 22CS814 - Optimization Techniques
- 22CS816 - Simulation and Modeling
- 22CSXXX - Natural Language Processing
- 22CSXXX - MLops

COURSE CONTENTS

I SEM & II SEM

22CS801 ADVANCED DATA STRUCTURES



Source: https://www.baeldung.com/cs/advanced-data-structures

Hours Per Week :

L	T	P	C
2	2	2	4

PREREQUISITE KNOWLEDGE: Data structures.

COURSE DESCRIPTION AND OBJECTIVES:

Advanced data structures are one of the most important disciplines of data science since they are used for storing, organizing, and managing data and information to make it more efficient, easier to access, and modify. The objective of this course is to improve students' skills in designing data structures and algorithms for various solutions, and problem design in large systems and applications such as databases, information retrieval systems, bioinformatics, and geographic information systems.

MODULE-1

UNIT-1 **8L+8T+8P=24 Hours**

HASHING, EXTERNAL SORTING & PRIORITY QUEUE

Hashing: Introduction, Hash Function, Collision Resolution Technique: Separate Chaining, Open Addressing.

External Sorting: Merge Sort.

Priority: Priority Queues: Introduction, Types of priority queues, implementing max priority queue and min priority queue.

UNIT-2 **8L+8T+8P=24 Hours**

TREES

Trees: Splay tree, Red Black Tree (RBT), Operations on RBT, M-way search tree and operations, Segment tree.

PRACTICES:

- Implementation of Hash Table.
- Implementation of Separate Chaining and Linear Probing
- Implementation of merge sort by taking a large file as input and sort the file.
- Implement different operations on Priority Queue. i.e. adding an element, removing an element, size of the priority queue, printing the queue, and top element of the queue.
- Implementation of a Splay tree operation.
- Implementation of a Red-Black tree operation.
- Implementation of M Way Search tree.
- You are given a tree with n vertices (numbered 1, 2, …, n) and an integer k.

A subtree is defined as a connected subgraph of the tree. That is, a subtree is another tree that can be obtained by removing some (possibly none) vertices and all edges incident to those vertices from T.

A subset S of vertices is called good if every subtree containing all the nodes in S has at least k nodes.

Find the size of the smallest good subset.

Input:

The first line contains a single integer T, the number of test cases. The descriptions of test cases follow.

The first line of each test case contains two integers, n, and k.

The next n-1 lines each contain two integers u, v, denoting an edge between vertices u and v of the tree..

SKILLS:

- ✓ *Analyse the data structure required for various applications.*
- ✓ *Usage of trees, graphs, heaps and tries.*
- ✓ *Know various pattern searching algorithms.*

Output:

For each test case print in a single line, the minimum size of a good subset.

MODULE-2

UNIT-1 **8L+8T+8P=24 Hours**

HEAPS

Heaps: Introduction, types of heaps, heap implementation with an array, heap sort, technique to sort elements, Mergeable heaps, Application of heap.

UNIT-2 **8L+8T+8P=24 Hours**

PATTERN SEARCHING & TRIES

Pattern Searching: Pattern matching algorithms -Brute force, the Boyer —Moore algorithm, Robin-Karp algorithm, Knuth-Morris-Pratt algorithm.

Tries: Standard Tries, Compressed Tries, Suffix tries.

PRACTICES:

- You are given N numbers you can decrease each number by 10 % or K kg whichever is more. If the number, you choose is less than K you can make it zero. Your task is to minimize the sum of the number as small as possible by performing N operations only. [Note: Use Priority Queue].

  Example:

  Input:

  N = 2

  k = 10

  arr = 100 15

  Output:

  95

  Explanation:

  You will remove 10 from 100

  then once again remove 10 90.
- Given a Binary Heap of size N in an array arr[]. Write a program to calculate the height of the Heap.

  Input: N = 6

  arr = {1, 3, 6, 5, 9, 8}

  Output: 2
- Given two binary max heaps as arrays, merge the given heaps to form a new max heap.

  Example:

  Input :

  n = 4 m = 3

  a[] = {10, 5, 6, 2},

  b[] = {12, 7, 9}

  Output :

  {12, 10, 9, 2, 5, 7, 6}
- In Doraland, people have unique Identity Numbers called D-id. Doraemon owns the most popular gadget shop in Doraland. Since his gadgets are in high demand and he has only K gadgets left he has decided to sell his gadgets to his most frequent customers only. N customers visit his shop and D-id of each customer is given in an array array [ ]. In case two or more people have visited his shop the same number of time he gives priority to the customer with higher D-id. Find the D-ids of people he sells his K gadgets to.

Example:

Input:

N = 6

array[] = {1, 1, 1, 2, 2, 3}

K = 2

Output:

1 2

Explanation:

Customers with D-id 1 and 2 are most frequent.

- You are given q queries of two types:
  X: Append value X into an array.

  X K: You are required to print the Kth minimum XOR of X with the current array.

  You have to make a new array whose ith element is current_array[i]X. Then sort it and print the Kth element.

  Input format

  The first line contains q (1d"qd"100000).

  Next q lines contain the types of queries. (1 or 2) If type is 1, then it contains X (1d"Xd"10e18).

  If type is 2, then it contains X (1d"Xd"10e18) and K (Kd"current array size).

  Output format

  Print the number in the second type of query. (Note: Use Tries)
- Given an array of strings arr[] of size n and given s a string str and an integer k. The task is to find the count of strings in arr[] whose prefix of length k matches with the k length prefix of str. [Note: Use Tries]

  Input:

  n = 6

  arr[] = {"abba", "abbb", "abbc", "abbd", "abaa", "abca"}

  str = "abbg"

  k = 3

  Output: 4

  Explanation:

  "abba", "abbb", "abbc" and "abbd" are the matching strings.
- Given an array of integers of size N find minimum xor of any 2 elements. [Note: Use Tries]

  Input:

  N = 3

  arr[] = {9,5,3}

  Output: 6

  Explanation:

  There are 3 pairs –

  9^5 = 12

  5^3 = 6

  9^3 = 10

  Therefore output is 6.

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Module No.	Mapping with POs
1	Implement external sorting and priority queue	Apply	1	1,2
2	Implement different types of trees and apply them to problem solutions.	Apply	1	1,2,3
3	Apply the concepts of advanced Trees for solving problems effectively.	Apply	1	1,2,3
4	Analyze how efficient pattern matching principles can be used in the design and implementation of qualitative research.	Analyse	2	1,2,3,4
5	Analyze the given scenario and choose the appropriate Data Structure for solving problems.	Analyse	2	1,2,3,4

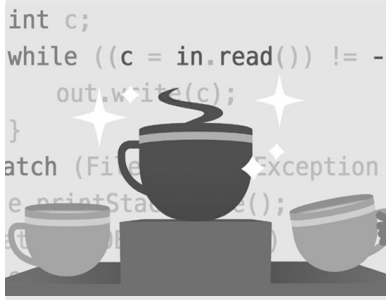
TEXT BOOKS:

1. Mark Allen Weiss, Data Structures and Algorithm Analysis in C++, 2nd Edition, Pearson, 2004.
2. T.Cormen, R.Rivest, C.Stein, C.Leiserson,“Introduction to Algorithms”, , PHI publication, Second Edition, 2004, ISBN 81-203-2141-3.

REFERENCE BOOKS:

1. Mark de Berg, Otfried Cheong, Marc van Kreveld, Computational Geometry: Algorithms and Applications:, Springer; 3rd Edition. 2008.
2. Sahani, Anderson freed, Horowitz, Fundamentals of DATA STRUCTURES in C, Silicon Pr; 2nd Edition, August 1, 2007.

22CS802 ADVANCED JAVA PROGRAMMING



Source: https://www.kpl.gov/catalog/item/?i=ent://LYNDA/0/LYNDA:103608

Hours Per Week :

L	T	P	C
2	2	2	4

PREREQUISITE KNOWLEDGE: Programming JAVA.

COURSE DESCRIPTION AND OBJECTIVES:

Advanced Java programming covers the standard concepts such as database connectivity, web-services, web application development etc. It is specially designed to develop web-based, network-centric or enterprise applications. It simplifies the complexity of building an n-tier application.

MODULE-1

UNIT-1 **8L+8T+8P=24 Hours**

INTRODUCTION TO JAVA WEB FRAME WORKS

Introduction to Java Web frameworks: What is Framework in Java, Advantages of Frameworks, Popular Java Web Frameworks: Spring, Spring Boot, Hibernate, Java Server Faces, Google Web Toolkit etc., Comparison of the web frameworks. Environmental setup to use Web Frameworks.

UNIT-2 **8L+8T+8P=24 Hours**

ORM AND HIBERNATE

ORM: What is Object Relational Mapping, How ORM Works, Features of ORM, Advantages, Java ORM- Hibernate, JAVA Persistence API(JPA), ORM implementation.

Hibernate: Overview of Hibernate, Hibernate Architecture, Hibernate Mapping Types, Hibernate O/R Mapping, Hibernate Annotation, Hibernate Query Language.

PRACTICES:

- Installation and Environmental setup to work with java Web Frameworks.
- Create a simple application of hibernate using XML.
- Create a maven based hibernate application using annotation
- Develop a web application using hibernate framework.
- Using the web application to perform database access with hibernate framework.

MODULE-2

UNIT-1 **6L+6T+6P=18 Hours**

STRUTS

Struts: Struts framework, Struts features, Model 1vs Model 2 (MVC) Architecture, Core components of Struts, Configuration of Struts, Creating a Struts Application.

UNIT-2 **10L+10T+10P=30 Hours**

SPRING AND SPRING BOOT

Spring: Overview of Spring, Spring Architecture, Inversion of Control (IOC) and Dependency Injection, XML Configuration on Spring, Creating a Spring Application. Spring MVC, flow of Spring Web MVC, Spring Web MVC Framework Example.

Spring Boot: Overview of Spring Boot, Spring vs Spring MVC vs Spring Boot, SB architecture, Components of SB, Creating Spring Boot Application.

SKILLS:

- ✓ *To know the importance of java web frameworks to develop enterprise java applications.*
- ✓ *To experience developing a web application using MVC architecture.*
- ✓ *Design and develop CRUD applications using different java frameworks.*
- ✓ *To derive hands on experience of developing enterprise applications using advanced web frameworks like Spring and Spring Boot.*

PRACTICES:

- Develop a simple web application using struts.
- Develop a CRUD (Create Read Update Delete) Application using spring.
- Implement Spring MVC Pagination, which is used to display a large number of records in different parts. Display 10, 20 or 50 records in one page.
- Using Spring MVC upload a file.
- Create a self-contained HTTP application that uses embedded server like Tomcat using Spring Boot.

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Module No.	Mapping with POs
1	Implement web frameworks to develop web applications using java.	Apply	1	1,2,3
2	Design java web applications using MVC architecture using web frameworks like hibernate.	Create	1	1,2,3
3	Implement java application to interact with database using struts.	Apply	2	1,2,3
4	Design secure transaction-based web applications using Spring framework.	Create	2	1,2,3,4

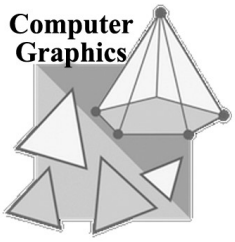
TEXT BOOKS:

1. James Keogh, “Complete Reference J2EE”, 9th Edition, McGraw Hill, Education, 2002.
2. Jeff Linwood and Dave Minter, “Beginning Hibernate”, 2nd Edition, Après publication 2010.

REFERENCES BOOKS:

1. Sharanam Shah, Vaishali Shah. “Structs 2 for Beginners”, 3rd Edition, Arizona Business Alliance, 2014.
2. Craig walls, “Spring in Action”, 5th Edition, Manning Publication, 2018.
3. https://struts.apache.org/getting-started/index.html.
4. https://www.javatpoint.com/hibernate-tutorial.
5. https://www.dineshonjava.com/spring-tutorial/.

22CS803 COMPUTER GRAPHICS



Source: https://www.javatpoint.com/computer-graphics-tutorial

Hours Per Week :

L	T	P	C
2	2	2	4

PREREQUISITE KNOWLEDGE: Basic Logical Thinking and Problem Solving Ability.

COURSE DESCRIPTION AND OBJECTIVES:

This course provides a comprehensive introduction to computer graphics leading to understanding the contemporary terminology and algorithms of computer graphics. To make the students learn the basic principles of visualization. To give an introduction to 2D and 3D modelling and animation.

MODULE-1

UNIT-1 **8L+8T+8P=24 Hours**

OVERVIEW OF GRAPHICS SYSTEM INTRODUCTION

Application areas of Computer Graphics; Video-display devices: raster-scan systems and random scan systems; Graphics primitives: display devices, primitive devices; Filled area primitives: scan line polygon fill algorithm, boundary-fill and flood-fill algorithms.

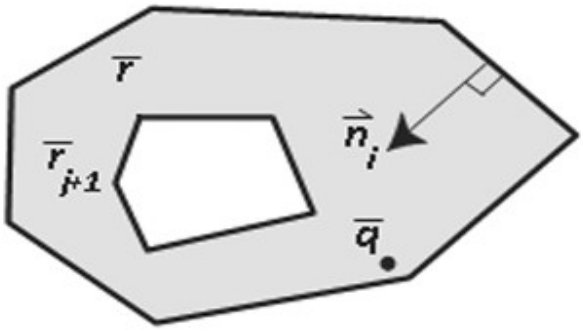
UNIT-2 **8L+8T+8P=24 Hours**

2D GEOMETRIC TRANSFORMATIONS AND VIEWING BASIC TRANSFORMATION

Geometric Transformations, matrix representations, and homogeneous coordinates. Inverse transformations, 2DComposite transformations, other 2D transformations, raster methods for geometric transformations; 2-D Viewing: The viewing pipeline, window to viewport coordinate transformation, Cohen-Sutherland and Cyrus-beck line clipping algorithms, Sutherland –Hodgeman polygon clipping algorithm.

PRACTICES:

- **Points & Polygons:** Suppose the vertices of a convex polygon are $\bar{p}1, \ldots, \bar{p}N$, given in counter-clockwise order. Let the coordinates of vertex $\bar{p}i$ be (xi ,yi).



Describe an algorithm that can tell whether a 2D point $\bar{q}$ is inside the gray-shaded region in the figure. You should assume that the polygons defined by vertices $\bar{p}1, \ldots, \bar{p}N$ and $\bar{r}1, \ldots, \bar{r}M$, respectively, are both convex. Hint: Each edge is contained in an infinite line. Each infinite line divides the 2D plane into two half-planes, the left half-plane and the right half-plane (here left/right means left/right with respect to a counter-clockwise direction of traversal of the vertices). The key insight you should use is that the interior of a convex polygon is the intersection of the left half-planes of all the edges of the polygon.

SKILLS:

- ✓ *Learn various clipping algorithms.*
- ✓ *Compare 2-D and 3_D transformations.*
- ✓ *Learn various surface detection methods*

- Transformations & Commutativity: We say that the 2D transformations f() and g() commute if and only if f(g($\bar{}$p)) = g(f($\bar{}$p)) for all points p$\bar{}$ ∈ R 2 . For each of the four cases below, where f() and g() are homographies, prove whether or not they commute:

  a) Both f() and g() are arbitrary homographies.

  b) One is an arbitrary rotation and the other an arbitrary translation.

  c) One is an arbitrary translation and the other is a non-uniform scaling.

  d) One is an arbitrary rotation and the other is a reflection. In each case, your solution can either be a derivation that proves/disproves commutativity or, if f() and g() do not commute, a specific counter-example.
- Contrast the implementation of Display File/Frame Buffer for a Random Scan System & Raster Scan System. Which type of system shall offer more consistent refresh rate and why?
- Determine the most appropriate pixels that will be plotted when Bresenham's algorithm is used to draw a line joining the points (10,20) and (20,30)
- Suppose a system with 8 inches by 10 inches video monitor that can display 100 pixels per inch. If memory is organized as one byte words, the starting frame buffer address is 0 and each pixel is assigned one byte of storage in memory, what is the frame buffer address of pixel with screen coordinates (x,y) ? Also, determine the total amount of memory consumed by the frame buffer.
- Consider a raster monitor of resolution 640*480 pixels. A scanning is used with horizontal retrace time of 4 micro seconds and vertical retrace time of 20 micro seconds respectively. Calculate the time available to display a pixel for both cases of (i) non-interlaced and (ii) interlaced. Assume a scan rate of 50 frames.

MODULE-2

UNIT-1 **8L+8T+8P=24 Hours**

3D GEOMETRIC TRANSFORMATIONS

3-D Object representation, 3-D geometry primitives, transformations, projection clipping. Interaction: Hardware input devices handling algorithms, Event handling echoing, Interactive techniques,3-D Geometric transformations, Translation, rotation, scaling, reflection and shear transformations, composite transformations, 3D viewing: Viewing pipeline, viewing coordinates, general projection transformation and clipping.

UNIT-2 **8L+8T+8P=24 Hours**

VISIBLE SURFACE DETECTION METHODS

Back-face detection, depth buffer, A-buffer, Z-buffer, scan-line Illumination Models, and Surface rendering Methods: Basic illumination models, polygon rendering method.

PRACTICES:

- Matrix implementation is a powerful technique used for computer graphics. For the concerns stated in section II, matrix for 3D space is represented as a 4X4 array instead of 3X3 array.
- Each operation of object such as translation and rotation can be represented by one Matrix. By multiplying the matrix, a vector/point can be transformed by the operation. For example, P = (x, y, z, 1) is a point's coordinator before the operation, and M = (1 0 0 a) (0 1 0 b)(0 0 1 c) (0 0 0 1) is the matrix representation of translating the object a blocks right, b blocks up and c blocks forward, then after the translating, the point's coordinator will become P' = P*M is the coordinator of the point after the operation.
- A sequence of operation can also be composed to be one operation. For example, matrix M1 represent first operation, matrix M2 represent second operation, the matrix M = M1*M2 will represent this sequence of 2 operations. If P = (x, y, z, 1) is the coordinator of the point before perform the sequence of the operations, then P' = P * M will be the coordinator of the point after the perform the operation1 then perform the operation2.

Following are matrix representation for some operations:

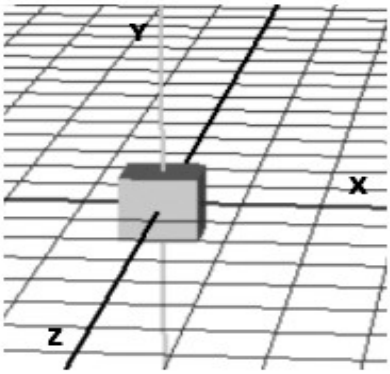
- Translate object by V, where V is the vector:

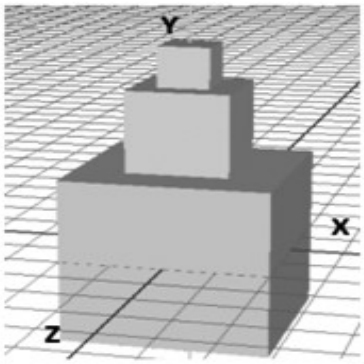
  M = (1 0 0 V[0]) (0 1 0 V[1]) (0 0 1 V[2]) (0 0 0 1)
- Rotate object along X axis by d degree:

  M = (1 0 0 0) (0 cos(d) -sin(d) 0) (0 sind(d) cos(d) 0) (0 0 0 1)
- Rotate object along Y axis by d degree:

  M = (cos(d) 0 sin(d) 0) (0 1 0 0) (-sin(d) 0 cos(d) 0) (0 0 0 1)
- Rotate object along Z axis by d degree:

  M = (cos(d) -sin(d) 0 0) (sin(d) cos(d) 0 0) (0 0 1 0) (0 0 0 1)
- Start with a unit cube centered at the origin. Use two more cubes like this one and obtain the model shown in the right image, where each new cube is twice the size of the previous one

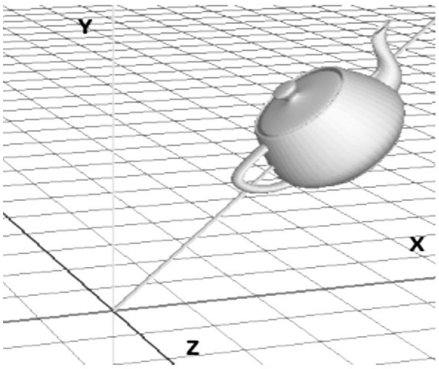


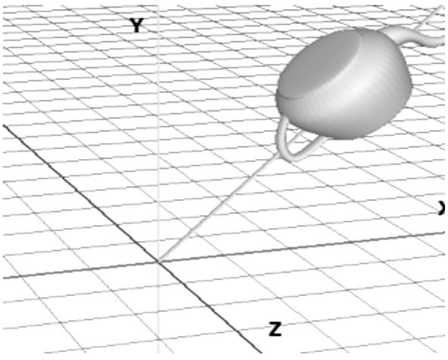


Below list the transformations in the order in which they were applied

1) scaling 2) Translation 3) Rotation

- There are situations in which rotations about axes other than the three coordinate axes are useful. In the next two activities, you will discover how a rotation about an arbitrary axis can be broken down into a series of rotations about the three coordinate axes.
- The yellow axis passing through the teapot in the image below is parallel to the vector (1, 1, 0). Determine a sequence of coordinate axis rotations that result in the teapot rotating 180 degrees about the yellow axis, as shown in the right image.





Write the sequence of rotations in the order in which you applied them. Hint: use a rotation that aligns the yellow axis with the x-axis, then rotate the object about the xaxis, then use the reverse rotation that brings the yellow axis back to original position.

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Module No.	Mapping with POs
1	Implement of various graphics systems along with the algorithms used for these devices for drawing.	Apply	1	1, 2
2	Demonstrate 2D graphics and algorithms including line drawing, polygon filling, clipping, and transformations.	Apply	1	1, 2
3	Demonstrate Concepts and techniques used in 3D computer graphics and basic about animation.	Apply	2	1, 2, 3
4	Implement Surface detection methods to get a realistic screen image.	Apply	2	1, 2,3

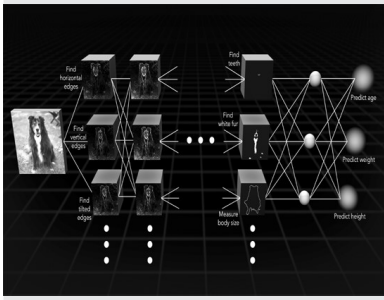
TEXT BOOKS:

1. Donald Hearn and M. Pauline Baker, “Computer Graphics C version”, Pearson education, Second Edition, 2014.
2. Zhigang Xiang, Roy Plastock, “Computer Graphics”, Schaum’s outlines, Second Edition, Tata McGraw Hill Edition, 2000.

REFERENCE BOOKS:

1. Foley, VanDam, Feiner and Hughes, “Computer Graphics Principles &Practice in C”, Pearson Education. 2nd Edition, 1996.
2. David F Rogers, “Procedural elements for Computer Graphics”, Tata McGraw Hill, 2nd Edition., 1988.
3. Neuman and Sproul, “Principles of Interactive Computer Graphics, Tata McGraw Hill”, 2nd Edition., 1978.
4. Shalini, Govil-Pai, “Principles of Computer Graphics”, Springer. 1st Edition, 2006.
5. Steven Harrington, “Computer Graphics”, TMH, 2nd Edition, 1987.

22CS804 DEEP LEARNING



Source: https://www.symmetrymagazine.org/sites/default/files/images/standard/neural_network_visual_final.jpg

Hours Per Week :

L	T	P	C
3	0	2	4

PREREQUISITE KNOWLEDGE: Machine learning, Python programming.

COURSE DESCRIPTION AND OBJECTIVES:

This course offers sufficient details required to understand the basic building blocks of various deep learning based models. Especially, focuses on different types of neural network models like feed forward neural networks, convolutional neural networks, recurrent neural networks, and deep auto encoders. During this course the students build, train, and evaluate deep neural network models for various applications in image, text, and speech domains. In addition, throughout this course students will be able to understand hyper parameter tuning and other best-practices to be followed while training deep neural network models.

MODULE-1

UNIT-1 **12L+0T+8P=20 Hours**

EVOLUTION OF DEEP NEURAL NETWORKS

Deep Learning Intuition: History of Deep Learning, what is Deep Learning? Applications of Deep Learning.

Neural Network Basics: McCulloch–Pitts neuron, Perceptron learning rule, Perceptron convergence theorem, Sigmoidal neuron, Multi-layer feed forward neural network, back propagation algorithm, Gradient descent method, Stochastic gradient descent method. Shallow Neural Networks and Deep Neural Networks.

REGULARIZATION and OPTIMIZATION for training Deep Models: Optimization methods - Adagrad, Adadelta, RMSProp, Adam; Regularization Methods-Dropout, Drop connect, Batch normalization; Activation functions - Linear, sigmoid, sigmoid, ReLU and variations of ReLU; Losss Function, Improving the training process – Dataset Augmentation, Noise Robustness, Weight Initialization methods, Early stopping, Parameter sharing and tying, bagging and other ensemble methods;

UNIT-2 **12L+0T+8P=20 Hours**

CONVOLUTIONAL NEURAL NETWORKS

Convolutional Neural Networks (CNNs): Foundations of Convolutional Neural Network, Popular Deep **CNN Models:** LeNet, AlexNet, VGGNet, ResNet, Google Net and other architectures.

INSTRUCTIONS FOR PRACTICES:

- Practice Assignments can be implemented using the Keras / Tensorflow APIs of Python
- Relevant data sets can be downloaded from standard repositories such as Kaggle/UCI or can be developed by the students.

PRACTICES:

- Implement Logistic regression With Neural Network Mindset.
  - logistic regression classifier for classification.
  - Plot the loss over each epoch.
  - Plot the accuracy over each epoch.
  - Report final Accuracy.

SKILLS:

- ✓ *Developing Vision and text based applications*
- ✓ *Hyperparameter Tuning of a deep Neural network model.*
- ✓ *Tensor Flow/ Keras tool usage for neural network implementation.*

- Implement Shallow Neural Network model:
  - Implement a binary classification neural network with a single and multiple hidden layers.
  - Implement a Multi-class classification neural network with a single and multiple hidden layers.
  - Vary the number of neurons at suitable layers.
- Hyper parameter Tuning of a Neural Network model implemented for hand-written digit classification:
  - Vary the type of activation functions.
  - Choose suitable Loss functions.
  - Vary the number of neurons at suitable layers.
  - Vary Weight Initialization methods.
  - Save the Best Model and load the saved model.
- Building a Deep Neural Network:
  - Implement a multi-class classification neural network with number of layers of your choice.
  - Include Batch Normalization layers.
  - Vary Optimization methods.
  - Add drop out layers.
- Convolutional Neural Network Models.
  - Design a Convolutional neural network with the layers of your choice
  - Compare the performance by changing the
  - Kernel size
  - Number of feature maps at each convolutional layer
  - Stride.
  - Padding.
  - Number of fully connected layers.
- Visualization of CNN Models.
  - Design a Convolutional Neural Network Model for image classification.
  - Plot Model Architecture.
  - Visualize feature maps after training of CNN.
  - Visualize class activation maps.

MODULE-2

UNIT-1 **14L+0T+10P=24 Hours**

DEEP UNSUPERVISED LEARNING

Transfer learning Approaches: Deep Pre-trained architectures- AlexNet, VGG16, VGG19, ResNEt. Use deep Convolutional architectures for feature extraction and fine-tuning tasks.

Deep Unsupervised Learning: Autoencoders- Under complete Autoencoders, regularized auto encoders, Representation power, layer size and depth, stochastic encoders and decoders, Denoising auto-encoders, Sparse auto encoder, Contractive auto-encoders

UNIT-2 **10L+0T+6P=16 Hours**

RECURRENT NEURAL NETWORKS

Architecture of an RNN, unfolding of an RNN, Backpropagation through time, Long short term memory (LSTM), Gated recurrent units, Applications- Text Classification, Sentiment Analysis.

PRACTICES:

- Using Deep pre-trained CNN model for feature extraction:
  - Extract features from the FC1 of VGG network.
  - Train any traditional ML model like SVM for classification.
  - Repeat the above by considering FC2 of VGG for feature extraction.

- Fine-tuning Deep pre-trained CNN for Classification:
  - Fine-tune VGG network for the task under consideration.
  - Check the performance by making.
  - all the layers trainable.
  - freezing the initial layers.
  - freezing the entire network except the final layer.
- Design MLFFNN with 3-level stacked autoencoder based pre-training for Black and white image data, Display features extracted by different levels of stacked autoencoder at the end of pre-training.
- Sentiment Analysis
  - Pre-process the text.
  - Convert the text into word embeddings.
  - Implement the classification network using LSTMs/ GRUs.
  - Pre-process the text.
  - Convert the text into word embeddings.
  - Implement the classification network using LSTMs/ GRUs.

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Module No.	Mapping with POs
1	Implementation of Deep learning models to solve various real-time problems.	Apply	1,2	1, 3
2	Analyse performance of a deep network and tune its capacity and hyper parameters.	Analyze	1	2
3	Leveraging tools to Build deep networks and apply them for real word tasks.	Apply	1,2	1,5
4	Developing core components for deep learning algorithms.	Design	1,2	3

TEXT BOOKS:

1. Ian Good Fellow and Yoshua Bengio and Aaron, “Deep Learning”, 1st Edition, MIT Press, 2016.
2. Charu C Aggarwal “Neural Networks and Deep learning”, Springer International Publishing, 2018.

REFERENCE BOOKS:

1. Francois Chollet, “Deep learning with python”, 1st edition, Manning Publications, 2017.
2. S. Haykin, “Neural Networks and Learning Machines”, 3rd edition, Prentice Hall of India, 2011.
3. Josh Patterson and Adam Gibson, “Deep Learning: A Practitioner’s Approach”, 1st Edition, O’Reilly, 2017.
4. Satish Kumar, “Neural Networks, A Classroom Approach”, Tata McGraw-Hill, 2007.



Source: https://www.electrosoft-inc.com/resources/digital-forensics

22CS805 DIGITAL FORENSICS

Hours Per Week :

L	T	P	C
3	0	2	4

PREREQUISITE KNOWLEDGE: Cyber Security and Cyber Laws, Cryptography and Network Security.

COURSE DESCRIPTION AND OBJECTIVES:

This course focuses on understanding forensic terminologies and approaches along with variety of tools used for digital forensic investigations. The objective of this course is to understand digital forensics and its usage in solving computer crimes. By end of the course, students will be able to identify improper usage of computer systems and legal concepts in digital forensic investigation stages.

MODULE-1

UNIT-1 **12L+0T+8P=20 Hours**

INTRODUCTION

Computer forensics fundamentals, Benefits of forensics, computer crimes, computer forensics evidence and courts, legal concerns and private issues, Steps of computer forensics.

Understanding Computing Investigations: Procedure for corporate High-Tech investigations, understanding data recovery workstation and software, conducting and investigations.

UNIT-2 **12L+0T+8P=20 Hours**

DATA ACQUISITION

Understanding storage formats and digital evidence, determining the best acquisition method, acquisition tools, validating data acquisitions, performing RAID data acquisitions, remote network acquisition tools, other forensics acquisitions tools.

PRACTICES:

- Recover Deleted Files using Forensics Tools.
- Study the steps for hiding and extract any text file behind an image file/ Audio file using Command Prompt.
- Extract Exchangeable image file format (EXIF) Data from Image Files using Exifreader
- Software How to make the forensic image of the hard drive using EnCase Forensics.
- Restoring the Evidence Image using EnCase Forensics.

MODULE-2

UNIT-1 **12L+0T+8P=20 Hours**

PROCESSING CRIMES AND INCIDENT SCENES

Securing a computer incident or crime, seizing digital evidence at scene, storing digital evidence, obtaining digital hash, reviewing case.

UNIT-2 **12L+0T+8P=20 Hours**

CURRENT COMPUTER FORENSICS TOOLS

Software, hardware tools, validating and testing forensic software, addressing data-hiding techniques, performing remote acquisitions, E-Mail investigations- investigating email crime and violations, understanding E-Mail servers, specialized E-Mail forensics tool.

PRACTICES:

- Study of Computer Forensics and different tools used for forensic investigation.
- Live Forensics Case Investigation using Autopsy.
- Collect Email Evidence in Victim PC.
- Extracting Browser Artifacts.
- View Last Activity of Your PC.
- Find Last Connected USB on your system (USB Forensics).
- Comparison of two Files for forensics investigation by Compare IT software.

SKILLS:

- ✓ *Identify sources and methods of system intrusion.*
- ✓ *Create bitwise images of hard drives for forensic analysis.*
- ✓ *Collecting evidences from log files.*

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

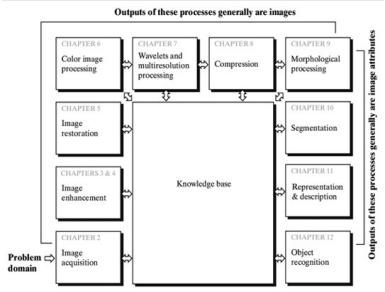
CO No.	Course Outcomes	Blooms Level	Module No.	Mapping with POs
1	Apply basic legal concepts related to digital forensics and evidence collection.	Apply	1	1,2,3
2	Analyze various digital forensics frameworks and its usage to solve crimes.	Analyse	1	1,2,3
3	Analyze artifacts like logs, packet captures, and registry.	Analyse	2	1,2,3
4	Demonstrate the ability to use forensic tools.	Create	2	1,2,5
5	Design and develop various forensic applications using variety of tools to carryout forensic investigation.	Analyse	2	1,2,3

TEXT BOOKS:

1. Warren G. Kruse II and Jay G. Heiser, “Computer Forensics: Incident Response Essentials”,Addison Wesley, 2020.
2. Nelson, B, Phillips, A, Enfinger, F, Stuart, C., “Guide to Computer Forensics and Investigations, 6th edition, Thomson Course Technology, 2020, ISBN: 0-619-21706-5.

REFERENCE BOOKS:

1. Vacca, J, Computer Forensics: Computer Crime Scene Investigation, 2nd Ed, Charles River Media, 2015, ISBN: 1-58450-389.



Source: https://www.researchgate.net/figure/Fundamental-steps-in-digital-image-processing_fig10_333856607

22CS806 DIGITAL IMAGE PROCESSING

Hours Per Week :

L	T	P	C
2	2	2	4

PREREQUISITE KNOWLEDGE: Probability and Statistics.

COURSE DESCRIPTION AND OBJECTIVES:

This course focuses on imparting knowledge about the aspects of Image Processing and its applications. The main objective of the course is to learn digital image fundamentals, image transforms, image enhancement, restoration and compression, morphological image processing, representation and description.

MODULE-1

UNIT-1 **6L+6T+6P=18 Hours**

FUNDAMENTALS OF IMAGE PROCESSING

Fundamental steps in digital image processing, Components of image processing system, A simple image formation model, Image sampling and quantization, Basic relationships between pixels, Introduction to Fourier Transform and DFT – properties of 2D Fourier Transform, FFT.

UNIT-2 **10L+10T+10P=30 Hours**

IMAGE ENHANCEMENT IN THE SPATIAL AND FREQUENCY DOMAINS

Basic gray - level transformations, Histogram processing, Basics of spatial filtering, Smoothing spatial filters, Sharpening spatial filters, The basics of filtering in the frequency domain, Image smoothing in frequency domain filters, Image sharpening in frequency domain filters.

Image Segmentation: Fundamentals, Point, Line and edge detection, Thresholding, Region-based segmentation, Segmentation using morphological watersheds, The use of motion in segmentation.

PRACTICES:

- Develop a module to enhance the image by using image arithmetic and logical operations.
- Develop a module for an image enhancement using kernel operations.
- Develop a module for gray level slicing with and without background.
- Develop a module for image enhancement using histogram equalization.
- Develop a module to filter an image using low pass & high pass filter in spatial domain. Compare the performance of both filters.
- Develop a module for smooth an image using low pass & high pass filters in frequency domain. Compare the performance of both filters.
- Develop a module for detecting lines & edges in an image.
- Develop a module for segmenting region of interest.

MODULE-2

UNIT-1 **8L+8T+8P=24 Hours**

IIMAGE RESTORATION

A model of image degradation/restoration, Noise models, inverse filtering, wiener filtering, Constrained Least Squares Filtering, Geometric Mean Filter.

Image Compression: Fundamentals, Huffman coding, Golomb coding, LZW coding, Run-length coding

UNIT-2 **8L+8T+8P=24 Hours**

MORPHOLOGICAL IMAGE PROCESSING

Erosion, Dilation, Opening, Closing, The hit-or-miss transformation; Basic morphological algorithms - boundary extraction, hole filling, extraction of connected components, thinning, thickening, skeletons, pruning.

Feature Extraction: Background, Boundary preprocessing, Boundary Feature Descriptors, Region Feature Descriptors, Principle Components as feature descriptors, Whole-image features.

PRACTICES:

- Develop a module to perform add & removal of salt and pepper noise. Compute PSNR & MSE and check the impact before and after removal of noise.
- Develop a module to remove noise using average filter and median filter. Compute PSNR & MSE before and after removal of noise.
- Develop a module for image compression and decompression.
- Develop a module for morphological image operations - erosion, dilation, opening & closing.
- Develop a module for morphological image operations - hit-or-miss transformation.
- Develop a module for morphological image operations - thinning, thickening
- Develop a module for extracting boundary features of an image.
- Develop a module for extracting features of an image using GLCM.

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Module No.	Mapping with POs
1	Understand the fundamental concepts of a digital image processing system.	Under-stand	1	1,2
2	Learn different techniques employed for the en-hancement of images.	Analyse	1	1,2,3,5,12
3	Employ image segmentation and representation techniques to extract region of interest.	Apply	1	1,2,3,5,12
4	Learn different causes for image degradation and overview of image restoration techniques.	Evaluate	2	1,2,3,5,12
5	Apply various compression techniques to reduce image size and morphological operations to extract features.	Apply	2	1,2,3,5,12
6	Learn different feature extraction techniques for image analysis and recognition.	Apply	2	1,2,3,5,12

TEXT BOOKS:

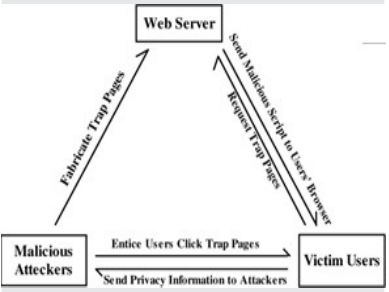
1. Rafeal C Gonzalez and Richard E.Woods, “Digital Image Processing”, 4th edition, Pearson Education/ PHI, 2018.
2. Rafeal C Gonzalez and Richard E.Woods, “Digital Image Processing using MATLAB”, 4th edition, Pearson Education/ PHI, 2020.

REFERENCE BOOKS:

1. Milan Sonka, Vaclav Hlavac and Roger Boyle, “Image Processing, Analysis and Machine Vision”, 4th Edition, Cengage, 2015.
2. Alasdair McAndrew, “Introduction to Digital Image Processing with Matlab”, Thomson Course Technology, 2004 Course Technology Press, Boston, MA, United States, 2004.
3. William K. Prat, “Digital Image Processing”, 4th Edition, Wiley-Interscience, A John Wiley & Sons, Inc., Publication, 2007.

SKILLS:

- ✓ *Apply knowledge of science and engineering principles to image related problems.*
- ✓ *Undertake image problem identification and formulate solutions.*
- ✓ *Implement algorithms for enhancement, restoration, compression etc.*



Source: https://www.intechopen.com/chapters/37306

22CS807 WEB AND DATABASE SECURITY

Hours Per Week :

L	T	P	C
3	0	2	4

PREREQUISITE KNOWLEDGE: Cryptography and Network Security, Web Technologies and Database Systems.

COURSE DESCRIPTION AND OBJECTIVES:

This course is imperative for understanding the fundamental security principles of the web and Data base security. The course provides an overview of the most common attacks, and illustrates fundamental countermeasures that every web application should implement. In essence, this course offers you the knowledge and skills to build better and more secure applications and to identify risks and vulnerabilities in operating systems from a database perspective.

MODULE-1

UNIT-1 **12L+0T+8P=20 Hours**

WEB SECURITY FUNDAMENTALS

Introduction to the web security landscape, and an overview of the most relevant threats. Understanding the security model of the web, and the recent evolution towards client-centric security.

UNIT-2 **12L+0T+8P=20 Hours**

SECURING THE COMMUNICATION CHANNEL

Understanding the dangers of an insecure communication channel. Practical advice on deploying HTTPS, and dealing with the impact on your application. Insights into the latest evolutions for HTTPS deployments.

Preventing Unauthorized Access: Understanding the interplay between authentication, authorization and session management. Practical ways to secure the authentication process, prevent authorization bypasses and harden session management mechanisms.

PRACTICES:

- Explore various tools for DOS attacks. (For Ex: SSPing, Land Exploit)
- Explore various tools for DDOS attacks. (For Ex: Trinoo, TFN)
- Identify various computer based social engineering ways to acquire sensitive information or inappropriate access privileges by an outsider
- Set up IPSEC under LINUX
- Implement a code to simulate buffer overflow attack.

MODULE-2

UNIT-1 **12L+0T+8P=20 Hours**

SECURELY HANDLING UNTRUSTED DATA

Investigation of injection attacks over time, understanding the cause behind both server-side and client-side injection attacks. Execution of common injection attacks, and implementation of various defenses.

UNIT-2 **12L+0T+8P=20 Hours**

ADMINISTRATION OF USERS

Introduction-Authentication-Creating Users, SQL Server User-Removing, Modifying Users-Default, Remote Users-Database Links-Linked Servers-Remote Servers-Practices for Administrators and Managers-Best Practices Profiles, Password Policies.

Privileges and Roles: Introduction-Defining and Using Profiles-Designing and Implementing Password Policies-Granting and Revoking User Privileges-Creating, Assigning and Revoking User Roles-Best Practice.

Database Application Security Models: Introduction-Types of Users-Security Models-Application Types-Application Security Models-Data Encryption.

SKILLS:

- ✓ *Investigation of injection attacks over time.*
- ✓ *Execution of common injection attacks and their defenses.*
- ✓ *Implement privacy preserving data mining algorithms.*

PRACTICES:

- Explore Web Application Vulnerabilities.
- Implement Web based Password Cracking Techniques.
- Implement SQL Injection attack.
- Implement XSS attack.

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

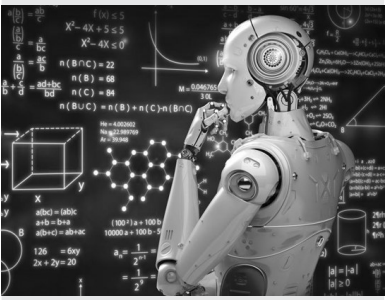
CO No.	Course Outcomes	Blooms Level	Module No.	Mapping with POs
1	Analyze the cause behind both server-side and client-side injection attacks.	Analyze	1	1,2,,3
2	Analyze the privileges and roles of the users.	Analyze	2	1,2,3
3	Analyze common attacks and countermeasures.	Analyze	2	1,2,3
4	Implement administration policies for users, database security models.	Evaluate	2	1,2,3

TEXT BOOKS:

1. Bing Liu, “Web Data Mining”, 2nd edition, Spinger, 2011.
2. Hassan A. Afyouni, “Database Security and Auditing”, Third Edition, Cengage Learning, 2013.

REFERENCE BOOKS:

1. Jiawei Han, Micheline Kamber and Jian Pei, “Data Mining: Concepts and Techniques”, 3rdedition, Morgan Kaufmann Publishers, 2011.
2. Bing Liu, “Sentiment analysis and opinion mining”, 2nd edition, Morgan & Claypool Publishers, 2012.
3. Jure Leskovec, Anand Raja Raman and Jeffrey D Ullman, “Mining of Massive Datasets”, 5thedition, Stanford University, 2014.Networks, Springer.
4. Frank Adelstein, Sandeep K.S. Gupta, Golden G. Richard III, and Loren Schwiebert. Fundamentals of Mobile and Pervasive Computing, McGraw-Hill Professional, 2015.



Source: https://www.forbes.com/sites/kalevleetaru/2019/01/15/why-machine-learning-needs-semantics-not-just-statistics/

22CS808 MACHINE LEARNING

Hours Per Week :

L	T	P	C
3	0	2	4

PREREQUISITE KNOWLEDGE: Probability and Linear algebra, Python programming.

COURSE DESCRIPTION AND OBJECTIVES:

This course provides a broad introduction to various machine learning concepts including Supervised learning (parametric/non-parametric algorithms, support vector machines, kernels, neural networks) and Unsupervised learning (clustering, dimensionality reduction) methods. Students will get an understanding of various challenges of Machine Learning and will be able to decide on model complexity. Numerous case studies introduced in this course allow the students to apply machine-learning algorithms in computer vision, medical imaging, audio, and text domains. Laboratory experiments of this course will introduce students to advanced Machine Learning Python libraries such as Scikit-Learn, Matplotlib, and many other recent ML-related APIs. The course is designed such that the students get enough hands-on experience with a major focus on the practical implementation of theoretical concepts.

MODULE-1

UNIT-1 **14L+0T+8P=22 Hours**

INTRODUCTION

What is machine learning? Machine learning applications; Types of Learning: Supervised learning; Un-supervised learning; Reinforcement learning, Applications of Probability and statistics.

Model Training Essentials: Re-sampling methods: Bias–Variance Trade-off. Hypothesis Testing and Variable Selection, Sub sampling and Upsampling, SMOTE; Cross Validation (validation set, Leave-One-Cut (LOO), k-fold strategies) and bootstrap; Evaluation measures-Error functions, Confusion Matrix, Accuracy, Precision and Recall, F1 Score.

Regression Analysis: Linear Regression, Simple and Multiple Linear Regression, Polynomial Regression, Logistic Regression, Multi nominal Regression. Ordinary Least Squares Method, Model Shrinkage-Ridge, and LASSO regression.

UNIT-2 **10L+0T+8P=18 Hours**

FEATURE SELECTION

Feature Selection Strategies: Problem statement and Uses, Filter methods, Wrapper methods, Embedded methods. Branch and bound algorithm, Sequential forward/backward selection algorithms.

Dimensionality Reduction: Singular value decomposition, matrix factorization, Linear discriminant analysis, Principal components analysis.

PRACTICES:

- Apply the following tasks to any given dataset:
  a) Load and visualize data.
  b) Check out and replace missing values.
  c) Encode the Categorical data.
  d) Splitting the dataset into Training and Test set.
  e) Splitting the dataset into k-folds.
  f) Feature scaling.

- House price prediction:
  a) Create a model that predicts a continuous value (price) from input features (square footage, number of bedrooms and bathrooms).
  b) Implement a univariate Model using Least Squares and plot best-fit line.
  c) Implement a multivariate Model using Least Squares and plot best-fit line.
  d) Retrieve model error and model coefficients.
  e) Observe Variance Inflation Factor (VIF).
  f) Implement Ridge regression model.
  g) Implement LASSO regression model.
  h) Report your observations on the above models for house prediction.
- Heart disease prediction:
  a) Implement a logistic regression model to predict whether an individual is suffering from heart disease or not.
  b) Evaluate and compare model performance using the following validation approaches:
  i. Validation set approach.
  ii. K-fold cross validation.
  iii. Stratified K-fold cross validation.
  iv. LOO strategy.
  c) Plot Confusion matrix.
  d) Report performance of the model in terms of the following metrics:
  i. Accuracy.
  ii. Precision-Recall.
  iii. F1 Score.
  e) Report your observations and explain when to use what type of measures.
- Implement the Polynomial Regression algorithm to fit data points. Select the appropriate data set for your experiment and draw graphs.
- Working with imbalanced datasets:
  a) Load an imbalanced dataset and visualize imbalance in the data as a bar plot.
  b) Implement KNN model for classification.
  c) Balance the dataset using:
  i. Random Over sampling.
  ii. Random Under sampling.
  iii. SMOTE.
  d) Implement KNN model for classifying data balanced in the above steps.
  e) Report your observations on the performance of models trained using balanced and imbalanced data.
- Perform effective feature selection in a given dataset using any one of the feature selection techniques.
- Dimension Reduction:
  a) Load a dataset and Implement Bayes classification model.
  b) Apply dimension reduction using:
  i. Principal Component Analysis
  ii. Linear Discriminant Analysis
  c) Apply the model on data with reduced dimension.
  d) Compare and contrast model performance in each case.

SKILLS:

- ✓ *Statistical data analysis.*
- ✓ *Classify / Cluster data.*
- ✓ *Tool usage for developing ML applications.*

MODULE-2

UNIT-1 **16L+0T+8P=24 Hours**

CLASSIFICATION

Classification: Binary, Multi-class and Multi-label Classification; K-Nearest Neighbours, Support Vector Machines, Decision Trees, The Naïve Bayes' Classifier, Class Imbalance, Perceptron ANN model.

Ensemble Methods: Ensemble Learning Model Combination Schemes, Voting, Error-Correcting Output Codes, Bagging: Random Forest Trees, Boosting: Adaboost, Stacking.

UNIT-2 **8L+0T+8P=16 Hours**

CLUSTERING

Clustering: Different distance functions and similarity measures, K-means clustering, Medoids, Hierarchical Clustering-Single linkage and Complete linkage clustering, Graph based Clustering -MST, DBSCAN, Spectral clustering.

PRACTICES:

- Implement and demonstrate the FIND-S algorithm for finding the most specific hypothesis based on a given set of training data samples. Read the training data from a .CSV file.
- Implement the naïve Bayesian classifier for a sample training data set stored as a.csv file. Compute the accuracy of the classifier, considering few test data sets.
- Assuming a set of spam or not-spam mails that need to be classified, use the naïve Bayesian classifier model to perform this task. Calculate the accuracy, precision, and recall for your data set.
- Implement k-Nearest Neighbor algorithm to classify the iris data set. Print both correct and wrong predictions. Python ML library classes can be used for this problem.
- Demonstrate the working of the decision tree-based ID3 algorithm. Use an appropriate data set for building the decision tree and apply this knowledge to classify a new sample?
- Build a model using SVM with different kernels.
- Implement and build models using the following Ensemble techniques.
  a) Bagging.
  b) Boosting: Adaboost, Stacking.
- Build a model to perform Clustering using K-means after applying PCA and determining the value of K using the Elbow method.
- Unsupervised Modeling:
  a) Cluster the data using the following models:
  i. Spectral Clustering.
  ii. K-medoids.
  iii. DBSCAN.
  iv. Hierarchical Clustering.
  b) Compare and contrast model performance in each case.

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Module No.	Mapping with POs
1	Apply a wide variety of learning algorithms such as Probabilistic, Discriminative and Generative algorithms for a given application.	Apply	1, 2	1
2	Design an end-to-end Machine-learning model to realize solutions for real-world problems.	Apply	1	3
3	Implement various machine learning models using advanced ML tools.	Create	1, 2	5
4	Analyze and evaluate the performance of various machine learning models approaches on different kinds of data.	Analyze	2	2

TEXT BOOKS:

1. Ethem Alpaydin, “Introduction to Machine Learning”, 3rd Edition, The MIT Press, 2014.
2. Flach, Peter. “Machine learning: the art and science of algorithms that make sense of data”. Cambridge University Press, 2012.

REFERENCE BOOKS:

1. Murphy, Kevin P. Machine learning: a probabilistic perspective. MIT press, 2012.
2. Aurélien Géron, “Hands-on Machine Learning with Scikit Learn and Tensor Flow”, O’reilly, 2017.
3. Gareth James, Daniela Witten, Trevor Hastie, and Robert Tibshirani, “An Introduction to Statistical Learning with Applications in R”, Springer, 2013. (ISLR).



Source: https://www.educba.com/mobile-ad-hoc-network/

22CS809 MOBILE AD-HOC NETWORKS

Hours Per Week :

L	T	P	C
3	0	2	4

PREREQUISITE KNOWLEDGE: Computer networks.

COURSE DESCRIPTION AND OBJECTIVES:

This course focuses on major aspects of ad hoc networking, from design through performance issues to application requirements. It starts with the design issues and challenges associated with implementations of ad hoc network applications. This includes mobility, disconnections, and battery power consumption. The course provides a detailed treatment of proactive, reactive, and hybrid routing protocols in mobile wireless networks. It also covers the IEEE 802.11 Wireless LAN and discusses their characteristics and operations. Through activities, the course gives students hands-on experience in designing a mobile ad hoc network using the NS2 simulator.

MODULE-1

UNIT-1 **8L+0T+6P=14 Hours**

INTRODUCTION

Introduction to ad-hoc networks-definition, characteristics, features, applications; Characteristics of the wireless channel; Ad-hoc mobility models-indoor and outdoor models.

UNIT-2 **16L+0T+10P=26 Hours**

MEDIUM ACCESS PROTOCOLS

MAC protocols- design issues, goals and classification; Contention-based protocols – with reservation, without reservation; Scheduling algorithms; Protocols using directional antennas; IEEE standards - 802.11a, 802.11b, 802.11g, 802.15; HIPERLAN.

Network Protocols: Routing protocols - design issues, goals, and classification; Proactive Vs reactive routing; Unicast routing algorithms; Multicast routing algorithms; Hybrid routing algorithm; Energy-aware routing algorithm; Hierarchical routing; QoS aware routing.

PRACTICES:

- Installation of NS-2 and basics of TCL scripting.
- TCL script for
  a) computing the arithmetic operations on two operands.
  b) finding the given number is prime or not using functions.
  c) finding the factorial value of a given number.
- Set the node property and routing protocol in the same MANET scenario.
- Analyse the performance of the MANET.
- Develop MAC Protocol using any suitable Network Simulator for MANETs to send the packet without any contention through wireless link using the following MAC protocols (CSMA/CA (802.11)). Analyze its performance with increasing node density and mobility.
- Simulate MANET environment using suitable Network Simulator and test with various mobility model such as Random walk, Random waypoint and Group mobility. Analyze throughput, PDR and delay with respect to different mobility models.

MODULE-2

UNIT-1 **12L+0T+8P=20 Hours**

END-END DELIVERY AND SECURITY

Transport layer - issues in designing, transport layer classification, ad-hoc transport protocols; Security issues in ad-hoc networks - issues and challenges, network security attacks; Secure routing protocols.

UNIT-2 **12L+0T+8P=20 Hours**

CROSS LAYER DESIGN

Cross layer design - need for cross layer design, cross layer optimization; Parameter optimization techniques; Cross layer cautionary perspective; Integration of adhoc with mobile IP networks.

SKILLS:

- ✓ *Evaluate various routing protocols.*
- ✓ *Analyse the performance of MAC protocols for Ad-hoc networks.*
- ✓ *Analyse the performance of Network protocols for Ad-hoc networks.*

PRACTICES:

- Create CBR traffic over UDP and TCP.
- Write an awk script that takes data from trace file and give the report for performance metrics such as packet delivery ratio, and throughput.
- Implement Transport Control Protocol in Sensor Network.
- Design and Implementation of Security algorithm for Wireless networks (b)Implementation of security protocol for mobile network.

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Module No.	Mapping with POs
1	Summarize the protocols used at the MAC layer and scheduling mechanisms to express the mathematical properties.	Evaluate	1	1, 12
2	Apply proactive and reactive routing algorithms to find optimal paths.	Apply	1	1, 2, 5, 12
3	Analyze types of routing protocols used for unicast and multicast routing.	Analyse	1	1, 2, 5, 12
4	Compare the performance of various routing protocols in ad-hoc networks.	Analyse	2	1, 2
5	Develop the network security solution and routing mechanism.	Apply	2	1, 2, 12

TEXT BOOKS:

1. C. Siva Ram Murthy and B. S. Manoj, “Ad hoc Wireless Networks Architecture and Protocols”, 2nd Edition, Pearson Edition, 2007.
2. Charles E. Perkins, “Ad hoc Networking”, 1st Edition, Addison – Wesley, 2000.

REFERENCE BOOKS:

1. Stefano Basagni, Marco Conti, Silvia Giordano and Ivan stojmenovic, “Mobile ad-hoc networking”, Wiley-IEEE press, 1st Edition, 2004.
2. Mohammad Ilyas, “The Handbook of Adhoc Wireless Networks”, 1st Edition, CRC press, 2002.
3. T. Camp, J. Boleng and V. Davies “A Survey of Mobility Models for Ad Hoc Network Research” Wireless Commun. and Mobile Comp., Special Issue on Mobile Ad Hoc Net working Research, Trends and Applications, vol. 2, no. 5, 2002, pp. 483–502.
4. A survey of integrating IP mobility protocols and Mobile Ad hoc networks, Fekri M. Abduljalil and Shrikant K. Bodhe, IEEE communication Survey and tutorials, v no.1, 2007.
5. V. T. Raisinhani and S.Iyer “Cross Layer Design Optimization in Wireless Protocol Stacks” Comp. Communication, Vol 27 no. 8, 2004.



Source: https://www.tatvasoft.com.au/blog/mobile-application-development-methodology/

22CS810 MOBILE APPLICATION DEVELOPMENT

Hours Per Week :

L	T	P	C
2	0	4	4

PREREQUISITE KNOWLEDGE: OOPs through Java, DDL & DML Commands – DBMS..

COURSE DESCRIPTION AND OBJECTIVES:

This course guides the student in designing and building a mobile application using Android™. The main objective of this course is to let the student learn basic Android programming concepts while building a variety of apps, starting with basic to making use of advanced concepts.

MODULE–1

UNIT-1 **8L+0T+16P=24 Hours**

INTRODUCTION

Introduction to Mobile Application Development- Mobile Applications and Device Platforms, Alternatives for Building Mobile Apps; Introduction to Android, Android versions, Android Architecture.

Application Development Process- Developers Workflow basics, Installing the Android SDK Tools; Anatomy of an Android Application.

Basic Building blocks - Activities, Services, Broadcast Receivers & Content providers; Intents & Fragments.

View Group- Layout: Linear Layout, Relative Layout, Frame Layout, Grid Layout, constraint Layout, Table Layout, and Absolute Layout.

UNIT-2 **8L+0T+16P=24 Hours**

VIEWS

Views: Basic Views; Picker Views- Time Picker View, Data Picker View; List Views – List View, Spinner View; Scroll View.

Activities: Creating an activity, Understanding the activity life cycle using Log and Toast, applying styles and themes to an activity, and hiding the activity title.

Linking Activities using Intents: Introduction to Intents and its types with examples, passing data between activities with intents, Activity Navigation- Implement up navigation with parent activities.

Fragments: Introduction to Fragment, the life cycle of a fragment, Adding fragments dynamically, Interaction between fragments.

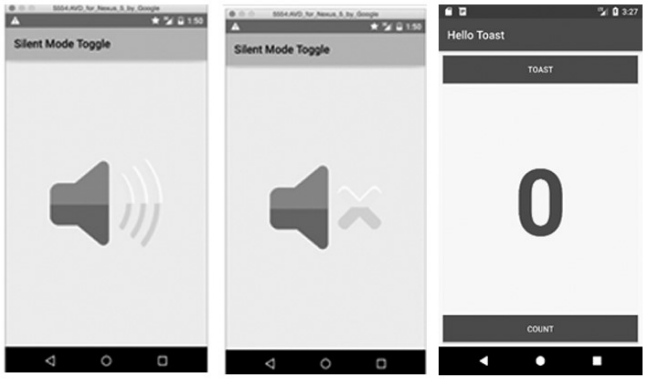
PRACTICES:

- Setting up Android Studio:
  a) Installing Android Studio
  b) Select an empty activity to simulate the “Welcome App” Using Android Studio.
  c) Exploring the interface of the Android Studio to understand the Project Structure.
- Develop an Android application using controls like Button, TextView, and EditText for designing a calculator having basic functionality like Addition, Subtraction, Multiplication, and Division.
- **Design the HelloToast app:** The HelloToast app consists of two Button elements and one Text View. When the user taps the first Button, it displays a short message (a Toast) on the screen. Tapping the second Button increases a “click” counter displayed in the TextView, which starts at zero.

SKILLS:

- ✓ *Design mobile applications for user requirements.*
- ✓ *Use of suitable advanced components to design mobile apps.*
- ✓ *Utilization of activities, intents, layouts, and views for content.*

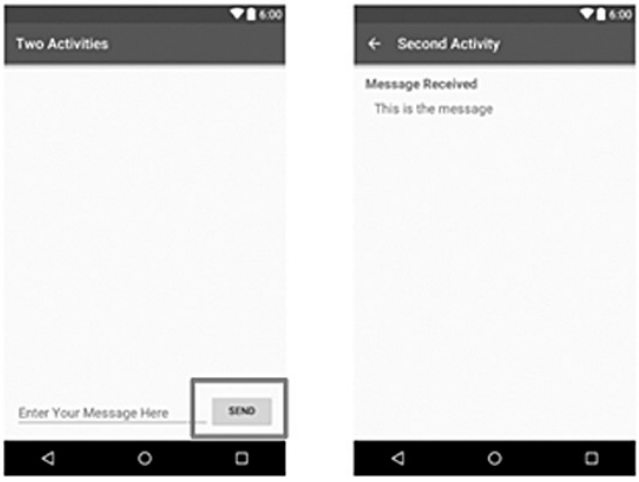
- **Design Silent Model Toggle application:** This app allows the user to toggle the ringer mode on the phone by simply pressing a button.



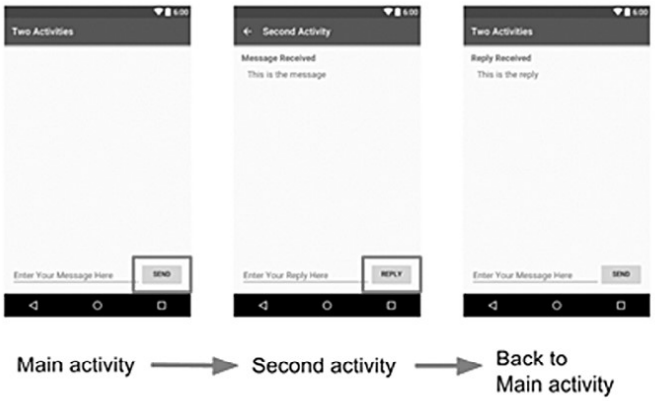
- In this assignment, students will create and build an app called Two Activities. Students will build the app in four stages.

a) In the first stage, you create an app whose main activity contains one button, Send. When the user clicks this button, your main activity uses an intent to start the second activity.

b) In the second stage, you add an EditText view to the main activity. The user enters a message and clicks Send. The main activity uses an intent to start the second activity and sends the user's message to the second activity. The second activity displays the message received.



c) In the final stage of creating the Two Activities app, you add an EditText and a Reply button to the second activity. The user can now type a reply message and tap Reply, and the reply is displayed on the main activity. At this point, you use an intent to pass the reply from the second activity to the main activity.



d) Implement all the Activity lifecycle call back methods to print messages to logcat when those methods are invoked. These log messages will allow you to see when the Activity lifecycle changes state, and how those lifecycle state changes affect your app as it runs.

- **Design an application with implicit intents:** Create a new app with one Activity and three options for actions: open a website, open a location on a map, and share a snippet of text. All the text fields are editable (EditText) but contain default values.



- **Design Droid Café:** In this practical, the student will create and build a new app starting with the Basic Activity template that imitates a dessert-ordering app. The user can tap an image to perform an action-in this case, display a Toast message-as shown in the figure below. The user can also tap a shopping cart button to proceed to the next Activity.
- **Experiment with the android:** inputType attribute for EditText elements. You add EditText elements for a person's name and address and use attributes to define single-line and multiple-line elements that make suggestions as you enter text. You also add an EditText that shows a numeric keypad for entering a phone number.
- Other types of input controls include interactive elements that provide user choices. You add radio buttons to Droid Cafe for choosing only one delivery option from several options. You also offer a spinner input control for selecting the label (Home, Work, Other, Custom) for the phone number.



MODULE–2

UNIT-1 **8L+0T+16P=24 Hours**

CREATING A FEATURE-RICH APPLICATION

Creating a Feature-Rich Application: Display Orientation – Anchor Views, resizing and repositioning Views, Managing changes to Screen Orientation; Notifications; Action bar; Dialog box; Adapters- Array Adapters and Base Adapters; Recycler View.

UNIT-2 **8L+0T+16P=24 Hours**

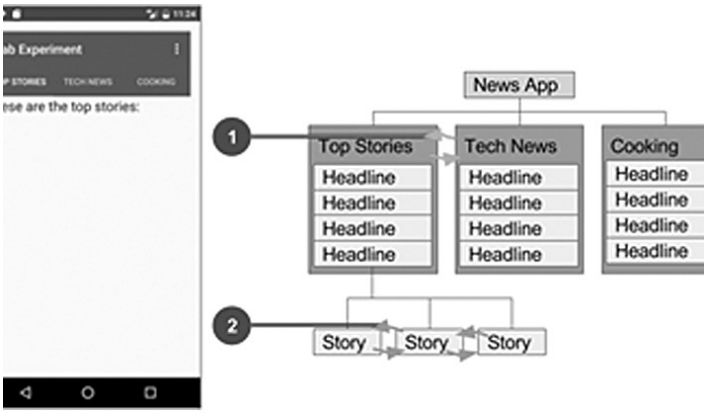
SQLITE DATABASE

SQLite Database – Creating the database, Dealing with CRUD;

Firebase-Getting Started with Firebase, Add Firebase to your Android project, Firebase database-Introduction to Firebase database, set up Firebase Real-time Database for Android, Read and Write Data on Android; Publish the App in Play store.

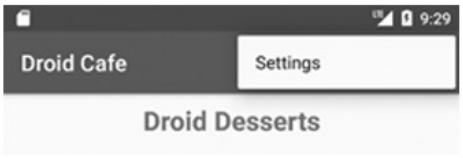
PRACTICES:

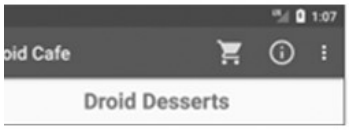
- Design an application to keep data when the user rotates the device, and when the screen is rotated: When the user rotates the device, Android will normally destroy and re-create the current Activity. You want to keep some data across this cycle, but all the fields in your Activity are lost during it.
- Create a Splash Screen for the existing project- Droid Café from Module-1.
- Design a News App- Consider the following screen as reference:



NOTE: Use Recycle View to display the news under each category.

- Adding more features to Droid Café: In the previous assignments, you created an app called Droid Café, using the Basic Activity template. This template also provides a skeletal options menu in the app bar at the top of the screen.
  a) Update that menu option as shown in the following images:
  b) Add notification option: The app must notify the user when the user places the order.
- Provide user authentication for the Droid Café using Firebase – Authentication or SQLite.







- Save all the user preferences in the Firebase Real time Database to fetch whenever required.

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Module No.	Mapping with POs
1	Apply views, intents, and fragments to an existing application.	Apply	1	2
2	Evaluate an existing app to enrich it with new features.	Evaluate	2	2,3
3	Analyse methods for storing, sharing, and retrieving data in an Android app.	Analyse	2	5
4	Design and publish a mobile app in the play store with a database forgiven real-time scenarios using modern tools- Android Studio, and Firebase.	Create	2	5,10

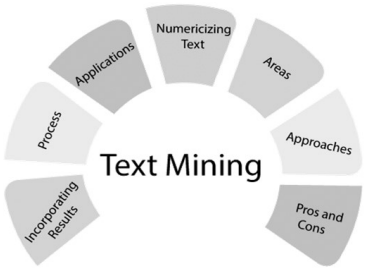
TEXT BOOKS:

1. John Horton “Android Programming for Beginners: Build in-depth, full-featured Android apps starting from zero programming experience”, 3rd Edition, 2021.
2. Wei-Meng Lee, “Beginning Android Application Development”, 1st Edition, John Wiley & Sons, 2012.

REFERENCE BOOKS:

1. Michael Burton,” Android App Development for Dummies “, 3rd Edition, A Wiley Brand, 2020.
2. Dawn Griffiths & David Griffiths, “Headfirst Android Development A Brain-Friendly Guide” 2nd Edition, O’Reilly, 2015.
3. https://aws.amazon.com/mobile/mobile-application-development/
4. https://google-developer-training.github.io/android-developer-fundamentals-course-concepts/.

22CS811 TEXT MINING



Source: https://www.javatpoint.com/text-data-mining

Hours Per Week :

L	T	P	C
3	0	2	4

PREREQUISITE KNOWLEDGE: Data analysis skills, Database, Machine learning or Deep learning algorithms..

COURSE DESCRIPTION AND OBJECTIVES:

This course introduces the fundamental concepts and techniques used in Text Processing. Students will gain an in-depth understanding of the computational properties of natural languages and the commonly used algorithms for processing linguistic information. The course examines models and algorithms used in both the traditional symbolic and the more recent statistical approaches.

MODULE-1

UNIT-1 **10L+0P+6P=16 Hours**

INTRODUCTION

NLP tasks in syntax, semantics, and pragmatics. Applications such as information extraction, question answering, and machine translation. The role of machine learning. Brief history of the field.

Regular Expressions, Text Normalization, Edit Distance: Regular Expressions, words, Corpora, Text Normalization and Minimum edit distance.

UNIT-2 **14L+0P+10P=24 Hours**

N-GRAMS, VECTOR SEMANTICS AND EMBEDDING

N-Gram Language Models: N-Grams, Evaluating Language Models, Generalization and Zeros, Smoothing, Kneser-Ney Smoothing, Huge Language Models and Stupid Back off.

Vector Semantics and Embeddings: Lexical semantics, Vector semantics, words and vectors, cosine for measuring similarity, TF-IDF: weighing terms in the vector, Point wise Mutual Information, Applications of TF/ IDP and PPMI vector models, word 2 vec, visualizing embeddings, semantic properties of embeddings, bias and embeddings, evaluating vector models.

PRACTICES:

- Perform basic text pre-processing using the following approaches:
  - Stop word Elimination.
  - Removal of Special Characters.
  - Stemming.
  - Lemmatization.
  - N-gram Tokenization.
- Design and develop a text classification model with various machine learning algorithms over the following feature extraction methods and compare their performance.
  - TF-IDF.
  - Word2 Vec.
- Perform Exploratory Data Analysis using following:
  - Word Cloud.
  - Frequent Word Detection.
  - Keyword Extraction based on ranking.

SKILLS:

- ✓ *Exploratory Data analysis.*
- ✓ *Opinion Mining/ Sentiment Analysis using various deep Learning models.*
- ✓ *Text Summarization and Categorization etc.*

MODULE-2

UNIT-1 **12L+0P+8P=20 Hours**

SEQUENCE LABELING FOR PARTS OF SPEECH AND NAMED ENTITIES

Part-of-Speech Tagging, Named Entities and Named Entity Tagging, HMM Part-of-Speech Tagging, Conditional Random Fields, Evaluation of Named Entity Recognition.

UNIT-2 **12L+0P+8P=20 Hours**

SEQUENCE AND TRANSLATION PROCESSING

Deep Learning Architectures for Sequence Processing: Language Models Revisited, Self-Attention **Networks:** Transformers.

Machine Translation and Encoder-Decoder Models: The Encoder-Decoder Model, Encoder-Decoder with RNNs, Attention, Beam Search, Encoder-Decoder with Transformers.

PRACTICES:

- Perform Named Entity Recognition to extract required entities from a given unstructured text using NLTK.
- Perform POS Tagging with HMM and also optimize the performance of HMM with Viterbi.
- Design and develop a text classification model using Latent Dirichlet allocation and compare its performance with TF-IDF and Word2Vec.
- Perform text classification using following methods and compare their performance in terms of various evaluation metrics such as Accuracy, Precision, Recall and F-Score.
  - CNN
  - LSTM
  - GRU
  - Encoder-Decoder
  - Transformers
- Design and develop text classification model using Attention.

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Module No.	Mapping with POs
1	Understand and apply the usage of regular expressions to process the raw text.	Apply	1	1
2	Apply various approaches to sentiment analysis using Machine Learning methods.	Apply	1,2	1
3	Analyze the vector semantics and embedding in the representation of the text.	Analyze	1	2
4	Design and development various statistical approaches to machine translation.	Evaluate	1,2	3,4,5

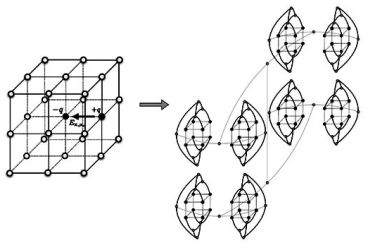
TEXT BOOK:

1. Daniel Jurafsky and James HMartin, “Speech and Language processing: An introduction to Natural Language Processing, Computational Linguistics and speech Recognition”, 3rd Edition, 2020.

REFERENCE BOOKS:

1. Steven Bird, Ewan Klein, Edward Lopper, “Natural Language Processing with Python”, 1st Edition, O’Reilly Publishers, 2009.
2. Nitin Indurkhya, Fred J. Damerau, “Handbook of Natural Language Processing”, 2nd Edition, CRC Publishers, 2010.

22CS812 NUMERICAL ALGORITHMS



Source: https://pasquans.eu/new-numerical-algorithms-for-old-problems/

Hours Per Week :

L	T	P	C
3	2	0	4

PREREQUISITE KNOWLEDGE: Basics of integration, Differentiation and Polynomials.

COURSE DESCRIPTION AND OBJECTIVES:

The goal of this course is to build a base for Numerical methods, which are the basic algorithms underpinning computer predictions in modern systems science. Such methods include techniques for simple optimisation, interpolation from the known to the unknown, linear algebra underlying systems of equations, ordinary differential equations to simulate systems, and stochastic simulation under random influences.

MODULE-1

UNIT-1 **12L+8T+0P=20 Hours**

ROOT FINDING METHODS, SYSTEM OF LINEAR EQUATIONS AND INTERPOLATION

Absolute error, order of Convergence, Geometrical Description, method of successive approximation, Bisection method, Regula - Falsi method, Newton's Method. Gauss Seidal method, Crouts method, Triangularization method, Relaxation method.

Interpolation - Finite differences, interpolation, Newton's forward and backward difference formulae, Newton's divided difference and Lagrange's formulae (with proof), errors in interpolation formula.

UNIT-2 **12L+8T+0P=20 Hours**

APPLICATIONS

Finding positive, negative and real root of algebraic and transcendental equation, Solution of simultaneous linear algebraic equation. nth difference of a polynomial, finding missing terms in a sequence, sum of n terms in a series, finding polynomial using a given set of data, estimated values of a function inside and outside the given intervals of data.

PRACTICES:

- Finding positive, negative and real root of algebraic and transcendental equation using any of your known programming knowledge.
- Solution of simultaneous linear algebraic equation equation using discrete methods and implement using C.
- By python program estimate the values of a function inside and outside the given intervals of data.

MODULE-2

UNIT-1 **12L+8T+0P=20 Hours**

NUMERICAL DIFFERENTIATION, INTEGRATION AND DIFFERENTIAL EQUATION

Numerical Differentiation: Newton's forward and Backward formulas to compute up to second order differentiation of a function.

Numerical Integration: Trapezoidal and Simpson's 1/3 and 3/8 rules.

ODE: Picard's approximation, Milne's Predictor Corrector formulas.

PDE: Liebman's Iteration Process, Bender Schmidth.

SKILLS:

- ✓ *Analyze the types and occurrence of roots.*
- ✓ *Interpolate the unknown values of function.*
- ✓ *Develop a difference equation.*
- ✓ *Gain the knowledge to solve an ODE numerically.*
- ✓ *Gain the knowledge to solve an ODE numerically.*

UNIT-2 **12L+8T+0P=20 Hours**

APPLICATIONS

Finding maxima and minima of a function, population growth, acceleration, area bounded by the curve, Solution of ODE, Solution of Elliptic, Parabolic and Hyperbolic PDE.

PRACTICES:

- Finding maxima and minima of a function.
- Solve ODE numerically and plot the curve.
- Classify the PDE.
- Solve PDE numerically and plot the curve.
- Developing difference equations from ODE and PDE.

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Module No.	Mapping with POs
1	Apply numerical methods to find roots.	Apply	1	1, 2, 4, 5, 9, 10, 12
2	Evaluate the unknown values using interpolation.	Apply	1	1, 2, 5, 9, 10
3	Develop a finite difference scheme.	Analyze	2	1, 2, 3, 5, 9, 10
4	Apply numerical methods to solve ODE and PDE and analyse graphically.	Analyze	2	1, 2, 5, 9, 10, 12

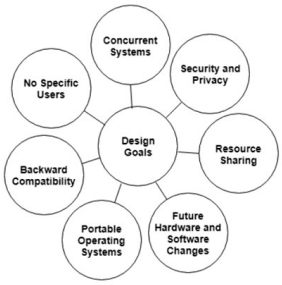
TEXT BOOKS:

1. S. S. Sastry, “Introductory methods of numerical analysis”, 5th ed, PHI learning, 2012.
2. M K Jain, “Numerical Methods for Scientific and Engineering Computation”, New Age internarnational, 8th Ed. 2022.

REFERENCE BOOKS:

1. P Kandasamy, “Numerical Methods”, S Chand, 2nd ed, 2015.
2. C. Ray Wylie, Louis C. Barrett: “Advanced Engineering Mathematics”, 6” Edition, 2. McGraw-Hill Book Co., New York, 1995.
3. Gupta C.B., Singh S.R. and Mukesh Kumar: “Engineering Mathematics for Semester I & II”, Mc-Graw Hill Education (India) Pvt. Ltd., 2015.

22CS813 OPERATING SYSTEM DESIGN



Source: https://www.tutorialspoint.com.cach3.com/operating-system-design-goals.html

Hours Per Week :

L	T	P	C
3	0	2	4

PREREQUISITE KNOWLEDGE: Knowledge of computers fundamentals, Computer organization and Digital logic design..

COURSE DESCRIPTION AND OBJECTIVES:

This course aims at concepts and principles of Operating Systems, its overall responsibility in acting as an interface between the system's hardware components and the user. Further, it also helps students to understand the different scheduling policies, process synchronization mechanisms, deadlock handling mechanisms and memory management techniques.

MODULE-1

UNIT-1 **12L+0T+8P=20 Hours**

INTRODUCTION & PROCESSES

Introduction to Operating System: Introduction, Operating System Concepts, System Calls, Operating System Structure.

Processes: Introduction, Inter Process Communication, Classical IPC Problems, Scheduling, Implementation of MINIX 3, System Task in MINIX 3 and Clock Task in MINIX 3.

UNIT-2 **12L+0T+8P=20 Hours**

INPUT/OUTPUT

Input/Output: Principles of Hardware and Software, Deadlock- Principles, Ostrich Algorithm, Detection and Recovery, Deadlock Prevention and Avoidance; Input/output in MINIX 3, Block Devices in MINIX 3.

PRACTICES:

- Implementation of new process creation and its communications.
- Implement of thread creation and deletion.
- Implementation of FCFS scheduling.
- Implementation of SJF and RR Scheduling.
- Implementation of producer consumer problem.
- Implementation of Banker's algorithm for Deadlock avoidance.
- A MINIX file whose owner has UID = 12 and GID = 1 has mode rwxr-x - Another user with UID = 6, GID = 1 tries to execute the file. What will happen?
- In MINIX 3 if user 2 links to a file owned by user 1, then user 1 removes the file, what happens when user 2 tries to read the file?
- Write a program (or series of programs) to test all the MINIX 3 system calls. For each call, try various sets of parameters, including some incorrect ones, to see if they are detected.
- Suppose that a computer can execute 1 billion instructions/sec and that a system call takes 1000 instructions, including the trap and all the context switching. How many system calls can the computer execute per second and still have half the CPU capacity for running application code?
- Write a shell script that produces a file of sequential numbers by reading the last number in the file, adding 1 to it, and then appending to the file. Run one instance of the script in the background and one in the foreground, each accessing the same file. How long does it take before a race condition manifests itself? What is the critical section? Modify the script to prevent the race (Hint: use ln file file.lock to lock the data file).

SKILLS:

- ✓ *Know the concepts of Processes scheduling and File Systems.*
- ✓ *Identification of different disk scheduling methodologies.*
- ✓ *Interpret UNIX Commands, Shell basics, and shell environments*

- Show how counting semaphores (i.e., semaphores that can hold an arbitrarily large value) can be implemented using only binary semaphores and ordinary machine instructions.
- A fast food restaurant has four kinds of employees: (1) order takers, who take customer's orders; (2) cooks, who prepare the food; (3) packaging specialists, who stuff the food into bags; and (4) cashiers, who give the bags to customers and take their money. Each employee can be regarded as a communicating sequential process. What form of interprocess communication do they use? Relate this model to processes in MINIX 3.
- Five batch jobs A through E, arrive at a computer center at almost the same time. They have estimated running times of 10, 6, 2, 4, and 8 minutes. Their (externally determined) priorities are 3, 5, 2, 1, and 4, respectively, with 5 being the highest priority. For each of the following scheduling algorithms, determine the mean process turnaround time. Ignore process switching overhead.

  a) Round robin.

  b) Priority scheduling.

  c) First-come, first-served (run in order 10, 6, 2, 4, 8).

  d) Shortest job first.

  For (a), assume that the system is multi programmed, and that each job gets its fair share of the CPU. For (b) through (d) assume that only one job at a time runs, until it finishes. All jobs are completely CPU bound.
- Solve the dining philosopher's problem using monitors instead of semaphores.
- The banker's algorithm is being run in a system with m resource classes and n processes. In the limit of large m and n, the number of operations that must be performed to check a state for safety is proportional to ma and nb. What are the values of a and b?

MODULE-2

UNIT-1 **12L+0T+8P=20 Hours**

MEMORY MANAGEMENT

Memory Management: Basic concept of memory management, Swapping, Virtual Memory, Page Replacement Algorithms, Design issues for Paging systems, Segmentation, Implementation of MINIX 3 Process Manager.

UNIT-2 **12L+0T+8P=20 Hours**

FILE SYSTEMS

File Systems: Files, Directories, File System Implementation- layout, Implementing files, Disk Space Management, File System Reliability, File system Performance, Security, Implementation of MINIX 3 File System.

PRACTICES:

- Consider a swapping system in which memory consists of the following hole sizes in memory order: 10 KB, 4 KB, 20 KB, 18 KB, 7 KB, 9 KB, 12 KB, and 15 KB. Which hole is taken for successive segment requests of (a) 12 KB (b) 10 KB (c) 9 KB for first fit? Now repeat the question for best fit, worst fit, and next fit.
- A machine has a 32-bit address space and an 8 KB page. The page table is entirely in hardware, with one 32-bit word per entry. When a process starts, the page table is copied to the hardware from memory, at one word every 100 nsec. If each process runs for 100 msec (including the time to load the page table), what fraction of the CPU time is devoted to loading the page tables?
- A computer with a 32-bit address uses a two-level page table. Virtual addresses are split into a 9-bit top-level page table field, an 11-bit second-level page table field, and an offset. How large are the pages and how many are there in the address space?
- Suppose that a 32-bit virtual address is broken up into four fields, a, b, c, and d. The first three are used for a three-level page table system. The fourth field, d, is the offset. Does the number of pages depend on the sizes of all four fields? If not, which ones matter and which ones do not?

- A computer has four page frames. The time of loading, time of last access, and the R and M bits for each page are as shown below (the times are in clock ticks):

Page	Loaded	Last ref.	R	M
0	126	279	0	0
1	230	260	1	0
2	120	272	1	1
3	160	280	1	1

  Which page will NRU replace?

  Which page will FIFO replace?

  Which page will LRU replace?

  Which page will second chance replace?

- A small computer has four page frames. At the first clock tick, the R bits are 0111 (page 0 is 0, the rest are 1). At subsequent clock ticks, the values are 1011, 1010, 1101, 0010, 1010, 1100, and 0001. If the aging algorithm is used with an 8-bit counter, give the values of the four counters after the last tick.
- Free disk space can be kept track of using a free list or a bitmap. Disk addresses require D bits. For a disk with B blocks, F of which are free, state the condition under which the free list uses less space than the bitmap. For D having the value 16 bits, express your answer as a percentage of the disk space that must be free.
- A disk has 4000 cylinders, each with 8 tracks of 512 blocks. A seek takes 1 msec per cylinder moved. If no attempt is made to put the blocks of a file close to each other, two blocks that are logically consecutive (i.e., follow one another in the file) will require an average seek, which takes 5 msec. If, however, the operating system makes an attempt to cluster related blocks, the mean interblock distance can be reduced to 2 cylinders and the seek time reduced to 100 microsec. How long does it take to read a 100 block file in both cases, if the rotational latency is 10 msec and the transfer time is 20 microsec per block?
- Write a pair of programs, in C or as shell scripts, to send and receive a message by a covert channel on a MINIX 3 system. Hint: A permission bit can be seen even when a file is otherwise inaccessible, and the sleep command or system call is guaranteed to delay for a fixed time, set by its argument. Measure the data rate on an idle system. Then create an artificially heavy load by starting up numerous different background processes and measure the data rate again.
- Implement immediate files in MINIX 3, that is small files actually stored in the i-node itself, thus saving a disk access to retrieve them.
- Assume that you have a page-reference string for a process with m frames (initially all empty). The page-reference string has length p, and n distinct page numbers occur in it.

  a) What is a lower bound on the number of page faults?

  b) What is an upper bound on the number of page faults?

- Consider the following page-replacement algorithms. Rank these algorithms on a five- point scale from “bad” to “perfect” according to their page-fault rate. Separate those algorithms that suffer from Belady’s anomaly from those that do not.

  a) LRU replacement.

  b) FIFO replacement.

  c) Optimal replacement.

  d) Second-chance replacement.

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Module No.	Mapping with POs
1	Classify the basic concepts of operating system.	Analyze	1	1
2	Apply the concepts of process scheduling algorithms and process synchronization techniques to derive the efficiency of resource utilization.	Apply	1	1, 2, 3, 5, 12
3	Synthesize the concepts of I/O management, file system implementation and problems related to Deadlock.	Analyze	1,2	1,2,12
4	Design the various memory management schemes for a given scenario.	Create	2	3,5
5	Apply the concepts of file system interface and implementation.	Apply	2	2,5

TEXT BOOKS:

1. Andrew S. Tanenbaum, Albert S.Woodhull. The MINIX Book Operating Systems Design and Implementation, Pearson Publicastion, 3rd Edition, June 2006.
2. Abraham Silberschatz, Peter Baer Galvin and Greg Gagne, “Operating System Concepts”, 9th edition, John Wiley & Sons Inc, 2013.

REFERENCE BOOKS:

1. Richard. Stevens and Stephen A Rago, “Advanced Programming in the Unix Environment”, 3rd edition, Addison-Wesley, 2013.
2. William Stallings, “Operating Systems-Internals and Design principles” PHI, 7th Edition, 2012.
3. Gary J. Nutt. Addison-Wesley , “Operating Systems: A Modern Perspective”, Aug 2001, 2nd Edition.
4. B.A. Forouzan& R.F. Giberg, —Unix and shell Programming‖, Thomson, First Edition, New Delhi, 2003.

22CS814 OPTIMIZATION TECHNIQUES



Source: https://oworkers.com/8-process-optimization-techniques-how-to-get-started/

Hours Per Week :

L	T	P	C
3	2	0	4

PREREQUISITE KNOWLEDGE: Probability & Linear algebra.

COURSE DESCRIPTION AND OBJECTIVES:

This course introduces the fundamental concepts of Optimization Techniques and to make the learners aware of the importance of optimizations in real scenarios. The most important objective is to provide the concepts of various classical and modern methods for constrained and unconstrained problems in both single and multivariable. Finally, the basic idea behind the evolutionary algorithms like Ant Colony Optimization, Particle Swarm Optimization could be discussed for further study.

MODULE-1

UNIT-1 **12L+8T+0P=20 Hours**

INTRODUCTION TO CLASSICAL METHODS AND LINEAR PROGRAMMING PROBLEMS TERMINOLOGY

Design Variables – Constraints, Objective Function - Problem Formulation, Linear Programming Problem - Simplex method - Concept of Duality.

General Transportation Problem: The transportation table finding in initial basic feasible solution, North-West corner method, Least cost method, Row minima method, Column minima method.

UNIT-2 **12L+8T+0P=20 Hours**

SINGLE VARIABLE OPTIMIZATION

Problems Optimality Criterion - Bracketing Method - Region Elimination Methods -Interval Halving Method - Fibonacci Search Method - Golden Section Method. Gradient Based Methods: Newton -Raphson Method - Bisection Method - Application to Root finding.

Multi Variable Optimization, Algorithms Optimality: Criteria - Unidirectional Search. Direct Search Methods: Hooke -Jeeves pattern search method. Gradient Based Methods: Cauchy's Steepest Descent Method - Newton's method, Quasi-Newton methods, L-BFGS.

PRACTICES:

A company makes two products (X and Y) using two machines (A and B). Each unit of X that is produced requires 50 minutes processing time on machine A and 30 minutes processing time on machine B. Each unit of Y that is produced requires 24 minutes processing time on machine A and 33 minutes processing time on machine B. At the start of the current week there are 30 units of X and 90 units of Y in stock. Available processing time on machine A is forecast to be 40 hours and on machine B is forecast to be 35 hours. The demand for X in the current week is forecast to be 75 units and for Y is forecast to be 95 units. Company policy is to maximise the combined sum of the units of X and the units of Y in stock at the end of the week. Formulate the problem of deciding how much of each product to make in the current week as a linear program. Solve this linear program graphically.

SKILLS:

- ✓ *Project scheduling process skills including defining project activities, and estimation of time and resources*
- ✓ *Resource optimization skills to adjust the project schedule as per the demand and supply issues of project resources*
- ✓ *Feasibility Analysis.*

Solve using the Simplex method the following problem:

Maximize Z = f(x,y) = 3x + 2y

subject to: $2x + y \leq 18$

$2x + 3y \leq 42$

$3x + y \leq 24$

$x \geq 0 , y \geq 0$

Luminous lamps have three factories - F1, F2, and F3 with production capacity 30, 50, and 20 units per week respectively. These units are to be shipped to four warehouses W1, W2, W3, and W4 with requirement of 20, 40, 30, and 10 units per week respectively. The transportation costs (in Rs.) per unit between factories and warehouses are given below. Find an initial basic feasible solution of the given transportation problem using northwest corner rule.

Factory	**Warehouse**				**Supply**
	W1	W2	W3	W4	
F1	1	2	1	4	30
F2	3	3	2	1	50
F3	4	2	5	9	20
Demand	20	40	30	10	

- A mobile phone manufacturing company has three branches located in three different regions, say Jaipur, Udaipur and Mumbai. The company has to transport mobile phones to three destinations, say Kanpur, Pune and Delhi. The availability from Jaipur, Udaipur and Mumbai is 40, 60 and 70 units respectively. The demand at Kanpur, Pune and Delhi are 70, 40 and 60 respectively. The transportation cost is shown in the matrix below (in Rs). Use the Least Cost method to find a basic feasible solution (BFS).

		DESTINATIONS			
		Kanpur	**Pune**	**Delhi**	**Supply**
Sources	Jaipur	4	5	1	40
	Udaipur	3	4	3	60
	Mumbai	6	2	8	70
	Demand	**70**	**40**	**60**	**170**

- Find Solution using Row minima method.

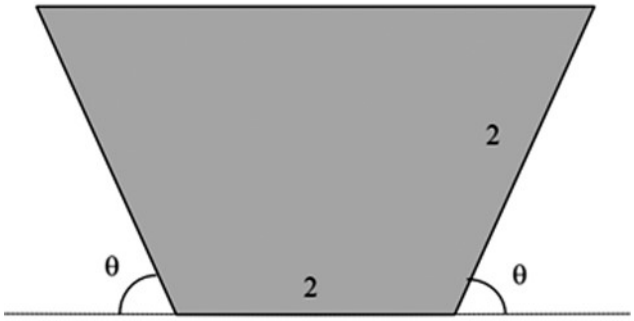
	D1	**D2**	**D3**	**D4**	**Supply**
S1	11	13	17	14	250
S2	16	18	14	10	300
S3	21	24	13	10	400
Demand	200	225	275	250	

- Find Solution using Column minima method.

	D1	D2	D3	D4	Supply
S1	11	13	17	14	250
S2	16	18	14	10	300
S3	21	24	13	10	400
Demand	200	225	275	250	

- Find the positive root of the following equation by (Bisection) method,
  $f(x)=\sin(x/2\pi)-\cos^2 x$
- Consider figure below. The cross-sectional area A of a gutter with equal base and edge length of 2 is given by:
  $A = 4\sin\theta\,(1+\cos\theta)$

Using an initial interval of $[0,\pi/2]$, find the interval after 3 iterations. Use an initial interval $\varepsilon = 0.2$.



MODULE-2

UNIT-1 **12L+8T+0P=20 Hours**

STOCHASTIC OPTIMIZATION PROBLEMS

Notion of regret, online to batch conversion, Methods offering vanishing regret - OGD, EG, OMD Convex Sets and Functions: Affine and convex sets, convexity preserving operations, separating and supporting hyper-planes, generalized inequalities, Operations preserving convexity, conjugate function, Quasi-convex functions, Log-concave and log-convex functions, Convexity with respect to generalized inequalities

UNIT-2 **12L+8T+0P=20 Hours**

NON-CONVEX OPTIMIZATION PROBLEMS

Applications - sparse recovery, affine rank minimization, low-rank matrix completion, Convex approaches - relaxation-based methods, Non-convex approaches - projected gradient descent, alternating minimization

Algorithms: Unconstrained problems, equality constrained problems

PRACTICES:

- Show that each convex cone is indeed a convex set.
- Consider the linear system $0 \le x_i \le 1$ for $i = 1, \ldots, n$ and let P denote the solution set. Explain how to solve a linear programming problem $\max\{c^T x : x \in P\}$.
  What if the linear system was $a_i \le x_i \le b_i$ for $i = 1, \ldots, n$. Here we assume $a_i \le b_i$ for each i.
- Can projected gradient descent (PGD) be used to obtain a stationary solution?
- Show that strong smoothness does not imply convexity by constructing a nonconvex function f : R
- p → R that is 1-SS.

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Module No.	Mapping with POs
1	Understand the classical, linear programming and transportation problem along with game theory.	Understand	1	1
2	Analyze real world problems around you.	Analyze	1	1, 2
3	Apply optimization to real-world problems.	Apply	1	1, 2, 3
4	Implement dynamic programming in real world complex problems.	Implement	2	3, 4, 11, 12

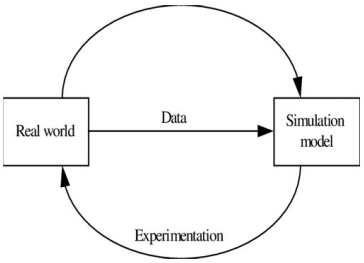
TEXT BOOKS:

1. Kanti Swarup, Man Mohan and P.K.Gupta, “Operations Research”, Sultan Chand &Sons, 2005.
2. S. S. Rao, “Engineering Optimization Theory and Practice”, 4th Edition, Wiley Publishers, 2009.

REFERENCE BOOKS:

1. M.C. Bhuvaneswari, “Application of Evolutionary Algorithms for Multi-Objective Optimization in VLSI and Embedded Systems”, Spinger, 2014.
2. Ashlock D, “Evolutionary Computation for Modeling and Optimization”, Springer, 2006.
3. Kalyanmoy Deb, “Optimization for Engineering Design”, Prentice Hall of India, 2nd Edition, 2012.
4. A. Ravindran and K.M. Rogsdell, G.V. Reklaites, “Engineering Optimization: Methods and Applications”,Wiley, 2nd Edition, 2006.
5. E. J. Haug and J.S. Arora, “Applied Optimal Design”, Wiley Publishers,1979.

22CS816 SIMULATION AND MODELLING



Source: https://www.researchgate.net/figure/The-Simulation-Modelling-Process-Simple-Outline_fig1_3834396

Hours Per Week :

L	T	P	C
3	0	2	4

PREREQUISITE KNOWLEDGE: Probability and statistics, OOPS.

COURSE DESCRIPTION AND OBJECTIVES:

This course introduces discrete-event simulation techniques including model design and development, comparison to analytical models, input data preparation, random number generation, output statistical analysis, and model validation and evaluate the performance of real-world systems by analyzing the output of the model under various conditions.

MODULE-1

UNIT-1 **8L+0T+6P=14 Hours**

INTRODUCTION TO SIMULATION, GENERAL PRINCIPLES

When simulation is the appropriate tool and when it is not appropriate, Advantages and disadvantages of Simulation, Areas of application, Systems and system Environment, Components of a System-Discrete and continuous systems, Model of a system, Types of Models, Discrete-Event System Simulation, Steps in a Simulation Study, Simulation Examples.

General Principles, Simulation Software: Concepts in Discrete-Event Simulation, The Event-Scheduling/Time-Advance Algorithm, World Views, Manual simulation Using Event Scheduling, List processing, Basic properties, Operations-Using Arrays, Dynamic Allocation, Linked Lists-Simulation in Java -Simulation in GPSS.

UNIT-2 **16L+0T+10P=26 Hours**

MATHEMATICAL AND STASTICAL MODELS

Statistical Models in Simulation: Review of terminology, concepts, Useful statistical models, Discrete Distributions, Continuous Distributions, Poisson Process, Empirical distributions.

Queuing Models: Characteristics of queuing Systems, Queuing notation, Long-run measures of performance of queuing Systems, Steady-state behavior of M/G/1 queue, Networks of queues, Rough-cut modeling: An illustration

PRACTICES:

- Discuss the system components of a grocery store, healthcare system, material dispatching system.
- Discuss the steps in simulation study for the above given systems.
- A simulation is to be conducted of cooking a porridge to discover at what time a person should start order to have the it on the table by 7:00 P.M. Read a recipe for preparing a dinner or ask a friend or relative for the recipe. As best you can, trace what you understand to be needed, in the data-collection phase of the simulation process, include each step in the recipe. What are the events, activities, and state variables in this system?
- The daily demand for a product is found to follow the distribution as

SKILLS:

- ✓ *Interpret the model and apply the results to resolve critical issues in a real world environment.*
- ✓ *Analyse the Simulation models using input analyzer, and output analyzer.*
- ✓ *Verify and Validate of simulation model.*

Demand	Probability
10	0.25
11	0.35
12	0.30
13	0.10

Determine the total demand for the next 10 days.

- Students are arriving at the college office at the rate of one every 6 ± 2 minutes to pay the fees. They hand over the forms to one of the two clerks available and it takes I 0 ± 2 minutes for the clerk to verify each form. Then the forms are sent to a single cashier who takes 6 ± I minute per form. Simulate the system for 100 hours and determine the

  (a) utilization of each clerk (b) utilization of the cashier (c) average time required to process a form (clerk + cashier).

- A patient arrives at the Emergency Room at Hello-Hospital about every 40 ± 19 minutes. Each patient will be treated by either Doctor Slipup or Doctor Gutcut. Twenty percent of the patients are classified as NIA (need immediate attention) and the test as CW (can wait). NIA patients are given the highest priority (3), see a doctor as soon as possible for 40 ± 37 minutes, but then iheir priority is reduced to 2 and they wait until a doctor is free again, when they receive further treatment for 30 ± 25 minutes and are then discharged. CW patients initially receive the priority 1 and are treated (when their torn comes) for 15 ± 14 minutes; their priority is then increased to 2, they wait again until a doctor is free and receive I 0 ± 8 minutes of final treatment, and are then discharged. Simulate for 20 days of continuous operation, 24 hours per day. Precede this by a 2-day initialization period to load the system with patients. Report conditions at times 0 days, 2 days, and 22 days. Does a 2-day initialization appear long enough to load the system to a level reasonably close to steady-state conditions? (a) Measure the average and maximum queue length of NIA patients from anival to first seeing a doctor. What percent do not have to wait at all? Also tabulate and plot the distribution of this initial waiting time for NIA patients. What percent wait less than 5 minutes before seeing a doctor? (b) Tabulate and plot the distribution of total time in system for all patients. Estimate the 90% quantile-that is, 90% of the patients spend less than x amount of time in the system. Estimate x. (c) Tabulate and plot the distribution of remaining time in system from after the first treatment to discharge, for all patients. Estimate the 90% quantile. (Note: Most simulation packages provide the facility to automatically tabulate the distribution of any specified variable).

- Using Excel, generate 12 columns, each with 250 values, using the formula = RAND(). In cell Ml, place the formula = SUM(Al:Ll)-6 and copy it to the 249 cells below M1 in column M.

a) Compute descriptive statistics about the data in that column, including minimum value, maximum value, mean, median, and standard deviation.

b) Tabulate the values with 9 bins: the first bin will include all values less than or equal to -3.5; the next six bins are of width one; the last bin will include all values greater than 3.5.

c) Does the histogram resemble any distribution with which you are familiar? If so, what is its name? Hint l: Use FREQUENCY in Excel to form bins.

- Of the orders a job shop receives, 25% are welding jobs and 75% are machining jobs. What is the probability that (a) half of the next five jobs will be machining jobs? (b) the next four jobs will be welding jobs?
- Students' arrival at a university library follows Poisson with a mean of 20 per hour. Determine (a) the probability that there are 50 arrivals in the next I hour. (b) the probability that no student arrives in the next I hour. (c) the probability that there are 75 arrivals in the next 2 hours.
- The cars arriving at a gas station is Poisson distributed with a mean of IO per minute. Determine the number of pumps to be installed if the ftrm wants to have 50% of arriving cars as zero entries (i.e., cars serviced without waiting).
- Given the following distributions, Normal (10, 4) Triangular (4, 10, 16) Uniform (4, 16) find the probability that 6 < X< 8 for each of the distributions.
- Vehicles pass through a toll gate at a rate of 90 per hour. The average time to pass through the gate is 36 seconds. The arrival rate and service rate follow Poisson distribution. There is

a complaint that the vehicles wait for a long duration. The authorities are willing to install one more gate to reduce the average time to pass through to 30 seconds, if the idle time of the toll gate is less than 10% and the present average queue length at the gate is more than five vehicles. Check whether the installation of the second gate is justified.

MODULE-2

UNIT-1 **6L+0T+10P=26 Hours**

RANDOM NUMBERS AND INPUT MODELLING

Random-Number Generation, Random-Variate Generation: Properties of random numbers, Generation of pseudo-random numbers, Techniques for generating random numbers, Tests for Random Numbers, Random- Variate Generation, Inverse transform technique, Acceptance-Rejection technique, Special properties.

Input Modeling: Data Collection, Identifying the distribution with data, Parameter Estimation, Goodness of Fit Tests, Fitting a non-stationary Poisson process, Selecting input models without data, Multi-variate and Time-Series input models.

UNIT-2 **8L+0T+6P=14 Hours**

ANALYSIS OF SIMULATION DATA

Verification and Validation of simulation models: Optimization of simulation Models: Model Building, Verification, Validation, Verification of simulation models, Calibration, Validation of models, Optimization, Optimization via Simulation

Output Analysis for A Single Model: Types of simulations with Respect to Output analysis, Stochastic Nature of Output Data, Measures of Performance and their Estimation, Output Analysis for Terminating Simulations, Output analysis for steady-State Simulations.

PRACTICES:

- Develop the triangular random-variate generator with range (0, 12) and mode 5.
- In an inventory system, the lead time is found to follow uniform distribution with mean 10 days and half width 3 days. Generate five lead times.
- Write a computer program to generate exponential random variates for a given mean value. Generate 1000 values and verify the variates generated using chi-square test.
- In a college library, collect the following in formation at the books return counter: arrival of students for returning books service time taken by the counter clerk 305 Consolidate the data collected and verify whether it follows any standard distribution.
- Draw the pdf of normal distribution with J.l. = 6, a= 3
- The following data represent the time to perform transactions in a bank, measured in minutes: 0.740, 1.28, 1 .46, 2.36, 0.354, 0.750, 0.912, 4.44, O.l l4, 3.08, 3.24, 1. 10, 1.59, 1.47, 1.17, 1.27, 9. 12, 1 1.5, 2.42, 1.77. Develop an input model for these data.
- A simulation model. of a job shop was developed to investigate diff erent schedulliJg rules. To validate the model, the scheduling rule currently used was incorporated into the model and the resulting output was compared agamst observed system behavior. By searching the previous year's database records it was estimated that the average number of jobs in the shop was 22.5 on a given day. Seven independent replications of the model were run, each of 30 days' duration, with the following results for average number of Jobs in the shop: 18.9 22.0 19.4 22.1 19.8 21.9 20.2 (a) Develop and conduct a stastical test to evaluate whether model output is consistent with system behavior. Use the level of significance a= 0.05. · (b) What is the power of this test if a difference of two jobs is viewed as critical? What sample size is needed to guarantee a power of 0.8 or higher? (Use a= 0.05.)

SKILLS:

- ✓ *Interpret the model and apply the results to resolve critical issues in a real world environment.*
- ✓ *Analyse the Simulation models using input analyzer, and output analyzer.*
- ✓ *Verify and Validate of simulation model.*

- Consider the following inventory system: (a) Whenever the inventory level falls to or below 10 units, an order is placed. Only one order can be outstanding at a time. (b) The size of each order is Q; Maintaining an inventory costs $0.50 per day per item in inventory. Placing an order incurs a fixed cost, $10.00. (c) Lead time is distributed in accordance with a discrete uniform distribution between zero and 5 days. (d) If a demand occurs during a period when the inventory level is zero, the sale is lost at a cost of $2.00 per unit. (e) The number of customers each day is given by the following distribution:

No of customers per day	Probability
1	0.23
2	0.41
3	0.22
4	0.14

f) The demand on the part of each customer is Poisson distributed with a mean of 3 units.

g) For simplicity, assume that an·demands occur at noon and that all orders are placed immediately thereafter. Assume further that orders are received at 5:00 P.M., or after the demand that occurred on that day. Consider the poi icy having Q = 20 .. Make five independent replications, each of length 100 days, and compute a 90% confidence interval for long-run mean daily cost. Investigate the effect of initial inventory level and existence of an outstanding order on the estimate of mean daily cost. Begin with an initial inventory of Q + I 0 and no outstanding orders.

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Module No.	Mapping with POs
1	Analyze the steps in design of various simulation models.	Analyze	1	2
2	Conceptualize real world situations related to systems development decisions, originating from source requirements and goals. Construct model for a given set of data and motivate its validity.	Apply	1	1, 4
3	Generate and test random number variates and apply them to develop simulation models.	Apply	2	1
4	Analyze output data produced by a model and test validity of the model.	Analyze	2	2

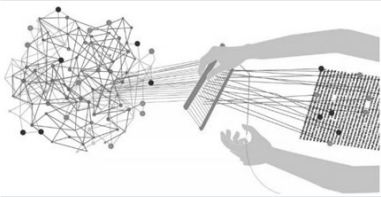
TEXT BOOKS:

1. Jerry Banks, John S. Carson II, Barry L. Nelson, David M. Nicol, “Discrete-Event System Simulation”, 5th Edition, Pearson Education © 2013.
2. Zeigler B.P. Praehofer. H. and Kim I.G. “Theory of modeling and simulation”, 2nd Edition Academic press 2000.

REFERENCE BOOKS:

1. Averill M. Law, “Simulation Modeling and Analysis”, 4th Edition, Tata McGraw-Hill, 2007.
2. Lawrence M. Leemis, Stephen K. Park, “Discrete – Event Simulation: A First Course”, Pearson Education, 2006.

22CSXX6 NATURAL LANGUAGE PROCESSING



Source: https://www.google.com/search?newwindow=1&sca_esv=8dc90ac1bcc54bd2&udm=2&fbs=ADc_l-aN0CWEZBOHjofHoaMMDiKpaEWjvZ2Py1XXV8d8KvlI3p-ML-906rRL_m6h4jR-tdCeKlwp94h-QiJ4lJfObsqU79yRFgWBtc5FGpXu1cRl7WMsuLbm9eXF2miyzQA_QJ-KLUyb79lO5lIA-PPacG7YkvHaSyBedy6hthtWjStMQiZ2RD9OtPsgS3IsqZY70yeSJQNP8JwOLf8cwuCDlzkjkYbxQ&q=NATURAL+LANGUAGE+PROCESSING+images&sa=X&ved=2ahU-

Hours Per Week :

L	T	P	C
3	0	2	4

PREREQUISITE KNOWLEDGE: Programming using Python, Data Structures, Probability and Statistics, Machine Learning Basics, Database Management Systems, and Basic Concepts of Text Mining and Linear Algebra.

COURSE DESCRIPTION AND OBJECTIVES:

This course provides a comprehensive understanding of Natural Language Processing (NLP) by introducing the fundamental concepts of linguistic analysis, text preprocessing, statistical language modeling, sequence labeling, vector semantics, and modern contextual word representations. It enables students to develop practical skills in tokenization, morphological analysis, N-gram modeling, POS tagging, Named Entity Recognition, word embeddings, and transformer-based models such as BERT and GPT using Python-based implementations. The course emphasizes the design and development of intelligent NLP applications including text analytics, chatbots, sentiment analysis, and language understanding systems for real-world AI applications.

MODULE-1

UNIT-1 **12L+0T+6P=18 Hours**

FOUNDATIONS OF NATURAL LANGUAGE PROCESSING

Introduction to NLP: What is Natural Language Processing? Neighbors of NLP, Origins and Use cases, Themes of NLP, Language and Knowledge, The Challenges of NLP, Language and Grammar, Processing Indian Languages, Components of NLP systems, Phases of processing, The problem of ambiguity, and Typical NL Tasks.

UNIT-2 **12L+0T+6P=18 Hours**

MORPHOLOGY AND TEXT PRE-PROCESSING

Finding the Structure of Words: Words and Their Components, Lexemes, Morphemes, and Morphology, Morphological Typology (General Typology), Problems in morphological processing.

Text Pre-processing Pipeline: Tokenization, Word Tokenization vs. Sentence Tokenization, Filtering, Stop-word removal and Filtering Stop words, Normalization, Stemming and Lemmatization, Regular Expressions.

PRACTICES:

TEXT PROCESSING & LEXICAL ANALYSIS

- Implement a Python program for Natural Language Processing that demonstrates the components and phases of NLP systems, explains the relationship between NLP, AI, and Linguistics, and analyzes the role of NLP in real-world applications.
- Develop a Python application to identify and resolve ambiguity in Natural Language Processing by implementing lexical and syntactic ambiguity detection techniques and comparing semantic and pragmatic ambiguities using suitable examples.
- Create a Python-based mini NLP system that performs sentence classification, chatbot-based query handling, and sentiment analysis using standard NLP libraries and preprocessing techniques.
- Implement a Python program for morphological analysis that identifies lexemes and morphemes, demonstrates inflectional and derivational morphology, and analyzes different morphological typologies of words.
- Develop a Python application for text tokenization that performs word tokenization, sentence

SKILLS:

- ✓ *Analyze the types and occurrence of roots.*
- ✓ *Interpolate the unknown values of function.*
- ✓ *Develop a difference equation.*
- ✓ *Gain the knowledge to solve an ODE numerically.*
- ✓ *Gain the knowledge to solve an ODE numerically.*

tokenization, and comparative analysis of both techniques using sample textual datasets.

- Create a Python program using Regular Expressions for Natural Language Processing tasks such as extracting email IDs, URLs, and phone numbers, performing pattern matching, and validating textual data.
- Develop a complete Python-based text pre-processing pipeline that performs filtering, stop-word removal, normalization, and pre-processing impact analysis on NLP tasks.
- Implement a Python application for stemming and lemmatization that demonstrates stemming algorithms, WordNet-based lemmatization, and comparative analysis of processed outputs for different textual inputs.

MODULE-2

UNIT-1 **12L+0T+6P=18 Hours**

FORMAL LANGUAGE MODELING AND SEQUENCE LABELLING

Statistical Modeling: Unigrams, Bigrams, and N-grams, N-gram Language Models, Construction and evaluation of models, Model Optimization, Smoothing techniques and Entropy, Text Comparison algorithms.

Tagging Parts of Speech: Part-of-Speech Tagging, Structural Parsing, chinking, Named Entity Recognition (NER), Hidden Markov Models (HMM), Forward and Viterbi algorithms, and Expectation-Maximization (EM) training, Conditional Random Fields (CRF).

UNIT-2 **12L+0T+6P=18 Hours**

VECTOR SEMANTICS AND WORD REPRESENTATIONS

Traditional Sparse Representations: Embedding, One-Hot Encoding, Term Frequency-Inverse Document Frequency (TF-IDF), Pointwise Mutual Information (PMI).

Static Dense Vectors: Word2Vec, Skip-gram, Continuous Bag of Words (CBOW), GloVe, FastText.

Contextualized Word Representations: (Embeddings from Language Models) ELMo, (Bidirectional Encoder Representations from Transformers) BERT, (Generative Pre-trained Transformer) GPT.

PRACTICES:

- Implement a Python program for statistical language modeling that demonstrates unigrams, bigrams, and n-grams, constructs and evaluates N-gram language models, and analyzes their applications in Natural Language Processing.
- Develop a Python application for language model optimization that performs smoothing techniques, entropy calculation, and probability estimation to improve the performance of statistical language models.
- Create a Python-based text comparison system that implements similarity measurement algorithms, compares multiple textual documents, and analyzes text matching accuracy using NLP techniques.
- Implement a Python program for Part-of-Speech tagging that performs token tagging, syntactic analysis, and structural parsing of sentences using standard NLP libraries.
- Develop a Python application for chunking and Named Entity Recognition (NER) that identifies phrase structures, extracts named entities from textual data, and evaluates the effectiveness of entity recognition techniques.
- Create a Python-based sequence labelling system using Hidden Markov Models (HMM) that implements forward and Viterbi algorithms for probabilistic sequence prediction and tagging tasks.
- Implement a Python application for Expectation-Maximization (EM) training and Conditional Random Fields (CRF) that performs sequence labeling, parameter optimization, and comparative analysis of probabilistic models.
- Develop a Python program for traditional sparse vector representations that demonstrates One-Hot Encoding, TF-IDF, Pointwise Mutual Information (PMI), and embedding generation for textual datasets.

- Create a Python-based word embedding system that implements Word2Vec, Skip-gram, Continuous Bag of Words (CBOW), GloVe, and FastText models for generating dense vector representations of words.
- Implement a comprehensive Python application for contextualized word representations that demonstrates ELMo, BERT, and GPT embeddings, performs semantic analysis on textual data, and compares contextual embeddings with traditional word representation techniques.

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Module No.	Mapping with POs
1	Understand the foundations of Natural Language Processing, linguistic structures, ambiguity handling, and text preprocessing techniques for analyzing natural language data.	Understand	1	1,2
2	Apply morphological analysis, tokenization, normalization, stemming, lemmatization, and regular expression techniques for text preprocessing and NLP tasks.	Apply	1	1, 2, 5
3	Analyze statistical language models, N-gram models, smoothing techniques, and sequence labeling methods such as POS tagging, HMM, NER, and CRF.	Analyze	2	1,2,4
4	Evaluate vector semantics and word representation techniques including TF-IDF, Word2Vec, CBOW, GloVe, FastText, and contextual embeddings such as BERT and GPT.	Evaluate	2	1,2,3,5
5	Develop real-world Natural Language Processing applications using statistical, embedding-based, and transformer-based NLP models for intelligent text processing systems.	Create	2	1,2,3,5

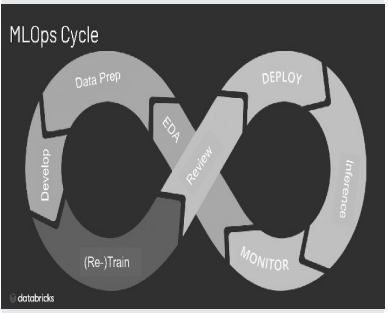
TEXT BOOKS:

1. Daniel Jurafsky and James H. Martin, Speech and Language Processing: An Introduction to Natural Language Processing, Computational Linguistics and Speech Recognition, 3rd edition, Pearson, 2022.
2. Christopher D. Manning and Hinrich Schütze, Foundations of Statistical Natural Language Processing, MIT Press, 1999.

REFERENCE BOOKS:

1. Steven Bird, Ewan Klein, and Edward Loper, Natural Language Processing with Python – Analyzing Text with the Natural Language Toolkit, 1st edition, O'Reilly Media, 2009.
2. Nitin Indurkhya, Fred J. Damerau, "Handbook of Natural Language Processing", 2nd Edition, CRC Publishers, 2010.
3. Jacob Eisenstein, Introduction to Natural Language Processing (Adaptive Computation and Machine Learning series), Kindle Edition, MIT Press, 2019.
4. Anders Sogaard, Ivan Vulic, Sebastian Ruder, Manaal Faruqui , Cross-Lingual Word Embeddings (Synthesis Lectures on Human Language Technologies), Morgan & Claypool Publishers , 2019.

.



https://www.google.com/search?newwindow=1&sca_esv=9170600a296bad62&udm=2&fbs=ADc_l-aN0CWEZBOHjofHoaMMDiKpaEWjvZ2Py1XXV8d8KvlI3p-ML-906rRL_m6h4jR-tdAeyw6pOVABma0FfM0NmtARdcuGnZVoiSrEQCt2b10nLmqaIq-f5xasgNuw3lO57iddeJHudkWbt3y-GkUgAwSrlIOEh4y0QHta-lHKrOMGa3x8dLHObxl-GgmXGELkj9eaIaxR7VC-a_Ap23LNHRcpEvkl6Q&q=MLops+images&sa=X&ved=2ahUKEwj6vJ-b-7CUAxUT2

22CSXX6 MLops

Hours Per Week :

L	T	P	C
3	0	2	4

PREREQUISITE KNOWLEDGE: Python Programming, Data Handling and Visualization, Basics of Intelligent Application Development.

COURSE DESCRIPTION AND OBJECTIVES:

This course introduces students to the lifecycle of intelligent applications, focusing on building, deploying, and maintaining scalable AI systems. It covers experiment tracking, pipeline orchestration, deployment strategies, monitoring, and governance, including modern practices for large-scale AI systems such as LLMs.

MODULE-1

UNIT-1 **12L+0T+6P=18 Hours**

INTRODUCTION TO AI SYSTEM LIFECYCLE & MLOPSS

Overview of AI system lifecycle and need for lifecycle management (T1, pp. 1–35), Comparison: Traditional Software vs DevOps vs AI workflows(T3, pp. 10–25), Key challenges in deploying intelligent models, MLOps maturity levels and workflow (T2, pp. 5–30), Overview of tools and ecosystem (R1, pp. 1–20), Running example: Real-world dataset. Environment setup.

UNIT-2 **12L+0T+6P=18 Hours**

EXPERIMENT TRACKING, DATA & MODEL MANAGEMENT

Data engineering basics and data pipelines (R4, pp. 50–110), Data versioning and dataset management (T2, pp. 60–85), Feature stores: Concept, benefits, role in AI applications, Feast overview (T1, pp. 120–150), Experiment tracking: MLflow introduction, Logging parameters, metrics (T2, pp. 90–130), Model management: Saving, loading, model registry (R2, pp. 140–170), Hyperparameter tuning basics (R5, pp. 180–210).

PRACTICES:

TEXT PROCESSING & LEXICAL ANALYSIS

- Apply the following tasks to any given dataset (e.g., house price / student / health dataset):
  a. Define the problem statement (classification/regression).
  b. Design the complete AI lifecycle:
  - Data collection
  - Preprocessing
  - Model training
  - Deployment
  - Monitoring

  c. Represent the lifecycle using a workflow diagram.
  d. Identify challenges at each stage.
  - Data Pipeline Implementation
- Apply the following tasks:
  a. Load a dataset and perform preprocessing:
  o Handle missing values
  o Encode categorical variables

- o Normalize/scale data

b. Split dataset into training and testing sets.

c. Automate preprocessing steps using a pipeline.

d. Save processed dataset for reuse.

e. Compare results before and after preprocessing.

- Feature Engineering & Feature Store Simulation:

Apply the following tasks:

a. Load dataset and create new features:

- o Derived attributes
- o Aggregated features

b. Store features separately (simulate feature store).

c. Reuse stored features in another experiment.

d. Compare model performance with and without engineered features.

e. Report importance of feature reuse in AI systems.

- Model Management & Registry

Apply the following:

a. Train at least two models on the same dataset.

b. Save models using serialization techniques.

c. Register models in model registry.

d. Select best model for deployment.

e. Load model and perform predictions on new data.

- Hyperparameter Tuning and Optimization

Perform the following tasks:

a. Train a baseline model.

b. Apply hyperparameter tuning:

- o Grid search OR
- o Random search

c. Compare performance before and after tuning.

d. Visualize results (graphs/tables).

e. Report impact of tuning on model accuracy.

MODULE-2

UNIT-1 **12L+0T+6P=18 Hours**

MODEL DEPLOYMENT & PIPELINE ORCHESTRATION

ML pipelines and workflow orchestration, Converting notebooks into pipelines, Orchestration tools overview (T2, pp. 150–190), CI/CD concepts for AI systems (T3, pp. 120–150), Deployment strategies: Online (API-based), Batch processing, Streaming systems (T1, pp. 200–240), Model serving techniques (R2, pp. 200–230), Containerization basics (Docker intro) (R4, pp. 250–280).

UNIT-2 **12L+0T+6P=18 Hours**

MONITORING, PERFORMANCE & OPTIMIZATION

Monitoring ML systems in production (T1, pp. 250–280), Logging and error tracking (R3, pp. 60–90), Model performance metrics tracking (R5, pp. 300–330), Tools overview: Prometheus, Grafana, Evidently, Performance optimization: Scaling strategies, Resource optimization (T3, pp. 200–230). Security in AI systems (MLSecOps overview) (R3, pp. 120–150), Governance: Explainability, Bias detection (R5, pp. 350–380), Ethical considerations (T1, pp. 300–330).

PRACTICES:

- Model Deployment as API
- Apply the following tasks:
  a. Train a model on any dataset.
  b. Convert the trained model into a REST API using Flask/FastAPI.
  c. Create endpoints for:
  - Prediction
  - Health check

  d. Test API using Postman or browser.
  e. Deploy the API locally and document results.
- End-to-End ML Pipeline Implementation
  a. Load dataset and preprocess data.
  b. Train model and save it.
  c. Create pipeline for:
  - Data preprocessing
  - Training
  - Prediction

  d. Convert notebook into reusable pipeline script.
  e. Execute pipeline automatically and report outputs.
- Batch Prediction System
  a. Load a large dataset.
  b. Apply trained model for batch predictions.
  c. Store predictions in:
  - CSV file OR
  - Database

  d. Compare batch vs real-time prediction scenarios.
  e. Report observations.
- CI/CD Pipeline Design for AI Systems
  a. Create version-controlled repository (Git).
  b. Simulate code update scenario.
  c. Design CI/CD pipeline:
  - Code commit
  - Model retraining
  - Deployment

  d. Use GitHub Actions (or conceptual workflow).
  e. Report automation benefits.
- Logging & Error Tracking System
  a. Design logging mechanism for:
  - Input data
  - Predictions
  - Errors

  b. Implement logging in deployed API.
  c. Simulate failure cases.
  d. Analyze logs to identify issues.
  e. Report improvements.

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Module No.	Mapping with POs
1	Understand the AI system lifecycle, MLOps concepts, and the role of lifecycle management in intelligent application development.	Understand	1	1,2
2	Apply data engineering, experiment tracking, feature management, and model management techniques for building reliable AI workflows.	Apply	1	2, 3, 5
3	Develop and deploy machine learning applications using pipelines, APIs, CI/CD practices, and containerization tools.	Apply	2	3,5,9
4	Analyze deployed AI systems using monitoring, performance optimization, explasinability, and responsible AI practices.	Analyze	2	4,5,811

TEXT BOOKS:

1. "Designing Machine Learning Systems" – Chip Huyen Designing Machine Learning Systems
2. "Practical MLOps" – Noah Gift Practical MLOps
3. "Machine Learning Engineering" – Andriy Burkov Machine Learning Engineering.

REFERENCE BOOKS:

1. "Introducing MLOps" – Mark Treveil Introducing MLOps.
2. "Building Machine Learning Powered Applications" – Emmanuel Ameisen ML Powered Applications.
3. "Reliable Machine Learning" – Cathy Chen Reliable Machine Learning.
4. "Fundamentals of Data Engineering" – Joe Reis Fundamentals of Data Engineering.
5. "Hands-On Machine Learning with Scikit-Learn, Keras & TensorFlow" – Aurélien Géron Hands-On Machine Learning.

.

HONOURS

COMPUTER SCIENCE AND ENGINEERING

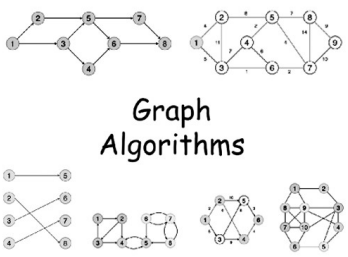
B.Tech.

22CS951	-	Advanced Graph Algorithms
22CS952	-	Biometrics
22CS953	-	Parallel and Distributed Computing
22CS954	-	Internet of Things
22CS955	-	Wireless Sensor Networks
22CS956	-	Capstone Project

COURSE CONTENTS

I SEM & II SEM

22CS951 ADVANCED GRAPH ALGORITHMS



Source: https://towardsdatascience.com/10-graph-algorithms-visually-explained-e57faa1336f3

Hours Per Week :

L	T	P	C
3	0	2	4

PREREQUISITE KNOWLEDGE: Basic Logical Thinking and Problem Solving Ability.

COURSE DESCRIPTION AND OBJECTIVES:

The course will cover some traditional discrete approaches to various graph problems, especially flow problems, and then contrast these approaches with modern, asymptotically faster methods based on combining convex optimization with spectral and combinatorial graph theory.

MODULE-1

UNIT-1 **12L+0T+8P=20 Hours**

INTRODUCTION

Introduction to Optimization, Convex Geometry, Linear Algebra, Convexity and Second Derivatives, Gradient Descent and Acceleration.

UNIT-2 **12L+0T+8P=20 Hours**

SPECTRAL GRAPH THEORY

Introduction to Spectral Graph Theory, Effective Resistance, Gaussian Elimination as Optimization, Additive Perspective on Gaussian Elimination.

PRACTICES:

- Implement the gradient descent optimization algorithm with Nesterov Momentum.
- Assume that S is subset of R^n is a convex set and that the function f : S->R is convex. Suppose that x1, …, xn belongs to S and theta_1,…,theta_n >= 0 with theta_1,…,theta_n = 1. Prove that f(theta_1x1 + _ _ _ + theta_nxn) <= theta_1f(x1) + …. + theta_nf(xn)

MODULE-2

UNIT-1 **12L+0T+8P=20 Hours**

RANDOM MATRIX CONCENTRATION

Introduction to Random Matrix Concentration, Spectral Graph Sparsification, Laplacian Linear Equations, Classical Algorithms for Maximum Flow

UNIT-2 **12L+0T+8P=20 Hours**

SEPARATING HYPERPLANE THEOREM

Separating Hyperplane Theorem, Langrange Multipliers, and Convex Duality, Karush-Kuhn-Tucker Conditions, Fenchel Conjugates, Newton's Method.

SKILLS:

- ✓ *Develop a deeper understanding of fundamental phenomena in optimization.*
- ✓ *Deep dive into modern approaches to graph algorithms using convex optimization techniques.*
- ✓ *Central techniques in the development of graph algorithms including graph decomposition techniques, oblivious routing etc.*

PRACTICES:

- Show that the Ford-Fulkerson algorithm may not terminate; moreover, it may converge a value not equal to the value of the maximum flow.
- An electric company is setting up a power plant in a foreign country and it has to plan its capacity. The peak period demand for power is given by p1 = 400 − q1 and the off-peak is given by p2 = 380 − q2. The variable cost to is 20 per unit (paid in both markets) and capacity costs 10 per unit which is only paid once and is used in both periods.
  - Write down the lagrangian and Kuhn-Tucker conditions for this problem
  - Find the optimal outputs and capacity for this problem.
  - How much of the capacity is paid for by each market (i.e. what are the values of λ1 and λ2)?
  - Now suppose capacity cost is 30 per unit (paid only once). Find quantities, capacity and how much of the capacity is paid for by each market (i.e. λ1 and λ2)?

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Module No.	Mapping with POs
1	Analyze key concepts in optimization such as first and second-order optimization, convex duality, multiplicative weights and dual-based methods, acceleration, preconditioning, and non-Euclidean optimization.	Analyze	1	1, 2, 9, 10, 12
2	Design convex optimization through the lens of graph algorithms	Create	1	1,2,3,9
3	Apply the central techniques in the development of graph algorithms including graph decomposition techniques, sparsification, oblivious routing, and spectral and combinatorial preconditioning.	Apply	2	1, 2, 9, 10, 12

TEXT BOOKS:

1. Boyd, Stephen, Stephen P. Boyd, and Lieven Vandenberghe. Convex optimization. Cambridge university press, 2004.
2. Cartan, Henri. Differential calculus. Hermann, 1983.

REFERENCES:

1. Tarjan, Robert Endre. Data structures and network algorithms. Society for industrial and Applied Mathematics, 1983.
2. Cook, William J., et al. “Combinatorial optimization.” Oberwolfach Reports 5.4 (2009): 2875-2942.
3. Rockafellar, R. Tyrrell. Convex analysis. Vol. 18. Princeton university press, 1970.

22CS952 BIOMETRICS



Source: https://cyberhoot.com/cybrary/biometrics/

Hours Per Week :

L	T	P	C
3	0	2	4

PREREQUISITE KNOWLEDGE: Engineering Mathematics, Probability and Statistics, Image Processing.

COURSE DESCRIPTION AND OBJECTIVES:

This course is a foundation for design and implementation of biometric systems which helps in providing authentication, identification of persons. This can be performed using the biometric traits like finger print, face, iris etc. It is mainly used in security applications.

MODULE-1

UNIT-1 **8L+0T+6P=14 Hours**

INTRODUCTION

Introduction: Biometric Systems, Biometric Functionalities, Biometric System Errors. The Design Cycle of Biometric Systems, Applications of Biometric Systems.

UNIT-2 **16L+0T+10P=26 Hours**

FINGERPRINT AND FACE RECOGNITION SYSTEMS

Finger Print Recognition: Introduction, Friction Ridge Pattern, Fingerprint Acquisition, Feature Extraction, Matching, Fingerprint Indexing, Fingerprint Synthesis, Palmprint.

Face Recognition: Introduction, Image Acquisition, Face Detection, Feature Extraction and Matching.

PRACTICES:

- The minutiae-based representation and matching algorithms.
- Collect various data sets of different traits and analyse the usage of these data sets.
- For a given fingerprint, identify the minutia points by binarization and thinning of ridges.
- From a fingerprint image, extract the singularity points.
- Identify the inter subject and intra subject variations for given finger prints.
- Implementation of ViolaJones object detection algorithm.
- Perform feature extraction by using
  a. Principal component analysis.
  b. Linear Discriminant analysis

MODULE-2

UNIT-1 **10L+0T+8P=18 Hours**

IRIS RECOGNITION

Iris recognition: Introduction, Design of Iris Recognition System, Iris Segmentation, Iris Normalization, Iris Encoding and Matching, Iris Quality.

UNIT-2 **14L+0T+8P=22 Hours**

MULTI-BIOMETRICS

Multi-biometrics: Introduction, Sources of Multiple Evidence, Acquisition and Processing Architecture, Fusion Levels.

SKILLS:

- ✓ *Identify the datasets to be used in various applications.*
- ✓ *Usage of multi-biometrics to enhance security.*
- ✓ *Design of fingerprint, Iris, face detection systems.*

PRACTICES:

- Implementation of Iris segmentation.
- Generation of Iris code.
- Taking a biometric trait, performing.
  a. Sensor-level fusion.
  b. Feature-level fusion.
- Working with feature normalization for the fusion of two heterogeneous feature vectors.
- Score level fusion using various classifiers.

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Module No.	Mapping with POs
1	Analyze the steps in design of various biometric system, functionalities, traits and metrics used to measure their performance.	Analyze	1	2
2	Design of biometric systems depending on the choice of using the trait like fingerprint, face, iris etc.	Create	1,2	1, 3, 6
3	Applying the multimodal biometric traits and fusion levels in various applications.	Apply	2	1,6
4	Analyze the usage of various biometric traits in real time applications.	Analyze	2	2,6

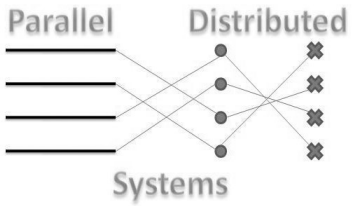
TEXT BOOKS:

1. Anil K. Jain, Arun A. Ross, Karthik Nanda kumar, “Introduction to Biometrics”, Springer, 2011.
2. N. V. Boulgouris , Konstantinos N. Plataniotis , Evangelia Micheli-Tzanakou “Biometrics: Theory, Methods, and Applications”, Wiley, 2009

REFERENCE BOOKS:

1. Samir Nanavati, Michael Thieme, Raj Nanavati, “Biometrics – Identity Verification in a Networked World”, WILEY, 2002.
2. John D. Woodward, John D.Woodward, Jr.Noicholas M.Orlans Peter T. Hig, “Biometrics-The Ultimate Reference”, DreamTech Press, 2003.
3. Julian Ashbourn, “Biometrics: Advanced Identity Verification The Complete Guide” Springer, 2020.

22CS953 PARALLEL AND DISTRIBUTED COMPUTING



Source: http://uceou.edu/PDS/About%20PDS.html

Hours Per Week :

L	T	P	C
3	2	0	4

PREREQUISITE KNOWLEDGE: Fundamental of programming and operating system, computer networking, and computer Organization..

COURSE DESCRIPTION AND OBJECTIVES:

This course covers the challenges faced in constructing parallel and distributed applications, Various implementation techniques, paradigms, architectures and parallel algorithms. And current trends in parallel and distributed computing like Open MP, POSIX Threads, Apache Hadoop (DFS).

MODULE-1

UNIT-1 **12L+8T+0P=20 Hours**

INTRODUCTION TO PARALLEL COMPUTING

The idea of Parallelism, Power and potential of parallelism, examining sequential and parallel programs, Scope and issues of parallel and distributed computing, Goals of parallelism, Parallelism and concurrency using multiple instructions streams.

Parallel Architecture: Pipeline architecture, Array processor, Multi-processor architecture, Systolic architecture, Dataflow architecture, Architectural classification schemes, Memory access classification, Memory Issues: Shared vs. distributed, Symmetric multiprocessing (SMP), SIMD, Vector processing, GPU co-processing, Flynn's Taxonomy, Instruction Level support for parallel programming, Multiprocessor caches and Cache Coherence, Non-Uniform Memory Access (NUMA).

UNIT-2 **12L+8T+0P=20 Hours**

PARALLEL ALGORITHM DESIGN PRINCIPLES AND PROGRAMMING

Need for communication and coordination/synchronization, Scheduling and contention, Independence and partitioning, Task- Based Decomposition, Data Parallel Decomposition, Characteristics of task and interaction, Load balancing, Data Management, parallel algorithm models, Sources of overhead in parallel programs, Performance metrics for parallel algorithm implementations, Parallel algorithmic patterns like divide and conquer, Map and Reduce, Specific algorithms like parallel Merge Sort, Parallel graph Algorithms.

PRACTICES:

- Identify Multiple Instruction Single Data, or MISD. How would an MISD system work? Give an example.
- Suppose a shared-memory system uses snooping cache coherence and write-back caches. Also suppose that core 0 has the variable x in its cache, and it executes the assignment x = 5. Finally suppose that core 1 doesn't have x in its cache, and after core 0's update to x, core 1 tries to execute y = x. What value will be assigned to y? Why?
- Consider a simplified version of bucket-sort. You are given an array A of n random integers in the range [1...r] as input. The output data consist of r buckets, such that at the end of the algorithm, Bucket i contains indices of all the elements in A that are equal to i.
    - Describe a decomposition based on partitioning the input data (i.e., the array A) and an appropriate mapping onto p processes. Describe briefly how the resulting parallel algorithm would work.
    - Describe a decomposition based on partitioning the output data (i.e., the set of r buckets) and an appropriate mapping onto p processes. Describe briefly how the resulting parallel algorithm would work.

SKILLS:

- ✓ *Recognize parallelism in computational problems.*
- ✓ *Know different parallel systems and their classification.*
- ✓ *Design parallel algorithms for different applications.*
- ✓ *Compare replication schemes with respect to performance, availability and consistency concerns.*
- ✓ *Design, implement, and debug distributed systems.*
- ✓ *Implement parallel algorithms using MPI and OpenMP environments.*
- ✓ *Element parallel algorithms using MPI and OpenMP environments 3.0 3.5*
- ✓ *AVERAGE.*

- Consider seven tasks with running times of 1, 2, 3, 4, 5, 5, and 10 units, respectively. Assuming that it does not take any time to assign work to a process, compute the best- and worst-case speedup for a centralized scheme for dynamic mapping with two processes.

MODULE-2

UNIT-1 **12L+8T+0P=20 Hours**

INTRODUCTION TO DISTRIBUTED SYSTEMS

Goals of the Distributed Systems, Relation to parallel systems, synchronous versus asynchronous execution, design issues and challenges, Types of Distributed Systems, Distributed System Models, Hardware and software concepts related to distributed systems, middleware models.

Distributed Computing and Communication design principles: A Model of distributed executions, Models of communication networks, Global state of distributed system, Models of process communication. Communication and Coordination: Shared Memory, Consistency, Atomicity, Message- 08 Passing, Consensus, Conditional Actions, Critical Paths, Scalability, and cache coherence in multiprocessor systems, synchronization mechanism.

UNIT-2 **12L+8T+0P=20 Hours**

PARALLEL AND DISTRIBUTED PROGRAMMING FRAMEWORKS

Overview of CUDA, Open MP, POSIX Threads, Apache Hadoop (DFS), and current trends in parallel and distributed computing.

PRACTICES:

- Give five types of hardware resource and five types of data or software resource that can usefully be shared. Give examples of their sharing as it occurs in practice in distributed systems.
- The host computers used in peer-to-peer systems are often simply desktop computers in users' offices or homes. What are the implications of this for the availability and security of any shared data objects that they hold and to what extent can any weaknesses be overcome through the use of replication?
- Consider two communication services for use in asynchronous distributed systems. In service A, messages may be lost, duplicated or delayed and checksums apply only to headers. In service B, messages may be lost, delayed or delivered too fast for the recipient to handle them, but those that are delivered arrive with the correct contents. Describe the classes of failure exhibited by each service. Classify their failures according to their effects on the properties of validity and integrity. Can service B be described as a reliable communication service?
- Illustrate distributed design through a substantial case study, examining in detail the design of the Google infrastructure, a platform and associated middleware that supports both Google search and a set of associated web services and applications including Google Apps.
- Implementation of the parallel algorithms (on a PC-cluster under Linux platform). The programs will be based on POSIX Thread, MPI programming, Hadoop, Apache Spark etc.

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Module No.	Mapping with POs
1	Apply the fundamentals of parallel and parallel computing including architectures and paradigms.	Apply	1	1
2	Analyse the various design principles of parallel algorithms.	Analyse	1	2
3	Learn the intricacies of distributed programming.	Under-stand	2	1
4	Develop and execute basic parallel and distributed applications using basic programming models and tools.	create	2	5

TEXT BOOKS:

1. DISTRIBUTED SYSTEMS Concepts and Design Fifth Edition. George Coulouris. Cambridge University. Jean Dollimore formerly of Queen Mary, University of London.
2. Distributed Systems Principles and Paradigms Andrew S. Tanenbaum Maarten Van Steen, 3rd Edition, 2017.

REFERENCE LINKS:

1. Introduction To Parallel Programming, Peter S. Pacheco University of San Francisco.
2. Introduction To Parallel Processing, M.Sasikumar, Dinesh Shikhare and P. Ravi Prakash, Randy Chow, T. Johnson, Distributed Operating Systems and Algorithms, Addison Wesley.
3. Ian Foster: Designing and Building Parallel Programs – Concepts and tools for Parallel Software Engineering, Pearson Publisher, 1st Edition, 2019.
4. Parallel Programming in C with MPI and Open MP Michael J.Quinn, McGrawHill Higher Education.
5. https://hpc.llnl.gov/training/tutorials/introduction-parallel-computing-tutorial.
6. https://www.geeksforgeeks.org/introduction-to-parallel-computing/.
7. https://nptel.ac.in/.
8. https://www.coursera.org/.



Source: https://toolsense.io/glossary/iot/

22CS954 INTERNET OF THINGS

Hours Per Week :

L	T	P	C
3	0	2	4

PREREQUISITE KNOWLEDGE: Computer networks.

COURSE DESCRIPTION AND OBJECTIVES:

Internet of Things (IoT) is presently a hot technology worldwide. Government, academia, and industry are involved in different aspects of research, implementation, and business with IoT. IoT cuts across different application domain verticals ranging from civilian to defense sectors. These domains include agriculture, space, healthcare, manufacturing, construction, water, and mining, which are presently transitioning their legacy infrastructure to support a IoT. To introduce the terminology, technology and its applications, to introduce the concept of M2M (machine to machine) with necessary protocols.

MODULE-1

UNIT-1 **12L+0T+8P=20 Hours**

INTERNET OF THINGS FUNDAMENTALS

Introduction to Internet of Things; Physical design & Functional Block of IoT, Device architectures, CoreIoT Functional Stack; Resource constrained devices; Sensors and Components; IoT Enabling Technologies. Societal Benefits of IoT (Domain Specific), Risks, Privacy, and Security.

Network And Communication Protocols: Network Components; Internet Structure, Wireless Protocols; IoT Communication Model & APIs, Wireless Protocol Stack, IoT levels.

UNIT-2 **12L+0T+8P=20 Hours**

IOT AND M2M

Software defined networks, Network function virtualization, difference between SDN and NFV for IoT Basics of IoT System Management with NETCONF-YANG: SDN; NFV; Need for IOT Systems Management; SNMP-NETCONF, YANG; IOT Systems management with NETCONF-YANG.

PRACTICES:

- Identify different Sensors and IoT devices
- Identify the Components in Raspberry pi, Arduino, and UNO boards
- Examine IoT levels with any one domain specific application like home automation, weather monitoring system etc.
- Design the Network Configuration and System Management with IoT devices using NETCONF-YANG.
- Design the Network Configuration and System Management with IoT devices using SNMP-NETCONF.

MODULE-2

UNIT-1 **12L+0T+8P=20 Hours**

INTRODUCTION TO SYSTEMS DESIGN & DEVELOPMENT

IoT system building blocks, Arduino, Node MCU– Board details, IDE programming; Raspberry Pi-Model and Interfaces, Platform: Axonize, Blynk IoT platform, Fogwing.

UNIT-2 **12L+0T+8P=20 Hours**

PROGRAMMING AND CASE STUDY

Embedded C vs Python; Operating systems for constrained devices; Domain Specific IoT Application, Task Support IoT Example: The Refrigerator, Weather Monitoring System – Case study- Design, Programming and Execution.

PRACTICES:

- Demonstration and study of Raspberry Pi board, GPIO Pins and familiarity of various sensors.
- Demonstration and study of other Hardware board of IoT such as Arduino Uno and NodeMCU.
- Design and Implementation of controlling LED-using Python in Raspberry Pi board.
- Design and Implementation of sensing light through LDR using Python in Raspberry Pi board.
- Design and Implementation to find obstacles through sensor using Python in Raspberry Pi board.
- Design and Implementation of sensing and display temperature using Python in Raspberry Pi board.
- Design and Implementation of detecting noise through microphone sensor using Python in Raspberry Pi board.
- Design and Implementation of output devices through relay using Python in Raspberry Pi board.
- Design and Implementation of vibration sensor using Python in Raspberry Pi board.
- Design and Implementation of uploading sensor data into cloud using Python.

SKILLS:

- ✓ *Sensor Identification and IoT system design.*
- ✓ *Sensor data analysis.*
- ✓ *Tool usage for developing IoT applications.*

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

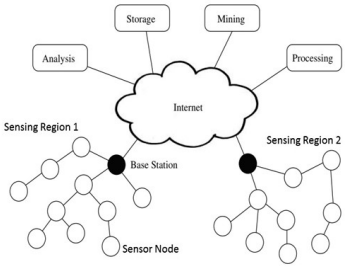
CO No.	Course Outcomes	Blooms Level	Module No.	Mapping with POs
1	Illustrate the impact and challenges posed by IoT networks leading to new architectural models.	Analyze	1	4, 6
2	Design an end-to-end Machine-learning model to realize solutions for real-world problems.	Design	1	3
3	Apply various machine-learning models to develop IoT applications.	Apply	2	1
4	Compare and contrast the deployment of smart objects and the technologies to connect them to network.	Evaluate	2	4

TEXT BOOKS:

1. Arshdeep Bahga and Vijay Madisetti “Internet of Things: A Hands-on Approach”, Universities Press, 2015.
2. Rajkumar Buyya and Amir Vahid Dastjerdi “Internet of Things: Principles and Paradigms”, Morgan Kaufmann; 1st Edition, May 25, 2016.

REFERENCE BOOKS:

1. Matt Richardson & Shawn Wallace “Getting Started with Raspberry Pi”, O’Reilly (SPD), 2014, ISBN: 9789350239759.
2. Waltenegus Dargie, Christian Poellabauer, “Fundamentals of Wireless Sensor Networks: Theory and Practice”.
3. Beginning Sensor networks with Arduino and Raspberry Pi – Charles Bell, A press, and 2013.



Source: https://www.electronicshub.org/wireless-sensor-networks-wsn/

22CS955 WIRELESS SENSOR NETWORKS

Hours Per Week :

L	T	P	C
3	0	2	4

PREREQUISITE KNOWLEDGE: Computer networks.

COURSE DESCRIPTION AND OBJECTIVES:

In this course we will provide an introduction to Wireless Sensor Networks (WSN) and cover latest topics in WSNs. The goal of this course is to give an overview of fundamental problems in the area of WSNs. We will discuss existing solutions for some of these problems. Data aggregation, information dissemination, security issues, power management, localization, topology control, routing, and security, are the topics will be covered in this course.

MODULE-1

UNIT-1 **10L+0T+6P=16 Hours**

INTRODUCTION

Introduction to Wireless Sensor Networks: Background of Sensor Network, Motivations, Performance metrics, Design factors, Sensor node hardware's and software's.

WSN Architecture: Traditional layered stack, roles and challenges, Enabling technologies in WSN, Applications of WSN, Physical layer and transceiver design considerations in WSNs.

UNIT-2 **14L+0T+10P=24 Hours**

MAC PROTOCOLS FOR WSN

Medium Access Control Protocols for WSN: Introduction, Fundamentals of MAC Protocols, Contention-Free, Contention-Based and Hybrid MAC Protocols, Data aggregation and fusion, Distributed data bases.

Localization: Global location (GPS-based) and relative location (Beacon-based). Localization methods: anchor-free, anchor-based, range-free, range-based. Clustering in WSN, Types of clustering.

PRACTICES:

- Implement different network topologies in WSN using NS2 Simulator/Arduino boards.
- Implement traffic signaling using Arduino boards.
- Establish communication between the two motes with Wi-Fi, XBee, modules on arduino and raspberry pi.
- Create cluster formation with m number of motes in WSN by using Arduino and raspberry pi.
- Collect the sensor geographical location information using Raspberry pi.

MODULE-2

UNIT-1 **10L+0T+6P=16 Hours**

ROUTING PROTOCOLS

Routing protocols for WSN: Introduction, Routing Challenges and Design Issues in WSN, Flooding and its variants, LEACH, Location-based protocols and energy-aware routing.

Transport Control Protocols for WSN: Feasibility of Using TCP or UDP for WSNs, TCP Design Issues, Existing TCPs in WSN: CODA, ESRT, RMST, PSFQ, GARUDA, ATP, Problems with TCP, Performance of TCP.

UNIT-2 **14L+0T+10P=24 Hours**

SECURITY

Security: Fundamentals, Security challenges in WSN, Security Attacks, Protocols and Mechanisms for Security, IEEE 802.15.4 and ZigBee Security, Sensor Network programming, Node-Centric Programming: nesC Language, TinyGALS, Sensor Network Simulators: Network Simulator Tools and Environments.

PRACTICES:

- Implement transmission between mobile nodes based on TCP and CBR traffic in WSN nodes using NS2 simulator.
- Implement a Low Energy Adaptive Hierarchy protocol using Simulation Tool.
- Implement different attack and its preventions in WSN using Arduino boards.

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Module No.	Mapping with POs
1	Design a wireless sensor network for given sensor data using microcontroller, transceiver, middle-ware and operating system.	Create	1	3
2	Evaluate the performance of schedule based and random Medium Access Control protocols for power consumption, fairness, channel utilization and control packet overhead.	Analyze	1	1, 2, 3
3	Evaluate the performance of low energy and geo-graphic routing protocols for power consumption, scalability and latency parameters.	Analyze	2	1, 2, 3
4	Implement solutions to real world problems using various sensors and arduino boards.	Create	2	3

TEXT BOOKS:

1. Dargie, Waltenegus, and Christian Poellabauer. Fundamentals of wireless sensor networks: theory and practice. John Wiley & Sons, 2010.
2. Kazem Sohraby, Daniel Minoli and Taieb Znati, “Wireless Sensor Networks Technology, Protocols, and Applications”, John Wiley & Sons, 2007.

REFERENCE BOOKS:

1. Holger Karl and Andreas Willig, “Protocols and Architectures for Wireless Sensor Networks”, John Wiley & Sons, Ltd, 2005.
2. Zhao and L. Guibas, “Wireless Sensor Networks”, Morgan Kaufmann, San Francisco, 2004.
3. C. S. Raghavendra, K.M.Shivalingam and T.Znati, “Wireless Sensor Networks”, Springer, New York, 2004.
4. Anna Hac, “Wireless Sensor Network Designs”, John Wiley & Sons, 2004.

SKILLS:

- ✓ *To Know the fundamentals of wireless sensor networks and its application to critical real time scenarios.*
- ✓ *To study the various protocols at various layers and its differences with traditional protocols.*
- ✓ *To know the issues pertaining to sensor networks and the challenges involved in managing a sensor network.*